

Seminar on Algebraic Geometry 3
Exposé VI: Generalities on Algebraic Groups and Group Schemes
English Translation, Native Diagrams, and Reference-v2 Successor

SGA 3 English reconstruction project

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Reconstruction note

The corrected Polo–Gille Exposé-VI PDFs control the French text, numbering, formulas, notes, and diagram appearance. Because no editable Polo–Gille source was publicly available when this reconstruction began, formulas were reconstructed in $\text{\LaTeX}$ during Loop 1. All 58 source-bound diagrams are now faithful native $\text{\LaTeX}$ reconstructions. This standalone Exposé-VI successor also adds the convention-v2 reference layer; its closure is asserted only by the accompanying PASS controls. It does not claim completion of SGA 3 as a whole.

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Exposé VI_A

Generalities on Algebraic Groups

by P. Gabriel

⁰Throughout this chapter, A will denote an Artinian local ring with residue field k . A group scheme G over $\mathrm{Spec} A$ will be called simply an A -group. This A -group is said to be *locally of finite type* if its underlying scheme is locally of finite type over A ; it is said to be *algebraic* if its underlying scheme is of finite type over A .

0. Preliminary Remarks

0.1. Let us first consider a group scheme G over an arbitrary scheme S . We call the structural morphism

$$\mu : G \times_S G \longrightarrow G$$

multiplication, and we call the morphism $c : G \rightarrow G$, defined by $c(T)(x) = x^{-1}$ for every scheme T over S and every $x \in G(T)$, *inversion*. If U and V are subsets of the underlying set of G , we denote by $U \cdot V$ the image under multiplication of the subset of $G \times_S G$ consisting of those points whose first projection belongs to U and whose second projection belongs to V . Likewise, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and let

$$\sigma : G \times_S G \longrightarrow G \times_S G$$

be the morphism with components pr_1 and μ . For every S -scheme T , the map $\sigma(T)$ is

$$(x, y) \longmapsto (x, xy);$$

it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S , pr_2 , and hence μ , are flat morphisms; when G is smooth over S , pr_2 , and hence μ , are smooth morphisms, and so on.

0.2. From now on we assume that S is the spectrum of an Artinian local ring A with residue field k . We denote by $(\mathbf{Sch}/k)_{\mathrm{red}}$ the category of reduced schemes over k . For every scheme X over A , the reduced scheme X_{red} is an object of $(\mathbf{Sch}/k)_{\mathrm{red}}$, and the functor

$$X \longmapsto X_{\mathrm{red}}$$

is right adjoint to the inclusion of $(\mathbf{Sch}/k)_{\mathrm{red}}$ into $\mathbf{Sch}/A$. It follows that, for every A -group G , G_{red} is a group in the category $(\mathbf{Sch}/k)_{\mathrm{red}}$.¹ In other words, for every reduced k -scheme

⁰Editor's note: version of October 13, 2024.

¹Editor's note: the sentences that follow have been added.

T , $G_{\text{red}}(T)$ is endowed, functorially in T , with a group structure. One must bear in mind that G_{red} is not necessarily a k -group, because its “multiplication” is only a morphism

$$(G_{\text{red}} \times_k G_{\text{red}})_{\text{red}} \longrightarrow G_{\text{red}}^2.$$

³ However, if k is a perfect field, the inclusion of $(\mathbf{Sch}/k)_{\text{red}}$ into $\mathbf{Sch}/k$ commutes with products. Groups in $(\mathbf{Sch}/k)_{\text{red}}$ may therefore be identified with group schemes over k whose underlying schemes are reduced. In this case, if G is a k -group, G_{red} is a group subscheme of G ; in general, this subgroup is not normal in G .

For example, if k is a field of characteristic 3, the constant group $(\mathbb{Z}/2\mathbb{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbb{Z}/3\mathbb{Z})$ (cf. Exposé I, 4.1 and 4.4). If G denotes the semidirect product of $D_k(\mathbb{Z}/3\mathbb{Z})$ by $(\mathbb{Z}/2\mathbb{Z})_k$ defined by this action, then G_{red} is identified with $(\mathbb{Z}/2\mathbb{Z})_k$ and is not normal in G .⁴

Let k be an arbitrary field, let $k^{p^{-\infty}}$ be its perfect closure, and let H be a group in $(\mathbf{Sch}/k)_{\text{red}}$. Then

$$(H \otimes_k k^{p^{-\infty}})_{\text{red}}$$

is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ have the same underlying topological space, groups in $(\mathbf{Sch}/k)_{\text{red}}$ share with k -groups certain topological properties that are invariant under extension of the base field. For example, it will follow from 0.3 and the preceding remarks that every group in $(\mathbf{Sch}/k)_{\text{red}}$ is separated.

In what follows, we shall encounter groups in $(\mathbf{Sch}/k)_{\text{red}}$ whenever we deal with a nonempty locally closed subset U of an A -group G such that

$$U \cdot U = U \quad \text{and} \quad U^{-1} = U.$$

Indeed, the reduced subscheme of G defined by U is then a group in $(\mathbf{Sch}/k)_{\text{red}}$.

0.3. *An A -group G is always separated, because the unit section $e : \text{Spec } A \rightarrow G$ is a closed immersion.* Indeed, let x be the unique point of $\text{Spec } A$, and let $\eta : G \rightarrow \text{Spec } A$ be the structural morphism. Since $\eta \circ e = \text{id}_{\text{Spec } A}$, for every affine open $U = \text{Spec } B$ of G containing $e(x)$, the homomorphism $B \rightarrow A$ has a section and is therefore surjective. It follows that e is a closed immersion.⁵

The diagonal in $G \times_A G$ is identified with the functor from $(\mathbf{Sch}/A)^\circ$ to **Sets** which associates to every A -scheme S the inverse image of the identity element of $G(S)$ under the map

$$\varphi(S) : G(S) \times G(S) \longrightarrow G(S), \quad (x, y) \longmapsto x \cdot y^{-1}.$$

We therefore have the following cartesian square. Since the diagonal morphism is obtained from a closed immersion by base change, it is itself a closed immersion.

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\varphi} & G \\ \text{diagonal morphism} \uparrow & & \uparrow \text{unit section} \\ G & \longrightarrow & \text{Spec } A \end{array}$$

²Editor’s note: for examples of group schemes G over a nonperfect field k for which G_{red} is not a group scheme over k , see 1.3.2 below.

³Editor’s note: the remainder of 0.2, as well as 0.3, has been slightly modified.

⁴Editor’s note: for an analogous example with G connected, one may take, when $\text{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbb{G}_{m,k}$.

⁵Editor’s note: this argument is valid for every local ring A of dimension zero, and shows that, if S is a discrete scheme, every S -group is separated; cf. VI_B, 5.2. On the other hand (cf. VI_B, 5.6.4), if S contains a nonisolated closed point s , the S -scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S , although it carries an S -group structure.

0.4.⁶ Let G be an A -scheme. We shall say that a point g of G is *strictly rational over A* if there exists an A -morphism

$$s : \operatorname{Spec} A \longrightarrow G$$

which sends the unique point of $\operatorname{Spec} A$ to g ; that is, if the homomorphism $A \rightarrow \mathcal{O}_{G,g}$ admits a retraction. One then has $\kappa(g) = k$, so g is a closed point of G . Indeed, if B is the ring of an affine open neighborhood of g , and if $\mathfrak{p}$ is the prime ideal of B corresponding to g , then $B/\mathfrak{p} \subset k$ is a finite A -algebra, hence an integral Artinian ring, and therefore a field.

Suppose henceforth that G is an A -group. Such a morphism $s : \operatorname{Spec} A \rightarrow G$ defines an automorphism r_s of the scheme G over A , called *right translation by s* . For every morphism $\pi : S \rightarrow \operatorname{Spec} A$, the automorphism $r_s(\pi)$ of $G(S)$ is defined by

$$x \longmapsto x \cdot G(\pi)(s), \quad x \in G(S).$$

Similarly, ℓ_s denotes *left translation by s* , namely the automorphism of G defined by

$$\ell_s(\pi)(x) = G(\pi)(s) \cdot x, \quad x \in G(S).$$

The schemes $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k -group, and $s \otimes_A k$ depends only on g , not on s . Consequently, the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s \otimes k}$ and $\ell_{s \otimes k}$) depend only on g , not on s . Thus, if P is a subset of $\underline{G}$, we may write

$$r_g(P) \text{ or } P \cdot g \quad (\text{respectively, } \ell_g(P) \text{ or } g \cdot P)$$

instead of $r_s(P)$ (respectively, $\ell_s(P)$), in agreement with 0.1.

Remark 0.4.1. If g is a strictly rational point of G , and if $A \rightarrow A'$ is a morphism of Artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g , and g' is strictly rational over A' . Moreover, if P' denotes the inverse image of P in G' , then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).

Proposition 0.5.⁷ Let G be an A -group, and let U, V be two dense open subsets of G . Then $U \cdot V$, that is, the image of $U \times_A V$ under multiplication, is the entire underlying space of G .

Proof. Since G and $G \otimes_A k$ have the same underlying space, after replacing A by k and G by $G \otimes_A k$, if necessary, we may assume that $A = k$. Let $g \in G$, and set $K = \kappa(g)$. Then left translation ℓ_g is an automorphism of G_K . Since the projection $G_K \rightarrow G$ is open, U_K and V_K are dense open subsets of G_K , as is the image of V_K under ℓ_g . Consequently there exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K .

Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L -points of G_L induced by g and v . Then

$$g_L \cdot v_L = u'$$

is a point of G_L above u , and therefore

$$g_L = u' \cdot v_L^{-1}$$

lies above $U \cdot V$. Thus $g \in U \cdot V$, proving the proposition. $\square$

⁶Editor's note: the original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

⁷Editor's note: the original stated this result under the hypothesis that G be locally of finite type over A . Since the result is useful in the general case, and since the proof is essentially the same, it has been stated and proved here in full generality. It will be used several times below.

Corollary 0.5.1.⁸ *If G is an irreducible A -group, then G is quasi-compact.*

Proof. Let U be a nonempty affine open subset of G . Then U is dense in G , so by 0.5 the morphism

$$\mu : U \times_A U \longrightarrow G$$

is surjective. Hence G is quasi-compact, since $U \times_A U$ is. $\square$

Corollary 0.5.2.⁹ *Let G be an A -group scheme, and let H be a group subscheme of G over A . Then H is closed.*

Proof. Let k' be the perfect closure of the residue field k of A . Since the underlying topological spaces of G and H are unchanged under the base change $A \rightarrow k \rightarrow k'$, we may assume that $A = k$ is a perfect field. We may then assume that G and H are reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H . Then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. The morphisms

$$H \longrightarrow \operatorname{Spec} k \quad \text{and} \quad \overline{H} \longrightarrow \operatorname{Spec} k$$

are universally open. Since H is dense in $\overline{H}$, $H \times H$ is dense in $\overline{H} \times H$, and $\overline{H} \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $H \times H$ is dense in $\overline{H} \times \overline{H}$. Consequently

$$\mu(\overline{H} \times \overline{H}) \subset \overline{H}.$$

Since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism

$$\mu' : \overline{H} \times \overline{H} \longrightarrow \overline{H}.$$

Now let $g \in \overline{H}$, and put $K = \kappa(g)$. The projection $\overline{H}_K \rightarrow \overline{H}$ is open, so H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$. Hence there exist $u, v \in H_K$ such that $\ell_g(v) = u$. As in the proof of 0.5, it follows that g belongs to $H \cdot H = H$. Therefore $\overline{H} = H$. $\square$

1. Local properties of an A -group locally of finite type

We shall first see that, if G is locally of finite type and flat over A , one can “make every closed point of G strictly rational” by means of a finite flat extension of the base.

1.1. Unless expressly stated otherwise, from now on we assume that G is an A -group locally of finite type. When A is a field k , we shall obtain in Exposé VII_B very precise results on the local rings of G .¹⁰ Here we content ourselves with a few elementary results.

Proposition 1.1.1. *Let x be a point of an A -group G locally of finite type and flat over A . Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there is a system of parameters $a_1, \dots, a_n$ of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is a finite flat A -module (hence finite free).*

We first suppose that A is equal to its residue field k . It then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and we may restrict to the case in which x is a closed point (cf. EGA 0_{IV}, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By SGA 1, I, § 9, this is equivalent to saying that, for every finite extension K of the base field k and every point $\overline{y}$ of $\overline{G} = G \otimes_k K$ lying above y , the local

⁸Editor’s note: this corollary has been inserted here; cf. 2.4 and 2.6.2 below.

⁹Editor’s note: this corollary, used implicitly in VIII, 6.7, has been added; see also VI_B, 6.2.5.

¹⁰Editor’s note: for example, they are always complete intersections; see Exposé VII_B, 5.5.1. Moreover, if $\operatorname{char}(k) = 0$, then G is smooth (Exposé VI_B, 1.6.1; see also Exposé VII_B, 3.3.1).

ring $\mathcal{O}_{\bar{G}, \bar{y}}$ is Cohen–Macaulay. If K has been chosen sufficiently large, that is, if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\bar{y}$ is (strictly) rational over K , as is every point $\bar{x}$ of $\bar{G}$ lying above x .¹¹ Since the automorphism

$$r_{\bar{x}} \circ (r_{\bar{y}})^{-1}$$

sends $\bar{y}$ to $\bar{x}$, it follows that $\mathcal{O}_{\bar{G}, \bar{x}}$, and hence $\mathcal{O}_{G, x}$ (SGA 1, I, § 9), are Cohen–Macaulay.

Now let A again be arbitrary. The preceding argument applies to $k \otimes_A G$, so that $k \otimes_A \mathcal{O}_{G, x}$ is Cohen–Macaulay. If $a_1, \dots, a_n$ is a sequence of elements of $\mathcal{O}_{G, x}$ whose image in $k \otimes_A \mathcal{O}_{G, x}$ is a system of parameters, it follows from SGA 1, IV, 5.7, or EGA 0_{IV}, 15.1.16, that $a_1, \dots, a_n$ is an $\mathcal{O}_{G, x}$ -regular sequence and that $\mathcal{O}_{G, x}/(a_1, \dots, a_n)$ is finite and flat (hence finite and free) over A .

Lemma 1.1.2. *Every nonempty scheme X , locally of finite type over an Artinian ring A , contains a closed point x whose local ring is Cohen–Macaulay.*

We may plainly suppose that X is affine, with algebra B , and argue by induction on $\dim X$ (the assertion is clear if X is discrete, since all its local rings are then Artinian). Since B is of finite type over A , if $\dim B > 0$, then B contains an element a that is not invertible and is not a zero divisor.¹² The closed subscheme

$$X' = \operatorname{Spec} B/(a)$$

of X then has dimension strictly less than $\dim X$, and by the induction hypothesis it contains a closed point x such that $\mathcal{O}_{X', x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X', x} = \mathcal{O}_{X, x}/(a)$, and a is neither invertible nor a zero divisor in $\mathcal{O}_{X, x}$, the ring $\mathcal{O}_{X, x}$ is Cohen–Macaulay (see also EGA IV₂, 6.11.3).

Proposition 1.2. *Let A be an Artinian local ring, let G be an A -group locally of finite type and flat over A , and let x be a closed point of G . Then there exists a local A -algebra A' , finite and free over A , such that every point of $G \otimes_A A'$ lying above x is strictly rational over A' .*

¹³ Indeed, let k_1 be a finite normal extension of k containing the residue field $\kappa(x)$ of x . By Lemma V.4.1.2, there exists a local A -algebra A_1 , finite and free over A , with residue field k_1 . In this case (cf. Editor’s note 11), all the points $g_1, \dots, g_n$ of $G \otimes_A A_1$ lying above $x \in G$ have k_1 as residue field; that is, $g_1, \dots, g_n$ are rational over A_1 in the sense of Exposé V, § 4.e).

Let $B_1, \dots, B_n$ be the local rings of $g_1, \dots, g_n$. By 1.1.1, the rings $B_1, \dots, B_n$ have quotients $B'_1, \dots, B'_n$ that are Artinian and finite and free over A_1 . Set

$$A' = B'_1 \otimes_{A_1} \cdots \otimes_{A_1} B'_n.$$

Then A' is local, finite, and free over A_1 , and, for each $i = 1, \dots, n$, there are surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

¹¹Editor’s note: indeed, the hypothesis on K implies that, for every extension L of K , every k -morphism $\kappa(x) \rightarrow L$ (respectively, $\kappa(y) \rightarrow L$) factors through K . Consequently, all the points of $G \otimes_k K$ lying above x or y have K as residue field; that is, they are (strictly) rational over K .

¹²Editor’s note: indeed, B is a Noetherian Jacobson ring (cf. [BAC], V, § 3.4). If every noninvertible element is a zero divisor, then every prime ideal is an associated prime ideal of B ; in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B . It follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B , and hence that $\dim B = 0$.

¹³Editor’s note: the original has been expanded in what follows.

the second induced by the multiplication map

$$B'_i \otimes_{A_1} B'_i \rightarrow B'_i.$$

Consequently, A' has the required property.

1.3. Let e be the unit element (or origin) of G , that is, the image of the unique point of $\mathrm{Spec} A$ under the unit section $\mathrm{Spec} A \rightarrow G$. By definition, e is strictly rational over A .

Proposition 1.3.1.¹⁴ *Let G be a group locally of finite type and flat over an Artinian ring A , and let K (respectively, $\bar{k}$) be the perfect closure (respectively, an algebraic closure) of the residue field k of A .*

- (1) *For every closed point x of $\bar{G} = G \otimes_k \bar{k}$, the local rings $\mathcal{O}_{\bar{G},e}$ and $\mathcal{O}_{\bar{G},x}$ are isomorphic. In particular, the tangent spaces $T_e \bar{G}$ and $T_x \bar{G}$ have the same dimension.*
- (2) *The following assertions are equivalent:*
 - (i) *$G \otimes_A K$ is reduced.*
 - (i bis) *$\mathcal{O}_{G,e} \otimes_A K$ is reduced.*
 - (ii) *G is smooth over A .*
 - (ii bis) *G is smooth over A at the origin.*

Proof. (1) Let x be a closed point of $\bar{G}$. There is exactly one $\bar{k}$ -morphism

$$s : \mathrm{Spec} \bar{k} \rightarrow \bar{G}$$

whose image is x . Right translation r_s then induces an isomorphism from

$$\mathcal{O}_{\bar{G},e} = \mathcal{O}_{G,e} \otimes_k \bar{k}$$

onto $\mathcal{O}_{\bar{G},x}$, which proves (1).

We now prove (2). By SGA 1, II.2.1, we reduce at once to the case in which A is a field ($A = k$). The implications

$$(i) \Rightarrow (i \text{ bis}), \quad (ii) \Rightarrow (ii \text{ bis}), \quad (ii) \Rightarrow (i), \quad (ii \text{ bis}) \Rightarrow (i \text{ bis})$$

are evident, so it remains only to prove that (i bis) implies (ii).

Now (i bis) implies that $\mathcal{O}_{G,e} \otimes_k \bar{k}$ is reduced. Hence, by (1), $\mathcal{O}_{\bar{G},x}$ is reduced for every closed point x of $\bar{G}$, and therefore $\bar{G}$ is reduced. Since $\bar{G}$ is locally of finite type over $\bar{k}$, there is at least one closed point y such that $\mathcal{O}_{\bar{G},y}$ is regular. By (1), the local rings at all closed points of $\bar{G}$ are isomorphic to $\mathcal{O}_{\bar{G},e}$; thus all these local rings are regular. Consequently, $\bar{G}$ is smooth over $\bar{k}$, and hence G is smooth over k .

¹⁵ We may now give the examples below, pointed out by M. Raynaud, of group schemes G over a nonperfect field k for which G_{red} is not a k -group scheme.

Examples 1.3.2. Let k be a nonperfect field of characteristic $p > 0$, let $t \in k - k^p$, let $\bar{k}$ be an algebraic closure of k , and let $\alpha \in \bar{k}$ satisfy $\alpha^p = t$.

¹⁴Editor's note: part (1), which is useful in Examples 1.3.2 below, has been added.

¹⁵Editor's note: the following sentence and Examples 1.3.2 have been added.

- (1) Consider the additive group $\mathbb{G}_{a,k} = \operatorname{Spec} k[X]$, and let G be the group subscheme, finite over k , defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\text{red}} = \operatorname{Spec} k[X]/X(X^{p(p-1)} - t)$$

is étale at the origin. If it were a group scheme over k , it would be smooth (by 1.3.1); but G_{red} is not geometrically reduced, so it is not a group scheme over k .

- (2) Consider

$$\mathbb{G}_{a,k}^4 = \operatorname{Spec} k[X, Y, U, V],$$

and let G be the group subscheme defined by the ideal I generated by the additive polynomials

$$P = X^p - tY^p \quad \text{and} \quad Q = U^p - tV^p.$$

Then G is two-dimensional and irreducible, since

$$(G_{\bar{k}})_{\text{red}} \simeq \operatorname{Spec} \bar{k}[Y, V]$$

is irreducible.

Let $A = k[X, Y, U, V]$, and let $\mathfrak{m}$ be its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in

$$\Omega_{A/k}^1 \otimes_A (A/\mathfrak{m}),$$

viewed as linear forms on the tangent space

$$k^4 = T_0 \mathbb{G}_{a,k}^4.$$

We shall show that the subspace $E = T_0 G_{\text{red}}$ equals k^4 . Otherwise, there would be a linear form

$$f = ax + by + a'u + b'v, \quad a, b, a', b' \in k,$$

not all of whose coefficients are zero, that vanishes on E . Recall that the formation of $\Omega_{A/k}^1$, and hence that of tangent spaces, commutes with base change (cf. EGA IV₄, 16.4.5), and identify f with its image in $(\bar{k}^4)^*$. Since

$$(G_{\bar{k}})_{\text{red}} \subset (G_{\text{red}})_{\bar{k}},$$

the form f vanishes on the subspace $T_0(G_{\bar{k}})_{\text{red}}$ of $\bar{k}^4$, which is defined by the equations

$$g_1 = x - \alpha y \quad \text{and} \quad g_2 = u - \alpha v.$$

Thus

$$f = \lambda g_1 + \mu g_2, \quad \lambda, \mu \in \bar{k}.$$

But $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$. This contradiction shows that $E = k^4$, and hence

$$T_0(G_{\text{red}})_{\bar{k}} = \bar{k}^4.$$

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, since

$$R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p).$$

Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\text{red}})_{\bar{k}}$ is contained in the hyperplane H of $\bar{k}^4$ with equation

$$\alpha dV + dX - dU - \alpha dY = 0,$$

and therefore has dimension at most 3.¹⁶ It follows from part (1) of 1.3.1 that G_{red} is not a group scheme over k .

¹⁶Editor's note: in fact, it is easy to see that $\sqrt{I}$ is generated by P, Q, R at every point other than 0, so $F = H$.

2. Connected components of an A -group locally of finite type

2.1. Let us first consider an arbitrary A -group G , and let G' be the connected component of the origin e in G . This connected component is plainly closed, so we may identify it with the reduced closed subscheme of G having G' as its underlying space.¹⁷

Proposition 2.1.1. *For every extension K of the residue field k of A , the underlying space of $G' \otimes_A K$ is the connected component of the origin in the K -group $G \otimes_A K$; in other words, G' is geometrically connected.*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the unit element of G , that image is contained in G' . Thus $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from the connectedness of $G' \otimes_A K$, which is proved in the following lemma.

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k . If X contains a rational point, then $X \times_k Y$ is connected.*

We give a direct proof below of this result from EGA IV₂ (4.5.8 and 4.5.14).

Suppose first that Y is nonempty, connected, and affine, with algebra B . Then $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$ -algebra

$$\mathcal{B} = \mathcal{O}_X \otimes_k B.$$

We wish to show that every nonempty subset U of $X \times_k Y$ that is both open and closed equals $X \times_k Y$. Now U is affine over X , and its affine $\mathcal{O}_X$ -algebra is a direct factor of $\mathcal{B}$. It follows from Lemma 2.1.3 below that the image of U in X is open and closed, and hence is all of X . In particular, this image contains a rational point x of X , so U meets the inverse image of x in $X \times_k Y$. Since this inverse image is isomorphic to Y , and hence connected, U contains the whole inverse image. The same conclusion would hold for the complement of U in $X \times_k Y$ if U were different from $X \times_k Y$, which is absurd.

If Y is now an arbitrary k -scheme, the preceding argument shows that the fibers of the canonical projection

$$X \times_k Y \longrightarrow Y$$

are connected. If x is a rational point of X , all these fibers meet the subscheme $\{x\} \times_k Y$, which is itself connected. This proves the assertion.

Lemma 2.1.3. *Let X be a scheme, and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$ -algebra that is locally¹⁸ a direct factor of a free $\mathcal{O}_X$ -module. Then the image of $\text{Spec } \mathcal{A}$ in X is open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, then $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$ -module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of $\text{Spec } \mathcal{A}$ at x , which is affine with algebra

$$\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

is nonempty. Thus the image of $\text{Spec } \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s , and hence $\mathcal{A}$, vanish in a neighborhood of x . Therefore the support of $\mathcal{A}$ is closed. On the other hand, we may

¹⁷Editor's note: we shall see below (2.2) that G' is a group in the category $(\mathbf{Sch}/k)_{\text{red}}$.

¹⁸Editor's note: the word "locally" has been added. The reference to Kaplansky is *Projective modules*, Ann. of Math. **68** (1958), 372–377; see also [BAC], § II.3, Ex. 3.

assume that $\mathcal{A}$ is a direct factor of the direct sum $\mathcal{L}$ of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.¹⁹ For each β , let φ^β denote the restriction to $\mathcal{A}$ of the canonical projection

$$\mathcal{L} = \bigoplus_{\alpha} \mathcal{O}_X^\alpha \longrightarrow \mathcal{O}_X^\beta.$$

For $x \in X$, let $\mathcal{A}_x^*$ denote the dual module

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}).$$

Since $\mathcal{A}_x$ is a direct factor of $\mathcal{L}_x$, the restriction map

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathcal{O}_{X,x}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, then, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\varphi \in \mathcal{A}_x^*$ such that $\varphi(a) = 1$. There is therefore a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$,

$$\varphi(u) = \sum_{\alpha} \xi^\alpha \varphi^\alpha(u);$$

this sum is finite because $\varphi^\alpha(u) = 0$ except for finitely many α . Applying this to $u = a$, we find $\alpha_1, \dots, \alpha_n$ such that

$$1 = \varphi(a) = \xi^{\alpha_1} \varphi^{\alpha_1}(a) + \dots + \xi^{\alpha_n} \varphi^{\alpha_n}(a).$$

There is then an open neighborhood U of x on which a and the ξ^{α_i} come from sections

$$\tilde{a} \in \mathcal{A}(U) \quad \text{and} \quad \tilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U).$$

The equality

$$\sum_i \tilde{\xi}^{\alpha_i} \varphi^{\alpha_i}(\tilde{a}) = 1 \quad \text{on } U$$

shows that $\tilde{a}_y \neq 0$ for every $y \in U$. (“The support of a projective module is open.”)

2.2. The notation remains that of 2.1. It is clear that G' is a reduced k -scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\mathrm{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism

$$\mu : G \times_A G \longrightarrow G$$

induces a morphism

$$\mu' : (G' \times_k G')_{\mathrm{red}} \longrightarrow G'$$

that makes G' a group in $(\mathbf{Sch}/k)_{\mathrm{red}}$.

2.2.bis.²⁰ Recall (cf. Exposé V) that, if P is a scheme, $\underline{P}$ denotes the underlying topological space of P . We define an A -subfunctor G^0 of G by setting, for every A -scheme S ,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \rightarrow G$ be the inversion morphism. Since $c(\underline{G'}) = \underline{G'}$, we have $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, then $u \boxtimes v$ sends $\underline{S}$ into the subspace of $\underline{G \times_A G}$ consisting of the points whose two projections belong to $\underline{G'}$. This subspace is identified with the underlying space of $G' \times_A G'$, which is connected by Lemma 2.1.2.

¹⁹Editor’s note: the original has been expanded in what follows.

²⁰Editor’s note: this paragraph has been added in order to make the connection with Exposé VI_B, § 3.

Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into $\underline{G}'$. This proves that G^0 is a group subfunctor of G over A .

If the connected component of e is open in G , then the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a group subscheme of G ; this subscheme will also be denoted by G^0 . In this case, with the notation of 2.1,

$$G' = (G^0)_{\text{red}},$$

and the topological spaces $\underline{G}'$ and $\underline{G}^0$ coincide.²¹

2.3. In accordance with our convention in 1.1, we again assume from now on that G is locally of finite type over A . Then G is locally Noetherian, hence locally connected;²² thus every connected component of G is open.

We shall write G^0 for the subscheme induced by G on the connected component G' of the unit element. By 2.2.bis, G^0 is a group subscheme of G , which we shall call the *neutral component* of G . For every A -scheme S , therefore,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G}^0 = \underline{G}'\}.$$

Let G^α be an arbitrary connected component of G , and let

$$\nu^\alpha : G^\alpha \times_A G^0 \longrightarrow G$$

be the morphism defined by

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\mathbf{Sch}/_A)$, $g \in G^\alpha(S)$, and $\gamma \in G^0(S)$. If e is the origin of G , the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the trivial morphism. Since $G^\alpha \times_A G^0$ is connected by 2.1.2, it follows that ν^α factors through G^0 . Thus, for every A -scheme S , $G^0(S)$ is a normal subgroup of $G(S)$. We have proved the following proposition.²³

Proposition 2.3.1. *Let G be an A -group locally of finite type. Then the neutral component G^0 is an open normal group subscheme of G .*

Proposition 2.4. *Let G be an A -group locally of finite type.*

- (i) G^0 is irreducible, and $G^0 \otimes_A k$ is geometrically irreducible over k .
- (ii) G^0 is quasi-compact, and hence is of finite type over A .

²⁴ *Proof.* (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to prove the second assertion. Let $\bar{k}$ be an algebraic closure of k . By 2.2, $(G \otimes_A \bar{k})_{\text{red}}$ is a reduced $\bar{k}$ -group locally of finite type, and hence is smooth over $\bar{k}$ by 1.3.1. A fortiori, the local rings of $(G \otimes_A \bar{k})_{\text{red}}$ are integral domains. Therefore,²⁵ since $G \otimes_A \bar{k}$ is locally Noetherian, the connected components of $G \otimes_A \bar{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \bar{k}$ (cf. 2.1.1) is irreducible.

²¹Editor's note: in this exposé, the notation G^0 is reserved for the case in which the connected component of e is open; in Exposé VI_B, § 3, this connected component will be denoted by $\underline{G}^0$ in every case. This is a slight abuse of notation, but it is compatible with the preceding discussion when the connected component is the underlying topological space of an open group subscheme G^0 of G .

²²Editor's note: indeed, in a Noetherian space there are only finitely many connected components, and each is therefore open; see also EGA I, 6.1.9.

²³Editor's note: the number 2.3.1 has been added in order to emphasize this statement. Moreover, G^0 is even a characteristic subgroup of G ; see 2.6.5(ii).

²⁴Editor's note: in the statement, " G^0 is geometrically irreducible" has been replaced by " $G^0 \otimes_A k$ is geometrically irreducible over k ," and the proof has been expanded.

²⁵Editor's note: the original has been expanded here, and the reference to EGA I, 6.1.10, has been added.

(ii) We now show that G^0 is of finite type over A . Since G^0 is locally of finite type over A , it suffices to prove that G^0 is quasi-compact. Since G^0 is irreducible, this follows from 0.5.1.

Corollary 2.4.1. *Every connected component of G is irreducible,²⁶ of finite type over A ,²⁷ and of the same dimension as G^0 .*

Indeed, we may assume that A equals its residue field k . Let C be a connected component of G , let x be a closed point of C , let $\kappa(x)$ be the residue field of x , and let k' be a finite normal extension of k containing $\kappa(x)$. The canonical projection

$$\pi : C \otimes_k k' \longrightarrow C$$

is open and closed.²⁸ Consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \rightarrow C$ is surjective. Thus C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2); it follows that C' is the image of $G^0 \otimes_k k'$ under right translation r_y . Now $G^0 \otimes_k k'$ is of finite type over k' , by 2.4, and $\pi^{-1}(x)$ is finite, of cardinality at most $[k' : k]$. Consequently, $C \otimes_k k'$ is of finite type over k' , and hence C is of finite type over k .

Moreover, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same is true of C' ,²⁹ and hence also of C , because the projection $C' \rightarrow C$ is surjective.

Finally, we saw above that $C \otimes_k k'$ is the disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), it follows that C has the same dimension as G^0 . (Moreover, by EGA IV₂, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. This paragraph has been added. The results that follow appear in Exposé VI_B, but could (or should) have appeared in VI_A; it is useful to have them available already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an Artinian local ring, and let $k = A/\mathfrak{m}$ be its residue field.*

- (i) *If X is an A -scheme such that $X \otimes_A k$ is locally of finite type (respectively, of finite type) over k , then X has the same property over A .*
- (ii) *Let $u : X \rightarrow Y$ be a morphism of A -schemes. If $u \otimes_A k$ is an immersion (respectively, a closed immersion), then so is u .*

Proof. (i) Suppose that $X \otimes_A k$ is locally of finite type over k . Let $U = \operatorname{Spec} B$ be an affine open subset of X . By hypothesis, there are elements $x_1, \dots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k -algebra, and the “nilpotent Nakayama lemma” then shows that the x_i generate B as an A -algebra. This proves that X is locally of finite type over A . If, in addition, $X \otimes_A k$ is quasi-compact, then so is X , which has the same underlying topological space; hence X is of finite type over A . This proves (i).

We prove (ii). Suppose that $u \otimes_A k$ is an immersion (respectively, a closed immersion). Then u is a homeomorphism of X onto a locally closed (respectively, closed) subset of Y , and, for every $x \in X$, the ring homomorphism

$$\varphi_x : \mathcal{O}_{Y, u(x)} \longrightarrow \mathcal{O}_{X, x}$$

²⁶Editor’s note: one should bear in mind that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbb{R}$, the group $\mu_{3, \mathbb{R}}$, represented by $\mathbb{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \operatorname{Spec} \mathbb{R}$ and $C = \operatorname{Spec} \mathbb{R}[X]/(X^2 + X + 1)$; and $C \otimes_{\mathbb{R}} \mathbb{C}$ has two components.

²⁷Editor’s note: the assertion that follows has been added; cf. Exposé VI_B, 1.5.

²⁸Editor’s note: the original has been expanded in what follows.

²⁹Editor’s note: the original has been simplified here.

is such that $\varphi_x \otimes_A k$ is surjective. The nilpotent Nakayama lemma therefore shows that φ_x is surjective, so u is an immersion (respectively, a closed immersion).

Proposition 2.5.2. *Let A be an Artinian local ring with residue field k , and let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type.*

- (a) *The set $u(G)$ is closed in H , and its connected components are irreducible and all have the same dimension.*
- (b) *One has*

$$\dim G = \dim u(G) + \dim \operatorname{Ker}(u).$$

- (c) *If u is a monomorphism, then it is a closed immersion.*

Proof. By the preceding lemma, it suffices to prove the proposition when $A = k$. Moreover, the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, so we may assume that k is algebraically closed.

We prove (a). Let C be the reduced subscheme of H whose underlying topological space is $u(G)$. Since $u(G)$ is stable under the inversion morphism of H , so is C . Furthermore, $u : G \rightarrow C$ is quasi-compact and dominant. Thus, by EGA IV₂, 2.3.7, the same is true of

$$u \times_k \operatorname{id}_G \quad \text{and} \quad \operatorname{id}_H \times_k u,$$

and hence of their composite

$$u \times_k u : G \times_k G \longrightarrow C \times_k C.$$

Consequently, multiplication on H sends $C \times_k C$ into C , and therefore C is a group subscheme of H .

Replacing H by C , we are thus reduced to the case in which u is dominant. Then $G(k)$ is dense in H , and hence meets every connected component of H ; it therefore acts transitively on the set of those connected components. It is consequently enough to show that $u(G)$ contains H^0 . Replacing G by $u^{-1}(H^0)$, we may therefore suppose that $H = H^0$. In that case, 2.4 shows that H is irreducible and of finite type over k , and hence Noetherian. Moreover, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, and is therefore of finite type. By Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.5), $u(G)$ is consequently a constructible (and dense) subset of

$$H = \overline{u(G)},$$

and thus contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). By 0.5, we then have

$$H = U \cdot U \subset u(G),$$

whence $u(G) = H$. Together with 2.4.1, this proves (a).

We prove (b). Recall first that the functor $\operatorname{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H . Since u is of finite type, $\operatorname{Ker}(u)$ is of finite type over k . On the other hand, replacing H by the reduced closed subscheme $u(G)$, we may assume that u is surjective. Let u^0 denote the restriction of u to G^0 . Since G and $\operatorname{Ker}(u)$ are equidimensional, and since

$$\operatorname{Ker}(u)^0 \subset \operatorname{Ker}(u^0),$$

we are reduced to the case in which G , and hence also H , is irreducible.

By EGA IV₃, 9.2.6.2 and 10.6.1(ii), the set of $y \in H$ such that

$$\dim u^{-1}(y) = \dim G - \dim H$$

contains a nonempty open subset V . Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G , and therefore contains a closed point x of G , because G is a Jacobson scheme (cf. EGA IV₃, 10.4.8). Right translation r_x is an isomorphism from $\text{Ker}(u)$ onto $u^{-1}(u(x))$, whence

$$\dim \text{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

We prove (c), following [DG70], I, § 3.4. (Another proof is given in Exposé VI_B, 1.4.2.) Suppose that u is a monomorphism. If C is a connected component of G , there is a closed point $x \in G$ such that $C = r_x(G^0)$. If u_C (respectively, u^0) denotes the restriction of u to C (respectively, to G^0), then

$$u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}.$$

It therefore suffices to show that u^0 is a closed immersion. We may thus suppose that $G = G^0$, so that G is irreducible and of finite type over k .

Let ξ be the generic point of G . Then $\mathcal{O}_{G,\xi}$ is an Artinian local ring; denote its maximal ideal by $\mathfrak{m}$. Let $h = u(\xi)$, let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{H,h}$, and set

$$A = \mathcal{O}_{G,\xi} / \mathfrak{n} \mathcal{O}_{G,\xi}.$$

Since u is a monomorphism, so is the morphism

$$u_h : \text{Spec}(A) \longrightarrow \text{Spec}(\kappa(h))$$

obtained by base change. The multiplication morphism

$$A \otimes_{\kappa(h)} A \longrightarrow A$$

is therefore an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. Nakayama's lemma then shows that

$$\mathcal{O}_{H,h} \longrightarrow \mathcal{O}_{G,\xi}$$

is surjective, since $\mathfrak{n} \mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$ and is therefore nilpotent.

Let V be an affine open subset of H containing h , let $U \neq \emptyset$ be an affine open subset of G contained in $u^{-1}(V)$, and let

$$\varphi : \mathcal{O}_H(V) \longrightarrow \mathcal{O}_G(U)$$

be the homomorphism of k -algebras induced by u . Let $\mathfrak{p}$ be the prime ideal of $\mathcal{O}_G(U)$ corresponding to ξ ,* and set $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k , the k -algebra $\mathcal{O}_G(U)$ is generated by finitely many elements $a_1, \dots, a_n$. By the preceding argument, there are elements $b_1, \dots, b_n, s \in \mathcal{O}_H(V)$, with $s \notin \mathfrak{q}$, such that in $\mathcal{O}_G(U)_{\mathfrak{p}}$ one has

$$a_i/1 = \varphi(b_i)/\varphi(s).$$

There are therefore elements

$$t_1, \dots, t_n \in \mathcal{O}_G(U) - \mathfrak{p}$$

such that

$$t_i(a_i\varphi(s) - \varphi(b_i)) = 0.$$

Set

$$t = t_1 \cdots t_n \varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}.$$

*Translator's note: the French source prints η here; the generic point introduced immediately above is ξ .

The equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ then already hold in $\mathcal{O}_G(U)_t$. Since

$$t \in \mathcal{O}_G(U) = k[a_1, \dots, a_n],$$

there is an element $b \in \mathcal{O}_H(V)$ such that, for some $r \in \mathbb{N}$,

$$t/1 = \varphi(b)/\varphi(s)^r.$$

Consequently, φ induces a surjection

$$\mathcal{O}_H(V)_{sb} \longrightarrow \mathcal{O}_G(U)_t,$$

and hence u is a local immersion at ξ .

The open subset W of G consisting of the points at which u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y . To prove that $W = G$, it is enough to show that every closed point of G belongs to W . But every closed point x is the image of y under the translation $r_x \circ r_y^{-1}$, and hence belongs to W . Thus $W = G$, which proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H , respectively, such that $x \in U_x$ and u induces a closed immersion of U_x into V_x . Since U_x is dense in G , the subset $u(U_x)$ is dense in $u(G) \cap V_x$. But $u(U_x)$ is closed in V_x , so

$$u(U_x) = V_x.$$

Moreover, since u is injective, $U_x = u^{-1}(V_x)$. Thus u induces a closed immersion of G into the open subscheme of H covered by the V_x . Hence $u : G \rightarrow H$ is an immersion. We have already seen that $u(G)$ is closed in H , so u is a closed immersion.

Lemma 2.5.3.³⁰ *Let A be an Artinian local ring, let k be its residue field, let G be a flat A -group, and let X be an A -scheme equipped with a left action*

$$\mu : G \times_A X \longrightarrow X$$

of G and with a section $s_0 : \operatorname{Spec} A \rightarrow X$. (This is the case, for example, if G' is a second A -group and one is given a morphism of A -groups $G \rightarrow G'$.)

Let φ be the morphism

$$\mu \circ (\operatorname{id}_G \times s_0) : G = G \times_A A \longrightarrow X.$$

If φ is flat at a point g of G , then φ is flat.

Proof. Since G is flat over A , the fiberwise criterion for flatness (EGA IV₃, 11.3.10.2) shows that it is enough to prove that $\varphi \otimes_A k$ is flat. We may therefore suppose that $A = k$. In that case, giving s_0 is equivalent to giving a k -point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let $h \in G$; we show that φ is flat at h . Let K be an extension of k containing copies of both $\kappa(g)$ and $\kappa(h)$. We have a cartesian square

$$\begin{array}{ccc} G_K & \xrightarrow{\varphi_K} & X_K \\ \downarrow & & \downarrow \\ G & \xrightarrow{\varphi} & X \end{array}$$

³⁰Editor's note: no finiteness hypotheses are made in this statement; this will be useful later (cf. 6.2 and Exposé VI_B, § 12).

in which the two vertical arrows are faithfully flat. Thus, by Exposé V, 7.4(i), φ_K is flat at every point $g' \in G_K$ lying over g ; and, in order to show that φ is flat at h , it is enough to show that φ_K is flat at one point h' lying over h . We are therefore reduced to the case in which g and h are rational.

Set $u = hg^{-1}$, and let ℓ_u (respectively, μ_u) be left translation on G (respectively, on X) defined by u . Since $\varphi \circ \ell_u = \mu_u \circ \varphi$, we obtain a commutative square

$$\begin{array}{ccc} \mathcal{O}_{G,h} & \xrightarrow{\sim} & \mathcal{O}_{G,g} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,hx_0} & \xrightarrow{\sim} & \mathcal{O}_{X,gx_0} \end{array}$$

whose horizontal arrows are isomorphisms. Since the morphism

$$\mathcal{O}_{X,gx_0} \longrightarrow \mathcal{O}_{G,g}$$

is flat, the morphism

$$\mathcal{O}_{X,hx_0} \longrightarrow \mathcal{O}_{G,h}$$

is flat as well.

Proposition 2.5.4. *Let A be an Artinian local ring, let k be its residue field, let G be an A -group locally of finite type, and let $X \neq \emptyset$ be an A -scheme locally of finite type equipped with a left action of G . Suppose that the morphism*

$$\varphi : G \times_A X \longrightarrow X \times_A X, \quad (g, x) \longmapsto (gx, x),$$

defined on underlying sets by the displayed rule, is surjective. Then:

- (i) *The connected components of X are of finite type, irreducible, and all of the same dimension.*
- (ii) *More precisely, let $\bar{k}$ be an algebraic closure of k , and let x be a closed point of $X \otimes_A \bar{k}$. Its stabilizer F is a closed group subscheme of $G \otimes_A \bar{k}$, and the dimension of the irreducible components of X is*

$$\dim G - \dim F.$$

In view of Lemma 2.5.1, we may suppose that $A = k$. Suppose first that k is algebraically closed. Then G_{red} is a k -group locally of finite type; hence, replacing G by G_{red} and X by X_{red} , we may suppose that G and X are reduced.

Since $G \times_k X$ is locally of finite type over k , the morphism φ is locally of finite type (cf. EGA I, 6.6.6(v)), and is therefore locally of finite presentation because $X \times_k X$ is locally Noetherian. Let x be a rational point of X . The morphism

$$\varphi_x : G \longrightarrow X$$

obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X , then $\mathcal{O}_{X,\eta}$ is a field, since X is reduced; consequently, φ_x is flat at every point of G lying over η . By Lemma 2.5.3, φ_x is therefore flat. Thus $\varphi_x : G \rightarrow X$ is faithfully flat and locally of finite presentation, and hence open (cf. EGA IV₂, 2.4.6).

Since G^0 is open in G , irreducible, and quasi-compact (by 2.4), every orbit

$$G^0 x = \varphi_x(G^0),$$

as x ranges over the rational points of X , is an open, irreducible, and quasi-compact subset of X , and is therefore of finite type over k , since X is locally of finite type over k .

Every nonempty open subset of X contains a rational point, so these open subsets cover X . Furthermore, any two of them are either disjoint or equal. Indeed, if

$$\varphi_x(G^0) \cap \varphi_y(G^0)$$

is nonempty, it contains a rational point z . There are then rational points $g, h \in G^0$ such that

$$gx = z = hy,$$

whence $x = g^{-1}z$ and $y = h^{-1}z$, and consequently

$$\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0).$$

It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore at once the connected components and the irreducible components of X .

Finally, let x, y be two rational points of X . Since φ_x is surjective, there is a rational point $g \in G$ such that $y = gx$. Since G^0 is a normal subgroup of G , the orbit G^0y is the image of G^0x^* under translation ℓ_g on X ; hence G^0y and G^0x have the same dimension.

Moreover, by Exposé I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the following cartesian square:

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \varphi_x \\ \text{Spec } k & \longrightarrow & X \end{array}$$

Thus F is a k -group locally of finite type, while $F \cap G^0$ is a k -group of finite type containing F^0 . By 2.4.1, the schemes F and $F \cap G^0$ are equidimensional, of the same dimension as F^0 . Let

$$C = \varphi_x(G^0)$$

be the irreducible component of X containing x . Arguing as in the proof of part (b) of 2.5.2, we obtain

$$\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F.$$

In the general case, that is, when k is an arbitrary field, let $\bar{k}$ be an algebraic closure of k . Let C be a connected component of X , and let C' be a connected component of $C \otimes_k \bar{k}$. Then C' is a connected component of

$$X' = X \otimes_k \bar{k}.$$

The morphism $\pi : X' \rightarrow X$ is open (cf. EGA IV₂, 2.4.10), and, since it is integral, it is also closed; therefore $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, and hence is of finite type over k , because X is locally of finite type over k .

Finally, dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), so $\dim C = \dim C'$. Since all the irreducible components of X' have the same dimension, the same is true of those of X .

*Translator's note: the French source prints G^0 here; as the object being translated is an orbit in X , the intended source expression is G^0x .

2.6. Complements.— This paragraph has been added; it is taken from [Per75], II, §§ 1–2, with complements due to O. Gabber.³¹ It shows that the preceding results are valid for every group scheme G over a field k . (This will be used in Sections 5, 6, and 7 of Exposé VI_B.)

Fix a field k . We begin with the following lemma (loc. cit., II, 2.1.1), which does not appear explicitly in EGA IV₂, § 4.4 (although it can probably be read between the lines at the beginning of loc. cit., § 4.4.1).

Lemma 2.6.0. *Let X be an irreducible k -scheme, let K be an extension of k , and let X' be an irreducible component of X_K . The projection $X' \rightarrow X$ is surjective.*

Proof. Let B be a transcendence basis of K over k , and set $L = k(B) \subset K$. By EGA IV₂, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L . Hence $X' \rightarrow X_L$ is integral and dominant, and therefore surjective. Since $X_L \rightarrow X$ is surjective (loc. cit., 4.4.1), so is $X' \rightarrow X$.

For the rest of 2.6, fix a k -group scheme G and an action $\mu : G \times X \rightarrow X$ of G on a k -scheme X , satisfying the following condition:

$$\Phi : G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x) \quad \text{is surjective.} \quad (\star)$$

(This holds in particular when G acts on itself by left translations.) For short, we shall say: “Let X be a G -scheme satisfying $(\star)$.” Finally, write k' for the perfect closure of k .

Proposition 2.6.1. *Let X be a G -scheme satisfying $(\star)$.*

- (i) *X is geometrically pointwise irreducible over k : for every extension K of k , each point $x \in X_K$ belongs to a unique irreducible component of X_K .*
- (ii) *Every local ring of $(X_{k'})_{\text{red}}$ is normal.*
- (iii) *Let η be a maximal point of $(X_{k'})_{\text{red}}$, put $C = \overline{\{\eta\}}$, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme and is geometrically irreducible over L ; that is, $C \otimes_L K$ is irreducible for every extension K of L .*
- (iv) *In particular, if x is a rational point of X , then the irreducible component of X containing x is geometrically irreducible over k .*

Proof. (i) Since hypothesis $(\star)$ is preserved by every base change $k \rightarrow K$, it is enough to show that every $x \in X$ belongs to a unique irreducible component of X . Since $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is a universal homeomorphism, we may further suppose that k is perfect. We may then suppose that G and X are reduced.

Let η be a maximal point of X , and let z be any point of $Z = \overline{\{\eta\}}$. Since X is reduced, $\mathcal{O}_{X,\eta} = \kappa(\eta)$; and, since k is perfect, $\kappa(\eta) \otimes_k K$ is normal for every extension K of k (cf. EGA IV₂, 6.14.2). Thus every point of X_K lying above η is normal.

Since Φ is surjective, there is a point γ of $G \times X$ such that the two projections of $\Phi(\gamma)$ are z and η . Set $K = \kappa(\gamma)$. There are then rational points $g \in G_K$ and $\eta' \in X_K$, with η' lying above η , such that $z' = g\eta'$ lies above z . By the preceding paragraph, η' is a normal point of X_K , and therefore so is z' . Since $\pi : X_K \rightarrow X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV₂, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) now follows from (ii). Since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L , by EGA IV₂, 4.5.9.

³¹Editor’s note: these results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in Section 5 of Exposé VI_B.

Finally, if X has a rational point x , it follows from (iii) that $L = k$, and hence the irreducible component C containing x is geometrically irreducible over k .^{*} This can also be seen directly as follows (cf. [Per75], II, 2.1). Let C be the irreducible component of X containing x , and let K be an extension of k . The scheme X_K has a unique point x_K above x , and by (i), x_K belongs to a unique irreducible component C' of X_K . On the other hand, by 2.6.0, every irreducible component of C_K contains x_K , and therefore equals C' .

Notation.— Write provisionally C^0 for the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the identity element e .

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is set-theoretically stable under the group law; that is, $(C_{k'}^0)_{\text{red}}$ is a subgroup of $(G_{k'})_{\text{red}}$. Consequently, C^0 is quasi-compact.*

Proof. By 2.6.1, C^0 is geometrically irreducible over k , so $C^0 \times C^0$ is irreducible and

$$\nu(C^0 \times C^0) \subset C^0,$$

where ν denotes the morphism $(g, h) \mapsto gh^{-1}$. Since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism, the same conclusion holds for $C_{k'}^0$, and then for

$$H = (C_{k'}^0)_{\text{red}}.$$

Since $H \times_{k'} H$ is reduced, ν induces a morphism

$$H \times_{k'} H \longrightarrow H;$$

thus H is a subgroup of $(G_{k'})_{\text{red}}$. Consequently, by 0.5.1, H , and hence C^0 , is quasi-compact.

Recall 2.6.3.— Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is called *retrocompact* if the inclusion $E \hookrightarrow Y$ is quasi-compact. Moreover, by EGA IV₁, 1.9.5(v) and 1.10.1, if U is a retrocompact open subset of Y , its closure $\overline{U}$ is the union of the closures $\overline{\{y\}}$ of the points $y \in U$. Of course, it is enough to let y run through the maximal points of U , that is, the maximal points of Y contained in U .

Consequently, if Y is quasi-separated and η is a maximal point of Y , then the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$. Indeed, if $y \in Y - \overline{\{\eta\}}$, then y belongs to an affine open V not containing η . Since Y is quasi-separated, V is retrocompact, and therefore $\overline{V}$ is the union of the $\overline{\{\xi\}}$, for ξ running through the maximal points of Y belonging to V . Hence $\eta \notin \overline{V}$, so $Y - \overline{V}$ is a closed neighborhood of η not containing y .

Proposition 2.6.4. *Let X be a G -scheme satisfying $(\star)$, and let U be an open subset of X .*

- (i) *C^0U is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*
- (i') *Consequently, if X is irreducible, it is quasi-compact.*
- (ii) *If, moreover, U is retrocompact in X , then $C^0U = \overline{U}$, and this subset is both open and closed in X .*

Proof. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets

$$g \cdot U_{\kappa(g)} \subset X_{\kappa(g)},$$

and every projection $X_{\kappa(g)} \rightarrow X$ is open.

^{*}Translator's note: the French source prints X in this conclusion and then switches from the point x to undefined symbols e and e_K . The statement and argument require the component C and the points x and x_K , respectively.

To prove the second assertion of (i), we may replace k by k' (since $\operatorname{Spec} k' \rightarrow \operatorname{Spec} k$ is a universal homeomorphism), and therefore suppose that k is perfect. We may then suppose that G and X are reduced, and hence geometrically reduced.

Let η be a maximal point of X contained in U , and put $Z = \overline{\{\eta\}}$. Consider

$$\mu' : C^0 \times Z \longrightarrow X.$$

Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; write γ for its generic point. The morphism μ' sends the point $\varepsilon(\eta)$, where ε denotes the identity section of C^0 , to η . Hence γ is sent to a generization of η , and therefore to η . Thus μ' sends the underlying space of $C^0 \times Z$ into Z , and, since $C^0 \times Z$ is reduced, μ' factors through Z .

Now let $z \in Z$, and put $K = \kappa(z)$. The morphism

$$\mu_z : G_K \longrightarrow X_K, \quad h \longmapsto h \cdot z,$$

is surjective. Let α be a maximal point of X_K . The local ring $\mathcal{O}_{X_K, \alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α . It follows from Lemma 2.5.3 that μ_z is flat.

On the other hand, μ_z sends the generic point ω of C_K^0 to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$. Since μ_z is flat, there is a generization ξ of ω such that $\mu_z(\xi) = \beta$. But ω is a maximal point of G_K , so necessarily $\xi = \omega$. Thus $\mu_z(\omega) = \beta$, and β lies above η , because $Z_K \rightarrow Z$ is flat.

Put $L = \kappa(\omega)$, and let ω_L and z_L be the L -points obtained from ω and z . Then

$$\omega_L \cdot z_L = \beta'$$

is a point of Z_L lying above β , and hence

$$z_L = \omega_L^{-1} \cdot \beta' \in C_L^0 \cdot U_L.$$

Therefore $z \in C^0 \cdot U$, which proves (i).

Since C^0 is quasi-compact by 2.6.2, assertion (i') follows: if X is irreducible and U is a nonempty affine open subset, then $X = C^0 U$, so X is the image of the morphism $C^0 \times U \rightarrow X$, and is therefore quasi-compact. Finally, if U is retrocompact in X , then by 2.6.3, $\overline{U}$ is the union of the $\overline{\{\eta\}}$, where η runs through the maximal points of X contained in U . Hence $\overline{U} = C^0 U$. This proves (ii).

One then obtains the following result ([Per75], II, Th. 2.4; see also [Per76], Prop. 4.1.1).

Theorem 2.6.5. *Let k be a field, and let G be a k -group scheme.*

- (i) *There is a unique group subscheme G^0 of G , called the identity component of G , such that:*
 - (a) *the underlying space of G^0 is the irreducible component of the identity element;*
 - (b) *$G^0 \rightarrow G$ is a flat closed immersion; that is,*

$$\mathcal{O}_{G, g} = \mathcal{O}_{G^0, g} \quad \text{for every } g \in G^0.$$

- (ii) *Moreover, G^0 is quasi-compact, geometrically irreducible, and is a characteristic subgroup of G .*
- (iii) *If G is connected, then $G = G^0$.*

Proof. (i) Recall first that G is separated (0.3), and hence, a fortiori, quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0 . By 2.6.3 and 2.6.4,

$$\overline{U} = C^0 U$$

is both open and closed, and, set-theoretically,

$$C^0 = \overline{\{\omega\}}$$

is the intersection of these open-and-closed subsets.

As U runs through the affine open subsets containing ω , the G -schemes $\overline{U}$ form a projective system whose transition morphisms are affine, since they are closed immersions. We can therefore form its projective limit G^0 (cf. EGA IV₃, 8.2.2). Thus, for every affine open V of G , $G^0 \cap V$ is the spectrum of the algebra

$$\varinjlim_U \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) / \sum_U I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of

$$\mathcal{O}_G(V) \longrightarrow \mathcal{O}_G(V \cap \overline{U}).$$

It follows that the underlying space of G^0 is C^0 , and that $G^0 \rightarrow G$ is a closed immersion.

Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0, g}$ is the inductive limit, as V runs through the affine open subsets of G containing g , of the k -algebras

$$\mathcal{O}_{G^0}(V \cap G^0) = \varinjlim_U \mathcal{O}_G(V \cap \overline{U}).$$

This double inductive limit is identified with

$$\varinjlim_U \varinjlim_{g \in V \subset \overline{U}} \mathcal{O}_G(V) = \mathcal{O}_{G, g}.$$

Thus

$$\mathcal{O}_{G^0, g} = \mathcal{O}_{G, g}$$

(see also EGA IV₂, 5.13.3(ii)), and consequently $i : G^0 \rightarrow G$ is a flat closed immersion. Conversely, this condition implies that $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, so G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k -scheme and let ϕ be an automorphism of the S -group G_S . For every $s \in S$, ϕ_s sends $G_{\kappa(s)}^0$ into itself, and therefore $\phi(G_S^0) \subset G_S^0$. Moreover, the closed immersion

$$i_S : G_S^0 \hookrightarrow G_S$$

obtained from i by base change is flat, so

$$\mathcal{O}_{G_S, \phi(z)} = \mathcal{O}_{G_S^0, \phi(z)} \quad \text{for every } z \in G_S^0.$$

Hence $\phi \circ i_S$ factors through G_S^0 . This proves that G^0 is a characteristic subgroup of G , and proves (ii). Finally, (iii) is a special case of part (i) of the following proposition.³²

Proposition 2.6.6. *Let k be a field, and let G be a k -group acting on a k -scheme X in such a way that*

$$G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose that X is quasi-separated. Then:

³²Editor's note: this proposition, as well as the preceding results, was communicated to us by O. Gabber. It will be used to correct the proof of Theorem 5.3 in Exposé VI_B.

- (i) Every connected component C of X is irreducible.
- (ii) Let η be the generic point of C , and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme, is geometrically irreducible over L , and the morphism

$$G_L^0 \times_L C \longrightarrow C \times_L C$$

is surjective.

- (iii) In particular, if C contains a rational point x , then C is geometrically irreducible over k , and

$$\phi : G^0 \longrightarrow C, \quad g \longmapsto gx,$$

is surjective.

Proof. (i) Let C be a connected component of X , let Y be an irreducible component of X contained in C , let η be the generic point of Y , and let U be an affine open subset of X containing η . Since X is quasi-separated, U is retrocompact in X . Hence, by 2.6.4,

$$\overline{U} = G^0 U$$

is an open-and-closed subset of X which meets C , and therefore contains C . By 2.6.3, the intersection of the $\overline{U}$, as U runs through a fundamental system of affine open neighborhoods of η , is $\overline{\{\eta\}}$. It follows that

$$C = \overline{\{\eta\}},$$

which proves (i).

The first assertion of (ii), and also that of (iii), follows from 2.6.1. We first prove the second assertion of (iii). Let ω , respectively η , denote the generic point of G^0 , respectively C . Let $z \in C$, and set $K = \kappa(z)$. Since G^0 , respectively C , is geometrically irreducible over k , the scheme G_K^0 , respectively C_K , is irreducible; denote its generic point by ξ , respectively β . We saw in the proof of 2.6.4 that

$$\mu_z : G_K \longrightarrow C_K, \quad h \longmapsto h \cdot z,$$

sends ξ to β . Likewise, $\phi(\omega) = \eta$.

Put $L = \kappa(\xi)$, and let ξ_L and z_L be the L -points obtained from ξ and z . Then

$$\xi_L \cdot z_L = \beta'$$

is a point of C_L lying above $\beta \in C_K$, and hence also above $\eta \in C$. Consider the cartesian square

$$\begin{array}{ccc} G_L^0 & \xrightarrow{\varphi_L} & C_L \\ \pi_{G^0} \downarrow & & \downarrow \pi_C \\ G^0 & \xrightarrow{\varphi} & C \end{array}$$

Since

$$\phi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$$

(cf. EGA I, 3.4.8), there is a $g \in G_L^0$ such that $\phi_L(g) = \beta'$. Therefore

$$z_L = \xi_L^{-1} \cdot \phi_L(g) = \phi_L(\xi_L^{-1}g),$$

whence

$$\phi(\pi_{G^0}(\xi_L^{-1}g)) = \pi_C(z_L) = z.$$

This proves that ϕ is surjective.

We now prove the second assertion of (ii). It is enough to show that, for every $z \in C$, the morphism

$$\mu_z : G_L^0 \otimes_L \kappa(z) \longrightarrow C \otimes_L \kappa(z)$$

is surjective. But this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$.

Remark 2.6.7.— *Under the hypotheses of 2.6.6, if $C(k) = \emptyset$, the morphism*

$$G^0 \times_k C \longrightarrow C \times_k C$$

need not be surjective. For example, let $k = \mathbb{R}$ and $G = \{\pm 1\}_{\mathbb{R}}$. The G -torsor

$$X = \operatorname{Spec} \mathbb{R}[X]/(X^2 + 1)$$

is connected, but

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{R}} X$$

is not surjective. (Here $L = \mathbb{C}$, and

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{C}} X$$

is an isomorphism.)

3. Construction of quotients $F \backslash G$ (for G, F of finite type)

3.1.— Let A be an Artinian local ring, and let $u : F \rightarrow G$ be a homomorphism of A -groups. If

$$\mu : F \times_A F \longrightarrow F \quad \text{and} \quad \nu : G \times_A G \longrightarrow G$$

denote the multiplication morphisms, and if λ denotes the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G$$

recall that the *left quotient* $F \backslash G$ of G by F is the cokernel of the (Sch/ A)-groupoid G_* described below:

$$\begin{array}{ccccc} F \times_A F \times_A G & \xrightarrow{F \times \lambda} & & \xrightarrow{\lambda} & G \\ & \xrightarrow{\mu \times G} & F \times_A G & \xrightarrow{\text{pr}_2} & \\ & \xrightarrow{\text{pr}_{2,3}} & & & \end{array}$$

Here pr_2 and $\text{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$, respectively, onto the second factors. We shall say that G_* is the *groupoid with base G defined by u* (cf. Exposé V, § 2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A -morphism $F \rightarrow \operatorname{Spec} A$ is universally open (EGA IV₂, 2.4.9), pr_2 is an open morphism. The same therefore holds for λ , which is the composite of pr_2 with the automorphism σ of $F \times_A G$ defined by

$$\sigma(S)(x, y) = (x, u(S)(x) \cdot y),$$

where S is a variable A -scheme and $x \in F(S)$, $y \in G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A .

Let us also note, to conclude these preliminaries, that every A -morphism $s : \operatorname{Spec} A \rightarrow G$ defines an automorphism of the groupoid G_* which induces on G , $F \times_A G$, and $F \times_A F \times_A G$, respectively, the automorphisms

$$r_s, \quad \operatorname{id}_F \times_A r_s, \quad \operatorname{id}_F \times_A \operatorname{id}_F \times_A r_s.$$

We shall again write r_s for this automorphism of G_* , and shall say that r_s is the *right translation defined by s* (cf. 0.4).

Theorem 3.2. *Let F and G be flat groups locally of finite type over an Artinian local ring A . Let $u : F \rightarrow G$ be a quasi-compact homomorphism of A -groups whose kernel is finite over A . Then:*³³

- (i) *The left quotient $F \backslash G$ of G by F exists in (Sch/A) , and the sequence*

$$F \times_A G \begin{array}{c} \xrightarrow{\lambda} \\ \xrightarrow{\operatorname{pr}_2} \end{array} G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The canonical morphism $p : G \rightarrow F \backslash G$ is surjective and open, and $F \backslash G$ is a direct sum of schemes of finite type over A .*
- (ii') *More precisely, $X = F \backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A , irreducible, and all of dimension $\dim G - \dim F$.*

- (iii) *The canonical morphism*

$$F \times_A G \longrightarrow G \times_{(F \backslash G)} G$$

is surjective.

- (iv) *If u is a monomorphism,*³⁴ *then:*

- (a) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \operatorname{pr}_2)} G \times_{(F \backslash G)} G$$

is an isomorphism, and $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation.

- (a') *$F \backslash G$ represents the (fppf) quotient sheaf $\widetilde{F \backslash G}$, and $G \rightarrow F \backslash G$ is an F -torsor locally trivial for the (fppf) topology.*
- (b) *$F \backslash G$ is flat over A , and is smooth over A if G is so.*
- (c) *$u : F \rightarrow G$ is a closed immersion, and $F \backslash G$ is separated.*
- (d) *If, moreover, F is a normal subgroup of G , there is a unique A -group structure on $F \backslash G$ for which $p : G \rightarrow F \backslash G$ is a morphism of A -groups.*

In the proof of this theorem, A' will denote an A -algebra which is local, finite, and free over A . If R is a relation involving A' , we shall say that “ $R(A')$ is true when A' is sufficiently large” if there is an A -algebra A_1 , local, finite, and free over A , such that the relation $R(A')$ holds for every A -algebra A' which is local, finite, and free over A_1 .

³³Editor's note: we have added (ii') and expanded point (iv), taking account of the additions made in 2.5.2, 2.5.4, and Exposé V, 6.1.

³⁴Editor's note: in this case, the hypothesis that G be flat can be omitted; cf. subsection 3.3.

We shall first prove the theorem when F and G are of finite type over A .

3.2.1.— Suppose for a moment that every point of G has an open saturated neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. Exposé V, § 6). Then, by Exposé V, 6.1, assertions (i), (ii), (iii), and (iv)(a) hold, and $F \backslash G$ is of finite type over A .^{*} Moreover, under the hypothesis of (iv), since

$$G \longrightarrow F \backslash G$$

is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. Assertion (iv)(a') follows from (iv)(a), by Exposé IV, 3.4.3.1, 5.2.2, and 5.1.6. Finally, assertion (iv)(c) will be proved in 3.2.5, while (ii') and (iv)(d) will be proved in Section 5.

We now prove the following assertion:

every finite subset of $F \backslash G$ is contained in an affine open subset. (†)

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If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G , then $U \otimes_A A'$ is a quasi-section of the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A) -groupoid induced by G_* on U , then $U_* \otimes_A A'$ is identified with the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.³⁶

It follows from the proofs in Exposé V that the construction of the quotient

$$X = F \backslash G$$

commutes with a base extension $A \rightarrow A'$ of the kind considered here.³⁷

Let $x_1, \dots, x_n$ be points of $X = F \backslash G$, which we may suppose to be closed,³⁸ and let $g_1, \dots, g_n$ be closed points of G mapping to $x_1, \dots, x_n$. Let V be an everywhere dense affine open subset of X ,³⁹ and let U be its inverse image in G . By 1.2, there is an A -algebra A' , local, finite, and free over A , such that the points $g'_1, \dots, g'_p$ of

$$G' = G \otimes_A A'$$

^{*}Translator's note: the French source prints k here and again in editor's note 39. In this part of the proof the base is the Artinian local ring A , and "finite type over A " is the required statement.

³⁵Editor's note: let us point out that if $A = k$ is a field, every quasi-compact open subset of $F \backslash G$ is quasi-projective (a result due to Chow for smooth algebraic groups); cf. [Ray70], VI, 2.6. By contrast, over the Artinian local ring $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, there exist abelian A -schemes G which are not projective (loc. cit., XII, 4.2).

³⁶Editor's note: the preceding assertion is valid for every base change $A \rightarrow A'$.

³⁷Editor's note: this is explained in detail in 4.6 below. One must show that formation of direct images by the morphisms p , λ , and pr_2 commutes with flat base changes $A \rightarrow A'$. Since F and G are of finite type over the Artinian ring A , the morphisms f in question are all quasi-compact and quasi-separated, and

$$f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$$

(with the evident notation) follows from EGA IV₁, 1.7.21.

³⁸Editor's note: indeed, let $y_1, \dots, y_n$ be arbitrary points of X . Since X is of finite type over A , the closure of each y_i contains a closed point x_i , and every open subset containing x_i contains y_i .

³⁹Editor's note: such an open subset exists because X is of finite type over A : X has only finitely many irreducible components $C_1, \dots, C_p$, and it is enough, for each i , to choose a nonempty affine open subset contained in

$$C_i - \bigcup_{j \neq i} C_j.$$

(Here, moreover, one knows from (ii') that the C_i are disjoint.)

lying above $g_1, \dots, g_n$ are strictly rational over A' .⁴⁰ Since the morphisms $G' \rightarrow G$ and $G \rightarrow X$ are open,

$$U' = U \otimes_A A'$$

is dense in G' . Hence the open subset

$$\bigcap_{i=1}^p (U')^{-1} \cdot g'_i$$

is nonempty and therefore contains a closed point x . By 1.2 (and 0.4.1), after enlarging A' if necessary, we may suppose that x is strictly rational over A' . Since $x \in (U')^{-1} \cdot g'_i$, we have

$$g'_i \in U' \cdot x \quad (1 \leq i \leq p).$$

Write V' for the inverse image of V in

$$X' = X \otimes_A A'.$$

It is an affine open subset of X' , and is also the image of U' under the projection $G' \rightarrow X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x , of the quotient X' . Consequently,

$$V' \cdot x = r_x(V'),$$

the image of $U' \cdot x$ in X' , is an affine open subset of X' containing the images $x'_1, \dots, x'_p$ of $g'_1, \dots, g'_p$.

Consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \rightarrow X$:

$$X \otimes_A A' \otimes_A A' \begin{array}{c} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{array} X \otimes_A A' \longrightarrow X$$

Here d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A -algebra, say of rank n , d_0 and d_1 are finite and locally free of rank n . We may therefore apply the argument of Exposé V, 5.b, taken from the proof of SGA 1, VIII.7.6. It follows that $x'_1, \dots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \dots, x_n$, and is an affine open subset of X , by Exposé V, 4.1(ii).

3.2.2.— For every A -algebra A' , local, finite, and free over A , let $U(A')$ denote the set of points of $G \otimes_A A'$ having an open saturated neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is clear that $U(A')$ is saturated under the operations of $G(\text{Spec } A')$ on $G \otimes_A A'$. We shall show that, when A' is sufficiently large,

$$U(A') = G \otimes_A A'.$$

By Theorem V.8.1, $U(A)$ is nonempty, and therefore contains a closed point y . We argue by induction on

$$\dim(G - U(A)).$$

Let $g_1, \dots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there is an A -algebra A' , local, finite, and free over A , such that the points

$$g'_1, \dots, g'_p \quad (\text{respectively, } x = x_1, \dots, x_r)$$

⁴⁰Editor's note: the original has been expanded in what follows.

of $G' = G \otimes_A A'$, lying above $g_1, \dots, g_n$ (respectively, above y), are strictly rational over A' . Then $U(A')$ contains

$$(U(A) \otimes_A A') \cdot x^{-1} g'_i \quad \text{for every } i.$$

Consequently, $U(A')$ contains $g'_1, \dots, g'_p$, and

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis therefore gives an algebra A'' , local, finite, and free over A' , such that

$$U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''.$$

3.2.3.— We are now in a position to prove the existence of $F \backslash G$ when F and G are of finite type over A . Let A' be sufficiently large over A that $U(A')$ coincides with $G \otimes_A A'$ (cf. 3.2.2). Set

$$A'' = A' \otimes_A A',$$

and, for every A -scheme X , write X' and X'' for the fiber products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F' \backslash G'$ and $F'' \backslash G''$ exist, and we have the following commutative diagram, in which the first two rows and columns are exact:

$$(*) \quad \begin{array}{ccccc} F'' \times_{A''} G'' & \xrightarrow{\text{pr}_2''} & G'' & \xrightarrow{p''} & F'' \backslash G'' \\ \downarrow w_1 & \downarrow w_2 & \downarrow v_1 & \downarrow v_2 & \downarrow u_1 \downarrow u_2 \\ F' \times_{A'} G' & \xrightarrow{\text{pr}_2'} & G' & \xrightarrow{p'} & F' \backslash G' \\ \downarrow h & & \downarrow g & & \\ F \times_A G & \xrightarrow{\text{pr}_2} & G & & \end{array}$$

λ

In this diagram, pr_2' and λ' (respectively, pr_2'' and λ'') are obtained from pr_2 and λ by the evident base changes; the morphisms g and h are induced by the canonical injection $A \rightarrow A'$. We write p' and p'' for the canonical morphisms, while v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A'' .

Finally, formation of the quotient $F' \backslash G'$ commutes with the two base changes

$$f_1, f_2 : \text{Spec } A'' \rightrightarrows \text{Spec } A'.$$

Thus, if

$$\pi' : F' \backslash G' \longrightarrow \text{Spec } A'$$

denotes the structure morphism, there are canonical isomorphisms, for $i = 1, 2$,

$$\tau_i : F'' \backslash G'' \xrightarrow{\sim} (F' \backslash G') \times_{\pi', f_i} \text{Spec } A'',$$

and u_i is the composite of τ_i with the projection

$$(F' \backslash G') \times_{\pi', f_i} \text{Spec } A'' \longrightarrow F' \backslash G'.$$

Now, for a diagram of type $(*)$ in which the first two rows and columns are exact, one checks readily that $\text{Coker}(\text{pr}_2, \lambda)$ exists if and only if $\text{Coker}(u_1, u_2)$ exists, and that these two cokernels are then identified. The existence of $F \backslash G$ therefore follows from that of $\text{Coker}(u_1, u_2)$.

The compatibility of the formation of $F \backslash G$ with the base extensions considered here (cf. editor's note 37 in 3.2.1 and 4.6 below) shows that the composite

$$(F' \setminus G') \times_{\pi', f_1} \operatorname{Spec} A'' \xrightarrow{\tau_1^{-1}} F'' \setminus G'' \xrightarrow{\tau_2} (F' \setminus G') \times_{\pi', f_2} \operatorname{Spec} A''$$

is a descent datum on $F' \setminus G'$ relative to

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By 3.2.1 (†) and SGA 1, VIII.7.6, this descent datum is effective; that is, $\operatorname{Coker}(u_1, u_2)$ exists. (One could instead apply Exposé V, Theorem 4.1 directly.)

3.2.4.— To complete the proof of assertions (i), (ii), (iii), and (iv)(a) of 3.2 in the case where F and G are of finite type over A , it remains to study the quotient $F \setminus G$. By Exposé V, 6.1, assertions (ii), (iii), and (iv)(a) “become true” after the base change

$$f : \operatorname{Spec} A' \longrightarrow \operatorname{Spec} A.$$

By EGA IV₂, 2.6.1, 2.6.2, and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), namely that $F \setminus G$ is the cokernel of $(\operatorname{pr}_2, \lambda)$ in the category of all ringed spaces, it suffices to refer to Exposé V, §6(c).

3.2.5.—⁴¹ We now prove assertion (iv)(c) of 3.2, following the proof of Exposé VI_B, 9.2.1. Write

$$X = F \setminus G$$

and let d be the morphism

$$F \times_A G \longrightarrow G \times_A G$$

whose components are λ and pr_2 .

Since u is a closed immersion by 2.5.2, and since

$$d = \sigma \circ (u \times \operatorname{id}_G), \quad \sigma(x, y) = (xy, y),$$

where σ is an automorphism of $G \times_A G$, the morphism d is a closed immersion. On the other hand, by (iv)(a), the following square is Cartesian:

$$\begin{array}{ccc} F \times_A G & \xrightarrow{d} & G \times_A G \\ \downarrow & & \downarrow p \times p \\ X & \xrightarrow{\Delta_X} & X \times_A X \end{array}$$

Moreover, p , and hence also $p \times p$, is faithfully flat and locally of finite presentation. Therefore, by fppf descent, since d is a closed immersion, so is Δ_X ; that is, X is separated.

3.3.—⁴² In Theorem 3.2, the hypothesis that G be flat can be omitted when $u : F \rightarrow G$ is a monomorphism. This generalization is mentioned in Exposé VI_B, Remark 9.3(b), and also in [Ray67a], Example a)(i), p. 82. The proof, given in [An73], Theorem 4, follows from Theorem 3.2 and from the following theorem of Grothendieck (mentioned in [Ray67a], Theorem 1(ii), and proved in [DG70], § III.2, 7.1). If X is a scheme and R an equivalence relation on X , we write $\widetilde{X/R}$ for the (fppf) quotient sheaf of X by R (cf. Exposé IV, 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, let X be an A -scheme, and let*

$$d_0, d_1 : R \rightrightarrows X$$

⁴¹Editor’s note: we have added this paragraph.

⁴²Editor’s note: we have added this subsection.

be an A -equivalence relation on X , with each d_i faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X , defined by a nilpotent ideal, and let R_0 be the equivalence relation induced by R on X_0 . If the (fppf) quotient sheaf $\widetilde{X_0/R_0}$ is representable by an A -scheme, then so is $\widetilde{X/R}$.

For the proof, see [DG70], § III.2, 7.1. Let us now return to the case where A is an Artinian local ring. Let $u : F \rightarrow G$ be a quasi-compact morphism between A -groups locally of finite type, and suppose in addition that u is a monomorphism. Then u is a closed immersion by 2.5.2.

We may now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an Artinian local ring, let G be an A -group locally of finite type, and let F be a closed subgroup of G , flat over A .⁴³ Then:*

- (i) *The (fppf) quotient sheaf $\widetilde{F \backslash G}$ is representable by a separated A -scheme $F \backslash G$, locally of finite type; moreover, the sequence*

$$F \times_A G \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \text{pr}_2)} G \times_{(F \backslash G)} G$$

is an isomorphism, and $p : G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation; hence p is an F -torsor locally trivial for the (fppf) topology.

- (iii) *If G is flat (respectively, of finite type; respectively, smooth) over A , then so is $F \backslash G$.*

- (iv) *$X = F \backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A , irreducible, and all of dimension $\dim G - \dim F$.*

- (v) *If, in addition, F is a normal subgroup of G , there is a unique A -group structure on $F \backslash G$ for which $p : G \rightarrow F \backslash G$ is a morphism of A -groups.*

Assertions (i) and (ii) follow from 3.2 and 3.3.1. Since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1, and 17.7.7. Assertions (iv) and (v) will be proved in Section 5. Let us record at once the following corollary.

Corollary 3.3.3. *Let A be an Artinian local ring, let G be an A -group locally of finite type, and let H be a closed subgroup of G , flat over A . Write p for the morphism $G \rightarrow G/H$, and write λ (respectively, pr_1) for the morphism $G \times H \rightarrow G$ defined by $\lambda(g, h) = gh$ (respectively, the projection $G \times H \rightarrow G$). Then, for every open subset U of G/H ,*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \text{pr}_1\}.$$

In other words, $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A -scheme S and every $g \in G(S)$, $h \in H(S)$.

Indeed, $p : G \rightarrow G/H$ is faithfully flat and locally of finite presentation, hence a covering for the (fppf) topology; the result follows from Exposé IV, 3.3.3.2.

⁴³Editor's note: for an example in which F is not flat and $\widetilde{F \backslash G}$ is not representable, see [DG70], § III.3, no. 3.3.

4. Construction of quotients $F \backslash G$ (general case)

We now assume the hypotheses of Theorem 3.2, without requiring F and G to be of finite type over A .

4.1.— Let G^α be a connected component of G . We first show that the saturation

$$\mathcal{S}(G^\alpha)$$

⁴⁴ of G^α under the equivalence relation defined by the groupoid G_* is an open-and-closed subset of G ; in other words, it is a union of connected components of G .

This saturation is the image of $F \times_A G^\alpha$ under λ , and is therefore open in G (cf. § 3.1). Let k be the residue field of A , and let $\bar{k}$ be an algebraic closure of k . It remains to show that the image of

$$(F \times_A G^\alpha) \otimes_A k$$

under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, equivalently by SGA 1, VIII.4.4, that the image of

$$(F \times_A G^\alpha) \otimes_A \bar{k}$$

under $\lambda \otimes_A \bar{k}$ is closed in $G \otimes_A \bar{k}$. Since $G^\alpha \otimes_A \bar{k}$ is a union of finitely many connected components of $G \otimes_A \bar{k}$, we are reduced to the case where A is an algebraically closed field, which we now assume. In this case, $\mathcal{S}(G^\alpha)$ is the union of the images of G^α under the left translations $\ell_{u(x)}$, where x runs through the closed points of F . The assertion follows because these images are connected components of G .

4.2.— Take G^α in particular to be the connected component G^0 of the identity in G . Then $\mathcal{S}(G^0)$ plainly contains the image of F under u , which is precisely the equivalence class of the identity. On the other hand, if F^β is a connected component of F , then $F^\beta \times_A G^0$ is connected by 2.1.2, so its image under λ is contained in the connected component of $u(F^\beta)$ in G . In other words, $\mathcal{S}(G^0)$ is the union of the connected components that meet the image of F .

Let us also observe that the open subscheme of G whose underlying space is $\mathcal{S}(G^0)$ is a subgroup of G ; we continue to denote it by $\mathcal{S}(G^0)$. Indeed, inversion on G preserves the image of F and permutes the connected components of G that meet this image. It therefore suffices to prove that

$$\nu : G \times_A G \longrightarrow G$$

maps $\mathcal{S}(G^0) \times_A \mathcal{S}(G^0)$ into $\mathcal{S}(G^0)$. For this we may suppose that A is an algebraically closed field: with the notation of 4.1, $\mathcal{S}(G^0) \otimes_A \bar{k}$ is the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$.

If G^γ and G^δ are connected components of $\mathcal{S}(G^0)$, then $G^\gamma \times_A G^\delta$ is connected and its image under ν meets the image of F . Consequently,

$$\nu(G^\gamma \times_A G^\delta)$$

is contained in a connected component of G that meets $u(F)$.^{*}

4.3.— It follows from the preceding discussion that the groupoid G_* , with base G , defined by u , is the direct sum of the groupoids $\mathcal{S}(G_*^\alpha)$ induced by G_* on the various open-and-closed parts of G of the form $\mathcal{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of the groupoids $\mathcal{S}(G_*^\alpha)$, which we are thus led to study separately.

⁴⁴Editor's note: we have written $\mathcal{S}(G^\alpha)$, rather than $\overline{G^\alpha}$, for the saturation of G^α .

^{*}Translator's note: the French source prints $u(G^\gamma \times_A G^\delta)$ in this final sentence. The map defined and used throughout the paragraph is the multiplication $\nu : G \times_A G \rightarrow G$; the target therefore uses ν .

First consider the groupoid $\mathcal{S}(G_*^0)$ induced by G_* on $\mathcal{S}(G^0)$. It is clear that $\mathcal{S}(G_*^0)$ is the groupoid with base $\mathcal{S}(G^0)$ defined by the homomorphism

$$F \longrightarrow \mathcal{S}(G^0)$$

induced by u (§ 3.1). The cokernel whose existence we wish to prove is thus identified with $F \backslash \mathcal{S}(G^0)$. Consider, on the other hand, the groupoid

$$G_2^0 \begin{array}{c} \xrightarrow{\ell_2'} \\ \xrightarrow{\ell_1'} \\ \xrightarrow{\ell_0'} \end{array} G_1^0 \xrightarrow[\ell_0]{\ell_1} G_0^0 = G^0$$

induced by $\mathcal{S}(G_*^0)$ on G^0 . Referring to the explicit construction in Exposé V, § 3.b, the object denoted there by $Y_0 \times_{X_0} X_1$ is precisely $F \times_A G^0$. Hence G_1^0 is the inverse image of G^0 under the morphism

$$F \times_A G^0 \longrightarrow \mathcal{S}(G^0)$$

induced by λ .

I claim that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u . Indeed, if F^β is a connected component of F_0 , then $F^\beta \times_A G^0$ is connected by 2.1.2 and $\lambda(F^\beta \times_A G^0)$ is contained in G^0 .^{*} Conversely, if F^β is a connected component of F not contained in F_0 , then the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0 , then $\lambda(F^\beta \times_A G^0)$ does not meet G^0 .

It follows that the groupoid G_*^0 induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism

$$F_0 \longrightarrow G^0$$

induced by u . Since G^0 , and hence F_0 , are of finite type over A , Section 3 shows that G_*^0 has a cokernel, namely $F_0 \backslash G^0$.[†]

I now claim that $F_0 \backslash G^0$ is identified with $F \backslash \mathcal{S}(G^0)$. The proof is analogous to that of the first part of assertion (i) of Exposé V, Lemma 6.1. Consider the diagram

$$\mathcal{S}(G^0) \xleftarrow[v]{} F \times_A G^0 \xrightarrow{\text{pr}_2} G^0$$

where v is the morphism induced by λ . Since pr_2 has a section, it is a universal effective epimorphism. Consequently, $F_0 \backslash G^0$ coincides with $\text{Coker}(v_0, v_1)$, where

$$V_2 \begin{array}{c} \xrightarrow{v_2'} \\ \xrightarrow{v_1'} \\ \xrightarrow{v_0'} \end{array} V_1 \xrightarrow[v_0]{v_1} V = F \times_A G^0$$

is the inverse image under pr_2 of the groupoid G_*^0 (cf. Exposé V, § 3.a), and hence also the inverse image of $\mathcal{S}(G_*^0)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\text{inclusion}} F \times_A \mathcal{S}(G^0) \xrightarrow{\text{pr}_2} \mathcal{S}(G^0)$$

Similarly, since v is faithfully flat and quasi-compact, $F \backslash \mathcal{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathcal{S}(G_*^0)$ under the base change v . This inverse image is isomorphic

^{*}Translator's note: the French source prints $\lambda(F \times_A G^0)$ at this point. The statement concerns the connected component F^β just introduced, and the converse immediately below confirms the intended restriction to $F^\beta \times_A G^0$.

[†]Translator's note: the French source says "by paragraph 4." The finite-type quotient invoked here is the result of Section 3; Section 4 is the general-case construction currently in progress.

to V_* by Exposé V, § 3.c. It follows that the canonical inclusion of G_*^0 in $\mathcal{S}(G_*^0)$ induces an isomorphism

$$F_0 \backslash G^0 \xrightarrow{\sim} F \backslash \mathcal{S}(G^0).$$

Finally, formation of $F \backslash \mathcal{S}(G^0)$ commutes with finite locally free base changes, because the same holds for $F_0 \backslash G^0$ (cf. editor's note 37 and 4.6 below).

4.4.— It remains to construct the cokernel of the groupoid $\mathcal{S}(G_*^\alpha)$ when G^α is an arbitrary connected component of G . Let A' be a sufficiently large local A -algebra, finite and free over A (cf. 3.2). Then $G^\alpha \otimes_A A'$ is the union of finitely many connected components

$$C^1, \dots, C^n$$

of $G \otimes_A A'$, each of which has a strictly rational point. For every i , there is therefore a right translation r_i of $G \otimes_A A'$ which maps $G^0 \otimes_A A'$ onto C^i . This translation induces an isomorphism

$$\mathcal{S}(G_*^0) \otimes_A A' \xrightarrow{\sim} \mathcal{S}(C_*^i).$$

Consequently, the groupoid induced by $G_* \otimes_A A'$ on the saturation of C^i has a cokernel.

Since $\mathcal{S}(G_*^\alpha) \otimes_A A'$ is the direct sum of some of the groupoids $\mathcal{S}(C_*^i)$, it too has a cokernel. This cokernel is a direct sum of some number of copies of

$$(F_0 \otimes_A A') \backslash (G^0 \otimes_A A').$$

Thus every finite subset of it is contained in an affine open subset. Moreover, formation of this cokernel commutes with finite locally free extensions of the base (cf. editor's note 37 and 4.6 below). As in 3.2.3, it follows that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathcal{S}(G_*^\alpha)$.

4.5.— We have now constructed $F \backslash G$ and shown that it is a direct sum of schemes of finite type over A . The remaining assertions of Theorem 3.2 reduce directly to assertions about the groupoids $\mathcal{S}(G_*^\alpha)$. As in Exposé V, § 6, the second assertion in (i) follows from the first together with (ii) and (iii); it is therefore enough to prove (ii), (iii), and (iv)(a).

Since A' is a local A -algebra, finite and free over A , the morphism $A \rightarrow A'$ is faithfully flat and of finite presentation. By SGA 1, VIII (3.1, 4.6, 5.4), it is therefore enough to verify the corresponding assertions for the groupoid

$$\mathcal{S}(G_*^\alpha) \otimes_A A'.$$

This groupoid is isomorphic to a direct sum of finitely many copies of $\mathcal{S}(G_*^0) \otimes_A A'$ (cf. 4.4), so we are reduced to the groupoid $\mathcal{S}(G_*^0)$. For this last groupoid we continue to follow the proof given in Exposé V, § 6, as we began to do in 4.3.

4.6.— Let us conclude this section with some remarks concerning Lemma 6.1 and § 9(a) of Exposé V. Under the hypotheses and with the notation of Exposé V, § 9(a), we seek a condition under which formation of the cokernel of the (Sch/S) -groupoid X_* commutes with a base extension

$$\pi : S' \longrightarrow S.$$

The cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U' , respectively. We are therefore reduced to the case of a (Sch/S) -groupoid satisfying the hypotheses of Exposé V, Theorem 4.1.

⁴⁵Let Y be the cokernel of U_* , put

$$Y' = Y \times_S S',$$

⁴⁵Editor's note: we have added the following sentence.

and let Y_1 be the cokernel of U'_* ; we saw in Exposé V, § 9(a), that the canonical morphism

$$Y_1 \longrightarrow Y'$$

is a homeomorphism (indeed, a universal homeomorphism); we may thus identify Y_1 and Y' as topological spaces. If $p : U \rightarrow Y$ is the canonical morphism and $p' : U' \rightarrow Y'$ is obtained from it by base change, we want the following sequence of $\mathcal{O}_{Y'}$ -modules to be exact:

$$\mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1}). \quad (*)$$

⁴⁶Since we are under the hypotheses of Exposé V, Theorem 4.1, u_0 and u_1 are finite and locally free; moreover, p is integral by Exposé V, 4.1(ii). Thus p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S , EGA III₁, 1.4.15 (taking account of correction Err_{III} 25 in EGA III₂) identifies the sequence (*) with the inverse image of the sequence

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_U) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}), \quad (**)$$

which is exact.⁴⁷ An analogous argument applies when the groupoid X_* has quasi-sections “locally” (cf. the proof of Exposé V, Theorem 7.1). We therefore obtain:

Proposition 4.6.1. *Formation of the cokernel of X_* commutes with flat base extensions when X_* has quasi-sections locally.*

4.7.— Consider the groupoid G_* of Theorem 3.2, supposing temporarily that F and G are of finite type over A . By 3.2.2 there is a local A -algebra A' , finite and free over A , such that $G_* \otimes_A A'$ has quasi-sections “locally.” For every base extension $T \rightarrow \operatorname{Spec} A$, the sequence

$$(F'' \setminus G'') \times_{\operatorname{Spec} A} T \rightrightarrows (F' \setminus G') \times_{\operatorname{Spec} A} T \longrightarrow (F \setminus G) \times_{\operatorname{Spec} A} T$$

obtained from diagram (*) of 3.2.3 is exact.

If, in addition, T is flat over $\operatorname{Spec} A$, then 4.6 identifies

$$(F'' \setminus G'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (F' \setminus G') \times_{\operatorname{Spec} A} T$$

respectively with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

The diagram obtained from 3.2.3(*) by the base change $T \rightarrow \operatorname{Spec} A$ then shows that

$$(F \setminus G) \times_{\operatorname{Spec} A} T$$

is the cokernel of $G_* \times_{\operatorname{Spec} A} T$. The same argument applies in the general case, when F and G are merely locally of finite type over A . Thus:

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A -scheme T , the scheme*

$$(F \setminus G) \times_{\operatorname{Spec} A} T$$

is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.⁴⁸

⁴⁶Editor’s note: in what follows we have modified the original; the additional hypotheses imposed there on X_* are unnecessary.

⁴⁷Editor’s note: we have added both the following sentence and the numbering 4.6.1 below, in order to highlight the stated result.

⁴⁸Editor’s note: we have added the numbering 4.7.1 in order to highlight this result.

5. Connections with Exposé IV and consequences

5.1.—⁴⁹ We retain the notation of § 3 and the hypotheses of Theorem 3.2. We then have the following commutative diagram:

$$\begin{array}{ccc}
 F \times_A G \times_A G & \xrightarrow{F \times \nu} & F \times_A G \\
 \text{pr}_2 \times G \downarrow & \lambda \times G & \downarrow \text{pr}_2 \downarrow \lambda \\
 G \times_A G & \xrightarrow{\nu} & G \\
 p \times G \downarrow & & \downarrow p \\
 (F \backslash G) \times_A G & \xrightarrow{\rho} & F \backslash G
 \end{array}$$

It satisfies

$$\text{pr}_2 \circ (F \times \nu) = \nu \circ (\text{pr}_2 \times G), \quad \lambda \circ (F \times \nu) = \nu \circ (\lambda \times G).$$

Moreover, since G is flat over A , the left-hand vertical sequence is exact by 4.7. Hence ν induces a morphism of A -schemes

$$\rho : (F \backslash G) \times_A G \longrightarrow F \backslash G.$$

The morphism ρ makes G act on the right on $F \backslash G$, as is immediately checked; moreover, the canonical morphism $G \rightarrow F \backslash G$ commutes with the right actions of G on G and on $F \backslash G$.⁵⁰ This proves the first assertion of Theorem 3.2(ii'). By 2.5.4, the connected components of

$$X = F \backslash G$$

are therefore of finite type, irreducible, and all of the same dimension.

To compute this dimension, we may suppose that $A = k$ and that k is algebraically closed. By Exposé I, 2.3.3.1, the stabilizer of the k -point $p(e)$ is represented by the fiber

$$H = p^{-1}(p(e)).$$

Since $F \backslash G$ is the quotient of G by F in the category of ringed spaces, the underlying space of this fiber is $u(F)$. As $\ker(u)$ is finite, we obtain

$$\dim H = \dim u(F) = \dim F.$$

It follows from 2.5.4(ii) that

$$\dim X = \dim G - \dim F.$$

This proves Theorem 3.2(ii'), and hence also Theorem 3.3.2(iv).

5.2.— When the homomorphism of A -groups $u : F \rightarrow G$ is a monomorphism, the result of 5.1 may also be recovered from the results of Exposé IV. Indeed, the canonical morphism

$$p : G \longrightarrow F \backslash G$$

is faithfully flat and open by 3.2; it is therefore a covering for the (fpqc) topology by Exposé IV, 6.3.1, and we may apply Exposé IV, Corollaries 5.2.2 and 5.2.4.

⁴⁹Editor's note: we have changed the title of this section, which was called "Complements" in the original.

⁵⁰Editor's note: we have added what follows.

In particular, suppose, in addition to the hypotheses of 3.2, that u is the inclusion in G of a normal subgroup F . Then there is a unique A -group structure on $F \backslash G$ for which the canonical morphism

$$p : G \longrightarrow F \backslash G$$

is a homomorphism of A -groups.⁵¹ This proves Theorem 3.3.2(v).

5.3.— We now review several statements from Exposé IV.

5.3.1.— The statements of Exposé IV, 5.2.7 and 5.3.1 translate as follows. Let F and G be groups locally of finite type and flat over A , with F a closed normal subgroup of G . The maps

$$H \longmapsto F \backslash H, \quad H' \longmapsto H' \times_{(F \backslash G)} G$$

define a bijection between the flat A -subgroups of G containing F and the flat A -subgroups of $F \backslash G$. Under this bijection, the closed (respectively, normal) subgroups of G containing F correspond to the closed (respectively, normal) subgroups of $F \backslash G$.⁵²

5.3.2.— Exposé IV, Proposition 5.2.9, gives the following result. Let F, H, G be groups locally of finite type and flat over A , and suppose

$$F \subset H \subset G,$$

with F closed in G and normal in H . Then $F \backslash H$ acts freely on the left on $F \backslash G$, the quotient scheme

$$(F \backslash H) \backslash (F \backslash G)$$

exists, and there is a canonical isomorphism of G -schemes

$$(F \backslash H) \backslash (F \backslash G) \xrightarrow{\sim} H \backslash G.$$

5.3.3.— Finally, Exposé IV, 5.2.8, gives the following assertion. Let F, H, G be groups locally of finite type and flat over A . Suppose that F is contained in G , closed and normal in G , that H is contained in G , and that $F \cap H$ is flat over A . Let

$$F \times_A^\tau H$$

be the A -group whose underlying scheme is $F \times_A H$, with multiplication defined by

$$((x, h), (y, h')) \longmapsto (xhyh^{-1}, hh').$$

Likewise, let

$$u : H \cap F \longrightarrow F \times_A^\tau H$$

be the monomorphism $x \mapsto (x^{-1}, x)$, and let $F \cdot H$ be the quotient

$$(F \cap H) \backslash (F \times_A^\tau H).$$

Then there is a canonical isomorphism

$$F \backslash (F \cdot H) \xrightarrow{\sim} (F \cap H) \backslash H.$$

⁵¹Editor's note: we have added the following sentence.

⁵²Editor's note: in addition to the cited statements of Exposé IV, we use the fact that, since $G \rightarrow F \backslash G$ is faithfully flat, an A -subgroup H of G is flat over A if and only if $F \backslash H$ is.

5.4.—⁵³ Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that the kernel N of u is flat over A . In this case, by 3.3.2 and 5.2, the quotient A -group

$$C = N \backslash G$$

exists, and the morphism $p : G \rightarrow C$ is faithfully flat and locally of finite presentation. On the other hand, by Exposé IV, 5.2.6, u induces a monomorphism

$$v : C \longrightarrow H,$$

which is quasi-compact (because u is quasi-compact and $G \rightarrow C$ is surjective; cf. EGA IV₁, 1.1.3), and is therefore a closed immersion by 2.5.2. We have thus obtained the following proposition.

Proposition 5.4.1.— *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that $N = \ker u$ is flat over A . Then u admits the factorization*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ N \backslash G & & \end{array}$$

where p is faithfully flat and locally of finite presentation, and i is a closed immersion.

Suppose in addition that G is flat over A . Then, by 3.3.2, $C = N \backslash G$ is flat over A , and hence the quotient

$$X = C \backslash H$$

exists in $(\mathbf{Sch}/_A)$ and represents the quotient (fppf)-sheaf $\tilde{C} \backslash H$, while

$$q : H \longrightarrow X$$

is a C -torsor. Consequently, if

$$e : \operatorname{Spec} A \longrightarrow H$$

denotes the unit section of H , then v induces an isomorphism of (fppf)-sheaves between $\tilde{C}$ and the fiber product of q with

$$q \circ e : \operatorname{Spec} A \longrightarrow X,$$

which is represented by a closed subscheme of H . Consequently, v is an isomorphism of C onto a closed group subscheme K of H , equal to the stabilizer of the A -point $q \circ e$ of X .^{*} (This also gives another proof that every quasi-compact monomorphism $v : C \rightarrow H$ between A -groups locally of finite type is a closed immersion; cf. 2.5.2 and Exposé VI_B, 1.4.2.)

Suppose further that C is a normal subgroup of H . In this case, the A -group

$$\overline{H} = C \backslash H$$

is the cokernel, in the category of A -groups, of the morphism $u : G \rightarrow H$, and K is the kernel of the morphism

$$H \longrightarrow \overline{H}.$$

⁵³Editor's note: we have expanded the original in what follows; in particular, we have added Proposition 5.4.1.

^{*}Translator's note: the French source declares $e : \operatorname{Spec} A \rightarrow G$, although the displayed composition $q \circ e$ requires the unit section of H , and then prints "a closed group subscheme K of G ." Here $v : C \rightarrow H$, the displayed fiber product is represented inside H , and the next paragraph identifies K with $\ker(H \rightarrow \overline{H})$; accordingly, e is taken to be the unit section of H , and the intended ambient group of K is H .

When G and H are abelian A -groups, K is the image of u in the category of abelian A -groups, whereas

$$C = (\ker u) \backslash G$$

is the coimage of u . Taking into account the isomorphism

$$C \xrightarrow{\sim} K$$

just established, we obtain the following.⁵³

Theorem 5.4.2.— *Let k be a field. The category of commutative algebraic k -groups is abelian.*

Indeed, when k is a field, $\ker(u)$ is flat over k , whatever u may be.

⁵⁴ Note that the full subcategory of commutative affine algebraic k -groups is thick. Indeed, consider an exact sequence of commutative algebraic k -groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, then N is clearly affine, and G/N is also affine by a theorem of Chevalley; cf. Exposé VI_B, 11.17. Conversely, if N and G/N are affine, then G is also affine by Exposé VI_B, 9.2(viii). We therefore obtain the following corollary.

Corollary 5.4.3.— *Let k be a field. The category of commutative affine algebraic k -groups is abelian.*

Let us also point out that the category of all commutative affine k -groups (not necessarily of finite type) is abelian. This follows from Exposé VI_B, 11.17 and 11.18.2 (cf. [DG70], § III.3, 7.4); see also Exposé VII_B, 2.4.2 for a proof using formal groups.

5.5.— Let G be a group locally of finite type and flat over an Artinian local ring A . We know from 2.3 that the connected component G^0 of the identity is a normal open group subscheme of G , hence is also flat over A . It then follows from 3.2 and 5.2 that

$$G^0 \backslash G$$

is an A -group scheme, flat over A . Moreover, since every connected component G^α of G is saturated for the equivalence relation defined by G^0 , the quotient $G^0 \backslash G$ is the disjoint union of the $G^0 \backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the identity in $G^0 \backslash G$ is precisely

$$G^0 \backslash G^0 \simeq \operatorname{Spec} A,$$

and hence the morphism

$$G^0 \backslash G \longrightarrow \operatorname{Spec} A$$

is a local isomorphism at the identity. Consequently, $G^0 \backslash G$ is étale over $\operatorname{Spec} A$, by Exposé VI_B, 1.3.⁵⁵ We therefore obtain the following proposition (for assertion (ii), see [DG70], § II.5, 1.7–1.10).

Proposition 5.5.1.— *Let A be an Artinian local ring and G an A -group locally of finite type and flat.*

(i) $G^0 \backslash G$ is an étale A -group.

⁵³Editor's note: we have added the number 5.4.2 to this theorem.

⁵⁴Editor's note: we have added what follows.

⁵⁵Editor's note: we have expanded the preceding discussion from the original and added Proposition 5.5.1 in order to make this result explicit.

- (ii) Consequently, if $A = k$ is an algebraically closed field, $G^0 \backslash G$ is a constant k -group acting simply transitively on the set of connected components of G . Thus, if G is algebraic, $G^0 \backslash G$ is finite.

5.6.— Now let k be a perfect field and G a k -group locally of finite type. We saw in 0.2 that G_{red} is then a group subscheme of G . Moreover, the equivalence class of the identity of G under the left action of G_{red} on G is the whole underlying space of G . Theorem 3.2 therefore gives the following result.

Proposition 5.6.1.— *Let k be a perfect field and G a k -group locally of finite type. Then the k -scheme*

$$G_{\text{red}} \backslash G$$

is the spectrum of a finite local k -algebra with residue field k .

⁵⁶ Indeed, by 3.2, the quotient $G_{\text{red}} \backslash G$ has a single point, with residue field k , and is a k -scheme of finite type. It is therefore the spectrum of a finite-dimensional local k -algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2.— *Let $u : F \rightarrow G$ be a morphism between groups locally of finite type over a perfect field k . The following assertions are equivalent:*

- (i) *u is flat.*
(ii) *The morphism*

$$u^0 : F^0 \longrightarrow G^0$$

is dominant, and the morphism

$$v : F_{\text{red}} \backslash F \longrightarrow G_{\text{red}} \backslash G$$

induced by u is flat.

⁵⁷ Indeed, consider the following commutative diagram:

$$\begin{array}{ccc} F & \xrightarrow{p} & F_{\text{red}} \backslash F \\ u \downarrow & & \downarrow v \\ G & \xrightarrow{q} & G_{\text{red}} \backslash G \end{array}$$

where p and q denote the canonical projections. By 3.2(iv), p and q are faithfully flat. Consequently, if u is flat, then

$$q \circ u = v \circ p$$

is flat, and hence v is flat as well.

Conversely, suppose that v is flat and u^0 is dominant. Since u^0 is quasi-compact (the scheme F^0 is of finite type over k by 2.4, and hence Noetherian), it maps the generic point ξ of F^0 to the generic point η of G^0 . Let R be the finite local k -algebra such that

$$G_{\text{red}} \backslash G = \text{Spec } R,$$

and let $\mathfrak{m}$ be its maximal ideal. We have local homomorphisms of local rings

$$R \longrightarrow \mathcal{O}_{G, \eta} \longrightarrow \mathcal{O}_{F, \xi}.$$

Observe that the following square is Cartesian:

⁵⁶Editor's note: we have added the number 5.6.1, as well as the proof that follows.

⁵⁷Editor's note: we have expanded the proof of (ii) $\Rightarrow$ (i) and simplified the diagram below.

$$\begin{array}{ccc}
G_{\text{red}} & \longrightarrow & G \\
\downarrow & & \downarrow q \\
\text{Spec}(R/\mathfrak{m}) & \longrightarrow & \text{Spec}(R)
\end{array}$$

It follows that

$$\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta} \simeq \mathcal{O}_{G_{\text{red}},\eta} = \kappa(\eta),$$

so that

$$\mathcal{O}_{F,\xi}/\mathfrak{m}\mathcal{O}_{F,\xi}$$

is flat over $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R . The local criterion for flatness (cf. EGA IV₃, 11.3.10.2) therefore shows that $\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}$; in other words, u is flat at ξ . It follows from 2.5.3 that u is flat.

6. Complements on k -groups not necessarily of finite type

⁵⁸ Let us also record the following results, which will be useful in the addendum, Exposé VI_B, § 12. We fix a base field k .

Lemma 6.1.— *Let G be a k -group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G .*

Proof. Set $K = \kappa(x)$. Since the projection $G_K \rightarrow G$ sends maximal points to maximal points, we are reduced to the case in which x is rational. Left translation λ_x (respectively, right translation ρ_x) then gives a morphism

$$G \longrightarrow G \times G, \quad g \longmapsto (\lambda_x(g^{-1}), g)$$

(respectively, $g \longmapsto (g, \rho_x(g^{-1}))$), which induces an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (respectively, p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, then $p_1(u)$ and $p_2(u)$ are maximal points of G , and u has the required properties.

Corollary 6.2.— *Let $f : G \rightarrow H$ be a quasi-compact dominant morphism of k -groups.*

- (i) *f is surjective.*
- (ii) *If H is reduced, f is faithfully flat.*

Proof. Write μ_H (respectively, μ_G) for the multiplication of H (respectively, G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$, $\beta = p_2(u)$ are maximal points of H . Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV₁, 1.1.5). Hence there exists $v \in G \times G$ such that

$$(f \times f)(v) = u$$

(cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This proves that f is surjective.

Suppose in addition that H is reduced. Then $\mathcal{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \emptyset$. Thus f is flat at every point $\xi \in f^{-1}(\alpha)$, and consequently f is flat by Lemma 2.5.3.

⁵⁸Editor's note: we have added the results that follow, taken from [Per75]. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points."

Reminder 6.3.— Recall (cf. EGA IV₃, 11.10.1) that a morphism of schemes $f : X \rightarrow Y$ is said to be *schematically dominant* if it satisfies the following condition: for every open subset U of Y , if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \rightarrow U$ factors through Z , then $Z = U$. When f is quasi-compact and quasi-separated, this is equivalent to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3(iv), and EGA I, 9.5.8).

Proposition 6.4.— *Let $f : H \rightarrow G$ be a quasi-compact morphism of k -groups. Then the closed image of f is a group subscheme H' of G , and f factors as*

$$\begin{array}{ccc} H & \xrightarrow{f} & G \\ f' \downarrow & \nearrow i & \\ H' & & \end{array}$$

where f' is schematically dominant, quasi-compact, and surjective.

Proof. Since H is separated (0.3), f is quasi-compact and separated. Thus $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$ -module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal

$$\mathcal{I} = \ker(\mathcal{O}_G \rightarrow f_*(\mathcal{O}_H))$$

(cf. EGA I, § 9.5).

Write c_G, μ_G (respectively, c_H, μ_H) for the inversion and multiplication morphisms of G (respectively, H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence

$$H' \subset c_G(H') \quad \text{and therefore} \quad H' = c_G(H'),$$

since $c_G^2 = \text{id}_G$. Similarly,

$$f \circ \mu_H = \mu_G \circ (f \times f)$$

factors through H' , so $f \times f$ factors through the closed subscheme $\mu_G^{-1}(H')$ of $G \times G$.^{*} On the other hand, since formation of the closed image commutes with flat base change (EGA III, 1.4.15, and EGA IV₁, 1.7.21), the closed image of $f \times \text{id}_H$ (respectively, $\text{id}_{H'} \times f$) is $H' \times H$ (respectively, $H' \times H'$). Hence, by “transitivity of closed images” (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$. It is therefore contained in $\mu_G^{-1}(H')$; that is, the restriction of μ_G to $H' \times H'$ factors through H' . This proves that H' is a closed group subscheme of G . Write $i : H' \hookrightarrow G$ for the inclusion.

Now $f = i \circ f'$, where $f' : H \rightarrow H'$ is schematically dominant and quasi-compact (because f is quasi-compact and i is separated). By 6.2, f' is therefore surjective. This proves 6.4.

We can now state the following theorem ([Per75], V, 3.1 and 3.2; see also [Per76], 0.0 and 0.1).

Theorem 6.5 (D. Perrin).— *Let G be a quasi-compact k -group.*

- (i) *G is the projective limit of a filtered system (G_i) of k -groups of finite type (whose transition morphisms $u_{ij} : G_j \rightarrow G_i$ are affine for i sufficiently large), and the morphisms $G \rightarrow G_i$ are faithfully flat (and affine for i sufficiently large).*
- (ii) *Let H be a closed k -subgroup of G . Then the fpqc quotient sheaf $\widetilde{G/H}$ is a k -scheme in either of the following two cases:*

^{*}Translator’s note: in these two occurrences the French source prints π_H and π_G , although the immediately preceding sentence defines the multiplication morphisms as μ_H and μ_G . The multiplication symbols required by the identity are used here.

- (1) *the immersion $H \rightarrow G$ is of finite presentation; in this case G/H is of finite type over k ;*
- (2) *H is normal in G .*

For the proof of this theorem, which rests on several intermediate theorems, we refer to [Per75]. For the reader's convenience, however, we prove the two corollaries below; cf. [Per75], V, 3.3–3.4, or [Per76], 4.2.3–4.2.5.

Corollary 6.6.— *Let $f : G \rightarrow H$ be a quasi-compact morphism of k -groups.*

- (i) *If f is schematically dominant, then it is faithfully flat.*
- (ii) *This is so, in particular, if H is affine and the morphism*

$$f^\natural : \mathcal{O}(H) \longrightarrow \mathcal{O}(G)$$

is injective.

Proof. Suppose that f is schematically dominant. By 6.2(i), f is surjective, so by 2.5.3 it is enough to prove that f is flat at the identity element of G . Since

$$\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$$

(cf. 2.6.5), we may replace H by H^0 , and hence suppose that H is irreducible. Then H is quasi-compact (loc. cit.); since f is quasi-compact, G is also quasi-compact. By 6.5(i),

$$H = \varprojlim_i H_i,$$

where each H_i is an algebraic k -group. Let N_i be the kernel of $G \rightarrow H_i$; it is a closed normal k -subgroup of G . Moreover, since the identity section $\{e\} \rightarrow H_i$ is of finite presentation, the immersion $N_i \rightarrow G$ is of finite presentation. Hence, by 6.5(ii), the fpqc quotient G/N_i is an algebraic k -group. We then have a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ q_i \downarrow & & \downarrow p_i \\ G/N_i & \xrightarrow{f_i} & H_i \end{array}$$

in which f is schematically dominant and p_i, q_i are faithfully flat, hence schematically dominant. Thus

$$f_i \circ q_i = p_i \circ f$$

is schematically dominant, and therefore so is f_i . On the other hand, f_i is a monomorphism, hence a closed immersion because G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and consequently $G \rightarrow H_i$ is faithfully flat. By [BAC], I, § 2.7, Proposition 9, the morphism $G \rightarrow H$ is then flat; it is also surjective by 6.2(i), and is therefore faithfully flat. This proves (i). Assertion (ii) follows, because if H is affine and $f^\natural$ is injective, then the closed image of f is H , so f is schematically dominant (cf. 6.4).

Corollary 6.7.— *Let $u : G \rightarrow H$ be a morphism of k -groups and set $N = \ker(u)$. Suppose that u is quasi-compact.*

- (i) *The fpqc quotient sheaf $\widetilde{G/N}$ is represented by a k -group scheme G/N , and u factors as*

$$\begin{array}{ccc}
 G & \xrightarrow{u} & H \\
 p \downarrow & \nearrow i & \\
 G/N & &
 \end{array}$$

where p is faithfully flat and i is a closed immersion.

- (ii) In particular, if u is a monomorphism, then u is a closed immersion; and if u is schematically dominant, then u is faithfully flat.

Proof. (i) By 6.4 and 6.6, the closed image G' of u is a closed group subscheme of H , and

$$p : G \longrightarrow G'$$

is faithfully flat and quasi-compact.* Clearly $\ker(p) = \ker(u) = N$. Thus, by Exposé IV, 3.3.2.1 and 5.1.7, G' represents the fpqc quotient sheaf $\widehat{G/N}$.

(ii) The second assertion is contained in 6.6; let us prove the first. If u is a monomorphism, then so is p . Thus p is both a monomorphism and an effective epimorphism, and is therefore an isomorphism (cf. Exposé IV, 1.14). This proves 6.7.

Let us finally record the following corollaries (cf. [Per76], 4.2.6–4.2.8).

Corollary 6.8.— *The category of commutative quasi-compact k -group schemes is abelian.*

Taking 6.7 into account, the proof is analogous to that of 5.4.2.

Corollary 6.9.— *If $\text{char}(k) = 0$, every k -group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, then

$$\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}.$$

It is therefore enough to prove that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. We are thus reduced to the case in which G is connected, hence quasi-compact by 2.6.5. The result then follows from 6.5 and Cartier's theorem for algebraic groups (cf. Exposé VI_B, 1.6.1, or [DG70], § II.6, Theorem 1.1).

Corollary 6.10.— *Let G be a quasi-compact k -group, and suppose that k is algebraically closed.*

- (i) *If $f : G \rightarrow H$ is a faithfully flat morphism of k -groups, then the induced map $G(k) \rightarrow H(k)$ is surjective.*
- (ii) *The set of rational points is dense in G .*

For the proof, we refer to [Per75], V, 3.7 and 3.9.

Bibliography

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*Translator's note: the French source prints “a closed group subscheme of G ” at this point. Since $u : G \rightarrow H$, its closed image lies in H , as the displayed factorization also shows.

⁵⁹Editor's note: additional references cited in this Exposé.

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Exposé VI_B

Generalities on Group Schemes

by J.-E. Bertin

⁰ This Exposé, which does not correspond to any oral lecture in the seminar, is intended to bring together a number of commonly used technical results concerning group schemes.*

1. Morphisms of groups locally of finite type over a field

1.1.— Let A be an Artinian local ring, let G and H be two A -groups,² and let $u : G \rightarrow H$ be a morphism of A -groups. Then u induces a group morphism

$$u(A) : G(A) \longrightarrow H(A).$$

³ Since $H(A)$ acts on H by right translation, restriction along $u(A)$ defines an action of $G(A)$ on H . This action is compatible with u and with the action of $G(A)$ on G by right translation. Since $G(A)$ acts transitively on the strictly rational points of G (see Exposé VI_A, 0.4 for the definition), these points “all behave in the same way with respect to u .” The following properties spring from this observation.

Proposition 1.2.— *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type over A . Then the set $u(G)$ is closed in H , and*

$$\dim G = \dim u(G) + \dim \ker u.$$

4

Since u commutes with the inversion morphisms of G and H , the image $u(G)$ is invariant under inversion on H ; the same is therefore true of its closure $\overline{u(G)}$ in H . On the other hand, let L be the set of points of $H \times_A H$ whose two projections belong to $u(G)$. It is clear that L is the image of

$$u \times_A u : G \times_A G \longrightarrow H \times_A H.$$

The multiplication morphism of H consequently sends L into $u(G)$; in other words,

$$u(G) \cdot u(G) = u(G).$$

⁰Editor’s note: version of October 13, 2024.

*This Exposé was quite substantially revised after its mimeographed edition; in particular, Sections 10 and 11 were completely rewritten.⁽¹⁾

¹Editor’s note: and also Section 5; cf. Editor’s notes 34 and 46.

²Editor’s note: recall the convention adopted at the beginning of Exposé VI_A: an A -group is a group scheme over A (and every such scheme is separated; cf. Exposé VI_A, 0.3).

³Editor’s note: we have replaced “homomorphism” by “morphism.”

⁴Editor’s note: this statement also appears in Exposé VI_A, 2.5.2.

Lemma 1.2.1 below shows that the closure $\overline{L}$ of L in $H \times_A H$ is the set of points whose two projections belong to $\overline{u(G)}$. Hence

$$\overline{u(G)} \cdot \overline{u(G)} = \overline{u(G)}.$$

Thus the reduced subscheme of H whose underlying space is $\overline{u(G)}$ is naturally equipped with a group structure in the category $(\mathbf{Sch}/k)_{\text{red}}$, where k is the residue field of A (cf. Exposé VI_A, 0.2).

Let us prove the first assertion of 1.2. After replacing A by an algebraic closure of its residue field k , we may suppose that A is an algebraically closed field k (cf. EGA IV₂, 2.3.12). Replacing u by

$$u_{\text{red}} : G_{\text{red}} \longrightarrow H_{\text{red}},$$

we may suppose that G and H are reduced. In this case, as we have just seen, $u(G)$ is the underlying space of a reduced group subscheme of H ;* we may therefore suppose that u is dominant. Then $G(k)$ acts transitively on the set of connected components of H , and it is enough to prove that $u(G) \cap H^0$ is closed. We are thus reduced to the case in which H is connected, hence irreducible and of finite type (Exposé VI_A, 2.4). The morphism u is then of finite type, since it is quasi-compact and locally of finite type. As H is Noetherian, $u(G)$ is constructible (EGA IV₁, 1.8.5), and therefore contains an open subset V of H (EGA 0_{III}, 9.2.2). By Exposé VI_A, 0.5, we then have

$$H = V \cdot V \subset u(G) \cdot u(G) = u(G).$$

Let us prove the second assertion. Recall first that the functor $\ker u$ (cf. Exposé I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H . When u is locally of finite type, $\ker u$ is therefore locally of finite type over A . As above, this time using EGA IV₂, 4.1.4, we reduce to the case in which A is an algebraically closed field k .

We may further suppose that G and H are irreducible and of finite type and that u is dominant. Indeed, since k is algebraically closed, the connected components of G , on whose set $G(k)$ acts transitively, clearly all have the same dimension; and if u^0 denotes the restriction of u to G^0 , then

$$(\ker u)^0 \subset \ker u^0, \quad \dim \ker u^0 = \dim \ker u.$$

Likewise, u is then of finite type over k . If η is the generic point of H , then

$$\dim u^{-1}(\eta) = \dim G - \dim H$$

(EGA IV₃, 10.6.1(ii)). By EGA IV₃, 9.2.3 and 9.2.6, the set of $y \in H$ such that

$$\dim u^{-1}(y) = \dim u^{-1}(\eta)$$

contains a nonempty open subset V . Since u is dominant, $U = u^{-1}(V)$ is a nonempty open subset of G ; it contains a closed point x , because G is a Jacobson scheme (EGA IV₃, 10.4.7). Right translation r_x is an isomorphism from $\ker u$ onto $u^{-1}(u(x))$, and hence

$$\dim \ker u = \dim u^{-1}(u(x)) = \dim u^{-1}(\eta) = \dim G - \dim H.$$

*Translator's note: the French source prints "a reduced group subscheme of G " at this point. Since $u(G) \subset H$, and the preceding paragraph constructs the reduced subscheme inside H , the ambient group is H .

Lemma 1.2.1.— *Let $f : X' \rightarrow X$ and $g : Y' \rightarrow Y$ be two quasi-compact dominant morphisms of schemes over an Artinian local ring A . Then*

$$f \times_A g : X' \times_A Y' \longrightarrow X \times_A Y$$

is dominant (and quasi-compact).

Indeed,

$$f \times_A g = (\mathrm{id}_X \times_A g) \circ (f \times_A \mathrm{id}_{Y'}).$$

It is therefore enough to prove that $f \times_A \mathrm{id}_{Y'}$ and $\mathrm{id}_X \times_A g$ are dominant. For this purpose we may replace A by its residue field k . Then X and Y' are flat over $A = k$, and $f \times_A \mathrm{id}_{Y'}$ (respectively, $\mathrm{id}_X \times_A g$) is obtained from f (respectively, g) by the flat base change $Y' \rightarrow A$ (respectively, $X \rightarrow A$). It is therefore dominant (and quasi-compact) by EGA IV₂, 2.3.7.

Counterexample 1.2.2.— *Let k be a field of characteristic zero, let G be the constant k -group $\mathbb{Z}$, and let H be the additive k -group $\mathbb{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, then $u(G)$ is not closed in H .*

Proposition 1.3.—⁵ *Let A be an Artinian local ring with residue field k , let G be an A -group locally of finite type and flat, and let $u : G \rightarrow H$ be a morphism of A -groups. The following assertions are equivalent:*

- (i) *u is flat (respectively, quasi-finite, unramified, smooth, or étale) at some point of G ;⁶*
- (ii) *u is flat (respectively, quasi-finite, unramified, smooth, or étale).*

Proof. It is enough to prove that (i) implies (ii). We first establish the following lemma.

Lemma 1.3.0.— *Let $A \rightarrow B \rightarrow C$ be morphisms of commutative rings, and let $\mathfrak{n}$ be a nilpotent ideal of A . Suppose that $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite type.*

- (i) *Then C is a B -algebra of finite type.*
- (ii) *Moreover, if C is flat over A and $C/\mathfrak{n}C$ is a $(B/\mathfrak{n}B)$ -algebra of finite presentation, then C is a B -algebra of finite presentation.*

Indeed, let $x_1, \dots, x_n$ be elements of C whose images generate $C/\mathfrak{n}C$ as a $(B/\mathfrak{n}B)$ -algebra. By the nilpotent Nakayama lemma, the x_i generate C as a B -algebra. This proves (i). Let

$$\varphi : B[X_1, \dots, X_n] \longrightarrow C$$

be the resulting surjective morphism, and put $I = \ker \varphi$.

Suppose now that C is flat over A and that $C/\mathfrak{n}C$ is of finite presentation over $B/\mathfrak{n}B$. On the one hand, $I/\mathfrak{n}I$ identifies with the kernel of

$$\overline{\varphi} = \varphi \otimes_A (A/\mathfrak{n}).$$

⁵Editor's note: in the original, 1.3, 1.3.1, and 1.3.1.1 were stated with A a field. For completeness, we treat the case of an Artinian local ring, and for this purpose have added Lemma 1.3.0.

⁶Editor's note: recall that a morphism of schemes $f : X \rightarrow Y$ is said to be of finite type (respectively, of finite presentation or quasi-finite) at x if there are affine open subsets V , containing $f(x)$, and U , containing x , such that $f(U) \subset V$ and $\mathcal{O}_X(U)$ is an $\mathcal{O}_Y(V)$ -algebra of finite type (respectively, of finite presentation, or with the additional condition that $f^{-1}(f(x'))$ is finite for every $x' \in U$); see also Editor's note 40 of Exposé V. The morphism is locally quasi-finite (respectively, locally of finite type or locally of finite presentation) if it has the property at every point x . It is smooth (respectively, unramified or étale) at x if there is an open neighborhood U of x such that $f|_U : U \rightarrow Y$ is smooth (respectively, unramified or étale). These definitions show that the loci where f is of finite presentation, of finite type, quasi-finite, smooth, unramified, or étale are open in X . Moreover, because the flatness locus of a morphism locally of finite presentation is open (EGA IV₃, 11.3.1), the locus where f is both of finite presentation and flat is open as well. These facts will be used repeatedly below.

On the other hand, by EGA IV₁, 1.4.4, $\ker \bar{\varphi}$ is an ideal of finite type. Choose $P_1, \dots, P_s \in I$ whose images generate $I/\mathfrak{n}I$ as an ideal. The nilpotent Nakayama lemma shows that the P_i generate I . This proves (ii).

We return to the proof of 1.3. Let x be an arbitrary point of G . Since G is flat over A , the fiberwise flatness criterion, in the form EGA IV₃, 11.3.10.2, shows that u is flat at x if $u \otimes_A k$ is. Likewise, the preceding lemma shows that u is of finite type (respectively, of finite presentation) at x if $u \otimes_A k$ is. The remaining properties can then be checked on the fibers (cf. EGA IV₄, 17.4.1, 17.5.1, and 17.6.1 for the unramified, smooth, and étale cases). We are therefore reduced to the case $A = k$.

Now let x be a point of G at which one of the conditions in 1.3(i) holds. The properties under consideration are preserved by fpqc descent (cf. EGA IV, 2.5.1, 2.7.1, and 17.7.1). Thus, after replacing k by an algebraic closure of $\kappa(x)$, we reduce to the case in which k is algebraically closed and $x \in G(k)$.

Since G is a Jacobson scheme (cf. EGA IV₃, 10.4.7), and the set W of points of G at which u is flat (respectively, quasi-finite, unramified, smooth, or étale) is stable under generization⁷ (respectively, open), it is enough to show that every $y \in G(k)$ belongs to W . For such a point y , the translation

$$r_y \circ r_x^{-1}$$

sends x to y , so u has the required property at y . Thus $y \in W$, and the proof is complete.

Corollary 1.3.1.— *Let A be an Artinian local ring with residue field k , and let G be a flat A -group. The following assertions are equivalent:*⁸

- (i) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A at some point;*
- (ii) *G is locally quasi-finite (respectively, unramified, smooth, or étale) over A .*

⁹ *Proof.* If G satisfies one of the conditions in (i) at a point x , there is an open neighborhood U of x that is of finite type over A . Consequently, it is enough to apply 1.3 in the case where H is the unit A -group, taking the following lemma into account.

Lemma 1.3.1.1.— *Let A be an Artinian local ring and G an A -group. If G contains a nonempty open subset of finite type over A , then G is locally of finite type over A .*

Proof. By Lemma 1.3.0, we may suppose that A equals its residue field k . Moreover, by fpqc descent, we may suppose that k is algebraically closed (cf. EGA IV₂, 2.7.1). Let V be the open subset of G consisting of the points at which G is of finite type over k ; by hypothesis, $V \neq \emptyset$. Since G is a Jacobson scheme, V contains a closed point x , and to prove that $V = G$ it is enough to show that every closed point y of G belongs to V . For such a point y , the translation

$$r_y \circ r_x^{-1}$$

sends x to y , and hence $y \in V$.

Corollary 1.3.2.— *Let A be an Artinian local ring, and let $u : G \rightarrow H$ be a morphism between A -groups locally of finite type. The following assertions are equivalent:*

⁷Editor's note: the original stated that the flat locus W is open, which does not seem evident a priori.

⁸Editor's note: it is not enough to suppose that G is flat over A at one point. For example, suppose that $A \neq k$, let H be the constant A -group $(\mathbb{Z}/2\mathbb{Z})_A$, and let G be the closed A -subgroup of H whose nonidentity component is reduced. Then the structural morphism $G \rightarrow \operatorname{Spec} A$ is a local isomorphism at the identity point e , but is not flat at the point $g \neq e$.

⁹Editor's note: we have added the sentence that follows and simplified the proof of Lemma 1.3.1.1 by invoking EGA IV, 2.7.1 instead of loc. cit., 17.7.5.

- (i) u is universally open;
- (ii) u is open;
- (iii) u is open at some point of G ;
- (iv) the morphism $u^0 : G^0 \rightarrow H^0$ induced by u is dominant;
- (iv') u^0 is surjective;
- (v) there is a connected component C of G such that, if D denotes the connected component of H containing $u(C)$, the induced morphism $u' : C \rightarrow D$ is dominant.

Proof. The implications

$$(i) \implies (ii) \implies (iii) \quad \text{and} \quad (iv') \implies (iv) \implies (v)$$

are clear. Since G^0 is of finite type over A (Exposé VI_A, 2.4), and hence Noetherian, u^0 is quasi-compact. Thus $u^0(G^0)$ is closed in H^0 by 1.2, and therefore (iv) implies (iv'). On the other hand, G^0 (respectively, C) is open in G (Exposé VI_A, 2.3), while H^0 (respectively, D) is irreducible (Exposé VI_A, 2.4.1). Hence (ii) implies (iv) (respectively, (iii) implies (v)). It remains to prove that (v) implies (i).

We endow the open subset C (respectively, D) of G (respectively, H) with its induced scheme structure, and let u' denote the induced morphism $C \rightarrow D$. Let k be the residue field of A .¹⁰ Since the base change

$$\mathrm{Spec} k \longrightarrow \mathrm{Spec} A$$

is a universal homeomorphism, we may suppose that $A = k$. By hypothesis u' is dominant; and since C is of finite type over k (Exposé VI_A, 2.4.1), and hence Noetherian, u' is quasi-compact. By EGA IV₂, 2.3.7,

$$u' \otimes_k \bar{k}$$

is again quasi-compact and dominant, where $\bar{k}$ denotes an algebraic closure of k . Since $C \otimes_k \bar{k}$ is a union of connected components of $G \otimes_k \bar{k}$, the morphism

$$u \otimes_k \bar{k} : G \otimes_k \bar{k} \longrightarrow H \otimes_k \bar{k}$$

satisfies assertion (v). In view of EGA IV₂, 2.6.4, we are thus reduced to the case where $A = k$ is an algebraically closed field.

In this case we may further replace u by u_{red} , and are thereby reduced to the case where H is reduced. Let ξ (respectively, η) be the generic point of C (respectively, D). Since u' is quasi-compact and dominant,

$$u'(\xi) = \eta$$

(cf. EGA IV₁, 1.1.5). Since H is reduced, the local ring $\mathcal{O}_{H,\eta}$ is a field, and hence u' is flat at ξ .¹¹ It follows from 1.3 that u is flat. Moreover, since u is locally of finite type and H is locally Noetherian, u is locally of finite presentation. Therefore u is universally open by EGA IV₂, 2.4.6.

Proposition 1.4.— *Let A be an Artinian local ring, and let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent:*

¹⁰Editor's note: we have expanded the original in what follows.

¹¹Editor's note: the original invoked the generic flatness theorem (EGA IV₂, 6.9.1), which is not necessary here.

- (i) u is proper;
- (ii) there exists $h \in H$ such that the fiber $u^{-1}(h)$ is nonempty and proper over $\kappa(h)$;
- (iii) $\ker u$ is proper over A .

Proof. It is clear that (i) implies (iii), and that (iii) implies (ii). The hypotheses imply that u is of finite type; and since G is separated (Exposé VI_A, 0.3), u is separated as well (EGA I, 5.5.1). It remains to prove that (ii) implies that u is universally closed. We may therefore suppose that A equals its residue field k .¹² Let k' be an algebraic closure of $\kappa(h)$, let

$$u' : G' \longrightarrow H'$$

be obtained from u by base change, and let $h' \in H'$ be a point above h . Then

$$u'^{-1}(h') = u^{-1}(h) \times_{\kappa(h)} k'$$

is nonempty and proper, and it is enough to prove that u' is proper (EGA IV₂, 2.6.4). Thus we may suppose that k is algebraically closed and $h \in H(k)$.

By 1.2, $u(G)$ is then the underlying set of a closed reduced group subscheme of H . Since every closed immersion is proper (EGA II, 5.4.2), we may suppose that u is surjective and that H is reduced. The group $G(k)$ then acts transitively on the set of closed points of H . Consequently, for every closed point $y \in H$, the fiber $u^{-1}(y)$ is proper over $\kappa(y)$. By EGA IV₃, 9.6.1, the set of $y \in H$ for which $u^{-1}(y)$ is not proper over $\kappa(y)$ is locally constructible. It contains no closed point, and is therefore empty (cf. EGA IV₃, 10.3.1 and 10.4.7).

¹³ Now consider the generic point η of H^0 . By what precedes, the fiber

$$u^{-1}(\eta) = G \times_H \text{Spec } \kappa(\eta)$$

is proper over $\kappa(\eta)$. Since H is reduced, $\kappa(\eta) = \mathcal{O}_{H,\eta}$. The latter ring is the direct limit of the rings $\mathcal{O}_H(V)$, as V ranges over the nonempty open subsets of H^0 . It follows from EGA IV₃, 8.1.2(a) and 8.10.5(xii), that there exists a nonempty open subset $V \subset H^0$ such that the restriction of u over V is proper. The translates $g \cdot V$, for $g \in G(k)$, form an open covering of H ; and for every $g \in G(k)$, the restriction of u over $g \cdot V$ is proper. Hence u is proper (cf. EGA II, 5.4.1).

Corollary 1.4.1.— *Let A be an Artinian local ring, and let $u : G \rightarrow H$ be a morphism between A -groups locally of finite type. The following assertions are equivalent:*

- (i) u is locally quasi-finite;
- (ii) u is quasi-finite at some point;
- (iii) $\ker u$ is discrete;
- (iv) the restriction of u to each connected component of G is finite.

If u is quasi-compact, these assertions are also equivalent to

- (v) u is finite.

¹²Editor's note: we have added the following sentence.

¹³Editor's note: we have expanded the original in what follows.

It is clear that (iv) implies (iii), that (iii) implies (ii) (EGA I, 6.4.4), and that, when u is quasi-compact, (iv) and (v) are equivalent. We have already seen in 1.3 that (i) and (ii) are equivalent.

It remains to prove that (i) implies (iv). Let C be a connected component of G . Since C is of finite type over A (Exposé VI_A, 2.4.1) and G, H are separated (Exposé VI_A, 0.3),* EGA I, 5.5.1 and 6.3.4 show that the restriction u' of u to C is separated and of finite type.¹⁴ Its fibers are discrete, so u' is quasi-finite (cf. EGA II, 6.2.2). Since every quasi-finite proper morphism is finite (cf. EGA III₁, 4.4.2), it is enough to prove that u' is universally closed.

For this, we may suppose that A equals its residue field k . By fpqc descent (cf. EGA IV₂, 2.6.4), it is enough to prove that

$$u' \otimes_k \bar{k}$$

is universally closed, where $\bar{k}$ is an algebraic closure of k . Since C is of finite type over k , the scheme $C \otimes_k \bar{k}$ is the disjoint union of finitely many connected components $C'_1, \dots, C'_n$ of $G \otimes_k \bar{k}$, and it is enough to prove the assertion for each C'_i . We are thus reduced to the case in which k is algebraically closed.

Let g be a closed point of C , and let

$$u^0 : G^0 \longrightarrow H$$

be the restriction of u to G^0 . Then

$$u' = r_{u(g)} \circ u^0 \circ r_{g^{-1}},$$

where r_g denotes right translation by g . Thus, to prove that u' is proper, it is enough to prove that u^0 is proper. By hypothesis u is locally quasi-finite, so the nonempty fiber $\ker u$ is discrete. We have seen that u^0 is of finite type, hence its fiber $\ker u^0$ is finite (cf. EGA II, 6.2.2), and is therefore nonempty and proper. Proposition 1.4 now shows that u^0 is proper.

Corollary 1.4.2.— *Let A be an Artinian local ring, and let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type. The following assertions are equivalent.¹⁵*

- (i) u is a closed immersion;
- (ii) u is a monomorphism;
- (iii) $\ker u$ is trivial, that is, isomorphic to the unit A -group.

In particular, every group subscheme¹⁶ of H is closed.

* It is clear that (i) implies (ii). Considering the functors represented by G and H , one sees immediately that (ii) and (iii) are equivalent. Finally, if $\ker u$ is the unit A -group, it is a proper nonempty fiber. Proposition 1.4 therefore shows that u is a proper monomorphism;

*Translator's note: the French source cites Exposé VI_A, 0.2 here; the separation statement is 0.3, as also recalled in Editor's note 2 above.

¹⁴Editor's note: we have expanded the original in what follows.

¹⁵Editor's note: the hypothesis that G and H be locally of finite type may be removed, because [Per76], 4.2.4, states that every quasi-compact monomorphism $u : G \rightarrow H$ between group schemes over a field k is a closed immersion. See also [DG70], III.3.7.2(b), for the case in which G and H are affine; in that case every morphism $G \rightarrow H$ is affine (EGA II, 1.6.2(v)), hence quasi-compact.

¹⁶Editor's note: in this particular case, see also Exposé VI_A, 0.5.2, which holds without any finiteness hypothesis.

*Translator's note: the French source calls the trivial kernel the “unit k -group” both in the statement and in the proof. Here the base is the Artinian local ring A , and $\ker u$ is an A -group; the unit object required by the monomorphism criterion is therefore $\text{Spec } A$, the unit A -group.

it is of finite presentation because H is locally Noetherian (EGA IV₁, 1.6.1), and hence is a closed immersion (EGA IV₃, 8.11.5).

The final assertion follows because H is locally Noetherian, so every immersion $G \rightarrow H$ is quasi-compact (EGA I, 6.6.4).

Counterexample 1.4.3.— *Let k be a field of characteristic zero, let G be the constant k -group $\mathbb{Z}$, and let H be the additive k -group $\mathbb{G}_{a,k}$. Let $u : G \rightarrow H$ be a morphism of k -groups. If $u \neq 0$, then $\ker u = 0$, but u is not a closed immersion (cf. 1.2.2).*

We shall use below the following two results, which should have appeared in Exposé VI_A.

Lemma 1.5.— *Let k be a field and let G be a k -group locally of finite type. Then:*

- (i) *all the irreducible components of G have the same dimension;*
- (ii) *for every $g \in G$,*

$$\dim_g G = \dim G.$$

¹⁷ Assertion (i) has already been proved in Exposé VI_A, 2.4.1, and (ii) follows. Indeed, let $g \in G$, and let C be the connected component of G containing g . By definition (EGA 0_{IV}, 14.1.2), $\dim_g G$ is the infimum of the integers $\dim U$, where U ranges over the open neighborhoods of g . Hence

$$\dim_g G = \dim U_0$$

for some U_0 , which we may suppose is contained in C (because $\dim V \leq \dim U$ when $V \subset U$). Since C is irreducible and of finite type over k (Exposé VI_A, 2.4.1),

$$\dim U_0 = \dim C = \operatorname{trdeg}_k \kappa(\xi),$$

where ξ is the generic point of C (EGA IV₂, 5.2.1). Therefore

$$\dim_g G = \dim C = \dim G.$$

Proposition 1.6.—¹⁸ *Let S be a scheme of characteristic zero, and let G be an S -group scheme locally of finite presentation over S at every point of the identity section $\epsilon(S)$. The scheme G is smooth over S at every point of the identity section if and only if the $\mathcal{O}_S$ -module*

$$\omega_{G/S} = \epsilon^*(\Omega_{G/S}^1),$$

called the conormal module of the identity section of G , is locally free.

Recall that a scheme S is said to be of characteristic zero if the field $\kappa(x)$ has characteristic zero for every closed point $x \in S$.

Recall also (Exposé II, 4.11) that, if $\pi : G \rightarrow S$ is the structural morphism, then

$$\Omega_{G/S}^1 = \pi^*(\omega_{G/S}).$$

Thus it is equivalent to say that the $\mathcal{O}_S$ -module $\omega_{G/S}$ is locally free or that the $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is locally free.

¹⁷Editor's note: the implication (ii) $\Rightarrow$ (i) is a general fact (cf. EGA 0_{IV}, 14.1.6), while (i) $\Rightarrow$ (ii) follows from the fact that if X is an irreducible scheme of finite type over a field, then $\dim X = \dim U$ for every nonempty open subset U of X . The essential point here is therefore assertion (i), which was already established in an addition to Exposé VI_A, 2.4.1. We have modified the statement and proof accordingly.

¹⁸Editor's note: in the statement, we have replaced "along the identity section" by "at every point of the identity section." At the end of the proof, we have also made explicit the cited results from EGA IV₄.

If there is an open neighborhood U of $\epsilon(S)$ that is smooth over S , then by EGA IV₄, 17.2.3, $\Omega_{U/S}^1$ is locally free of finite type, and so is

$$\omega_{G/S} = \epsilon^*(\Omega_{U/S}^1).$$

Conversely, if $\omega_{G/S}$ is locally free, then so is

$$\Omega_{G/S}^1 = \pi^*(\omega_{G/S}).$$

Since S has characteristic zero, the Jacobian criterion (EGA IV₄, 16.12.2) shows that G is differentially smooth over S . It follows from EGA IV₄, 17.12.5, that G is smooth over S at every point of the identity section.

Corollary 1.6.1 (Cartier).— *Let k be a field of characteristic zero. Every k -group locally of finite type over k is smooth over k .*

Indeed, the k -module $\omega_{G/k}$ is then plainly free. By 1.6, G is smooth over k at the identity point e , and hence smooth over k by 1.3.1.

2. “Open properties” of groups and group morphisms locally of finite presentation

2.0.— In all that follows, S will denote an arbitrary scheme; an S -group scheme will be called an S -group. Given an S -group G , we shall denote by ϵ the identity section, by

$$c : G \longrightarrow G$$

the inversion morphism, and by

$$\mu : G \times_S G \longrightarrow G$$

the multiplication morphism. For every S -scheme X , we shall denote by π , or by π_X , the structural morphism

$$X \longrightarrow S.$$

Given a property $\mathcal{P}(u)$ of a morphism of S -schemes

$$u : X \longrightarrow Y,$$

we shall say that $\mathcal{P}(u)$ is *stable under base change* if, whenever u satisfies $\mathcal{P}(u)$, the same is true of the morphism $u_{(Y')}$, for every S -morphism $Y' \rightarrow Y$. We say that $\mathcal{P}(u)$ is *local in nature for the topology $\mathcal{T}$* (cf. Exposé IV, §§ 4 and 6) if $\mathcal{P}(u)$ satisfies the following two conditions:

- a) $\mathcal{P}(u)$ is stable under base change;
- b) whenever there is a family of S -morphisms $Y_i \rightarrow Y$, covering for the topology $\mathcal{T}$, such that each morphism $u_{(Y_i)}$ satisfies $\mathcal{P}(u)$, then u satisfies $\mathcal{P}(u)$.

Proposition 2.1.— *Let $\mathcal{P}(u)$ be a property of a morphism of S -schemes, local in nature for the fpqc topology. Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose that G is flat and universally open over S .*

Let W be the largest open subset of H over which u satisfies the property $\mathcal{P}(u)$, and let $V = u^{-1}(W)$. Then

$$U = \pi_G(V)$$

is open in S , and V is an open group subscheme of $G|_U$.

The existence of a largest open subset W of H over which u satisfies $\mathcal{P}(u)$ follows from the fact that $\mathcal{P}(u)$ is local in nature for the Zariski topology. Since π_G is universally open, $\pi_G(V)$ is an open subset U of S . It is enough to show that V is a group subscheme of $G|_U$. We may therefore suppose that $U = S$.

Set

$$G' = G \times_S V, \quad H' = H \times_S V, \quad V' = V \times_S V, \quad W' = W \times_S V,$$

and $u' = u|_{V'}$. Let W'_1 be the largest open subset of H' over which u' satisfies $\mathcal{P}(u)$. Since V is flat and universally open over S , the same is true of H' over H , and Lemma 2.1.1 below shows that

$$W'_1 = W'.$$

Now consider the automorphism a (respectively, b) of the V -scheme G' (respectively, H'), namely right translation by the inverse of the diagonal section δ (respectively, by the inverse of $u(\delta)$), defined by

$$a(g, v) = (g \cdot v^{-1}, v), \quad (\text{respectively, } b(h, v) = (h \cdot u(v^{-1}), v)),$$

for every morphism $T \rightarrow S$, every $g \in G(T)$, $v \in V(T)$, and $h \in H(T)$.^{*} It is clear that

$$u' \circ a = b \circ u',$$

which shows that W' is stable under b , hence that V' is stable under a , so that V is a group subscheme of G .

Lemma 2.1.1.— *Let $\mathcal{P}(u)$ be a property of an S -morphism u , local in nature for the fpqc topology. Consider a cartesian square of morphisms of S -schemes:*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

where g is flat and open. Let W (respectively, W'_1) be the largest open subset of Y (respectively, Y') over which f (respectively, $f' = f|_{Y'}$) satisfies $\mathcal{P}(u)$. Then

$$W'_1 = W \times_Y Y'.$$

Set $W' = W \times_Y Y'$. Since $\mathcal{P}(u)$ is stable under base change, we have

$$W' \subset W'_1.$$

Since g is open,

$$W_1 = g(W'_1)$$

is an open subset of Y . Set

$$V_1 = f^{-1}(W_1), \quad V'_1 = V_1 \times_{W_1} W'_1;$$

^{*}Translator's note: the French source omits b before (h, v) in the second displayed definition. The preceding sentence defines the automorphism b , and the identity immediately below requires the displayed map to be $b(h, v)$; the symbol is supplied here.

it is clear that $V'_1 = f'^{-1}(W'_1)$. Since g is flat and open, the morphism

$$W'_1 \longrightarrow W_1 = g(W'_1)$$

induced by g is faithfully flat and open, hence is a covering for the fpqc topology (cf. Exposé IV, 6.3.1(iv)). Since the morphism

$$V'_1 \longrightarrow W'_1$$

induced by f' satisfies $\mathcal{P}(u)$, the same is true of the morphism $V_1 \rightarrow W_1$ induced by f . Thus

$$W_1 \subset W, \quad W'_1 \subset g^{-1}(W_1) \subset g^{-1}(W) = W',$$

and therefore $W' = W'_1$.

Remark 2.1.2.— *A great many properties of a morphism are local in nature for the fpqc topology; among them are the properties of being flat, (universally) open, (locally) of finite type, of finite presentation, quasi-finite (cf. EGA IV₂, 2.5.1, 2.6.1, and 2.7.1), smooth, étale, and unramified (EGA IV₄, 17.7.3).*

The proof of 2.1 in fact uses only base changes by flat morphisms. The proposition therefore applies to a property satisfying condition b) of 2.0 relative to the fpqc topology and stable under base change by flat morphisms (for example, the property of being quasi-compact and dominant).

Of course, one can state an analogous proposition concerning properties local in nature for a topology $\mathcal{T}$ finer than the Zariski topology, the condition to be verified by G then being that π_G be universally open and covering for the topology $\mathcal{T}$.

In particular, if G is flat and locally of finite presentation over S , there is an analogous statement for properties stable under base change by flat morphisms locally of finite presentation and satisfying condition b) of 2.0 relative to the fpqc topology (for example, the properties of being regular, reduced, Cohen–Macaulay, and so on; cf. EGA IV₂, 6.8).

Proposition 2.2.— *Let G and H be two S -groups, and let $u : G \rightarrow H$ be a morphism of S -groups. Then.¹⁹*

- (i) *Suppose that G or H is flat over S , and that G or H is locally of finite presentation over S . Let V be the largest open subset of G such that the restriction of u to V is flat and locally of finite presentation (respectively, smooth, or étale). Then $U = \pi_V(V)$ is open in S , and V is an open group subscheme of $G|_U$.*
- (ii) *Suppose that G or H is universally open over S , and let V be the largest open subset of G such that the restriction of u to V is universally open. Then $U = \pi_V(V)$ is open in S , and V is an open group subscheme of $G|_U$.*

We first prove (i). We show that the restriction π_V of π_G to V is flat and locally of finite presentation:

- a) if π_G is flat (respectively, locally of finite presentation), then so is π_V ;
- b) if π_H is flat (respectively, locally of finite presentation), then so is π_V , as the composite of the restriction of u to V with π_H .

¹⁹Editor's note: in case (i), the open subset V consists of all the points of G at which u is smooth, respectively étale, respectively of finite presentation and flat; cf. editor's note 6. By contrast, in case (ii), V is the largest open subset contained in the set E of points of G at which u is universally open, but E is not necessarily open, as the following example shows (EGA IV₃, 14.1.3(i)). Let k be a field, let $H = S = \operatorname{Spec} A$, where $A = k[x]$, and let G be the S -group $\operatorname{Spec} A[y]/(xy)$. Then E is the identity section of G , which is not open.

Thus, in the four cases contemplated in the statement, π_V is flat and locally of finite presentation, hence universally open (EGA IV₂, 2.4.6). Set $U = \pi_V(V)$; then U is open in S . It is therefore enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set

$$G' = G \times_S V, \quad H' = H \times_S V, \quad V' = V \times_S V, \quad \text{and} \quad u' = u_{(V)}.$$

Since V is flat and locally of finite presentation over S , the same is true of H' over H . By EGA IV₄, 17.7.4, V' is then the largest open subset of G' such that the restriction of u' to V' is flat and locally of finite presentation (respectively, smooth, or étale). With the automorphisms a and b defined as in the proof of 2.1, it is clear that V' is stable under a , hence that V is a group subscheme of G .

We now prove (ii). The restriction π_V of π_G to V is universally open, either because π_G is universally open, or because it is the composite of the restriction of u to V with π_H , when π_H is universally open. Set $U = \pi_V(V)$; then U is open in S . It is enough to show that V is an open group subscheme of $G|_U$, and we may suppose that $U = S$.

Set, as before,

$$G' = G \times_S V, \quad H' = H \times_S V, \quad V' = V \times_S V, \quad \text{and} \quad u' = u_{(V)}.$$

Then $\pi_V : V \rightarrow S$ is surjective and universally open; the same is therefore true of $G' \rightarrow G$. Consequently, V' is the largest open subset of G' such that the restriction of u' to V' is universally open, by EGA IV₃, 14.3.4(i),(ii). With the automorphisms a and b defined as before, it is clear that V' is stable under a , hence that V is a group subscheme of G .

Corollary 2.3.— *Let G be an S -group, and let V be the largest open subset of G that is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S . Then $U = \pi_V(V)$ is open in S , and V is an open group subscheme of $G|_U$.*

It is enough to apply 2.2 when H is the identity S -group and u is the unique morphism of S -groups $G \rightarrow H$, since π_H is then an isomorphism and

$$\pi_G = \pi_H \circ u.$$

Corollary 2.4.— *Let G be an S -group. If there is a neighborhood X of the identity section having one (or more) of the following properties:*

X is flat and locally of finite presentation (respectively, smooth, étale, or universally open) over S ,

then there is an open group subscheme of G having the same properties.

It is enough to apply 2.3, observing that here, with the notation of 2.2, we have

$$\epsilon(S) \subset V,$$

and hence $U = S$.

Proposition 2.5.—²⁰ *Let $u : G \rightarrow H$ be a morphism of S -groups.*

²⁰Editor's note: we have expanded the statement and proof, and in (i) we have weakened the hypothesis on H , replacing "of finite presentation" by "of finite type."

- (i) Suppose that G (respectively, H) is of finite presentation and flat (respectively, of finite type) over S at the points of its identity section. Then the sets

$$U_{\text{plat}} \supset U_{\text{lis}} \supset U_{\text{ét.}},$$

formed by the points $s \in S$ such that u_s is flat (respectively, smooth, or étale), are open in S .

If, moreover, G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S , then the set V_{plat} (respectively, V_{lis} or $V_{\text{ét.}}$) of points of G at which u is flat (respectively, smooth, or étale) is the inverse image under π_G of U_{plat} (respectively, U_{lis} or $U_{\text{ét.}}$).

- (ii) Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$ (a condition satisfied if G is of finite type over S at the points of the identity section (1.3.1.1)), and that u is locally of finite type (respectively, locally of finite presentation) at the points of the identity section of G . Then the sets

$$U_{\text{l.q.f.}} \supset U_{\text{n.r.}},$$

formed by the $s \in S$ such that u_s is locally quasi-finite (respectively, unramified), are open in S .

If, moreover, u is locally of finite type (respectively, locally of finite presentation), then the set $V_{\text{l.q.f.}}$ (respectively, $V_{\text{n.r.}}$) of points of G at which u is quasi-finite (respectively, unramified) is the inverse image under π_G of $U_{\text{l.q.f.}}$ (respectively, $U_{\text{n.r.}}$).

We prove (i). First note (1.3.1.1) that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Let Y be the open subset of H consisting of the points at which π_H is of finite type, and let X_1 be the open subset of $u^{-1}(Y)$ consisting of the points at which π_G is of finite presentation. By EGA IV₃, 11.3.1, the set X of points of X_1 at which π_G is of finite presentation and flat is open in X_1 , hence in G .

Let

$$u_X : X \longrightarrow Y$$

be the morphism induced by u . Since Y (respectively, X) contains the identity section of H (respectively, G), the restriction π_X of π_G to X is surjective, and the morphism

$$S \longrightarrow X$$

induced by ϵ_G is an S -section of X .

Consider the sets

$$W_{\text{plat}} \supset W_{\text{lis}} \supset W_{\text{ét.}},$$

consisting of the points of X at which u_X is flat (respectively, smooth, or étale). It is clear that W_{lis} and $W_{\text{ét.}}$ are open in X ; cf. Editor's note 6. On the other hand, since π_Y is locally of finite type and

$$\pi_X = \pi_Y \circ u_X$$

is locally of finite presentation, EGA IV₁, 1.4.3(v), shows that u_X is locally of finite presentation. Consequently, the flatness locus W_X of u_X is open in X (EGA IV₃, 11.3.1).

Let $x \in X$ and set $s = \pi_X(x)$. Then, by EGA IV₂, 11.3.10,* combined with EGA IV₄, 17.5.1 (respectively, 17.6.1), x belongs to W_{plat} (respectively, W_{lis} or $W_{\text{ét.}}$) if and only if

*Translator's note: the source citation is retained. Section 11 occurs in EGA IV₃, not EGA IV₂; the cited result is EGA IV₃, 11.3.10.

u_s is flat (respectively, smooth, or étale) at the point x , or, equivalently by 1.3, if and only if u_s is flat (respectively, smooth, or étale).

Consequently, for

$$“\flat = \text{flat, smooth, or étale},”$$

we have

$$U_{\flat} = \epsilon_G^{-1}(W_{\flat}), \quad W_{\flat} = \pi_X^{-1}(U_{\flat}).$$

Since $W_{\flat}$ is open in X , hence in G , the first equality shows that $U_{\flat}$ is open in S .

The second assertion of (i) follows from what we have just seen, for then

$$Y = H, \quad X = G, \quad V_{\flat} = W_{\flat} = \pi_G^{-1}(U_{\flat}).$$

We prove assertion (ii). Let

$$V_{\text{l.t.f.}} \supset V_{\text{l.q.f.}}$$

be the open subsets of G consisting of the points at which u is of finite type (respectively, quasi-finite). Set

$$X = V_{\text{l.t.f.}},$$

and let u_X denote the restriction of u to X . By hypothesis, X contains $\epsilon(S)$. Let $x \in X$ and set $s = \pi_X(x)$.

If u is quasi-finite at x , then, by base change, u_s is quasi-finite at x ; and since G_s is assumed to be locally of finite type, u_s is locally quasi-finite by 1.3.1.

Conversely, if u_s is locally quasi-finite, then

$$u_X^{-1}(u_X(x)) \subset u_s^{-1}(u_s(x))$$

is a finite set. Since

$$u_X : X \longrightarrow H$$

is locally of finite type, Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3) shows that there is an open neighborhood W of x in X such that the fiber

$$u_X^{-1}(u_X(x'))$$

is discrete for every $x' \in W$. Hence u_X is quasi-finite at x , by EGA II, 6.2.2 (and EGA III₂, Err_{III} 20).

Consequently,

$$U_{\text{l.q.f.}} = \epsilon_G^{-1}(V_{\text{l.q.f.}}), \quad V_{\text{l.q.f.}} = \pi_X^{-1}(U_{\text{l.q.f.}}),$$

and the first equality shows that $U_{\text{l.q.f.}}$ is open in S .

If, moreover, u is locally of finite type over S , then $X = G$, and hence $V_{\text{l.q.f.}}$ is the inverse image under π_G of $U_{\text{l.q.f.}}$.^{*} This proves the first half of (ii).

Now consider the open subsets

$$V_{\text{l.p.f.}} \supset V_{\text{n.r.}},$$

consisting of the points at which u is of finite presentation (respectively, unramified), and suppose that

$$Z = V_{\text{l.p.f.}}$$

contains the identity section of G . Let $z \in Z$; set

$$s = \pi_Z(z), \quad h = u_Z(z).$$

^{*}Translator's note: in this sentence and in the analogous sentence for $V_{\text{n.r.}}$ below, the French source prints G in place of u . The statement assumes that u is locally of finite type (respectively, locally of finite presentation), and X and Z are precisely the corresponding loci for u ; the required u is used here.

If u is unramified at z , then, by base change, u_s is unramified at z ; and since G_s is assumed to be locally of finite type, u_s is unramified by 1.3.1.

Conversely, if u_s is unramified at the point z , then the fiber

$$u_s^{-1}(h) = u^{-1}(h)$$

is unramified over $\kappa(h)$ at the point z . Since Z is an open subset of G , the fiber

$$u_Z^{-1}(h) = u_s^{-1}(h) \cap Z$$

is unramified over $\kappa(h)$ at the point z . Since u_Z is locally of finite presentation, EGA IV₄, 17.4.1, implies that u_Z is unramified at the point z .

Thus

$$U_{\text{n.r.}} = \epsilon_G^{-1}(V_{\text{n.r.}}), \quad V_{\text{n.r.}} = \pi_Z^{-1}(U_{\text{n.r.}}),$$

and the first equality shows that $U_{\text{n.r.}}$ is open in S .

If, moreover, u is locally of finite presentation over S , then $Z = G$, and hence $V_{\text{n.r.}}$ is the inverse image under π_G of $U_{\text{n.r.}}$. This completes the proof of assertion (ii).

Corollary 2.6.— *Let $u : G \rightarrow H$ be a morphism of S -groups that is radicial (as is the case if u is a monomorphism; EGA I, 3.5.4). Suppose that G (respectively, H) is locally of finite presentation and flat (respectively, locally of finite type) over S . Then the set U of $s \in S$ such that u_s is an open immersion is open in S , and the restriction of u over U is an open immersion.*

By 2.5(i), the set U' of points $s \in S$ such that u_s is étale is open in S . Since u is radicial, so is u_s , for every $s \in S$; hence EGA IV₄, 17.9.1, gives

$$U = U',$$

which shows that U is open. Finally, by 2.5(i), the restriction of u over U is étale; since u is radicial, this restriction is an open immersion (EGA IV₄, 17.9.1).

Proposition 2.7.— *Let G be an S -group. The following conditions are equivalent:*

- (i) G is unramified over S at the points of the identity section;
- (ii) the identity section is an open immersion;
- (iii) G is of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is unramified over $\kappa(s)$.

If, moreover, G is locally of finite presentation over S , then each of the three preceding conditions is equivalent to the following:

- (iv) G is unramified over S .

The equivalence of assertions (i) and (ii) follows from the more general Lemma 2.7.1 below. Note (1.3.1.1) that either condition (i) or condition (iii) implies that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$. Thus, by EGA IV₄, 17.4.1, assertion (i) is equivalent to the assertion that, for every $s \in S$, G_s is unramified over $\kappa(s)$ at the point e_s , the identity of G_s , or, equivalently by 1.3.1, to the assertion that G_s is unramified over $\kappa(s)$. Consequently, assertions (i) and (iii) are equivalent. Finally, if G is locally of finite presentation over S , assertions (iii) and (iv) are equivalent (EGA IV₄, 17.4.1).

Lemma 2.7.1.— *Let G be an S -scheme equipped with a section ϵ . In order that G be unramified over S at the points of this section, it is necessary and sufficient that ϵ be an open immersion.*

The condition is necessary by EGA IV₄, 17.4.1, a)⇒b''). Conversely, if ϵ is an open immersion, then the restriction to $\epsilon(S)$ of the structural morphism

$$G \longrightarrow S$$

is an isomorphism; hence G is unramified over S at the points of $\epsilon(S)$.

Corollary 2.8.—²¹ *Let $u : G \rightarrow H$ be a morphism of S -groups. Suppose that, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$.*

a) *If u is locally of finite type, the following conditions are equivalent:*

- (i) *u is locally quasi-finite;*
- (ii) *for every $s \in S$, the morphism $u_s : G_s \rightarrow H_s$ is locally quasi-finite;*
- (iii) *$\text{Ker } u$ is locally quasi-finite over S ;*
- (iv) *the fibers of $\text{Ker } u$ are discrete.*

b) *If u is locally of finite presentation, the following conditions are equivalent:*

- (v) *u is unramified;*
- (vi) *for every $s \in S$, the morphism $u_s : G_s \rightarrow H_s$ is unramified;*
- (vii) *$\text{Ker } u$ is unramified;*
- (viii) *the identity section $S \rightarrow \text{Ker } u$ is an open immersion.*

The equivalences

$$(i) \iff (ii) \quad \text{and} \quad (v) \iff (vi)$$

follow from 2.5(ii), and Reminder 2.8.1 below shows that

$$(i) \implies (iii) \quad \text{and} \quad (v) \implies (vii).$$

For every $s \in S$, denote by e_s the identity element of H_s . Then (iii) (respectively, (vii)) implies that, for every $s \in S$,

$$(\text{Ker } u)_s = \text{Ker } u_s = u_s^{-1}(e_s)$$

is locally quasi-finite (respectively, unramified) over $\kappa(s) = \kappa(e_s)$. It follows that u_s is quasi-finite (respectively, unramified) at the identity point of G_s , and hence, by 1.3, that u_s is locally quasi-finite (respectively, unramified). Thus

$$(iii) \implies (ii) \quad \text{and} \quad (vii) \implies (vi).$$

Finally,

$$(ii) \iff (iv) \quad \text{by 1.4.1,} \quad (vii) \iff (viii) \quad \text{by 2.7.}$$

Reminder 2.8.1.— Recall (cf. Exposé I, 2.3.6.1) that, given a morphism $u : G \rightarrow H$ of S -groups, the *kernel of u* , denoted $\text{Ker } u$, is the group subfunctor of G defined, for every morphism $f : T \rightarrow S$, by

$$(\text{Ker } u)(T) = \{a \in G(T) \mid u \circ a = \epsilon_H \circ f\}.$$

²¹Editor's note: we have changed the presentation by separating the assertions concerning the "quasi-finite" case from those concerning the "unramified" case.

By loc. cit. (or EGA I, 4.4.1), this functor is represented by the S -group

$$G \times_H S = u^{-1}(\epsilon_H(S)),$$

which is denoted simply by $\text{Ker } u$. In particular, the structural morphism

$$\text{Ker } u \longrightarrow S$$

is obtained from u by base change.

Lemma 2.9.— *Let $\pi : X \rightarrow S$ be a morphism admitting an S -section ϵ .*

- (i) *If π is injective, then it is integral.^(*)*
- (ii) *If π is locally of finite type and, for every $s \in S$, π_s is an isomorphism, then π is an isomorphism.^(**)*

First observe that, by Lemma 2.9.1 below, π , being a homeomorphism, is an affine morphism.

If π is injective, then ϵ is surjective. Since ϵ is a surjective immersion, $\epsilon(S)$ is isomorphic to the closed subscheme of X defined by a nil ideal $\mathcal{I}$ of $\mathcal{O}_X$. Since ϵ is an S -section of π , there is a direct-sum decomposition

$$\mathcal{O}_X = \mathcal{O}_S \oplus \mathcal{I}$$

of $\mathcal{O}_S$ -modules. Since $\mathcal{I}$ is a nil ideal of $\mathcal{O}_X$, it is plainly integral over $\mathcal{O}_S$. Thus $\mathcal{O}_X$ is integral over $\mathcal{O}_S$, and π is integral.

Now suppose that π is locally of finite type. Since $\pi \circ \epsilon = \text{id}_S$, the section ϵ is locally of finite presentation (cf. EGA IV₁, 1.4.3(v)); hence $\mathcal{I}$ is an ideal of finite type in $\mathcal{O}_X$ (EGA IV₁, 1.4.1). For every $s \in S$, one has

$$\mathcal{O}_{X_s} = \kappa(s) \oplus \mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s).$$

By hypothesis, ϵ_s is an isomorphism, so

$$\mathcal{I} \otimes_{\mathcal{O}_S} \kappa(s) = 0$$

for every $s \in S$, and therefore, a fortiori,

$$\mathcal{I} \otimes_{\mathcal{O}_X} \kappa(x) = 0$$

for every $x \in X$. Nakayama's lemma now gives $\mathcal{I} = 0$, and consequently π is an isomorphism.

Lemma 2.9.1.— *Let $f : X \rightarrow Y$ be a morphism of schemes that is a homeomorphism. Then f is an affine morphism.^(***)*

It is enough to show that every point $y \in Y$ has an open neighborhood W such that the restriction of f over W is affine. Let $y \in Y$, and let V be an affine open neighborhood of y in Y . Set

$$V' = f^{-1}(V).$$

Then V' is an open neighborhood of $x = f^{-1}(y)$ in X . There is an affine open neighborhood W' of x in X contained in V' . Set

$$W = f(W').$$

^(*)This is also a special case of EGA IV₄, 18.12.11, since π is plainly a universal homeomorphism.

^(**)This is also an immediate consequence of EGA IV₄, 18.12.6.

^(***)Cf. EGA IV₄, 18.12.7.1 for a slightly more general result, proved by the same argument.

Then W is an open neighborhood of y in Y , contained in the affine scheme V , and therefore W is separated. Since W' is affine, the restriction of f over W is an affine morphism (EGA II, 1.2.3).

Corollary 2.10.— *Let G be an S -group locally of finite type. Suppose that, for every $s \in S$, G_s is the identity $\kappa(s)$ -group. Then G is the identity S -group.*

More generally:

Corollary 2.11.— *Let $f : X \rightarrow Y$ be an S -morphism locally of finite type. In order that f be a monomorphism, it is necessary and sufficient that f_s be a monomorphism for every $s \in S$.*

The condition is plainly necessary; we prove that it is sufficient. If f_s is a monomorphism for every $s \in S$, then, a fortiori, f_y is a monomorphism for every $y \in Y$. We may therefore assume that $Y = S$.

By EGA I, 5.3.8, to prove that f is a monomorphism it is enough to show that

$$\Delta_f : X \longrightarrow X \times_S X$$

is an isomorphism, or, equivalently, that the first projection

$$p : X \times_S X \longrightarrow X$$

is an isomorphism. If f_s is a monomorphism, EGA I, 5.3.8 likewise shows that the first projection

$$X_s \times_{\kappa(s)} X_s \longrightarrow X_s,$$

which is identified with p_s , is an isomorphism. But p has the S -section Δ_f ; hence Lemma 2.9 shows that, if p_s is an isomorphism for every $s \in S$, then p itself is an isomorphism.

Corollary 2.12.— *Let G be an S -scheme admitting an S -section ϵ , and suppose that the structural morphism $\pi : G \rightarrow S$ is closed. Let $s \in S$ be such that π is of finite presentation at $\epsilon(s)$, and such that*

$$\epsilon_s : \text{Spec}(\kappa(s)) \longrightarrow G_s$$

is an isomorphism (or, equivalently, such that

$$\pi_s : G_s \longrightarrow \text{Spec} \kappa(s)$$

is an isomorphism). Then there exists an open neighborhood U of s in S such that

$$\epsilon|_U : U \longrightarrow G \times_S U$$

is an isomorphism.

Let V be the set of points of G at which π is unramified. This set is known to be open (cf. Editor's note 6), and it contains $\epsilon(s)$. Hence

$$W = \epsilon^{-1}(V)$$

is an open subset of S containing s , and π is unramified at $\epsilon(t)$ for every $t \in W$. Since π is closed, the same is true of $\pi|_W$, and we may therefore suppose that $W = S$.

By 2.7.1,

$$X = G - \epsilon(S)$$

is then a closed subset of G that does not meet G_s . Since π is closed,

$$F = \pi(X)$$

is a closed subset of S that does not contain s . Set

$$U = S - F.$$

Then U is an open subset of S , and $\epsilon|_U$ is an isomorphism from U onto $G|_U$.

3. Identity component of a group locally of finite presentation

3.0.— Given a subset A (respectively, B) of an S -scheme X (respectively, Y), by abuse of notation

$$A \times_S B$$

will denote the subset of $X \times_S Y$ consisting of the points whose first projection belongs to A and whose second projection belongs to B .

Given a subset A of an S -group G , we shall say that A is *stable under the group law of G* if

$$c(A) \subset A \quad \text{and} \quad \mu(A \times_S A) \subset A.$$

Definition 3.1.— Let G be an S -functor in groups satisfying the following condition:

$$\text{for every } s \in S, \text{ the functor } G_s = G \otimes_S \kappa(s) \text{ is representable.} \quad (+)$$

We then denote by $\underline{G}_s^0$ the connected component of the identity element of the $\kappa(s)$ -group G_s .²² We define a sub- S -functor in groups of G , called the *identity component of G* and denoted by G^0 , by setting, for every morphism $T \rightarrow S$,

$$G^0(T) = \{u \in G(T) \mid \text{for every } s \in S, u_s(T_s) \subset \underline{G}_s^0\}.$$

Thus we have defined the functor

$$G \longmapsto G^0$$

from

$$(\widehat{\mathbf{Sch}}/S)\text{-gr.} \quad \text{to} \quad (\widehat{\mathbf{Sch}}/S)\text{-gr.}$$

Remark 3.2.— (i) Let G be an S -functor in groups satisfying condition $(+)$. Then

$$\mathrm{Lie}(G^0/S) = \mathrm{Lie}(G/S)$$

by Exposé II.²³ Indeed, for every S -scheme T , denote by I_T the T -scheme of dual numbers over T , and by

$$\tau : T \longrightarrow I_T$$

the zero section (cf. Exposé II, 2.1). By definition,

$$\mathrm{Lie}(G/S)(T) = \{u \in G(I_T) \mid u \circ \tau = e\},$$

where $e : T \rightarrow G$ denotes the composite of $T \rightarrow S$ with the identity section $S \rightarrow G$. Likewise,

$$\begin{aligned} \mathrm{Lie}(G^0/S)(T) &= \{u \in G^0(I_T) \mid u \circ \tau = e\} \\ &= \{u \in G(I_T) \mid u \circ \tau = e \text{ and } u_s((I_T)_s) \subset \underline{G}_s^0 \text{ for every } s \in S\}. \end{aligned}$$

For every $s \in S$, the schemes T_s and

$$(I_T)_s = I_{T_s}$$

have the same underlying set. Hence, if $u \in \mathrm{Lie}(G/S)(T)$, then

$$u_s(I_{T_s}) = e_s,$$

²²Editor's note: this is a slightly abusive notation, but it is compatible with the notation of Exposé VI_A, 2.3 when this connected component is the underlying topological space of an open group subscheme G^0 of G ; cf. Exposé VI_A, 2.2.bis.

²³Editor's note: we have added what follows.

where e_s denotes the identity point of G_s , and therefore $u \in \text{Lie}(G^0/S)(T)$. Thus the inclusion

$$\text{Lie}(G^0/S)(T) \subset \text{Lie}(G/S)(T)$$

is an equality for every T , and consequently

$$\text{Lie}(G^0/S) = \text{Lie}(G/S).$$

(ii) Let G and G' be two S -functors in groups satisfying (+). Then:

- a) if $G \subset G'$, then $G^0 \subset G'^0$;
- b) if $G \subset G'$ and $G'^0 \subset G$, then $G^0 = G'^0$;
- c) if, for every $s \in S$, G_s is locally of finite type over $\kappa(s)$, then G^0 satisfies condition (+), by Exposé VI_A, 2.3, and

$$(G^0)^0 = G^0.$$

Proposition 3.3.— *Let G be an S -functor in groups satisfying condition (+), and let S' be an S -scheme. Then*

$$(G \times_S S')^0 = G^0 \times_S S'.$$

In other words, the functor $G \mapsto G^0$ commutes with base change; that is, the following diagram is commutative:

$$\begin{array}{ccc} (\widehat{\text{Sch}}_S)\text{-gr.} & \xrightarrow{G \mapsto G^0} & (\widehat{\text{Sch}}_S)\text{-gr.} \\ \downarrow & & \downarrow \\ (\widehat{\text{Sch}}_{S'})\text{-gr.} & \longrightarrow & (\widehat{\text{Sch}}_{S'})\text{-gr.} \end{array}$$

It is enough to check that, for every $s' \in S'$, whose image in S we denote by s ,

$$\begin{aligned} ((G \times_S S') \otimes_{S'} \kappa(s'))^0 &= (G_s \otimes_{\kappa(s)} \kappa(s'))^0 \\ &= G_s^0 \otimes_{\kappa(s)} \kappa(s'). \end{aligned}$$

This follows from Exposé VI_A, 2.1.2. Note, for later use in 4.2, that we have not used the group structure of G_s , but only the fact that G_s^0 has a rational point, namely $e(s)$, and is therefore geometrically connected (see also EGA IV₂, 4.5.14).

Special Case 3.4.— Let G be an S -group scheme, and denote by $\underline{G}^0$ the subset of G equal to the union of the $\underline{G}_s^0$ as s runs through S . Then $\underline{G}^0$ is a subset of G stable under the group law of G (cf. 3.0), and, for every morphism $S' \rightarrow S$,

$$G^0(S') = \{u \in G(S') \mid u(S') \subset \underline{G}^0\}.$$

When $\underline{G}^0$ is an open subset of G , it is clear that G^0 is represented by the subscheme of G induced on the open subset $\underline{G}^0$.

Proposition 3.5.— *Let S be a quasi-compact and quasi-separated scheme, and let G be an S -group with fibers locally of finite type. Then there is a quasi-compact open subset U of G containing $\underline{G}^0$.*

Since the identity section e is an immersion, $e(S)$ is a quasi-compact subspace of G . Thus there is a quasi-compact open subset V of G containing $e(S)$. Since S is quasi-separated

and V is quasi-compact, V is quasi-compact over S (EGA IV₁, 1.2.4). Hence $V \times_S V$ is quasi-compact over S , and therefore quasi-compact. It follows that

$$V \cdot V = \mu(V \times_S V)$$

is quasi-compact. Set

$$V_s = V \cap G_s, \quad V_s^0 = V \cap \underline{G}_s^0.$$

Then V_s^0 is open in $\underline{G}_s^0$, and is dense in $\underline{G}_s^0$ because G_s^0 is irreducible (Exposé VI_A, 2.4). Therefore

$$V_s^0 \cdot V_s^0 = \underline{G}_s^0 \quad (\text{Exposé VI}_A, 0.5),$$

which shows that

$$V_s \cdot V_s \supset \underline{G}_s^0, \quad \text{and hence} \quad V \cdot V \supset \underline{G}^0.$$

Finally, since $V \cdot V$ is quasi-compact, there is a quasi-compact open subset U of G containing $V \cdot V$, and therefore containing $\underline{G}^0$.

Corollary 3.6.— *Let G be an S -group whose fibers are locally of finite type and connected. Then G is quasi-compact over S .*

The assertion is local on S (EGA I, 6.6.1), so we are reduced to the case where S is affine. By 3.5, there is then a quasi-compact open subset U of G containing

$$\underline{G}^0 = G.$$

Thus G is quasi-compact, and hence quasi-compact over the affine scheme S (EGA I, 6.6.4(v)).

Proposition 3.7.—²⁴ *Let G be an S -group locally of finite presentation. Then:*

- (i) $\underline{G}^0$ is ind-constructible in G .
- (ii) If, moreover, G is quasi-separated over S , and S is quasi-compact and quasi-separated, then $\underline{G}^0$ is constructible.
- (iii) Consequently, if G is quasi-separated over S , then $\underline{G}^0$ is locally constructible.

We first prove the first assertion. Since

$$\pi : G \longrightarrow S$$

is locally of finite presentation, given $s \in S$, there are an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s such that

$$\pi(U) \subset V$$

²⁴Editor's note: we recall the following definitions and results (cf. EGA 0_{III}, §9.1, and EGA IV₁, §§1.8 and 1.9). Let X be a topological space. (i) An open subset U of X is called *retrocompact* if the inclusion $U \hookrightarrow X$ is quasi-compact, i.e. if $U \cap V$ is quasi-compact for every quasi-compact open subset $V \subset X$. (ii) A subset C of X is called *constructible* if it is a finite union of subsets of the form $U - V$, where U and V are retrocompact open subsets of X . (iii) A subset L of X is called *locally constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $L \cap U$ is constructible in U . (If X is quasi-compact and quasi-separated, then L is constructible.) (iv) A subset I of X is called *ind-constructible* if, for every $x \in X$, there is an open neighborhood U of x in X such that $I \cap U$ is a union of locally constructible subsets of U . Now let $f : X \rightarrow Y$ be a morphism of schemes. By Chevalley's constructibility theorem (cf. EGA IV₁, 1.8.4 and 1.9.5(viii)), if f is of finite presentation (respectively, locally of finite presentation), then the image under f of every locally constructible subset is locally constructible (respectively, ind-constructible).

and the induced morphism

$$\pi' : U \longrightarrow V$$

is of finite presentation. Then

$$T = \epsilon^{-1}(U)$$

is an open subset of S . Set

$$W = (\pi')^{-1}(T), \quad \pi'' = \pi'|_W.$$

Thus $\pi'' : W \rightarrow T$ is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ .

For every $t \in T$, since $\underline{G}_t^0$ is irreducible (VI_A, 2.4), $W \cap \underline{G}_t^0$ is dense in $\underline{G}_t^0$, hence irreducible and therefore connected. It is thus the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. It follows from EGA IV₃, 9.7.12, that the union W^0 of the $W \cap \underline{G}_t^0$, for $t \in T$, is locally constructible in W . On the other hand, VI_A, 0.5, gives

$$\underline{G}^0 \cap \pi^{-1}(T) = W^0 \cdot W^0;$$

that is, $\underline{G}^0 \cap \pi^{-1}(T)$ is the image of

$$W^0 \times_T W^0$$

under the morphism

$$\mu'' : W \times_T W \longrightarrow \pi^{-1}(T)$$

induced by μ .²⁵ Since $W \times_T W$ (respectively, $\pi^{-1}(T)$) is of finite presentation (respectively, locally of finite presentation) over T , the morphism μ'' is locally of finite presentation and quasi-separated, by EGA IV₁, 1.4.3 and 1.2.2. If, moreover, $\pi^{-1}(T)$ is quasi-separated over T , then μ'' is quasi-compact (loc. cit., 1.2.4), hence of finite presentation. Since $W^0 \times_T W^0$ is locally constructible in $W \times_T W$ (because W^0 is locally constructible in W), Chevalley's constructibility theorem (loc. cit., 1.8.4 and 1.9.5(viii)) shows that $\underline{G}^0 \cap \pi^{-1}(T)$ is ind-constructible in $\pi^{-1}(T)$, and is locally constructible in $\pi^{-1}(T)$ if G , and hence $\pi^{-1}(T)$, is quasi-separated over T . This proves assertions (i) and (iii).

Now suppose that G is quasi-separated over S , and that S is quasi-compact and quasi-separated. By 3.5, there is a quasi-compact open subset U of G containing $\underline{G}^0$. Since G is quasi-separated over S , G is quasi-separated; hence the open subset U is retrocompact (EGA IV₁, 1.2.7), and it suffices to show that $\underline{G}^0$ is constructible in U (EGA 0_{III}, 9.1.8).

Moreover, U is quasi-compact, hence quasi-compact over S (EGA IV₁, 1.2.4), and is quasi-separated over S . Therefore the restriction of π to U is of finite presentation. By EGA IV₃, 9.7.12, $\underline{G}^0$ is locally constructible in U , hence constructible in U , since U is quasi-compact and quasi-separated (EGA IV₁, 1.8.1). This proves (ii), and it follows that, for every quasi-compact and quasi-separated open subset T of S (for example, every affine open subset of S),

$$\underline{G}^0 \cap \pi^{-1}(T)$$

is constructible.

²⁵Editor's note: we have expanded the original in what follows.

Corollary 3.8.— *Let S_0 be a quasi-compact and quasi-separated scheme, I an increasing filtered preordered set, $(\mathcal{A}_i)_{i \in I}$ a direct system of commutative quasi-coherent $\mathcal{O}_{S_0}$ -algebras,*

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \quad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \quad S = \operatorname{Spec} \mathcal{A}$$

(cf. EGA II, 1.3.1). *Let G be an S_0 -group scheme locally of finite presentation. Then the canonical map*

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is bijective.

Since G is locally of finite presentation over S_0 ,* the canonical map

$$\varinjlim_i G(S_i) \longrightarrow G(S)$$

is bijective, by EGA IV₂, 8.14.2(c). It follows immediately that the canonical map

$$\varinjlim_i G^0(S_i) \longrightarrow G^0(S)$$

is injective. We prove that it is surjective. Let

$$g \in G^0(S) \subset G(S).$$

There is an $i \in I$ such that g factors, by means of an element $g_i \in G(S_i)$, through $S \rightarrow S_i$. By hypothesis,

$$g^{-1}(\underline{G}^0) = S.$$

But $\underline{G}^0$ is ind-constructible in G , by 3.7, so

$$g_i^{-1}(\underline{G}^0)$$

is ind-constructible in S_i . It follows from EGA IV₂, 8.3.4, that there is an index $j \geq i$ such that

$$g_j^{-1}(\underline{G}^0) = S_j,$$

where g_j is the map induced by g_i under the base change $S_j \rightarrow S_i$. Thus $g_j \in G^0(S_j)$, and hence g comes from an element of

$$\varinjlim_i G^0(S_i).$$

Proposition 3.9.— *Let G be an S -group locally of finite presentation. Suppose that G^0 is representable. Then the canonical morphism*

$$i : G^0 \longrightarrow G$$

is an open immersion; moreover, G^0 is quasi-compact over S .

Since G^0 is a subfunctor of the functor G , the morphism i is a monomorphism, hence is radicial. From the definition of the functor G^0 , one checks immediately from the definition (EGA IV₄, 17.1.1) that i is formally étale (noting that G^0 is the image of i in G). Finally, by the characterization (EGA IV₃, 8.14.2(c)) of S -schemes locally of finite presentation in terms of the functors they represent, and by 3.8, G^0 is locally of finite presentation over S , since G is. Hence i is locally of finite presentation (cf. EGA IV₁, 1.4.3). It is therefore a radicial étale morphism, and hence an open immersion (EGA IV₄, 17.9.1).

The last assertion follows from 3.6.

Theorem 3.10.— *Let G be an S -group. The following conditions are equivalent:*

*Translator's note: the French source prints "over S " here. The scheme G was defined over S_0 , while S and the S_i are the test schemes of the direct system; the required base is therefore S_0 .

- (i) G is smooth over S at the points of the identity section.
- (ii) G is flat and locally of finite presentation over S at the points of the identity section, and, for every $s \in S$, G_s is smooth over $\kappa(s)$.
- (iii) There is an open group subscheme G' of G that is smooth over S .
- (iv) G^0 is representable by an open subscheme of G that is smooth over S .

It is clear that

$$(iv) \implies (iii) \implies (i),$$

and, by 1.3.1 and 2.4, (i) implies both (ii) and (iii). Moreover, (ii) implies (i) by EGA IV₄, 17.5.1.

It remains to prove that (iii) implies (iv). Lemma 3.10.1 below shows that G' contains $\underline{G^0}$ and that

$$G'^0 = G^0.$$

It therefore suffices to show that $\underline{G^0}$ is open in G , for we have already seen in 3.4 that G^0 will then be represented by the smooth group subscheme induced by G' on the open subset

$$\underline{G'^0} = \underline{G^0}.$$

We may thus suppose that $G' = G$.

To show that $\underline{G^0}$ is open, it suffices to show that every $s \in S$ has a neighborhood T in S such that

$$\underline{G^0} \cap \pi^{-1}(T)$$

is open in $\pi^{-1}(T)$. Let $s \in S$. Since $G = G'$, the morphism π is locally of finite presentation. As in the proof of 3.7, we may therefore construct an open subset T of S containing s , and an open subset W of G containing $\epsilon(s)$, such that the induced morphism

$$\pi'' : W \longrightarrow T$$

is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ . For every $t \in T$,

$$W \cap \underline{G_t^0}$$

is then the connected component of $(\pi'')^{-1}(t)$ containing $\epsilon''(t)$. Since π is smooth, so is π'' , which is therefore smooth and of finite presentation. By EGA IV₃, 15.6.5, the union W^0 of the $W \cap \underline{G_t^0}$, for $t \in T$, is open in W .

On the other hand, VI_A, 0.5, gives

$$W^0 \cdot W^0 = \underline{G^0} \cap \pi^{-1}(T),$$

and we must show that this is open in $\pi^{-1}(T)$. We may now suppose that $T = S$; it remains to show that $W^0 \cdot W^0$ is open in G . Since π is universally open, so is μ (VI_A, 0.1). Hence, since W^0 is open in G , so is

$$W^0 \cdot W^0 = \mu(W^0 \times_S W^0).$$

This result will be strengthened in 4.4.

Lemma 3.10.1.— *Let G be an S -group whose fibers are locally of finite type. Then every open group subscheme H of G contains $\underline{G}^0$ and satisfies*

$$G^0 = H^0.$$

Let $s \in S$, and set

$$G'_s = H_s \cap \underline{G}_s^0.$$

Then G'_s is an open subset of $\underline{G}_s^0$, and is dense in $\underline{G}_s^0$ since $\underline{G}_s^0$ is irreducible (VI_A, 2.4). Hence

$$G'_s \cdot G'_s = \underline{G}_s^0$$

by VI_A, 0.5, which shows that

$$\underline{G}_s^0 = G'_s \cdot G'_s \subset H_s \cdot H_s = H_s.$$

Thus $\underline{G}_s^0 = \underline{H}_s^0$ for every s , whence

$$\underline{G}^0 \subset H, \quad G^0 = H^0.$$

Proposition 3.11.—²⁶ *Let $u : G \rightarrow H$ be a morphism between S -groups locally of finite presentation. If u is flat, the induced map*

$$u^0 : G^0 \longrightarrow H^0$$

is surjective. The converse holds if G is flat over S and H has reduced fibers.

Suppose that u is flat. Then, for every $s \in S$, u_s is flat and locally of finite presentation, hence open (EGA IV₂, 2.4.6). Therefore

$$u_s^0 : \underline{G}_s^0 \longrightarrow \underline{H}_s^0$$

is surjective, by 1.3.2.²⁷ Thus the map

$$u^0 : G^0 \longrightarrow H^0$$

is surjective.

Conversely, suppose that $u^0 : G^0 \rightarrow H^0$ is surjective, that G is flat over S , and that H has reduced fibers. Then, for every $s \in S$, the morphism

$$u_s^0 : \underline{G}_s^0 \longrightarrow \underline{H}_s^0$$

is surjective, and hence is flat at every point lying above the generic point η of $\underline{H}_s^0$, since

$$\mathcal{O}_{\underline{H}_s^0, \eta}$$

is a field. It follows from 1.3 that u_s is flat. Therefore u is flat, by the fiberwise flatness criterion (EGA IV₃, 11.3.11).

²⁶Editor's note: compare VI_A, 5.6.

²⁷Editor's note: we have shortened the original by referring here to 1.3.2. We have also simplified the argument below by omitting a reference to the generic flatness theorem (EGA IV₂, 6.9.1), which is not needed here.

4. Dimension of the fibers of groups locally of finite presentation

Proposition 4.1.— *Let G be an S -scheme locally of finite type, equipped with an S -section ϵ , and suppose that, for every $s \in S$,*

$$\dim G_s = \dim_{\epsilon(s)} G_s$$

(as is the case when G is an S -group; 1.5).

(i) *The function*

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S .

(ii) *If, moreover, G is locally of finite presentation over S , this function is locally constructible.*

Let

$$\pi : G \longrightarrow S$$

be the structural morphism. Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3) asserts that the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x))$$

is upper semicontinuous on G . For every $s \in S$, however,

$$\dim G_s = \dim \pi^{-1}(s) = \dim_{\epsilon(s)} \pi^{-1}(\pi(\epsilon(s))).$$

Since the map $s \mapsto \epsilon(s)$ is continuous on S , the composite function

$$s \longmapsto \dim G_s$$

is upper semicontinuous on S .

Now suppose that G is locally of finite presentation over S . Arguing as above, to show that $s \mapsto \dim G_s$ is locally constructible, it is enough to show that the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x))$$

is locally constructible on G , which follows from EGA IV₃, 9.9.1.

Proposition 4.2.— *Let $\pi : G \rightarrow S$ be an S -scheme locally of finite presentation, equipped with an S -section ϵ , and satisfying the following two conditions (which hold when G is an S -group, by 1.5 and VI_A, 2.4.1):*

(a) *For every $s \in S$ and every $x \in G_s$, one has*

$$\dim G_s = \dim_x G_s$$

(or equivalently (1.5), for every $s \in S$, all the irreducible components of G_s have the same dimension).

(b) *For every $s \in S$, if $\underline{G}_s^0$ denotes the connected component of G_s containing $\epsilon(s)$, then $\underline{G}_s^0$ is geometrically irreducible.*

Let $s \in S$. The following conditions are equivalent:

- (i) G is universally open over S at the points of $\underline{G}_s^0$.
 (i bis) G is universally open over S at every point of a neighborhood of $\epsilon(s)$ in $\underline{G}_s^0$.²⁸

(ii) The function

$$t \mapsto \dim G_t$$

is constant in a neighborhood of s in S .

- (iii) $\underline{G}^0$ is “universally open over S at the points of $\underline{G}_s^0$,” that is, given $S' \rightarrow S$, $s' \in S'_s$, $g \in \underline{G}_{s'}^0$, and an open neighborhood V of g in $G' = G_{S'}$, then

$$\pi(V \cap \underline{G}'^0)$$

is an open neighborhood of s' in S' .²⁹

Clearly (i) implies (i bis). By EGA IV₃, 14.3.3.1(ii), the set of points of $\underline{G}_s^0$ at which π_G is universally open is closed in $\underline{G}_s^0$. Since $\underline{G}_s^0$ is irreducible, it follows that (i bis) implies (i).

We show that (i) implies (ii). Let T be the set of $t \in S$ such that

$$\dim G_t = \dim G_s.$$

By 4.1, T is locally constructible; hence, by EGA IV₁, 1.10.1, to show that T is a neighborhood of s , it is enough to show that every generalization s' of s belongs to T .

Let x be the generic point of $\underline{G}_s^0$, and let U be an open subset of G containing x . Since π_G is universally open at x , EGA IV₃, 14.3.13 gives, for every generalization s' of s ,

$$\dim(U \cap G_{s'}) \geq \dim_x(U \cap G_s).$$

In view of hypothesis (a), this implies

$$\dim G_{s'} \geq \dim G_s.$$

On the other hand, by 4.1 the function $s \mapsto \dim G_s$ is upper semicontinuous, so that

$$\dim G_{s'} \leq \dim G_s.$$

Thus $s' \in T$, proving that (i) implies (ii).

It is clear that (iii) implies (i); we prove that (ii) implies (iii). The dimension of the fibers is unchanged by extension of the base field, and the formation of $\underline{G}^0$ commutes with base change (see the proof of 3.3). We may therefore assume that $S' = S$ and $s' = s$. Moreover, every open subset V of G meeting $\underline{G}_s^0$ contains the generic point η of $\underline{G}_s^0$, so we may assume that $g = \eta$.

We may further suppose that S is affine. Let U be an affine open neighborhood of $\epsilon(s)$; it is then of finite presentation over S . Replacing S by $\epsilon^{-1}(U)$, and then U by $U \cap \pi^{-1}(S)$, we reduce to the case in which

$$\pi : U \longrightarrow S$$

is of finite presentation and admits $\epsilon : S \rightarrow U$ as a section. By EGA IV₃, 9.7.12, $\underline{G}^0 \cap U$ is constructible in U . Replacing V by an affine open subset contained in $V \cap U$, we find

²⁸Editor’s note: the original reads “at the points of $\epsilon(s)$.” It is not enough to suppose that G is universally open over S at $\epsilon(s)$, as the example given in editor’s note 19 shows. We have corrected the proof accordingly.

²⁹Editor’s note: we have added condition (iii), pointed out by O. Gabber; it will be useful later (5.7).

that $\underline{G}^0 \cap V$ is constructible in V . Since $\pi : V \rightarrow S$ is of finite presentation, Chevalley's constructibility theorem (EGA IV₁, 1.8.4) shows that

$$\pi(\underline{G}^0 \cap V)$$

is locally constructible in S .

Consequently, by *loc. cit.*, 1.10.1, to show that $\pi(\underline{G}^0 \cap V)$ is an open neighborhood of s , it is enough to show that, for every generalization t of s , there is a generalization ξ of η belonging to $\underline{G}^0$, and hence to $\underline{G}^0 \cap V$. The generic point ξ of $\underline{G}_t^0$ is such a generalization of η . Indeed, $\epsilon(s)$ belongs to the closure X of $\{\xi\}$ in G ; hence, by Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3),

$$\dim_{\epsilon(s)} X_s \geq \dim_{\xi} X_t.$$

On the other hand, hypothesis (ii) implies

$$\dim_{\xi} \underline{G}_t^0 = \dim_{\epsilon(s)} G_s.$$

It follows that one of the irreducible components of X_s containing $\epsilon(s)$ is equal to $\underline{G}_s^0$, whence

$$\eta \in \overline{\{\xi\}}.$$

This proves that (ii) implies (iii), and hence proves the proposition.

³⁰One can also prove that (ii) implies (i) as follows. Since π is locally of finite presentation, there exist an open subset U of G containing $\epsilon(s)$ and an open subset V of S containing s , such that $\pi(U) \subset V$ and the morphism

$$\pi' : U \longrightarrow V$$

induced by π is of finite presentation. Set

$$T = \epsilon^{-1}(U), \quad W = \pi'^{-1}(V) = U \cap \pi^{-1}(V).$$

The morphism

$$\pi'' : W \longrightarrow T$$

induced by π' is of finite presentation and admits as a section the morphism

$$\epsilon'' : T \longrightarrow W$$

induced by ϵ . Moreover, for every $t \in T$, since $\underline{G}_t^0$ is irreducible, $W \cap \underline{G}_t^0$ is dense in $\underline{G}_t^0$, hence irreducible and therefore connected. It is consequently the connected component of $\pi''^{-1}(t)$ containing $\epsilon''(t)$.

Since $W \cap \underline{G}_t^0$ is a dense open subset of $\underline{G}_t^0$, 1.5 and EGA IV₂, 5.2.1 give

$$\dim(W \cap \underline{G}_t^0) = \dim \underline{G}_t^0 = \dim G_t.$$

Thus the function

$$t \longmapsto \dim(W \cap \underline{G}_t^0)$$

is constant in a neighborhood of s in T . Finally, we show that $W \cap \underline{G}_t^0$ is geometrically irreducible for every $t \in T$. Let K be an extension of $\kappa(t)$. Then

$$(W \cap \underline{G}_t^0) \otimes_{\kappa(t)} K = (W \otimes_{\kappa(t)} K) \cap (\underline{G}_t^0 \otimes_{\kappa(t)} K)$$

³⁰Editor's note: the preceding argument was communicated to us by O. Gabber. We have also retained the proof of the implication (ii) $\Rightarrow$ (i) given in the original.

is a nonempty open subset of $\underline{G}_t^0 \otimes_{\kappa(t)} K$, and is therefore irreducible because $\underline{G}_t^0 \otimes_{\kappa(t)} K$ is irreducible.

We are now in the situation of EGA IV₃, 15.6.6(ii), which asserts that π'' , and hence π , is universally open at the points of $W \cap \underline{G}_s^0$. But, by EGA IV₃, 14.3.3.1(ii), the subset of G_s consisting of the points at which π is universally open is closed in G_s . Since it contains $W \cap \underline{G}_s^0$, it therefore contains its closure $\underline{G}_s^0$.

Corollary 4.3.— *Let G be a flat S -group locally of finite presentation. Then the function*

$$s \longmapsto \dim G_s$$

is locally constant on S .

This follows immediately from 4.2, since every flat morphism locally of finite presentation is universally open (EGA IV₂, 2.4.6).

Corollary 4.4.— *Let G be an S -group locally of finite presentation over S at the points of the identity section. Consider the following conditions:*

(i) *G is smooth over S at the points of the identity section (cf. 3.10).*

(ii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and the function*

$$s \longmapsto \dim G_s$$

is locally constant on S .

(iii) *For every $s \in S$, G_s is smooth over $\kappa(s)$, and there exists a neighborhood V of the identity section such that*

$$\pi_V : V \longrightarrow S$$

is universally open.

(iv) *For every $s \in S$, G_s^0 is smooth over $\kappa(s)$, and G^0 is representable by an open group subscheme of G , universally open over S .*

Then the following implications hold:

$$(i) \implies (ii) \iff (iii) \iff (iv).$$

If, moreover, S is reduced, then conditions (i)–(iv) are equivalent and imply that G^0 is smooth over S .³¹

Let us show that (i) implies (ii). For every $x \in G$, by 1.5,

$$\dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x)).$$

Consequently, by EGA IV₄, 17.10.2, the function

$$x \longmapsto \dim_x \pi^{-1}(\pi(x)) = \dim \pi^{-1}(\pi(x))$$

³¹Editor's note: the hypothesis that S be reduced cannot be omitted here. Indeed, let k be a field, let $S = \operatorname{Spec} A$, where $A = k[\delta]$ with $\delta^2 = 0$, and let G be the closed subgroup

$$\operatorname{Spec} A[X]/(\delta X)$$

of $\mathbb{G}_{a,S}$ (which assigns to every A -algebra R the subgroup of those $r \in R$ such that $\delta r = 0$). Then G is an S -group satisfying (ii)–(iv), but it is not flat, and hence not smooth, over S .

is continuous in a neighborhood of the identity section. Therefore the function $s \mapsto \dim G_s$ is continuous on S , and hence locally constant on S . On the other hand, for every $s \in S$, G_s is smooth over $\kappa(s)$, by 1.3.1.

Let us show that (ii) implies (iv). It is enough to show that $\underline{G}^0$ is open in G , for then, by 3.4, G^0 will be representable by the group subscheme induced on $\underline{G}^0$, and the properties of G^0 stated above will follow from 2.4 and 4.2. Given $s \in S$, construct, as in the proof of 3.10,

$$W, \quad T, \quad \pi'', \quad \epsilon'', \quad \text{and} \quad W^0.$$

Then, by EGA IV₃, 15.6.7, W^0 is open in W . On the other hand, under hypothesis 4.4(ii), it follows from 4.2 that π is universally open at every point of W^0 . Therefore, by VI_A, 0.1, μ is universally open at every point of

$$W^0 \times_S W^0,$$

which shows that $W^0 \cdot W^0$ is open; the proof is completed as in the proof of Theorem 3.10.

It is clear that

$$(iv) \implies (iii),$$

and (iii) $\implies$ (ii) follows by applying 4.2 to V .

Finally, suppose that (ii)–(iv) hold, and let us show that G^0 is smooth over S when S is reduced.³² For this, we may suppose that $G = G^0$. Then G is of finite presentation over S , by 5.5 below. Thus π_G is of finite presentation, has geometrically integral fibers, and has locally constant relative dimension on S . It follows from EGA IV₃, 15.6.7 that the morphism

$$G \times_S S_{\text{red}} \longrightarrow S_{\text{red}}$$

induced by π_G is flat. Hence π_G is flat when S is reduced. In that case, $G = G^0$ is smooth over S , by Theorem 3.10.

5. Separation of groups and homogeneous spaces

Proposition 5.1.— *For an S -group G to be separated, it is necessary and sufficient that the identity section of G be a closed immersion.*

The condition is necessary (EGA I, 5.4.6); it is sufficient by virtue of the following cartesian diagram (cf. VI_A, 0.3):

$$\begin{array}{ccc} G \times_S G & \xrightarrow{\mu \circ (\text{id}_G \times c)} & G \\ \Delta_{G/S} \uparrow & & \uparrow \epsilon \\ G & \xrightarrow{\pi} & S \end{array}.$$

Proposition 5.2.— *If S is discrete, every S -group is separated.*

Indeed,

$$S = \coprod_{s \in S} \text{Spec } \mathcal{O}_{S,s},$$

and, by EGA I, 5.5.5, it is enough to show that, for every $s \in S$,

$$G \times_S \text{Spec } \mathcal{O}_{S,s}$$

³²Editor's note: we have expanded the original in the argument that follows.

is separated. This follows from VI_A, 0.3, since $\mathcal{O}_{S,s}$ is a local ring of dimension 0.³³

Theorem 5.3.— *Let S be a scheme, let G be an S -group scheme locally of finite presentation over S and universally open over S in a neighborhood of the identity section, and let X be an S -scheme on which G acts in such a way that the morphism*

$$\Phi : G \times_S X \longrightarrow X \times_S X, \quad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose moreover that, for every $s \in S$:

- (i) *there exists an open subscheme U of X , separated over S , such that U_s is dense in X_s ;*
- (ii) *the fiber X_s is locally of finite type over $\kappa(s)$.³⁴*

Then X is separated over S .

Corollary 5.4.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that X has fibers locally of finite type and connected. Then:*

- (i) *X is separated over S .*
- (ii) *If there exists an open subset V of X , quasi-compact over S , which meets every nonempty fiber X_s ,³⁵ then X is quasi-compact over S .*

Proof. (i) Let $s \in S$ be such that $X_s \neq \emptyset$. The morphism

$$\Phi_s : G_s \times_{\kappa(s)} X_s \longrightarrow X_s \times_{\kappa(s)} X_s$$

obtained from Φ by base change is surjective. Since X_s is connected, it is irreducible by VI_A, 2.5.4. Thus, if U is an affine open subset of X such that U_s is nonempty, then U_s is dense in X_s , and the theorem applies.

To prove (ii), we may suppose that S is affine. Then V is quasi-compact and, by 3.5, there exists a quasi-compact open subset U of G containing $\underline{G}^0$. Let $s \in S$ be such that $X_s \neq \emptyset$. Then X_s is irreducible (VI_A, 2.6.6), and, since U_s contains $\underline{G}_s^0$, the morphism

$$U_s \times_{\kappa(s)} V_s \longrightarrow X_s, \quad (g, x) \longmapsto gx,$$

is surjective (VI_A, 2.6.4). Consequently the morphism

$$U \times_S V \longrightarrow X$$

is surjective, and X is therefore quasi-compact, since U and V are. As S was assumed affine, hence separated, it follows that X is quasi-compact over S (cf. EGA I, 6.6.4(v)).

Remark 5.4.1.—³⁶ We shall see in the course of the proof that the conclusion of Theorem 5.3 remains true if one assumes only:

³³Editor's note: we have replaced "Artinian" by "of dimension 0" and mentioned this generalization in VI_A, 0.3.

³⁴Editor's note: we have corrected the original by assuming that G is universally open over S in a neighborhood of the identity section and by adding hypothesis (ii); see below for examples, due to O. Gabber, showing that the statements 5.3 and 5.4 of the original are false without additional hypotheses. We also point out that Theorem 5.3 is a reworked version of Theorem 5.3A below, which appears in the 1965 edition of SGAD and is due to M. Raynaud; compare the notes (*) in Exposé X, 8.5 and 8.8.

Theorem 5.3A (Raynaud).— *Let G be an S -group locally of finite presentation, universally open over S , and with connected fibers. Then G is separated over S . More generally, every S -scheme X locally of finite presentation over S , equipped with an action of G such that the morphism $\Phi : G \times_S X \rightarrow X \times_S X$, $(g, x) \mapsto (gx, x)$, is surjective, is separated over S .*

³⁵Editor's note: without this hypothesis there is the following counterexample, pointed out by O. Gabber. Let $G = S$ be a local scheme with closed point s , such that $S^* = S - \{s\}$ is not quasi-compact, and let X be the disjoint union of $\{s\}$ and S^* .

³⁶Editor's note: we have added this remark; compare editor's note 34.

(i') there exists an open subscheme U of X , separated over S , such that U_s is dense in every irreducible component of X_s .

This follows from (i) if X_s locally has a finite number of irreducible components, as is the case under hypothesis (ii). We shall also see later that 5.4 remains true under the hypothesis that every fiber X_s is quasi-separated and connected.

Corollary 5.5.— *Let S be a scheme, and let G be an S -group scheme, locally of finite presentation, with connected fibers, and universally open over S . Then G is separated and of finite presentation over S .³⁷*

Indeed, by 3.6 and 5.4, G is quasi-compact and separated over S . Since G is locally of finite presentation over S , it is therefore of finite presentation over S .

5.6.— *Proof of Theorem 5.3.* Before proving 5.3, let us establish a few lemmas.

Lemma 5.6.0.—³⁸

(i) *Let $A \subset B$ be integral domains, with B integral over A , and let $\mathfrak{q} \in \operatorname{Spec}(B)$ be such that A is unibranch at $\mathfrak{p} = \mathfrak{q} \cap A$. Then the morphism*

$$\pi : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(A)$$

is open at $\mathfrak{q}$.

(ii) *Let X, Y be irreducible preschemes, let $f : X \rightarrow Y$ be a dominant morphism, and let $x \in X$ be a point at which f is quasi-finite, such that $y = f(x)$ is a unibranch point of Y . Then f is open at x . In particular, if*

$$Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$$

is a local prescheme with closed point y , then $f(U) = Y$ for every neighborhood U of x .

(i) Let K (respectively, L) be the fraction field of A (respectively, B), let A' be the normalization of A , and let B' be the subring of L generated by A' and B . Then B' is integral over A' . Set

$$Y = \operatorname{Spec}(A), \quad X = \operatorname{Spec}(B), \quad Y' = \operatorname{Spec}(A'), \quad X' = \operatorname{Spec}(B').$$

We then have the commutative diagram

$$\begin{array}{ccc} X' & \xrightarrow{\pi'} & Y' \\ \downarrow & & \downarrow \phi \\ X & \xrightarrow{\pi} & Y \end{array}$$

³⁷Editor's note: we point out here the following result ([Ray70a], VI, 2.5). If S is normal, G is smooth with connected fibers, and X is an (fppf) homogeneous G -space—that is, the morphisms $X \rightarrow S$ and $G \times_S X \rightarrow X \times_S X$ are coverings for the (fppf) topology—locally of finite type over S , then X is locally quasi-projective over S . In particular, G is quasi-projective over S . See also editor's note 35 in VI_A.

³⁸Editor's note: we have added this lemma, communicated by O. Gabber. It improves EGA IV₃, 14.4.1.2 and corrects the proof of *loc. cit.*, 14.4.1.3 without changing its hypotheses (compare the erratum (ErrIV, 38) in EGA IV₄).

in which all morphisms are integral and surjective.

Since A is unibranch at $\mathfrak{p}$, Y' has a unique point $\mathfrak{p}'$ above $\mathfrak{p}$. Consequently, if U is an open neighborhood of $\mathfrak{p}'$ in Y' , then

$$\phi(Y' - U)$$

is a closed subset not containing $\mathfrak{p}$, so that its open complement is contained in $\phi(U)$. This shows that $\phi : Y' \rightarrow Y$ is open at $\mathfrak{p}'$. It is therefore enough to show that π' is open, and we are reduced to the case where $A = A'$ is normal.

Let N be a quasi-Galois extension of K containing L , let

$$\tilde{X} = \operatorname{Spec}(\tilde{B}),$$

where $\tilde{B}$ is the integral closure of B in N , and let $G = \operatorname{Aut}(N/K)$. Since $\tilde{X} \rightarrow X$ is surjective, it is enough to show that

$$\tilde{\pi} : \tilde{X} \rightarrow Y$$

is open. Let U be an open subset of $\tilde{X}$, and set

$$U' = \bigcup_{g \in G} gU.$$

Because G acts transitively on the fibers of $\tilde{X} \rightarrow Y$ (cf. [BAC], Chap. V, §2.3, Prop. 6), one has

$$\tilde{\pi}(U) = \tilde{\pi}(U'),$$

and the latter is the open complement of the closed subset

$$\tilde{\pi}(\tilde{X} - U').$$

This proves (i).

(ii) We may suppose that Y and X are reduced. By Zariski's Main Theorem (cf. EGA IV₃, 8.12.9), there exist affine open neighborhoods U of x and $V = \operatorname{Spec}(A)$ of y , such that $f(U) \subset V$, and a factorization

$$\begin{array}{ccc} U & \xhookrightarrow{j} & V' \\ & \searrow f & \downarrow u \\ & & V \end{array},$$

where j is an open immersion and u is finite. Replacing V' by the closure of $j(U)$, we may suppose that V' is irreducible, hence that $V' = \operatorname{Spec}(B)$, where B is an integral A -algebra, finite over A . Moreover, since f is dominant, so is u , and the morphism $A \rightarrow B$ is therefore injective.

Since A is unibranch at y , it follows from (i) that u is open at $j(x)$, and hence that

$$f = u \circ j$$

is open at x . This proves the first assertion of (ii). The second follows because, if Y is a local prescheme with closed point y , every open subset containing y equals Y . In the case $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$, one may also use, instead of EGA IV₃, 8.12.9, the local form of Zariski's Main Theorem found, for example, in [Pes66] or [Ray70b], Chap. IV, Th. 1.

Lemma 5.6.1.0.—³⁹ Let $f : X \rightarrow S$ be a morphism locally of finite presentation, let $s \in S$, and let $x \in X_s$. Set $n = \dim_x(X_s)$, and let $q : \mathbb{A}_S^n \rightarrow S$ be the structure morphism.

Suppose that a quasi-finite morphism $u : X \rightarrow \mathbb{A}_S^n$ is given such that $f = q \circ u$, and suppose that f is universally open at the generic point z of an irreducible component $\mathfrak{z}$ of X_s , containing x and of dimension n . Then u is universally open at the point $x \in X$.

First observe that

(†) the hypotheses are preserved under every base change $\pi : S' \rightarrow S$ covering s

(that is, such that $\pi^{-1}(s) \neq \emptyset$). Indeed, let $S' \rightarrow S$ be such a morphism, let

$$\begin{array}{ccc} X' & \xrightarrow{u'} & \mathbb{A}_{S'}^n \\ & \searrow f' & \downarrow q' \\ & & S' \end{array}$$

be the diagram obtained by base change, and let s' be a point of S' above s and x' a point of $X'_{s'}$ above x . Since

$$X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s'),$$

the lifting of generizations and the invariance of dimension under field extension (EGA IV₂, 2.3.4(i) and 4.1.4) show that x' is contained in an irreducible component $\mathfrak{z}'$ of $X'_{s'}$ whose generic point z' lies above z , and

$$n \leq \dim \mathfrak{z}' \leq \dim_{x'} X'_{s'} \leq n.$$

Thus

$$\dim \mathfrak{z}' = n = \dim_{x'} X'_{s'}.$$

Since f is universally open at z , f' is universally open at z' .¹

Set $Y = \mathbb{A}_S^n$. By EGA IV₃, 14.3.3.1(i), to prove that u is universally open at x , it is enough to prove that, for every integer $r \geq 0$ and every point x' of

$$X' = X[T_1, \dots, T_r]$$

above x , the morphism

$$u' : X' \longrightarrow Y' = \mathbb{A}_S^n[T_1, \dots, T_r]$$

is open at x' . Now set

$$S' = S[T_1, \dots, T_r],$$

and let q' be the projection

$$Y' \simeq \mathbb{A}_{S'}^n \longrightarrow S'.$$

This is the situation obtained by the base change $S' \rightarrow S$. Thus, by the preceding observation, we are reduced to showing that u is open at x .

Set $y = u(x)$. Since u is locally of finite presentation (because f and q are; cf. EGA IV₁, 1.4.3), EGA IV₁, 1.10.3 shows that it is enough to prove

$$u(\operatorname{Spec} \mathcal{O}_{X,x}) = \operatorname{Spec} \mathcal{O}_{Y,y}.$$

³⁹Editor's note: we have made explicit this lemma, which is used in the proof of Lemma 5.6.1.

¹Translator's note: the French source prints x and x' , respectively, in this sentence. The stated hypothesis and the immediately preceding choice of generic points require z and z' .

For this purpose we may assume that S is affine and integral. Let $\pi : S' \rightarrow S$ be its normalization, and denote by $u' : X' \rightarrow Y'$ the morphism obtained from u by base change. Since

$$\pi_Y : Y' \longrightarrow Y$$

is integral and surjective, one has

$$\mathrm{Spec} \mathcal{O}_{Y,y} = \bigcup_{y'} \pi_Y(\mathrm{Spec} \mathcal{O}_{Y',y'}),$$

where the union ranges over all points y' of Y' above y . It is therefore enough to show that, for every such y' and every $x' \in X'_{y'}$ above x ,

$$u'(\mathrm{Spec} \mathcal{O}_{X',x'}) = \mathrm{Spec} \mathcal{O}_{Y',y'}.$$

Since the hypotheses are preserved under the base change $\pi : S' \rightarrow S$, we are thereby reduced to the case of S' ; that is, we may assume that S is integral and normal.

The hypotheses on f now imply, by EGA IV₃, 14.3.13, that there is an irreducible component Z of X , containing $\mathfrak{z}$ (and hence x), dominating S , and such that

$$\dim_z(Z_s) = n = \dim(Z_\eta),$$

where η is the generic point of S . Let ξ be the generic point of Z . Since u , and hence also

$$u_\eta : Z_\eta \longrightarrow \mathbb{A}_\eta^n,$$

is quasi-finite, the closure in $\mathbb{A}_\eta^n$ of the point

$$u_\eta(\xi) = u(\xi)$$

has dimension n . Thus $u(\xi)$ is the generic point of $\mathbb{A}_\eta^n$, which is also the generic point of $Y = \mathbb{A}_S^n$. Consequently, if g denotes the restriction of u to Z , then

$$g : Z \longrightarrow Y$$

is quasi-finite and dominant. Since Y is normal, Lemma 5.6.0 shows that g is open. Hence u is open at every point of Z , and in particular at x . Consequently,

$$u(\mathrm{Spec} \mathcal{O}_{X,x}) = \mathrm{Spec} \mathcal{O}_{Y,y},$$

which completes the proof of Lemma 5.6.1.0.

The following lemma advantageously replaces EGA IV₃, 14.5.10,⁴⁰ in that it is independent of noetherian hypotheses.

Lemma 5.6.1.— *Let S be a scheme, let $f : X \rightarrow S$ be an S -scheme locally of finite presentation, let s be a point of S , and let x be a closed point of X_s . Suppose that f is universally open at the generic point z of an irreducible component Z of X_s containing x , and that*

$$\dim_x(X_s) = \dim Z.$$

Then there exists a commutative diagram

⁴⁰Editor's note: we have corrected the original, which indicated 19.5.10.

$$\begin{array}{ccc}
X & \xrightarrow{f} & S \\
\uparrow h & \nearrow \phi & \uparrow w \\
S'' & \xrightarrow{\pi} & S'
\end{array} ,$$

where S' is affine, w is étale, π is finite, surjective, and of finite presentation, and $\phi^{-1}(s)$ consists of a single point s'' , such that $h(s'') = x$ and $\kappa(s'') = \kappa(x)$.⁴¹

Proof. First, we may suppose that $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$, where B is an A -algebra of finite presentation. Let $\mathfrak{p}$ and $\mathfrak{q}$ be the prime ideals of A and B corresponding respectively to s and x , so that

$$\mathfrak{p} = A \cap \mathfrak{q}.$$

Set $n = \dim_x(X_s)$, and choose $t_1, \dots, t_n \in B$ whose images in

$$\mathcal{O}_{X_s, x} = B_{\mathfrak{q}} / \mathfrak{p}B_{\mathfrak{q}}$$

form a system of parameters. Then

$$\mathcal{O}_{X_s, x} / (t_1, \dots, t_n)$$

is finite-dimensional over $\kappa(x)$, and hence also over $\kappa(s)$, because x is a closed point of the algebraic $\kappa(s)$ -scheme X_s . Consequently, the S -morphism

$$u : X \longrightarrow \mathbb{A}_S^n = \operatorname{Spec}(A[T_1, \dots, T_n]), \quad T_i \longmapsto t_i,$$

is of finite presentation, x is isolated in the fiber $u^{-1}(u(x))$, and

$$u(x) = \tau_0(s),$$

where $\tau_0 : S \rightarrow \mathbb{A}_S^n$ denotes the zero section of $\mathbb{A}_S^n \rightarrow S$, corresponding to the A -algebra homomorphism

$$A[T_1, \dots, T_n] \longrightarrow A, \quad T_i \longmapsto 0.$$

The set of points of X that are isolated in their fibers over $\mathbb{A}_S^n$ is open (EGA IV₃, 13.1.4). Thus, after shrinking X , we may suppose that u is quasi-finite and that $u^{-1}(u(x)) = \{x\}$. By Lemma 5.6.1.0, u is universally open at x .

Let

$$B_0 = B / (t_1, \dots, t_n), \quad X_0 = X \times_{\mathbb{A}_S^n} \tau_0(S) = \operatorname{Spec}(B_0),$$

and let $u_0 : X_0 \rightarrow S$ be obtained from u by the base change $\tau_0 : S \hookrightarrow \mathbb{A}_S^n$. Then u_0 is quasi-finite and of finite presentation, is universally open at x , and x is the unique point of X_0 above s .

Let A' be the henselization of the local ring

$$A_{\mathfrak{p}} = \mathcal{O}_{S, s},$$

and set

$$S' = \operatorname{Spec}(A'), \quad B'_0 = B_0 \otimes_A A', \quad X'_0 = X_0 \times_S S' = \operatorname{Spec}(B'_0).$$

The closed point s' of S' is the unique point of S' above s , and $\kappa(s') = \kappa(s)$. The scheme X'_0 has a unique point x' above s' ; moreover $\kappa(x') = \kappa(x)$, and x' is also the unique point of X'_0

⁴¹Editor's note: consequently, h induces a section σ of $X \times_S S'' \rightarrow S''$ such that $\sigma(s'')$ lies above x ; compare EGA IV₃, 14.5.10.

above both s and x . Since A' is henselian, EGA IV₄, 18.5.11 gives a disjoint decomposition into open and closed parts

$$X'_0 = V \amalg W, \quad V = \operatorname{Spec}(\mathcal{O}_{X'_0, x'}). \quad (*)$$

The local ring $\mathcal{O}_{X'_0, x'}$ is finite and of finite presentation over A' . Hence the restriction π of u'_0 to V is finite and of finite presentation. Moreover, because $u'_0 : X'_0 \rightarrow S'$ is open at x' ,

$$u'_0(V) = \operatorname{Spec}(\mathcal{O}_{S', s'}) = S',$$

so π is surjective.

This proves the assertion when $S = S'$. In general, A' is the filtered direct limit of A -subalgebras A_i that are étale over A and for which

$$S_i = \operatorname{Spec}(A_i)$$

has a unique point s_i above s , with $\kappa(s_i) = \kappa(s)$. We have

$$B'_0 = \varinjlim_i B_i, \quad B_i = B_0 \otimes_A A_i.$$

Set

$$X_i = \operatorname{Spec}(B_i) = X_0 \times_S S_i, \quad C = \mathcal{O}_{X'_0, x'}.$$

By $(*)$, one has

$$C \simeq B'_0 / f B'_0$$

for some idempotent $f \in B'_0$. There are an index i and an element $f_i \in B_i$ whose image in B'_0 is f . Put

$$C_i = B_i / f_i B_i, \quad V_i = \operatorname{Spec}(C_i).$$

Then

$$C = C_i \otimes_{A_i} A', \quad V = V_i \times_{S_i} S',$$

and C_i is an A_i -algebra of finite presentation, since the same is true of B_i . Consequently, the morphisms

$$\begin{array}{ccc} X'_0 & & \\ \uparrow h' & \searrow u'_0 & \\ V & \xrightarrow{\pi} & S' \end{array}$$

arise, by the base change $S' \rightarrow S_i$, from morphisms of finite presentation

$$\begin{array}{ccc} X_i & & \\ \uparrow h_i & \searrow u_i & \\ V_i & \xrightarrow{\pi_i} & S_i \end{array}$$

For every $j \geq i$, let

$$V_j = V_i \times_{S_i} S_j$$

and let $\pi_j : V_j \rightarrow S_j$ be the morphism of finite presentation obtained from π_i by base change. Since $\pi : V \rightarrow S'$ is finite and surjective, EGA IV₂, 8.10.5 shows that, for some j , the morphism $\pi_j : V_j \rightarrow S_j$ is finite and surjective. Then $w : S_j \rightarrow S$ is affine and étale, S_j has a unique point s_j above s , and V_j has a unique point x_j above s_j , since x' is the unique point of

$$V = V_j \times_{S_j} S'$$

above s' :

$$\begin{array}{ccccccc}
X_j & \longrightarrow & X_0 & \longrightarrow & X & & \\
\uparrow h_j & & \searrow u_j & & \searrow u_0 & \searrow u & \\
V_j & \xrightarrow{\pi_j} & S_j & \xrightarrow{w} & S & \xrightarrow{\tau_0} & \mathbb{A}_S^n
\end{array}$$

Thus x_j is the unique point of V_j above s , its image under $V_j \rightarrow X_j \rightarrow X$ is x , and

$$\kappa(x_j) = \kappa(x).$$

Lemma 5.6.2.0.—⁴² *Let k be a field and let G be a nonempty k -scheme of finite type.*

- (i) *Let K be an extension of k , and let W be a dense open subset of G_K . Denote by π the projection $G_K \rightarrow G$. Then*

$$U = \{g \in G \mid W \text{ contains every maximal point of } \pi^{-1}(g)\}$$

is a dense open subset of G . (N.B. If g is a closed point of G belonging to U , then $\pi^{-1}(g) \subset W$.)

- (ii) *Suppose in addition that G is geometrically irreducible. Let*

$$\mu : G \times X \longrightarrow Y$$

be a morphism of k -schemes, let x be a point of X , and let Ω be an open subset of Y such that

$$\mu_x^{-1}(\Omega_{\kappa(x)}) \neq \emptyset,$$

where μ_x denotes the morphism

$$G_{\kappa(x)} \longrightarrow Y_{\kappa(x)}, \quad g \longmapsto \mu(g, x).$$

Then

$$U = \{g \in G \mid \mu \text{ sends every maximal point of } g \times x \text{ into } \Omega\}$$

is a dense open subset of G . For every closed point g of G belonging to U , one has

$$\mu(g \times x) \subset \Omega,$$

and hence

$$\mu(g', x) \in \Omega_{\kappa(x)} \quad (\text{respectively, } \mu(g, x') \in \Omega_{\kappa(g)}),$$

for every point $g' \in G_{\kappa(x)}$ above g (respectively, $x' \in X_{\kappa(g)}$ above x).

- (iii) *Suppose that $G = H^0$, where H is a k -group scheme locally of finite type, acting on a nonempty k -scheme X in such a way that the morphism*

$$H \times_S X \longrightarrow X \times_S X, \quad (h, x) \longmapsto (hx, x)$$

is surjective. Let U be an open subset of X . Then:

- (a) *$G \cdot U$ is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .*

⁴²Editor's note: we have added this lemma, communicated by O. Gabber. It makes it possible to simplify the proof of 5.6.2 and to prove Theorem 5.3, as well as 5.4, in a more general form; see 5.7 and 5.8 below.

- (b) Suppose that U is dense in every irreducible component of X . Then, for every finite subset A of X , there exists a closed point $g \in G$ such that

$$g \cdot A' \subset U_{\kappa(g)},$$

where A' is the inverse image of A in $X_{\kappa(g)}$. In particular, the morphism

$$G \times U \longrightarrow X$$

is surjective.

Proof. (i) Let $\bar{k}$ be a separable closure of k , and let L be an extension of k containing a copy of $\bar{k}$ and of K . Denote by π_L (respectively, ϕ) the projection

$$G_L \longrightarrow G \quad (\text{respectively, } G_L \longrightarrow G_K).$$

For every $g \in G$, the map ϕ sends the maximal points of $\pi_L^{-1}(g)$ surjectively onto those of $\pi^{-1}(g)$. Consequently,

$$U = \{g \in G \mid W_L \text{ contains every maximal point of } \pi_L^{-1}(g)\}.$$

Thus, replacing K by L , we are reduced to the case in which K contains $\bar{k}$.

Since the projection

$$p : G_K \longrightarrow G_{\bar{k}}$$

is surjective and open, $V = p(W)$ is a dense open subset of $G_{\bar{k}}$. Moreover,

$$p^{-1}(g) = \text{Spec}(\kappa(g) \otimes_{\bar{k}} K)$$

is irreducible for every $g \in G_{\bar{k}}$ (cf. EGA IV₂, 4.3.3). Hence, for every $g \in V$, the generic point of $p^{-1}(g)$ belongs to W .

On the other hand, set

$$\mathcal{G} = \text{Gal}(\bar{k}/k).$$

The projection $q : G_{\bar{k}} \rightarrow G$ is surjective, and $\mathcal{G}$ acts transitively on the fibers of q . Therefore

$$U = q(V'),$$

where V' is the intersection of the $\mathcal{G}$ -conjugates of V .

Let Z be the closed subset $G_{\bar{k}} - V$, endowed with its reduced closed subscheme structure. Since $G_{\bar{k}}$, and hence Z , is of finite presentation over $\bar{k}$, the scheme Z is obtained by base change from a reduced closed subscheme

$$Z_1 \subset G \otimes_k k_1$$

for some finite Galois extension k_1 of k . Thus the $\mathcal{G}$ -conjugates of Z are finite in number, and their union is again a nowhere-dense closed subset F of $G_{\bar{k}}$. It follows that

$$V' = G_{\bar{k}} - F$$

is a dense open subset of $G_{\bar{k}}$. Since $q : G_{\bar{k}} \rightarrow G$ is surjective and open, $U = q(V')$ is therefore a dense open subset of G .

Moreover, for every closed point g of G , the fiber $\pi^{-1}(g)$ consists of finitely many closed points of G_K . Thus, if $g \in U$, then

$$\pi^{-1}(g) \subset W.$$

This proves (i), and (ii) follows from it.

Assertion (iii)(a) was proved in VI_A, 2.6.4. Finally, if U is dense in every irreducible component of X , then

$$X = G \cdot U.$$

Hence, for every $x \in X$,

$$\mu_x^{-1}(U_{\kappa(x)})$$

is a nonempty open subset of $G_{\kappa(x)}$, and (iii)(b) follows from (ii).

Corollary 5.6.2.— *Let S , G , and X be as in the preliminary hypotheses of 5.3, and let U be an open subset of X , let $s \in S$, and let A be a finite subset of X_s . Suppose that U_s is dense in X_s and that X_s is locally of finite type over $\kappa(s)$.⁴³*

Then there are morphisms $f : S'' \rightarrow S$ and $h : S'' \rightarrow G$, where f factors as a finite surjective morphism $S'' \rightarrow S'$ followed by an étale morphism $S' \rightarrow S$, such that the inverse image A'' of A in $X \times_S S''$ (i.e., in $X_s \times_{\text{Spec } \kappa(s)} S''_s$) is contained in

$$\ell_h^{-1}(U_{S''}),$$

where ℓ_h denotes the translation of $X_{S''} = X \times_S S''$ defined by the element $h \in G(S'')$.

Proof. Since X_s is locally of finite type over $\kappa(s)$, the connected components of X_s are open and irreducible (cf. VI_A, 2.5.4). Hence U_s is dense in every irreducible component of X_s .

By Lemma 5.6.2.0, there therefore exists a closed point

$$g \in G_s^0$$

such that

$$g \cdot a' \in U_s \otimes_{\kappa(s)} \kappa(g)$$

for every $a' \in X_{\kappa(g)}$ lying above a point of A .

By Lemma 5.6.1, there are morphisms $f : S'' \rightarrow S$ and $h : S'' \rightarrow G$, where f factors as a finite surjective morphism $S'' \rightarrow S'$ followed by an étale morphism $S' \rightarrow S$, such that $f^{-1}(s)$ consists of a single point s_0 , and such that

$$h(s_0) = g, \quad \kappa(s_0) = \kappa(g).$$

Denote by A'' the inverse image of A in

$$X_s \times_{\text{Spec } \kappa(s)} S''_s = X_s \otimes_{\kappa(s)} \kappa(s_0) = X_s \otimes_{\kappa(s)} \kappa(g).$$

Then the translation ℓ_h of $X_{S''}$ sends A'' into

$$U_s \otimes_{\kappa(s)} \kappa(s_0).$$

Lemma 5.6.3.— *Let X be an S -scheme. The following conditions are equivalent:*

- (i) *X is separated over S .*
- (ii) *For every S -scheme T , every section $\sigma : T \rightarrow X_T$ is a closed immersion.*
- (iii) *For every reduced S -scheme T , two S -morphisms $f_1, f_2 : T \rightarrow X$ that agree on a dense open subset U of T are equal.*

⁴³Editor's note: we have added the hypothesis on X_s and simplified (and corrected) the proof, taking account of the addition of 5.6.2.0. Moreover, the proof shows that the conclusion remains valid if one assumes only that U_s is dense in every irreducible component of X_s .

(iv) For every $s \in S$ and every pair of points x_1, x_2 of X_s , there exist a morphism

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subscheme V of $X_{S''}$, separated over S'' , such that:

- (a) $S' \rightarrow S$ is open, $S'' \rightarrow S'$ is closed and surjective, and $\phi^{-1}(s) \neq \emptyset$;
- (b) the inverse image of $\{x_1, x_2\}$ in $X_{S''}$ is contained in V .

(iv') For every $s \in S$, every pair of points $x_1, x_2 \in X_s$ is contained in an open subset V of X that is separated over S .⁴⁴

Proof. The implication (ii) $\Rightarrow$ (i) is clear (take $T = X$ and let σ be the diagonal section), as are

$$(i) \Rightarrow (iv') \Rightarrow (iv).$$

On the other hand, (i) $\Rightarrow$ (ii) by EGA I, 5.4.6.

(iii) $\Rightarrow$ (ii). Let $\sigma : T \rightarrow X_T$ be a section of $p : X_T \rightarrow T$. By EGA I, 5.3.13, σ is an immersion, that is, an isomorphism of T onto a locally closed subscheme E of X_T . To show that E is closed, we may suppose that T and E are reduced. Let $\bar{E}$ be the reduced closed subscheme of X_T whose underlying space is the closure of E , so that E is a dense open subscheme of $\bar{E}$. The immersion

$$i : \bar{E} \hookrightarrow X_T$$

and $\sigma \circ p \circ i$ agree on E , hence on $\bar{E}$ by hypothesis (iii). Thus every point of $\bar{E}$ belongs to $\sigma(T) = E$, and consequently $\bar{E} = E$.

(i) $\Rightarrow$ (iii). Suppose that X is separated over S , and let T be a reduced S -scheme and $f_1, f_2 : T \rightarrow X$ two S -morphisms that agree on a dense open subset U of T . Let

$$g : T \longrightarrow X \times_S X$$

be the morphism with components f_1 and f_2 . Since

$$D = \Delta_{X/S}(X)$$

is closed, its inverse image under g is a closed subset of T containing U , and is therefore equal to T . Since T is reduced, g factors through D (cf. EGA I, 5.2.2); consequently

$$f_1 = p_1 \circ g = p_2 \circ g = f_2.$$

(iv) $\Rightarrow$ (iii). Let T be a reduced S -scheme, and let $f_1, f_2 : T \rightarrow X$ be two S -morphisms that agree on a dense open subset U . Since T is reduced, in order to prove that $f_1 = f_2$, it is enough to prove that $f_1 = f_2$ set-theoretically. Indeed, suppose this has been established, let $t \in T$, let V be an affine open subset of X containing $f_1(t) = f_2(t)$, and let W be the

⁴⁴Editor's note: we have simplified the formulation of condition (iv) and added condition (iv'). We have also added the proof of the implication (i) $\Rightarrow$ (iii), which is used in the proof of (iv) $\Rightarrow$ (iii). Observe moreover that, if

$$T = \operatorname{Spec} B, \quad B = k[\varepsilon, x]/(\varepsilon^2, \varepsilon x)$$

(k a field) and

$$X = T_{\text{red}} = \operatorname{Spec} k[x],$$

then the morphisms

$$\phi_\lambda : T \longrightarrow X, \quad x \longmapsto x + \lambda \varepsilon \quad (\lambda \in k),$$

agree on the dense open subset $\operatorname{Spec} B_x$, but are not equal.

open neighborhood of t that is the inverse image of V under the continuous map underlying both f_1 and f_2 . Then the morphisms

$$f_i|_W : W \longrightarrow V$$

agree on the dense open subset $U \cap W$ of W . Since V is separated and W is reduced, this implies that $f_1|_W = f_2|_W$, and hence that $f_1 = f_2$.

Now let $t \in T$, let s be its image in S , and let us show that the points

$$x_1 = f_1(t), \quad x_2 = f_2(t)$$

of X_s are equal. Let

$$\phi : S'' \longrightarrow S' \longrightarrow S$$

and an open subset V of $X \times_S S''$ be as in (iv). Set

$$T' = T \times_S S', \quad T'' = T \times_S S'',$$

and denote by

$$g : T'' \longrightarrow T, \quad f_i'' : T'' \longrightarrow X_{S''} \quad (i = 1, 2)$$

the morphisms obtained by base change.

Since U is dense in T and $T' \rightarrow T$ is open, the inverse image U' of U in T' is dense in T' . Let U'' be the inverse image of U' in T'' , and let F be the reduced subscheme of T'' whose underlying space is $\overline{U''}$. Since $T'' \rightarrow T'$ is surjective and closed, the image of F contains U' and is closed, and is therefore equal to T' . Consequently,

$$F \cap g^{-1}(t)$$

contains a point u .

For $i = 1, 2$, let h_i be the restriction of f_i'' to F . Then $h_i(u)$ is a point of $X_{S''}$ above $f_i(t) = x_i$, and hence belongs to V , since V contains the inverse image of $\{x_1, x_2\}$ in $X_{S''}$.

Thus

$$W = h_1^{-1}(V) \cap h_2^{-1}(V)$$

is an open subset of F containing u , and the S'' -morphisms

$$h_i|_W : W \longrightarrow V$$

agree on the dense open subset $U'' \cap W$ of W . Since V is separated over S'' , we have

$$h_1|_W = h_2|_W.$$

It follows that $h_1(u) = h_2(u)$, and hence that $f_1(t) = f_2(t)$. This proves (iv) $\Rightarrow$ (iii).

Theorem 5.3 now follows from 5.6.2 and the implication (iv) $\Rightarrow$ (i) of 5.6.3.

Counterexample 5.6.4.— *Not every S -group G is separated. Let S be a scheme having a non-isolated closed point s , and let G be the scheme obtained by gluing two copies of S along the open subset $S - \{s\}$. It is easily seen that G is not separated over S , and that it carries a natural S -group structure whose fibers are all trivial, except for the fiber G_s , which is isomorphic to the two-element group $\mathbb{Z}/2\mathbb{Z}$.⁴⁵*

⁴⁵Editor's note: we reproduced this example in VI_A, 0.3, editor's note 5.

Theorem 5.7.—⁴⁶ *Let S be a scheme, let G be an S -group locally of finite presentation over S , such that the function*

$$s \longmapsto \dim G_s$$

is locally constant on S , let X be an S -scheme on which G acts, and let U be an open subset of X , separated over S . Then $\underline{G}^0 \cdot U$ is an open subset of X , separated over S .

Proof. Let

$$\mu : G \times_S X \longrightarrow X$$

denote the action of G on X . It is the composite of the automorphism

$$(g, x) \longmapsto (g, gx)$$

of $G \times_S X$ and the projection onto X . Since $G \times_S U$ is an open subset of $G \times_S X$, it follows from 4.2(iii) that

$$V = \underline{G}^0 \cdot U$$

is open in X .

Then, arguing as in the proof of 5.6.2, we deduce from the implication (iv) $\Rightarrow$ (i) of 5.6.3 that V is separated over S .

Corollary 5.7.1.— *Let S and G be as in 5.7, and let σ, τ be two S -sections of G^0 , that is, $\sigma, \tau \in G^0(S)$. Then the coincidence subscheme $S(\sigma, \tau) \subset S$ of σ and τ (that is, the inverse image of the diagonal of $G \times_S G$ under the morphism (σ, τ)) is closed.*

Indeed, for every $s \in S$, let U be an affine open subset of G containing $\epsilon(s)$. Then

$$V = \epsilon^{-1}(U)$$

is an open neighborhood of s in S , and, since $G^0 U$ is separated, $S(\sigma, \tau) \cap V$ is closed in V .

Remark 5.7.2.— Gabber points out that one can show the following: if S is henselian local, with closed point s , then the intersection of the open subsets $G^0 U$, where U ranges over the affine open neighborhoods of $\epsilon(s)$, is an open group subscheme of G , separated over S .

5.8. Complements

⁴⁷We begin by recalling Proposition VI_A, 2.6.6:

Proposition 5.8.1.— *Let k be a field, let G be a k -group locally of finite type acting on a k -scheme X in such a way that the morphism*

$$G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x),$$

is surjective. Suppose that X is quasi-separated. Then the connected components of X are irreducible.

⁴⁶Editor's note: on the one hand, we omitted Corollary 5.6.5, which repeated 5.5. On the other hand, the original stated Corollary 5.7.1 below as Remark 5.7, referring for its proof to 4.7, a number that does not exist in *Lecture Notes in Mathematics* 151 (but did appear in the 1965 edition of SGAD, whose numbers 4.5 and 4.6 became 5.6.1 and 5.6.2); we have added Theorem 5.7, communicated by O. Gabber, which makes the aforementioned statement of SGAD more precise.

⁴⁷Editor's note: we have added this paragraph of complements and counterexamples, all communicated by O. Gabber.

Corollary 5.8.2.— *Let S, G, X be as in the preliminary hypotheses of 5.3. Suppose moreover that every fiber X_s is quasi-separated and connected. Then X is separated and quasi-compact over S .*

Indeed, by the preceding proposition, every fiber X_s is irreducible, and the rest of the proof is identical to that of 5.4.

Example 5.8.3.— Fix an algebraically closed field k . Recall first that every “locally Boolean” topological space X , that is, every space having a basis of compact open subsets, becomes a k -scheme when equipped with the sheaf of locally constant k -valued functions (cf. [DG70], I, §1, 2.12). Such an X will be called a *locally Boolean k -scheme*.

We also note that every topological space E having a basis of separated open subsets—and likewise every scheme X —has a dense separated open subset. Indeed, every increasing union of separated open subsets is separated (for a scheme X , this follows from 5.6.3), so there is a maximal such open subset U . It must be dense: otherwise there would be a nonempty open subset V with $U \cap V = \emptyset$; the open set V would contain a nonempty separated open subset W , and

$$U \cup W$$

would still be separated, contradicting the maximality of U .

Now let C be the Cantor space, viewed as the underlying space of the group

$$(\mathbb{Z}/2\mathbb{Z})^{\mathbb{N}}$$

with its product topology. For every $p \in C$, let $C(p)$ be another copy of C , and let X be obtained by gluing each $C(p)$ to C along $C - \{p\}$. Then X is a nonseparated locally Boolean k -scheme, and C is a dense open subset of X .

Let G be the group of automorphisms of the k -scheme X , that is, of homeomorphisms of X . Then G acts transitively on X . Indeed, X is the union of C and, for each point $p \in C$, a second point p' . This second point is characterized as the unique $x \in X - \{p\}$ for which

$$(C - \{p\}) \cup \{x\}$$

is separated. Consequently, every automorphism ϕ of C extends to an automorphism ϕ_X of X satisfying

$$\phi_X(p') = \phi_X(p)'$$

for every p . Moreover, the map

$$\theta_p : X \longrightarrow X$$

that exchanges p and p' and fixes every other point is an automorphism of X . Finally, the automorphism group of C acts transitively on C : using the group structure of C , it is enough to consider translations.

Thus the discrete k -group G , which is locally of finite type, acts on the k -scheme X in such a way that

$$G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x),$$

is surjective, but X is not separated, even though C is a dense separated open subset.

Example 5.8.4.— Retain the notation of the preceding example. Using the description of C as the set of sequences

$$(u_n)_{n \in \mathbb{N}}, \quad u_n \in \mathbb{Z}/2\mathbb{Z},$$

one sees that C with one point p removed is homeomorphic to a countable disjoint union of copies of C . Indeed, using the group structure of C , for example, translation reduces us to

the case in which $p = 0$, the zero sequence. Then

$$C - \{0\} = \coprod_{i \in \mathbb{N}} C_i,$$

where

$$C_i = \{(u_n)_{n \in \mathbb{N}} \mid u_i = 1 \text{ and } u_j = 0 \text{ for } j < i\},$$

and every C_i is homeomorphic to C . It follows by induction on $|F|$ that, for every nonempty finite subset F of C , the space $C - F$ is homeomorphic to a countable disjoint union of copies of C , and hence to C with one point removed.

For each $F \subset C$ of cardinality 2, let $C(F)$ be another copy of C , denote by q_F the point 0 of $C(F)$, and choose a homeomorphism

$$\phi_F : C(F) - \{q_F\} \xrightarrow{\sim} C - F.$$

Let X' be obtained by gluing every $C(F)$ to C by means of ϕ_F . Then X' is a nonseparated locally Boolean k -scheme. Moreover, the construction shows that every locally constant function

$$f : X' \longrightarrow k$$

is constant. Indeed, if $x, y \in C$ and $F = \{x, y\}$, then every neighborhood of q_F meets every neighborhood of x and of y . Thus

$$f(x) = f(q_F) = f(y).$$

If $F' = \{z, t\}$, with $z, t \in C$, then likewise

$$f(q_{F'}) = f(z) = f(q_{\{z, x\}}) = f(x).$$

Consequently, X' is connected.

Every point $x \in X'$ also has a pointed neighborhood homeomorphic to $(C, 0)$. More precisely, for every F , fix a homeomorphism of pointed topological spaces

$$h_F : (C, 0) \xrightarrow{\sim} (C(F), q_F),$$

and let T be the group of translations of C . If $x = q_F$, use h_F ; if $x \in C$, the translation $t_x \in T$ is a homeomorphism

$$t_x : (C, 0) \xrightarrow{\sim} (C, x),$$

and it is the unique element of T with this property.¹

Let L be the free group generated by the h_F , and set

$$G = T * L,$$

the free product (coproduct) of T and L . For

$$h \in \{h_F\}_{|F|=2} \cup T,$$

let $\sigma(h)$ denote the corresponding generator of G , and let $S(h)$ and $B(h)$ be respectively the source and the target of h . It is convenient also to set

$$\sigma(h_F^{-1}) = \sigma(h_F)^{-1}, \quad S(h_F^{-1}) = B(h_F), \quad B(h_F^{-1}) = S(h_F),$$

¹Translator's note: in these two passages the French source prints X in place of the X' just constructed, and q_F in place of the homeomorphism h_F just defined. The English uses X' and h_F , respectively.

and to let E be the set consisting of the elements of T and all h_F, h_F^{-1} .

On the product space

$$P = G \times X',$$

where G carries the discrete topology, consider the equivalence relation generated by

$$(g\sigma(h), x) \sim (g, h(x)) \quad \text{when } x \in S(h),$$

for every $h \in E$, and let Z be the quotient space. Thus Z is obtained by gluing open subsets of the disjoint union

$$\coprod_{g \in G} \{g\} \times X'.$$

Consequently, the saturation $\tilde{\Omega}$ of every open subset Ω of P is open (cf. [BTop], I, §5.1, Example 2). Explicitly, since every open subset of P is the union of its intersections with the slices $\{g\} \times X'$, it is enough to consider $\{1\} \times W$, with W open in X' . Its saturation is the union of $\{1\} \times W$ and the sets

$$\begin{aligned} & \{\sigma(h_1)\} \times h_1^{-1}(W \cap B(h_1)), \\ & \{\sigma(h_1)\sigma(h_2)\} \times h_2^{-1}(h_1^{-1}(W \cap B(h_1)) \cap B(h_2)), \end{aligned}$$

and so on, for all finite sequences $h_1, \dots, h_n$ of elements of E . It is therefore open, and the projection

$$\pi : P \longrightarrow Z$$

is open.

Moreover, the word

$$\sigma(h_1) \cdots \sigma(h_n)$$

is reduced in G , unless one of the h_i is the identity of T , two consecutive h_i 's belong to T , or one of the pairs (h_i, h_{i+1}) is

$$(h_F, h_F^{-1}) \quad \text{or} \quad (h_F^{-1}, h_F).$$

Suppose, therefore, that $x \in X'$ and that

$$\beta = (\sigma(h_1) \cdots \sigma(h_n), h_n^{-1} \cdots h_1^{-1}(x))$$

belongs to $\{1\} \times X'$. We may then suppose that every h_i is a translation t_i . In that case,

$$\sigma(t_1 \cdots t_n) = 1$$

implies $t_1 \cdots t_n = 1$, and hence $\beta = (1, x)$. The equivalence relation is compatible with the action of G on $P = G \times X'$ by left translation on the first factor. It follows that, for every $g \in G$, the restriction of π to $\{g\} \times X'$ is injective.

Now let $z \in Z$ and choose a representative $(g, x) \in P$. By the preceding discussion,

$$U = \pi(\{g\} \times X')$$

is an open neighborhood of z , and the continuous map

$$\{g\} \times X' \longrightarrow U$$

induced by π is open and bijective, hence a homeomorphism. Thus Z is locally isomorphic to X' , and hence also to C ; it is therefore again a locally Boolean k -scheme.

Finally, G acts transitively on Z . Indeed, every $z \in Z$ is G -conjugate to an element of the form $\pi(1, x)$, so it is enough to show that every $(1, x) \in P$ is equivalent to some $(\sigma(h), 0)$. If $x \in C$, take $h = t_x$; if $x = q_F$, take $h = h_F$. Thus Z is a locally Boolean k -scheme endowed with a transitive action of the discrete group G , but Z is not separated.

6. Subfunctors and group subschemes^{*}

Definition 6.1.— (i) Let X be an S -functor (that is, a functor from $(\mathbf{Sch}/S)^\circ$ to $(\mathbf{Ens})$), let G be an S -functor in groups, and let u and v be two S -morphisms from X to G . The *transporter of u into v* , denoted $\underline{\text{Transp}}(u, v)$, is the sub- S -functor of G defined by

$$\begin{aligned} \underline{\text{Transp}}(u, v)(S') &= \{g \in G(S') \mid (\text{int } g) \circ u_{S'} = v_{S'}\} \\ &= \left\{ g \in G(S') \mid \begin{array}{l} g_{S''} u_{S''}(x) g_{S''}^{-1} = v_{S''}(x), \\ \text{for every } x \in X(S'') \text{ and every } S'' \rightarrow S' \end{array} \right\}. \end{aligned}$$

In particular, $\underline{\text{Transp}}(u, u)$ is a group sub- S -functor of G ; it is called the *centralizer of u* and is denoted $\underline{\text{Centr}}(u)$.

(ii) Let G be an S -functor in groups, and let X and Y be two sub- S -functors of G . The *transporter of X into Y* (respectively, the *strict transporter of X into Y*), denoted $\underline{\text{Transp}}_G(X, Y)$ (respectively, $\underline{\text{Transp}}_G^{\text{str}}(X, Y)$), is the sub- S -functor of G defined by

$$\begin{aligned} \underline{\text{Transp}}_G(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) \subset Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} \subset Y(S'') \text{ for every } S'' \rightarrow S'\}, \end{aligned}$$

and, respectively,

$$\begin{aligned} \underline{\text{Transp}}_G^{\text{str}}(X, Y)(S') &= \{g \in G(S') \mid (\text{int } g)(X_{S'}) = Y_{S'}\} \\ &= \{g \in G(S') \mid g_{S''} X(S'') g_{S''}^{-1} = Y(S'') \text{ for every } S'' \rightarrow S'\}. \end{aligned}$$

Note that

$$\underline{\text{Transp}}_G^{\text{str}}(X, Y) = \underline{\text{Transp}}_G(X, Y) \cap c(\underline{\text{Transp}}_G(Y, X)),$$

where c denotes the inversion morphism of G .⁴⁹

(iii) Let G be an S -functor in groups, let H be a sub- S -functor of G , and let $i : H \rightarrow G$ be the canonical S -morphism. The *centralizer* and *normalizer of H in G* are the following group sub- S -functors of G :

$$\underline{\text{Centr}}_G H = \underline{\text{Centr}}(i) = \underline{\text{Transp}}(i, i), \quad \underline{\text{Norm}}_G H = \underline{\text{Transp}}_G^{\text{str}}(H, H).$$

Finally, the *center of G* is the S -functor in groups

$$\underline{\text{Centr}}(\text{id}_G) = \underline{\text{Centr}}_G G;$$

it is denoted $\underline{\text{Centr}}G$.

^{*}On the same theme, see also the results of Exposé XI, §6, whose natural place would be in the present Exposé VI_B. There one finds, in particular, representability criteria for certain group subfunctors of a given group scheme. (48) Editor's note: we have inserted these results below (6.5.2–6.5.5).

⁴⁹Editor's note: if $u : X \rightarrow G$ and $v : Y \rightarrow G$ are two arbitrary morphisms of S -functors, let $u(X)$ be the image functor of u , defined by

$$u(X)(S') = u(S')(X(S')) \subset G(S');$$

it is a subfunctor of G , as is the image functor $v(Y)$. One may then consider the transporter of the image of u into the image of v , $\underline{\text{Transp}}_G(u(X), v(Y))$. Thus, in part (ii) of the definition, it is not necessary to restrict to subfunctors, that is, to the case in which u and v are monomorphisms. In the original, this restriction sometimes imposed repetitions in the hypotheses, such as: “Let $u, v : X \rightarrow G$ be morphisms of S -functors and let $w : H \rightarrow G$ and $w' : K \rightarrow G$ be monomorphisms; then $\underline{\text{Transp}}(u, v)$, $\underline{\text{Centr}}_G(w) = \underline{\text{Centr}}_G H$, and $\underline{\text{Transp}}_G(H, K)$ satisfy ...” Such repetitions can be avoided by considering $\underline{\text{Transp}}_G(u(X), v(Y))$. We have made such modifications in 6.2 and, later, in 10.11.

Remark 6.1.1.—⁵⁰ It follows from the definitions that the functors $\underline{\text{Transp}}(u, v)$ and $\underline{\text{Transp}}_G(X, Y)$ (and hence also $\underline{\text{Transp}}_G^{\text{str}}(X, Y)$) commute with base change: for every $S' \rightarrow S$, if G', X', Y', u', v' are obtained from G, X, Y, u, v by base change, then

$$\underline{\text{Transp}}(u, v)_{S'} = \underline{\text{Transp}}(u', v'), \quad \underline{\text{Transp}}(X, Y)_{S'} = \underline{\text{Transp}}(X', Y').$$

Proposition 6.2.—⁵¹ Let G be an S -group. For a subfunctor T of the functor G , consider the following property:

(+_f) for every $s \in S$, T_s is representable by a closed subscheme of G_s .

Let $u, w : X \rightarrow G$ and $v : Y \rightarrow G$ be morphisms of S -schemes. Then:

- (i) $\underline{\text{Transp}}(u, w)$ and $\underline{\text{Centr}}(u) = \underline{\text{Transp}}(u, u)$ satisfy condition (+_f).
- (ii) $\underline{\text{Transp}}_G(u(X), v(Y))$ and $\underline{\text{Norm}}_G v(Y)$ satisfy condition (+_f) if, for every $s \in S$, v_s is a closed immersion.
- (iii) $\underline{\text{Transp}}_G^{\text{str}}(X, Y)$ satisfies condition (+_f) if, for every $s \in S$, u_s and v_s are closed immersions.

This follows from Remark 6.1.1 and Corollary 6.2.5 below.⁵²

Definition 6.2.1.— Let $f : X \rightarrow S$ be a morphism of schemes. One says that f is *essentially free*, or that X is *essentially free over S* , if one can find a covering of S by affine open subsets S_i , for every i an affine S_i -scheme S'_i faithfully flat over S_i , and a covering $(X'_{ij})_j$ of

$$X'_i = X \times_S S'_i$$

by affine open subsets X'_{ij} , such that, for every (i, j) , the ring of X'_{ij} is a projective⁵³ module over the ring of S'_i .

Proposition 6.2.2.—

- a) If X is essentially free over S , it is flat over S ; the converse holds if S is Artinian.
- b) If S is the spectrum of a field, every S -scheme is essentially free over S .
- c) If X is essentially free over S , then $X' = X \times_S S'$ is essentially free over S' , for every $S' \rightarrow S$. The converse holds if $S' \rightarrow S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in a) the fact that a flat module over an Artinian local ring is free.⁵⁴

⁵⁰Editor's note: we have added this remark, which is used implicitly in Proposition 6.2.

⁵¹Editor's note: we have modified the statement as indicated in editor's note 49.

⁵²Editor's note: the original referred to the results of Exposé VIII, §6. For the reader's convenience, we have reproduced these results (with the exception of VIII, 6.3 and 6.8) in 6.2.1–6.2.5 below. Moreover, this had been suggested by A. Grothendieck in a note at the beginning of VIII.6: "The present section is independent of the theory of diagonalizable groups; its natural place would be in VI_B."

⁵³Editor's note: on the one hand, we have replaced the word "free" by "projective", as indicated in VIII, Remark 6.8. On the other hand, the notion of an essentially free S -scheme is closely related to the geometric notion of an S -scheme that is flat and pure, introduced and studied by M. Raynaud and L. Gruson ([RG71]); see the addition 6.9 below.

⁵⁴Editor's note: indeed, let $(A, \mathfrak{m})$ be an Artinian local ring, k its residue field, M an arbitrary A -module, and let $(x_i)_{i \in I}$ be elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and let $\varphi : F \rightarrow M$ be the A -morphism defined by $\varphi(e_i) = x_i$. Then $Q = \text{Coker } \varphi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose moreover that M is flat over A ; then $K = \text{Ker } \varphi$ satisfies $K \otimes_A k = 0$, that is, $K = \mathfrak{m}K$, whence $K = 0$.

The introduction of Definition 6.2.1 is justified by the following theorem.

Theorem 6.2.3.— *Let S be a scheme, let Z be an S -scheme that is essentially free, and let Y be a closed subscheme of Z . Consider the functor*

$$F = \prod_{Z/S} Y/Z : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Sets}$$

*defined by the following condition: $F(S') = \emptyset$ when $Y_{S'} \neq Z_{S'}$, and $F(S')$ is a singleton otherwise.*⁵⁵

- (i) *This functor is representable by a closed subscheme T of S .*
- (ii) *If, moreover, $Y \rightarrow Z$ is of finite presentation, then so is $T \rightarrow S$.*

Proof. First observe that the functor in question is a sheaf for the fpqc topology. Since $F(S')$ is either $\emptyset$ or $\{\text{pt}\}$ for every S' , this amounts to checking the following assertions. If (S_i) is an open covering of S and every

$$Y_{S_i} \longrightarrow Z_{S_i}$$

is an isomorphism, then $Y \rightarrow Z$ is an isomorphism; and if $S' \rightarrow S$ is faithfully flat and quasi-compact and

$$Y_{S'} \longrightarrow Z_{S'}$$

is an isomorphism, then again $Y \rightarrow Z$ is an isomorphism. The first assertion is clear, and the second follows from SGA 1, VIII, 5.4, or EGA IV₂, 2.7.1.

Moreover, by SGA 1, VIII, 1.9, faithfully flat and quasi-compact morphisms are effective descent morphisms for the fibered category of closed-immersion arrows. With the notation of 6.2.1, this allows us to restrict to the case $S = S'_i$.

Let (Z_j) be a covering of Z by affine open subsets such that

$$\mathcal{O}(Z_j)$$

is a projective module over $A = \mathcal{O}(S)$. Put

$$Y_j = Y \cap Z_j,$$

and let

$$F_j : (\mathbf{Sch}/S)^\circ \longrightarrow \mathbf{Sets}$$

be the functor defined in terms of (Z_j, Y_j) as F was defined in terms of (Z, Y) . It is a subfunctor of the final functor, and clearly

$$F = \bigcap_j F_j.$$

It is therefore enough to prove that every F_j is representable by a closed subscheme T_j of S , for then F is represented by the closed subscheme

$$T = \bigcap_j T_j$$

⁵⁵Editor's note: cf. Exposé II, §1, where this functor is denoted $\prod_{Z/S} Y$. For every $S' \rightarrow S$,

$$F(S') = \Gamma(Y_{S'}/Z_{S'}),$$

which here is equal to $\{\text{id}_{Z_{S'}}\}$ if $Y_{S'} = Z_{S'}$, and is empty otherwise. We have also added assertion (ii), which is used in the proof of 6.5.3.

of S .

We may consequently suppose that Z is itself affine, say

$$Z = \operatorname{Spec} B,$$

where B is a projective A -module. Let L be a free A -module with basis $(e_\lambda)_{\lambda \in \Lambda}$ of which B is a direct summand as an A -module, and let

$$\phi_\lambda : L \longrightarrow A$$

be the coordinate forms with respect to this basis. Let E be a set of generators for the ideal J of B defining the closed subscheme Y of Z , and let I be the ideal of A generated by the coordinates

$$\phi_\lambda(x), \quad x \in E, \quad \lambda \in \Lambda.$$

For every A -algebra C , the morphism

$$B \otimes_A C \longrightarrow (B/J) \otimes_A C$$

is an isomorphism if and only if the image of $x \otimes 1$ in $L \otimes_A C$ is zero for every $x \in E$. This, in turn, is equivalent to saying that the kernel of $A \rightarrow C$ contains the ideal I . Hence

$$T = V(I) = \operatorname{Spec}(A/I)$$

has the required property, proving (i).

Finally, if $Y \rightarrow Z$ is of finite presentation, the set E may be chosen finite. Then I is a finitely generated ideal of A , that is, the closed immersion $T \rightarrow S$ is of finite presentation. This proves (ii).

Examples 6.2.4.— Let us give important examples of functors which reduce to functors

$$\prod_{Z/S} Y/Z$$

of the type considered in 6.2.3, and for which it will be useful to have representability criteria later on. We denote by S a scheme, and by $X, Y, Z, \dots$ schemes over S .

a) Suppose given an S -morphism

$$q : X \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z), \tag{x}$$

(“ X acts on Y , with values in Z ”), that is, a morphism

$$r : X \times_S Y \longrightarrow Z. \tag{xx}$$

Consider a subscheme Z' of Z , and hence a monomorphism

$$\underline{\operatorname{Hom}}_S(Y, Z') \longrightarrow \underline{\operatorname{Hom}}_S(Y, Z),$$

which makes the first functor a subfunctor of the second. Let X' be the inverse image of this subfunctor under q . Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Z_T$$

factors through Z'_T .

This functor X' may be described as follows. Put

$$P = X \times_S Y,$$

and let P' be the inverse image of Z' under $r : P \rightarrow Z$. Then there is an evident isomorphism

$$X' \simeq \prod_{P/X} P'/P. \quad (\text{xxx})$$

It follows that, if Y is essentially free over S and Z' is closed in Z , the subfunctor X' of X is representable by a closed subscheme of X .⁵⁶ If, moreover, $Z' \rightarrow Z$ is of finite presentation, then so is $X' \rightarrow X$.

b) Suppose given two ways of letting X act on Y , with values in Z , that is, two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\text{Hom}}_S(Y, Z),$$

and put

$$X' = \text{Ker}(q_1, q_2).$$

Thus X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms

$$q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$$

are equal. The data of q_1, q_2 are equivalent to the data of a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Z \times_S Z),$$

or, equivalently, of a morphism

$$r : X \times_S Y \longrightarrow Z \times_S Z.$$

Put $U = Z \times_S Z$, and let U' be the diagonal subscheme of $Z \times_S Z$. Then X' is precisely the inverse image under q of the subfunctor

$$\underline{\text{Hom}}_S(Y, U') \subset \underline{\text{Hom}}_S(Y, U).$$

It can therefore be put in the form (xxx), with $P = X \times_S Y$ and P' the inverse image of the diagonal under r , that is, the kernel of

$$r_1, r_2 : X \times_S Y \rightrightarrows Z.$$

We are thus under the hypotheses of (a).

Consequently, if Y is essentially free over S and Z is separated over S , the subfunctor X' of X is representable by a closed subscheme of X .⁵⁶ If, moreover, $Z \rightarrow S$ is locally of finite type, then $X' \rightarrow X$ is of finite presentation.

c) Suppose given a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Y),$$

that is, “ X acts on Y .” Let X' be the “kernel” of this morphism: X' is the subfunctor of X for which $X'(T)$ is the set of $x \in X(T)$ such that

$$q(x) : Y_T \longrightarrow Y_T$$

⁵⁶Editor’s note: we have added the sentence that follows.

is the identity. This functor falls under (b), as one sees by introducing a second morphism

$$q' : X \longrightarrow \underline{\mathrm{Hom}}_S(Y, Y)$$

by letting X act trivially on Y . Hence, if Y is essentially free and separated over S , the subfunctor X' , the kernel of q , is representable by a closed subscheme of X .⁵⁶ If, moreover, $Y \rightarrow S$ is locally of finite type, then $X' \rightarrow X$ is of finite presentation.

d) Under the hypotheses of (c), consider the subfunctor Y' of Y “of the invariants under X .” Thus $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism

$$\bar{q}(y) : X_T \longrightarrow Y_T$$

is the constant T -morphism with value y . Introducing $\bar{q}'$ as in (c), and the morphisms corresponding to $\bar{q}$ and $\bar{q}'$,

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\mathrm{Hom}}_S(X, Y),$$

one sees that

$$Y' = \mathrm{Ker}(\bar{q}, \bar{q}').$$

It therefore falls again under (b), with the roles of X and Y reversed and with $Z = Y$.

Consequently, if X is essentially free over S and Y is separated over S , the subfunctor Y' of Y of the invariants under X is representable by a closed subscheme of Y .⁵⁶ If, moreover, $Y \rightarrow S$ is locally of finite type, then $Y' \rightarrow Y$ is of finite presentation.

e) Constructions of the kind described in the preceding examples occur especially often in group theory. Thus, when G is an S -group scheme acting on the S -scheme X ,

$$q : G \longrightarrow \underline{\mathrm{Aut}}_S(X),$$

the kernel of q , the subgroup of G acting trivially, is a closed subscheme of G , provided X is essentially free and separated over S (example (c)). The subobject X^G of invariants is a closed subscheme of X , provided G is essentially free over S and X is separated over S (example (d)).

Let Y, Z be subschemes of X , and consider the subfunctor $\underline{\mathrm{Transp}}_G(Y, Z)$ of G , the transporter of Y into Z . Its T -valued points, for T over S , are the elements $g \in G(T)$ such that the corresponding automorphism of X_T satisfies

$$g(Y_T) \subset Z_T,$$

that is, induces a morphism $Y_T \rightarrow X_T$ which factors through $Y_T \rightarrow Z_T$. Hence, if Y is essentially free over S and Z is closed in X , then $\underline{\mathrm{Transp}}_G(Y, Z)$ is a closed subscheme of G (example (a)).

One may also consider the strict transporter of Y into Z ,⁵⁷ whose T -valued points, for T over S , are the elements $g \in G(T)$ such that $g(Y_T) = Z_T$. This functor is precisely

$$\underline{\mathrm{Transp}}_G(Y, Z) \cap \sigma(\underline{\mathrm{Transp}}_G(Z, Y)),$$

where σ is the inversion morphism of G . Consequently, if Y and Z are essentially free over S and closed in X , the strict transporter of Y into Z is a closed subscheme of G .

An important case is $X = G$, with G acting on itself by inner automorphisms. If H is a subscheme of G , the strict transporter of H into H is also called the *normalizer of H in*

⁵⁷Editor's note: denoted $\underline{\mathrm{Transp}}_G^{\mathrm{str}}(Y, Z)$.

G , and is denoted $\underline{\text{Norm}}_G H$. Hence, if H is a closed group subscheme of G , essentially free over S , then $\underline{\text{Norm}}_G H$ is representable by a closed group subscheme of G .

Finally, let Z be a subscheme of G . Its centralizer $\underline{\text{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure of (d), when one regards Z as acting on G by the operations induced by those of G . Thus, if Z is essentially free over S and G is separated over S , then $\underline{\text{Centr}}_G(Z)$ is a closed group subscheme of G . In particular, if G is essentially free and separated over S , then the center C of G , which is none other than $\underline{\text{Centr}}_G(G)$, is a closed group subscheme of G .

When S is the spectrum of a field, 6.2.2(b) shows that the “essentially free” conditions in examples (a)–(e) above are automatically satisfied, leaving only the separation conditions. Recalling that a group scheme over a field is necessarily separated (VI_A, 0.3), we obtain, for example:

Corollary 6.2.5.— *Let G be a group scheme over a field k , and let Y, Y' be two subschemes of G . Then:*

- (i) *The centralizer of Y in G is a closed group subscheme of G ; in particular, this holds for the center $\underline{\text{Centr}}_G(G)$ of G .*
- (i') *More generally, if $u, v : X \rightarrow G$ are morphisms of schemes, then $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subscheme of G .*
- (ii) *If Y is closed, the transporter $\underline{\text{Transp}}_G(Y', Y)$ is a closed subscheme of G . If Y' is also closed, the same conclusion holds for $\underline{\text{Transp}}_G^{\text{str}}(Y', Y)$.*
- (iii) *For every group subscheme⁵⁸ H of G , $\underline{\text{Norm}}_G(H)$ is a closed group subscheme of G .*

Corollary 6.2.6.—⁵⁹ *Let k be a field and let G be a connected algebraic k -group. Then $\underline{\text{Centr}}_G(G)$ is representable by a closed group subscheme Z of G , and G/Z is an affine algebraic k -group.*

Proof. The first assertion is, of course, contained in 6.2.5(i), but we shall see that it also follows from the proof of the second assertion. Indeed, G acts by the adjoint representation on the finite-dimensional k -vector spaces

$$V_n = \mathcal{O}_{G,e} / \mathfrak{m}_e^{n+1},$$

where $\mathfrak{m}_e$ is the maximal ideal of $\mathcal{O}_{G,e}$. Let K_n be the kernel of the morphism

$$\rho_n : G \longrightarrow \text{GL}(V_n).$$

By VI_A, 5.4.1, ρ_n induces a closed immersion

$$G/K_n \hookrightarrow \text{GL}(V_n),$$

so every G/K_n is affine. Since G is Noetherian, the intersection K of the K_n is equal to one of the K_n ; hence G/K is affine.

On the other hand, if Z denotes the center of G , it is clear that

$$Z \subset K.$$

⁵⁸Editor's note: indeed, over a field k , every group subscheme of G is closed; see VI_A, 0.5.2. Moreover, 6.2.5 completes the insertion of the results taken from VIII, §6.

⁵⁹Editor's note: we have inserted this corollary here (cf. Exposé XII, 6.1), as it will be useful later. We return in 6.3 to the original text of Exposé VI_B.

We shall prove the reverse inclusion. Let $\widehat{\mathcal{O}}_{G,e}$ be the completion of $\mathcal{O}_{G,e}$ for the $\mathfrak{m}_e$ -adic topology, and put

$$\widehat{S} = \operatorname{Spec} \widehat{\mathcal{O}}_{G,e}, \quad S = \operatorname{Spec} \mathcal{O}_{G,e}.$$

Since $\widehat{S} \rightarrow S$ is faithfully flat, and since the two morphisms

$$K \times_k S \longrightarrow S, \quad (g, x) \longmapsto gxg^{-1} \quad \text{and} \quad (g, x) \longmapsto x$$

coincide after the base change $\widehat{S} \rightarrow S$, they coincide. Thus K acts trivially on $\mathcal{O}_{G,e}$.

By 6.2.4(e), the subobject G^K of points of G fixed by K , which is precisely $\underline{\operatorname{Centr}}_G(K)$, is a closed subscheme of G . It is therefore defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_G$. Since G^K contains

$$S = \operatorname{Spec} \mathcal{O}_{G,e}$$

and $\mathcal{I}$ is of finite type (because G is Noetherian), there is an open neighborhood U of e such that

$$\mathcal{I}|_U = 0.$$

The subgroup

$$G^K = \underline{\operatorname{Centr}}_G(K)$$

therefore contains U , and hence also $U \cdot U$, which is equal to G because G is irreducible (VI_A, 0.5). Consequently,

$$\underline{\operatorname{Centr}}_G(K) = G.$$

Thus $K \subset Z$, and therefore $Z = K$, as required.

Remark 6.3.— *Let k be an algebraically closed field, let G be a k -group, and let H be a group subscheme of G . Suppose that G and H are of finite type over k and are reduced. Then $\underline{\operatorname{Norm}}_G H$ (respectively, $\underline{\operatorname{Centr}}_G H$) is representable by a group subscheme of G , whose associated reduced subscheme is precisely the normalizer (respectively, centralizer) of H in G in the sense of the Bible.*

Proposition 6.4.— *Let G be an S -group, and let $u : X \rightarrow G$ be a monomorphism of S -schemes. Set*

$$T = \underline{\operatorname{Transp}}_G(X, X).$$

The following conditions are equivalent:

(i) *T is a group sub- S -functor of G .*

(ii)

$$T = \underline{\operatorname{Transp}}_G^{\operatorname{str}}(X, X) = \underline{\operatorname{Norm}}_G X.$$

These conditions hold in each of the following two cases:

a) *X is of finite presentation over S .*

b) *T is representable by a scheme of finite presentation over S .*

The equivalence of (i) and (ii) follows from the fact that, for every morphism $S' \rightarrow S$ and all $t, t' \in T(S')$, one has

$$tt' \in T(S'),$$

and from the identity

$$\underline{\operatorname{Transp}}_G^{\operatorname{str}}(X, X) = T \cap c(T)$$

(cf. 6.1(ii)).

Suppose first that a) holds. If $t \in T(S)$, then $\text{int}(t)$ is a monomorphism from X to itself, and therefore an S -automorphism of X by EGA IV₄, 17.9.6. Thus

$$t \in \underline{\text{Transp}}_G^{\text{str}}(X, X),$$

which proves the assertion in case a).

In case b), it is clear that

$$\mu(T \times_S T) \subset T,$$

and the assertion follows from the next lemma.

Lemma 6.4.2.—⁶⁰ *Let G be an S -scheme of finite presentation, equipped with an associative law (in the sense of EGA 0_{III}, 8.2.5). Suppose that, for every S -scheme S' and every $g \in G(S')$, right and left translation by g in the set $G(S')$ are injective, and suppose that $G(S) \neq \emptyset$. Then G is an S -group.*

Proof. It is enough to prove that $G(S')$ is a group for every S -scheme S' . The hypothesis immediately implies that right and left translation by every $g \in G(S')$ on $G_{S'}$ are S' -monomorphisms

$$G_{S'} \longrightarrow G_{S'}.$$

They are therefore S' -automorphisms, since G is of finite presentation over S (EGA IV₄, 17.9.6). Consequently, right and left translation by g in the set $G(S')$ are bijective, and the assertion reduces to the following lemma.

Lemma 6.4.3.— *Let G be a nonempty set with an associative law such that right and left translations are bijective. Then G is a group.*

The proof is left to the reader.

Definition 6.5.— *Let G be an S -group, let H be an S -functor, and let $u : H \rightarrow G$ be a monomorphism.*

- (i) The *connected centralizer of H in G* , denoted

$$\underline{\text{Centr}}_G^0 H,$$

is the identity component of the functor $\underline{\text{Centr}}_G H$ (cf. 3.1 and 6.1(iii)).

- (i') For every morphism $u : X \rightarrow G$, one similarly defines the functor

$$\underline{\text{Centr}}^0(u)$$

(cf. 6.2(iv)).

- (ii) When u_s is a closed immersion for every $s \in S$, the *connected normalizer of H in G* , denoted

$$\underline{\text{Norm}}_G^0 H,$$

is the identity component of the functor $\underline{\text{Norm}}_G H$.

N.B. By 1.4.2, the hypothesis in (ii) is satisfied when H is an S -group scheme, G and H are locally of finite type over S , and u is a quasi-compact morphism of S -groups.

Proposition 6.5.1.— *Let G be an S -group locally of finite presentation and quasi-separated over S , let H be a smooth S -group with connected fibers, and let $u : H \rightarrow G$ be a monomorphism. Let N be the normalizer of H in G (cf. 6.1). By 6.5.5 below, N is representable by a closed group subscheme of G , of finite presentation over G . Under these circumstances, the following conditions are equivalent:*

⁶⁰Editor's note: we have retained the numbering of the original; there is no number 6.4.1.

- (i) *The canonical morphism $H \rightarrow N$ is an open immersion.*
- (ii) $N^0 = H$ (cf. 3.10).
- (iii) *For every $s \in S$, one has $H_s = (N_s)^0$.*

Condition (i) implies (ii) by Lemma 3.10.1, since $H^0 = H$. It is also clear that (ii) implies (iii), for

$$H_s = (N^0)_s = (N_s)^0.$$

Finally, let us prove that (iii) implies (i). Since $H_s = (N^0)_s$ for every $s \in S$, the morphism

$$H_s \longrightarrow N_s$$

is an open immersion. Moreover, H and N are locally of finite presentation over S , and H is flat over S . Consequently, $H \rightarrow N$ is an open immersion (EGA IV₄, 17.9.5).

For the reader's convenience, the results 6.8–6.11 of Exposé XI have been included below.⁶¹

Theorem 6.5.2.— *Let X be a smooth scheme over S , with geometrically irreducible fibers.*⁶²

- (i) *For every closed subscheme Y of X , the functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme T of S .

- (ii) *Moreover, if $Y \rightarrow X$ is of finite presentation, then so is $T \rightarrow S$.*

Proof. Let $f : X \rightarrow S$ denote the structure morphism. Since f is faithfully flat and locally of finite presentation, it is a covering for the fpqc topology. Moreover,

$$T = \prod_{X/S} Y/X$$

is evidently a subsheaf of S for the fpqc topology. It follows that the representability of T by a closed subscheme of S is a question local on S for the fpqc topology,⁶³ and the same is true of the question whether T is of finite presentation over S (EGA IV₂, 2.7.1).

After the base change $S' \rightarrow S$ with $S' = X$, we are reduced to the case in which X admits a section

$$\epsilon : S \longrightarrow X.$$

We may also suppose that S is affine, and hence quasi-compact. We then have:

Corollary 6.5.3.— *Under the hypotheses of 6.5.2, suppose that S is quasi-compact, that $X \rightarrow S$ admits a section ϵ , and that $Y \rightarrow X$ is of finite presentation. Then there exists an integer $n \geq 0$ such that*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

⁶¹Editor's note: Sections 6.5.2–6.5.5, taken from Exposé XI, 6.8–6.11, have been inserted here. This was in fact suggested by A. Grothendieck in a note at the beginning of XI 6: “The present section does not use the results of Sections 3, 4, and 5 [of XI]; its natural place would be in VI_B.”

⁶²Editor's note: on the one hand, we have corrected the original by replacing “connected” with “irreducible” (see the proof). On the other hand, according to [RG71], I, 3.3.4(iii) and 4.1.1, it is enough to assume that X is flat over S , with geometrically irreducible fibers and without embedded components.

⁶³Editor's note: see, for example, the addition 1.7 in Exposé VIII.

where X_n is the n -th infinitesimal neighborhood of the immersion $\epsilon : S \rightarrow X$, and $Y_n = Y \cap X_n$.

When Y is of finite presentation over X , this corollary does indeed imply 6.5.2. Since X is smooth over S , X_n is finite and locally free over S , and is therefore, a fortiori, essentially free over S (see 6.2.1). Hence

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T of S , of finite presentation over S , by 6.2.3.

Let us first prove 6.5.3, and hence 6.5.2, when S is Noetherian. Put

$$T_n = \prod_{X_n/S} Y_n/X_n.$$

The T_n form a decreasing sequence of closed subschemes of S , and since S is Noetherian this sequence is stationary. Let

$$R = \bigcap_{n \geq 0} \prod_{X_n/S} Y_n/X_n$$

be their common value for all sufficiently large n . Clearly $T \subset R$, and it is enough to prove $R \subset T$.

After the base change $R \rightarrow S$, we are reduced to the case

$$R = S,$$

that is,

$$Y_n = X_n \quad \text{for every } n, \quad \text{or equivalently} \quad Y \supset X_n \quad \text{for every } n.$$

We must then prove $T = S$, that is, $Y = X$.

Now $Y \supset X_n$ for every n , together with the fact that X is locally Noetherian, implies that in a neighborhood of each point of $\epsilon(S)$, Y is an induced open subscheme of X .⁶⁴ Consequently, there exists an induced open subscheme U of X , containing $\epsilon(S)$, such that $U \subset Y$. By EGA IV₃, 11.10.10, since the fibers of X/S are integral, U is schematically dense in X . Since Y is a closed subscheme containing U , it follows that $Y = X$. This proves 6.5.3, and hence 6.5.2, when S is Noetherian.

The general case is reduced to the preceding one. For every $s \in S$, there exist an affine open neighborhood U of s and an affine open neighborhood V of $\epsilon(s)$ such that $f(V) \subset U$. Then $f(V)$ is an open neighborhood of s contained in U . Let S' be an affine open neighborhood of s contained in

$$\epsilon^{-1}(V) \cap f(V),$$

and put

$$X' = V \cap f^{-1}(S').$$

Then X' and S' are affine open subsets of X and S , respectively, and X'/S' admits a section. By the local nature of 6.5.2 and 6.5.3, we may suppose that $S = S'$.

Thus X' is an affine open subset of X containing $\epsilon(S)$. Since every fiber X_s is irreducible, X'_s is schematically dense in X_s . Consequently, by EGA IV₃, 11.10.10, X' is schematically dense in X ; likewise, after every base change $S_1 \rightarrow S$,

$$X' \times_S S_1$$

⁶⁴Editor's note: see, for example, the proof of 6.2.6.

is schematically dense in $X \times_S S_1$.

It follows that

$$\prod_{X/S} Y/X = \prod_{X'/S} Y'/X', \quad Y' = Y \cap X'.$$

This reduces us to the case $X = X'$, so we may suppose that both S and X are affine. Moreover, if

$$X = \operatorname{Spec} B$$

and J is the ideal of B defining Y , then J is the filtered colimit of its finitely generated subideals. Thus Y is the intersection of closed subschemes Y_i of X that are of finite presentation over X , and consequently

$$\prod_{X/S} Y/X = \bigcap_i \prod_{X/S} Y_i/X.$$

To prove 6.5.2, we are therefore reduced to the case in which Y is of finite presentation over X , with S and X affine.

In that case X and Y over S are obtained by base change $S \rightarrow S_0$ from an analogous situation X_0, Y_0 over a Noetherian scheme S_0 . This reduces us to the Noetherian case already treated, and completes the proof of 6.5.2 and 6.5.3.

Corollary 6.5.4.— *Let X be a smooth S -group scheme of finite presentation with connected fibers, let Y be a group scheme of finite presentation over S , and let $i : Y \rightarrow X$ be a monomorphism of S -group schemes.*

(i) *The functor*

$$\prod_{X/S} Y/X$$

is representable by a closed subscheme of S of finite presentation.

(ii) *If, moreover, S is quasi-compact, then for all sufficiently large n ,*

$$\prod_{X/S} Y/X = \prod_{X_n/S} Y_n/X_n,$$

where X_n denotes the n -th infinitesimal neighborhood of the unit section $\epsilon : S \rightarrow X$, and $Y_n = X_n \cap Y$.

The proof is essentially that of 6.5.3.⁶⁵ On the one hand, i is locally of finite presentation (EGA IV₁, 1.4.3). On the other hand, the unit sections of Y and X induce bijective immersions

$$S \longrightarrow Y_n \quad \text{and} \quad S \longrightarrow X_n,$$

and hence isomorphisms of S_{red} with $(Y_n)_{\text{red}}$ and $(X_n)_{\text{red}}$. Consequently, i_n is quasi-compact, hence of finite type, and one has the commutative diagram

$$\begin{array}{ccc} (Y_n)_{\text{red}} & \xrightarrow{\tau} & Y_n \\ & \searrow \sigma & \downarrow i_n \\ & & X_n \end{array}$$

⁶⁵Editor's note: the original has been expanded in what follows, and references to EGA IV have been added.

where σ and τ are closed immersions and τ is surjective. Since i_n is separated (being a monomorphism), it follows that i_n is proper (EGA II, 5.4.3). Thus i_n is a proper monomorphism of finite presentation, and therefore a closed immersion (EGA IV₃, 8.11.5). It follows from 6.2.3 that

$$\prod_{X_n/S} Y_n/X_n$$

is representable by a closed subscheme T_n of S , of finite presentation over S . It remains to prove the last assertion of 6.5.4 when S is assumed affine as well.

We again reduce immediately to the case where S is Noetherian. With the notation of the proof of 6.5.3, it is enough to prove $R = T$, or equivalently that

$$Y \supset X_n \quad \text{for every } n \quad \implies \quad Y = X.$$

The hypothesis implies that $i : Y \rightarrow X$ is étale at the points of the unit section of Y over S . Thus Y is smooth over S at those points, and the open subset U of points of Y at which Y is smooth over S is an induced open subgroup of Y (cf. 2.3). The morphism

$$\tau : U \longrightarrow X$$

is then an étale monomorphism by 2.5, and hence an open immersion. Since the fibers of X are connected and every open subgroup of an algebraic group is also closed, τ is a surjective open immersion, that is, an isomorphism. Therefore $U = X$, and a fortiori $Y = X$. This completes the proof of 6.5.4.

Proceeding as in 6.2.4(e), one deduces Corollary 6.5.5 from 6.5.4.

Corollary 6.5.5.— *Let G be an S -group scheme locally of finite type and quasi-separated over S , let H be a smooth group scheme of finite presentation over S with connected fibers, and let $i : H \rightarrow G$ be a monomorphism of S -groups. Then:*

- a) $\underline{\text{Centr}}_G(H)$ and $\underline{\text{Norm}}_G(H)$ are representable by closed subschemes of G , of finite presentation over G .
- a') Likewise, for every monomorphism $j : K \rightarrow G$ of S -group schemes that is of finite presentation, $\underline{\text{Transp}}_G(H, K)$ is representable by a closed subscheme of G , of finite presentation over G .
- b) If G is quasi-compact, there exists an integer $n \geq 0$ such that, if H_n denotes the n -th infinitesimal neighborhood of the unit section of H , then

$$\begin{aligned} \underline{\text{Centr}}_G(H) &= \underline{\text{Centr}}_G(H_n), & \underline{\text{Norm}}_G(H) &= \underline{\text{Norm}}_G(H_n), \\ \underline{\text{Transp}}_G(H, K) &= \underline{\text{Transp}}_G(H_n, K) = \underline{\text{Transp}}_G(H_n, K_n). \end{aligned}$$

*Proof.*⁶⁶ First observe that the hypothesis on G implies that the monomorphism $H \rightarrow G$ is of finite presentation (EGA IV₁, 1.2.4 and 1.4.3), as is the diagonal immersion

$$\Delta_{G/S} : G \longrightarrow G \times_S G$$

(loc. cit., 1.4.3.1). The case of $\underline{\text{Norm}}_G(H)$ therefore reduces to that of the transporter, by taking $H = K$.

⁶⁶Editor's note: the argument that follows has been expanded. This also completes the insertion of XI, 6.8–6.11; at 6.6 below we return to the original text of VI_B.

In view of 6.2.4(e), we apply 6.5.4 to the group scheme

$$X = H_G = H \times_S G$$

over the base scheme G , and to the following group subscheme Y .

In the case of $\underline{\text{Transp}}_G(H, K)$, take Y to be the inverse image of K_G under the morphism of G -groups

$$H \times_S G \longrightarrow G \times_S G, \quad (h, g) \longmapsto (ghg^{-1}, g).$$

In the case of $\underline{\text{Centr}}_G(H)$, take Y to be the inverse image of the diagonal subgroup

$$\Delta_{G/S}(G)_G$$

of $(G \times_S G)_G$ under the morphism of G -groups

$$H \times_S G \longrightarrow G \times_S G \times_S G, \quad (h, g) \longmapsto (h, ghg^{-1}, g).$$

Definition 6.6.— Let G be an S -functor in groups and let H be a group sub- S -functor of G . One says that H is *invariant* (respectively, *central*, respectively, *characteristic*) in G if

$$\underline{\text{Norm}}_G H = G$$

(respectively, if $\underline{\text{Centr}}_G H = G$, respectively, if for every S -scheme T and every automorphism

$$a \in \text{Aut}_{T\text{-gr}}(G_T)$$

one has $a(H_T) \subset H_T$). In other words, for every S -scheme T , the subgroup $H(T)$ of $G(T)$ is normal in $G(T)$ (respectively, central in $G(T)$, respectively, invariant under every automorphism of G_T).

N.B. If H is central (respectively, characteristic), then it is invariant.

Remark 6.7.— Let G and H be two S -groups and let $u : H \rightarrow G$ be a monomorphism. The group H is invariant (respectively, central) in G if and only if the morphism

$$\mu \circ (c \circ \text{pr}_2, \mu \circ (u \times \text{id}_G)) : H \times_S G \longrightarrow G$$

(defined by

$$(h, g) \longmapsto g^{-1}hg$$

for $g \in G(S')$ and $h \in H(S')$) factors through u (respectively, is equal to $u \circ \text{pr}_1$). The group H is characteristic in G if and only if, for every S -scheme T and every T -group automorphism a of G_T , the morphism $a \circ u(T)$ factors through $u(T)$.

Example 6.8.— Let G be an S -functor in groups. Then $\underline{\text{Centr}}G$ is characteristic and central. If G_s is representable for every $s \in S$, then G^0 is characteristic. This follows from the definitions and 3.3.

6.9.—⁶⁷ In [RG71], I, 3.3.3, the authors introduce the geometric notion of a *pure S -scheme*, which is local on S for the étale topology; we refer to *loc. cit.* for the precise definition. We merely record the following points:

- a) (*loc. cit.*, Theorem 3.3.5) If B is a flat, finitely presented A -algebra, then B is a projective A -module if and only if $\text{Spec}(B)$ is pure over $\text{Spec}(A)$.

⁶⁷Editor's note: this subsection has been added.

- b) Consequently, if $X \rightarrow S$ is locally of finite presentation, flat, and pure, then X is essentially free over S .
- c) (*loc. cit.*, 3.3.4(iii)) If $X \rightarrow S$ is locally of finite type, flat, has geometrically irreducible fibers, and has no embedded components, then X is pure over S .

Every group scheme locally of finite type over a field is Cohen–Macaulay (cf. VI_A, 1.1.1), and therefore has no embedded components. We obtain in particular:

- d) Every S -group scheme G that is locally of finite presentation, flat, and has connected fibers is pure over S , and hence essentially free over S .

All the statements of 6.2.4(e) may therefore be used with the results of (b) and (d) above taken into account.

7. Generated subgroups; commutator group

Throughout this section, k denotes a fixed field.

Proposition 7.1.— *Let G be a k -group, let $(X_i)_{i \in I}$ be a family of geometrically reduced k -schemes,⁶⁸ and, for every $i \in I$, let $f_i : X_i \rightarrow G$ be a k -morphism.*

- (i) *There exists a smallest closed k -group subscheme of G through which every f_i factors. It is denoted*

$$\Gamma_G((f_i)_{i \in I}).$$

It is a geometrically reduced k -scheme, and hence smooth when G is locally of finite type over k (1.3.1).

- (ii) *Put*

$$X = \coprod_{i \in I} X_i,$$

and let $f : X \rightarrow G$ be the morphism whose restriction to X_i is f_i , for every $i \in I$. Put

$$X^1 = X \amalg X,$$

and let $f^1 : X^1 \rightarrow G$ be the morphism whose restrictions to the two copies of X are f and $c \circ f$, respectively. For every $n > 1$, put

$$X^n = X^1 \times_k X^{n-1}, \quad f^n = \mu \circ (f^1 \times_k f^{n-1}) : X^n \rightarrow G.$$

Then $\Gamma_G((f_i)_{i \in I})$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n \geq 1} f^n(X^n).$$

- (iii) *For every k -scheme S , $\Gamma_G((f_i)_{i \in I})_S$ is the smallest closed group subscheme of G_S through which every*

$$f_{i,S} : (X_i)_S \rightarrow G_S$$

factors.

⁶⁸Editor's note: here and below, the little-used term “separable” has been replaced by the customary term “geometrically reduced”; see EGA IV₂, 4.6.2.

(iii') Moreover, $\Gamma_G((f_i)_{i \in I})_S$ is the smallest group subscheme of G_S , not necessarily closed, through which every $f_{i,S}$ factors.⁶⁹

First note that, to prove (i), (iii), and (iii'), after defining X and f as in (ii), we are reduced to the case in which I consists of a single element.

Let H be the reduced subscheme of G whose underlying set is

$$\bigcup_{n \geq 1} \overline{f^n(X^n)}.$$

The family of morphisms

$$f^n : X^n \longrightarrow H$$

is schematically dominant (cf. EGA IV₃, 11.10.4), and therefore every closed subscheme of G through which all the f^n factor also contains H . Moreover, by *loc. cit.*, 11.10.7, H is geometrically reduced. To prove (i) and (ii), it is therefore enough to show that H is a group subscheme of G , that is, that the restriction of c to H and the restriction of μ to $H \times_k H$ factor through the inclusion $H \rightarrow G$.

Since H is geometrically reduced, $H \times_k H$ is reduced, and it is enough to verify, set-theoretically, that

$$c(H) \subset H, \quad \mu(H \times_k H) \subset H.$$

By EGA IV₃, 11.10.6, the union of the

$$f^n_{(H)}(X^n \times_k H)$$

is schematically dense in $H \times_k H$. Similarly, for every $n \geq 1$, the union of the

$$f^m_{(X^n)}(X^n \times_k X^m), \quad m \geq 1,$$

is schematically dense in $X^n \times_k H$. It is therefore enough to prove

$$\mu\left(f^n_{(H)}\left(f^m_{(X^n)}(X^n \times_k X^m)\right)\right) \subset H$$

and $c(f^n(X^n)) \subset H$. But

$$\begin{aligned} \mu\left(f^n_{(H)}\left(f^m_{(X^n)}(X^n \times_k X^m)\right)\right) &= \mu((f^n \times_k f^m)(X^n \times_k X^m)) \\ &= f^{n+m}(X^{n+m}) \subset H. \end{aligned}$$

Also, since $c(f^1(X^1)) \subset f^1(X^1)$, for every n one has

$$c(f^n(X^n)) \subset f^n(X^n) \subset H.$$

This proves (i) and (ii).

We now prove (iii). Let G' be a closed group subscheme of G_S through which f_S factors. We must show that G' contains H_S , or equivalently that

$$H_S = H_S \times_{G_S} G'.$$

Put

$$H'_S = H_S \times_{G_S} G' = G' \cap H_S.$$

⁶⁹Editor's note: the original stated (iii') under the additional hypothesis that G be locally of finite type over k ; this hypothesis may be omitted in view of VI_A, 0.5.2.

Every f_S^n factors through both G' and H_S , and hence through H'_S . Since the family

$$f^n : X^n \longrightarrow H$$

is schematically dominant, EGA IV₃, 11.10.6 implies that the base-changed family

$$f_S^n : X_S^n \longrightarrow H_S$$

is schematically dominant as well. Because every f_S^n factors through H'_S , EGA IV₃, 11.10.1(c) gives

$$H'_S = H_S.$$

This proves (iii).

Finally, we show that H_S is the smallest group subscheme, not necessarily closed, of G_S through which f_S factors. Let G' be such a group subscheme and again put

$$H'_S = H_S \times_{G_S} G'.$$

We must show $H'_S = H_S$. By (iii), it is enough to show that H'_S is closed in H_S . It is enough in turn to show that H_S and H'_S have the same underlying set, and for this it suffices to show, for every $s \in S$, that

$$H'_s := H'_S \times_S \kappa(s) = H_s \times_{G_s} G'_s = H_s.$$

By VI_A, 0.5.2, the $\kappa(s)$ -group subscheme G'_s is closed in G_s . Hence H'_s is closed in H_s , and the preceding argument, applied to H'_s , H_s , and the f_s^n , shows that

$$H'_s = H_s.$$

This proves (iii') and completes the proof.

Corollary 7.1.1.—⁷⁰ *Let G be a k -group locally of finite type, let A and B be two k -subgroups of G that are smooth and of finite type, and let i_A (respectively, i_B) be the inclusion of A (respectively, B) into G . Suppose that B normalizes A . Then*

$$A \cdot B = \Gamma_G(i_A, i_B).$$

Proof. Let

$$H = A \rtimes B$$

be the semidirect product of A and B (cf. I, 2.3.5); it is a smooth k -group of finite type. The group morphism

$$u : H \longrightarrow G, \quad (a, b) \longmapsto ab,$$

is therefore quasi-compact. Hence, by 1.2,

$$u(H) = A \cdot B$$

is a closed reduced subscheme of G , and is a group in the category $(\text{Sch}/k)_{\text{red}}$. By EGA IV₃, 11.10.7, $A \cdot B = u(H)$ is geometrically reduced, since H is; it is therefore a closed subgroup of G . The evident inclusion

$$A \cdot B \subset \Gamma_G(i_A, i_B)$$

now proves the corollary.

⁷⁰Editor's note: this corollary has been added to point out this special case of 7.1.

Definitions and Remarks 7.2.—⁷¹

- (i) Let G be a k -group, let $(X_i)_{i \in I}$ be a family of geometrically reduced k -schemes, and, for every $i \in I$, let $f_i : X_i \rightarrow G$ be a k -morphism. The *closed group subscheme of G generated by the family $(f_i)_{i \in I}$* is the smallest closed group subscheme of G through which every f_i factors; in this subsection it is denoted

$$\Gamma_G((f_i)_{i \in I}).$$

If X is a subscheme of G , geometrically reduced over k , and $f : X \hookrightarrow G$ is the immersion, we write $\Gamma_G(X)$ instead of $\Gamma_G(f)$.

- (i') With the notation of 7.1(ii), we shall sometimes put

$$\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n).$$

Notice that $\Gamma'_G(f)$ is a subset of G stable under the group law, in the sense of 3.0.

- (ii) If X_1 and X_2 are geometrically reduced k -schemes and $f_1 : X_1 \rightarrow G$ and $f_2 : X_2 \rightarrow G$ are k -morphisms such that

$$f_1(X_1) = f_2(X_2)$$

as sets, then it is clear that

$$\Gamma_G(f_1) = \Gamma_G(f_2).$$

- (iii) Let E be a subset of G such that the reduced subscheme $\overline{E}$ of G is geometrically reduced. The *closed group subscheme of G generated by E* is the group subscheme

$$\Gamma_G(E) := \Gamma_G(i),$$

where $i : \overline{E} \hookrightarrow G$ is the inclusion of that reduced subscheme.

- (iv) Since every group subscheme of G is closed, by VI_A, 0.5.2,⁷² we shall speak of the “generated group subscheme” rather than the “generated closed group subscheme.”
- (v) Let X be a geometrically reduced k -scheme and let $f : X \rightarrow G$ be a k -morphism. Suppose that $f(X)$ contains the unit element e of G . Put

$$X'^1 = X \times_k X, \quad f'^1 = \mu \circ (f \times_k (c \circ f)),$$

and, for $n > 1$, put

$$X'^n = X'^1 \times_k X'^{n-1}, \quad f'^n = \mu \circ (f'^1 \times_k f'^{n-1}).$$

Then $\Gamma_G(f)$ is the reduced subscheme of G whose underlying space is the closure of

$$\bigcup_{n \geq 1} f'^n(X'^n).$$

Indeed, recall the notation of 7.1(ii):

$$X^1 = X \amalg X, \quad X^n = X^1 \times_k X^{n-1} \quad (n \geq 2).$$

⁷¹Editor's note: item (viii) of the original has been placed in (i'), and items (vi) and (vii) have been singled out as Corollary 7.2.1 and Definition 7.2.2 below.

⁷²Editor's note: the hypothesis that G be locally of finite type over k has been removed here.

The hypothesis $e \in f(X)$ gives, for every $n \geq 1$,

$$f^n(X^n) \subset f'^n(X'^n) \subset f^{2n}(X^{2n}).$$

Consequently, there exists an integer n such that $f^n(X^n) = \Gamma_G(f)$ if and only if there exists an integer m such that $f'^m(X'^m) = \Gamma_G(f)$.

From Remark 7.2(v) we deduce the following result.

Corollary 7.2.1.— *Let X be a geometrically reduced and geometrically connected k -scheme, and let $f : X \rightarrow G$ be a k -morphism such that $f(X)$ contains the unit element of G . Then the k -group $\Gamma_G(f)$ is connected, and hence irreducible.*

Proof. Each X'^n is connected, so the union of the

$$f'^n(X'^n),$$

all of which contain the unit element, is connected. Its closure $\Gamma_G(f)$ is therefore connected as well. It follows that $\Gamma_G(f)$ is irreducible, by VI_A, 2.4 when G , and hence also $\Gamma_G(f)$, is locally of finite type over k , and by VI_A, 2.6.5(iii) in the general case.

Definition 7.2.2.— Let G be a k -group.

- a) Let A and B be two geometrically reduced k -group subschemes of G . The *commutator group subscheme of A and B in G* , denoted

$$(A, B)_G \quad \text{or simply} \quad (A, B),$$

is the closed group subscheme of G generated by the morphism

$$\nu : A \times_k B \longrightarrow G$$

defined, for every k -scheme S , by

$$(a, b) \longmapsto aba^{-1}b^{-1}, \quad a \in A(S), \quad b \in B(S).$$

- b) Suppose that G is geometrically reduced over k . The *derived group of G* , denoted $\mathfrak{D}(G)$,⁷³ is the group

$$\mathfrak{D}(G) = (G, G).$$

N.B. The group G is commutative if and only if $\mathfrak{D}(G)$ is the unit k -group.

Reminders 7.3.0.—⁷⁴ Let $\varphi : Y \rightarrow Z$ be a morphism of S -schemes. Recall that the image presheaf $\varphi(Y)$ is the S -functor which assigns to every S' over S the subset

$$\varphi(Y(S')) \subset Z(S').$$

Suppose that T is a subscheme of Z . If the inclusion of presheaves

$$\varphi(Y) \hookrightarrow Z$$

factors through T , that is, if

$$\varphi \circ h \in T(S')$$

⁷³Editor's note: the notation $\text{Der}(G)$ of the original has been changed to avoid any risk of confusion with a space of derivations.

⁷⁴Editor's note: these reminders have been added, and the statement of 7.3 has consequently been modified and expanded.

for every S -morphism $h : S' \rightarrow Y$, then, set-theoretically, $\varphi(Y) \subset T$: take $h = \text{id}_Y$. Conversely, if Y is reduced and $\varphi(Y) \subset T$ set-theoretically, then φ factors through T .

Recall also that, by 6.2, if H is a closed k -group subscheme of G , then

$$\underline{\text{Centr}}_G H \quad \text{and} \quad \underline{\text{Norm}}_G H$$

are representable by closed k -group subschemes of G .

Corollary 7.3.— *Let G be a k -group, let X be a geometrically reduced k -scheme, and let $f : X \rightarrow G$ be a k -morphism.*

(i) *Let S be a k -scheme and let u be an endomorphism of the S -group G_S .*

(a) *If*

$$u(f_S(X_S)) \subset \Gamma_G(f)_S$$

set-theoretically, then the morphism

$$u : \Gamma_G(f)_S \longrightarrow G_S$$

factors through $\Gamma_G(f)_S$.

(b) *If u is an automorphism of the S -group G_S and*

$$u(f_S(X_S)) = f_S(X_S)$$

set-theoretically, then u induces an automorphism of $\Gamma_G(f)_S$. In particular, if $g \in G(S)$ satisfies

$$\text{int}(g)(f_S(X_S)) = f_S(X_S)$$

set-theoretically, then

$$g \in \underline{\text{Norm}}_{G_S}(\Gamma_G(f)_S)(S).$$

(c) *If $u \circ f_S = f_S$, then the restriction of u to the subgroup $\Gamma_G(f)_S$ of G_S is the identity. In particular, if $g \in G(S)$ satisfies*

$$\text{int}(g) \circ f_S = f_S,$$

then

$$g \in \underline{\text{Centr}}_{G_S}(\Gamma_G(f)_S)(S).$$

(ii) *Let H be a group subscheme of G . Then H centralizes (respectively, normalizes) $\Gamma_G(f)$ if and only if, for every k -scheme S and every $h \in H(S)$,*

$$\text{int}(h) \circ f_S = f_S$$

(respectively,

$$\text{int}(h)(f_S(X_S)) \subset \Gamma_G(f)_S$$

set-theoretically). Equivalently, for every $S' \rightarrow S$ and every $x \in X(S')$, the elements $h_{S'}$ and $f(x)$ of $G(S')$ commute (respectively,

$$h_{S'} f(x) h_{S'}^{-1} \in \Gamma_G(f)(S').$$

).

- (iii) In particular, let Y be a second geometrically reduced k -scheme and let $\varphi : Y \rightarrow G$ be a k -morphism. Suppose that, for every k -scheme S' , every $x \in X(S')$, and every $y \in Y(S')$, the elements $\varphi(y)$ and $f(x)$ of $G(S')$ commute (respectively,

$$\text{int}(\varphi(y))(f(x)) \in \Gamma_G(f)(S').$$

). Then $\Gamma_G(\varphi)$ is a k -subgroup of

$$\underline{\text{Centr}}_G(\Gamma_G(f)) \quad (\text{respectively, of}) \quad \underline{\text{Norm}}_G(\Gamma_G(f)).$$

- (iv) If g is a k -point of G , then the k -group $\Gamma_G(\{g\})$ is commutative.
 (v) Let A and B be two group subschemes of G , geometrically reduced over k . If A and B are invariant (respectively, characteristic), then so is (A, B) .

Proof. We first prove (i)(a). Put

$$\Gamma_S = \Gamma_G(f)_S.$$

Then $u^{-1}(\Gamma_S)$ is a closed S -group subscheme of G_S . By 7.1(iii), it is therefore enough to show that f_S factors through $u^{-1}(\Gamma_S)$. Since

$$u(f_S(X_S)) \subset \Gamma_S$$

and X_S is reduced, $u \circ f_S$ factors through Γ_S ; hence f_S factors through $u^{-1}(\Gamma_S)$. This proves (a).

The first assertion of (b) follows by applying (a) to u and u^{-1} . In fact, it is enough to assume

$$u(f_S(X_S)) \subset \Gamma_S \quad \text{and} \quad u^{-1}(f_S(X_S)) \subset \Gamma_S.$$

The second assertion is the special case $u = \text{int}(g)$.

We now prove (c). Consider the morphism of S -groups

$$G_S \longrightarrow G_S \times_S G_S, \quad g \longmapsto (u(g), g),$$

and let H be the inverse image of the diagonal. It is a group subscheme of G_S . Since G is separated over k , by VI_A, 0.3, G_S is separated over S , and hence H is closed in G_S . By hypothesis, f_S factors through H ; therefore H contains $\Gamma_G(f)_S$, by 7.1(iii). This proves the first assertion of (c), and the second is the special case $u = \text{int}(g)$.

We next prove (ii). Put

$$\Gamma = \Gamma_G(f)$$

and let $i : \Gamma \hookrightarrow G$ denote the inclusion. The group H is contained in

$$N = \underline{\text{Norm}}_G(\Gamma)$$

if and only if, for every k -scheme S and every $h \in H(S)$,

$$\text{int}(h)(\Gamma_S) \subset \Gamma_S.$$

Applying this condition to h and h^{-1} shows that the inclusion is then an equality. By (i)(a), this condition holds if and only if

$$\text{int}(h)(f_S(X_S)) \subset \Gamma_S.$$

Similarly, H is contained in

$$C = \underline{\text{Centr}}_G(\Gamma)$$

if and only if, for every k -scheme S and every $h \in H(S)$,

$$\text{int}(h) \circ i_S = i_S.$$

By (i)(c), this holds if and only if

$$\text{int}(h) \circ f_S = f_S.$$

This proves (ii).

We now prove (iii). In view of (ii), the hypothesis implies that φ factors through C (respectively, N). The latter is a closed subgroup of G , by 6.2, and therefore contains $\Gamma_G(\varphi)$, by 7.1(i).*

Assertion (iv) follows from (iii) by taking both f and φ to be the closed immersion

$$\text{Spec } k \hookrightarrow G$$

defined by g . Indeed, for every k -scheme S , the sets $X(S)$ and $Y(S)$ then each have a single element, which is sent by f , respectively by φ , to g .

Finally, we prove (v). Let

$$\nu : G \times_k G \longrightarrow G, \quad (g, h) \longmapsto ghg^{-1}h^{-1},$$

and let ν' be its restriction to $A \times_k B$. By 7.2.2,

$$(A, B) = \Gamma_G(\nu').$$

Let S be a k -scheme, and let u be an inner automorphism, respectively an arbitrary S -group automorphism, of G_S . Then

$$u \circ \nu_S = \nu_S \circ (u \times_S u).$$

The hypothesis implies that u induces automorphisms u_1 of A_S and u_2 of B_S . Consequently,

$$u(\nu'_S(A_S \times_S B_S)) = \nu'_S(u_1(A_S) \times_S u_2(B_S)) = \nu'_S(A_S \times_S B_S).$$

By (i)(b), u therefore induces an automorphism of $(A, B)_S$. This proves (v).

Proposition 7.4.— *Let G be a k -group locally of finite type, let X be a finite-type k -scheme that is geometrically reduced and geometrically connected, and let $f : X \rightarrow G$ be a k -morphism such that $f(X)$ contains the unit element e of G . Then, with the notation of 7.1(ii), there exists an integer N such that, set-theoretically,*

$$f^N(X^N) = \Gamma_G(f).$$

Proof. By 7.1(iii) and EGA IV₂, 2.6.1, we may assume that k is algebraically closed. By Corollary 7.2.1, we may assume that $G = G^0$. Finally, with the notation of 7.2(v), it is enough to show that there exists an integer N such that

$$f'^N(X'^N) = \Gamma_G(f).$$

First case. Suppose that X is irreducible. The closed irreducible subsets

$$\overline{f'^n(X'^n)}$$

*Translator's note: the French source prints $\Gamma_G(f)$ here. Since φ , rather than f , is the morphism shown to factor through C (respectively, N), 7.1(i) gives $\Gamma_G(\varphi)$, as required by the statement.

then form an increasing sequence in the space G , which is Noetherian because $G = G^0$ is of finite type over k (VI_A, 2.4). Hence this sequence is stationary, and there exists an integer m such that

$$\overline{f'^m(X'^m)} = \Gamma_G(f).$$

Moreover, because X and G are of finite type over k , the morphisms f'^n are of finite type over k . Consequently, $f'^m(X'^m)$ is constructible in G (EGA IV₁, 1.8.5), and therefore contains an open subset U of its closure $\Gamma_G(f)$ (EGA 0_{IV}, 9.2.3). By VI_A, 0.5, one then has

$$\Gamma_G(f) \subset U \cdot U \subset f'^{2m}(X'^{2m}) \subset \Gamma_G(f),$$

and hence

$$f'^{2m}(X'^{2m}) = \Gamma_G(f).$$

Second case. Suppose that X has exactly two irreducible components A_1 and A_2 . Since X is connected and k is algebraically closed, there exists

$$a \in (A_1 \cap A_2)(k).$$

The four irreducible subsets

$$A_i \times_k A_j \quad (i, j = 1, 2)$$

therefore cover $X \times_k X$, and the image of each under the morphism

$$f'^1 = \mu \circ (f \times_k (c \circ f))$$

contains e . Let $f'_{ij}{}^1$ denote the restriction of f'^1 to the reduced subscheme $A_i \times_k A_j$, and put

$$\begin{aligned} Y &= (A_1 \times_k A_1) \times_k (A_1 \times_k A_2) \times_k (A_2 \times_k A_2) \times_k (A_2 \times_k A_1), \\ g &= \mu \circ \left(\mu \circ (f'_{11}{}^1 \times_k f'_{12}{}^1) \times_k \mu \circ (f'_{22}{}^1 \times_k f'_{21}{}^1) \right). \end{aligned}$$

The scheme Y is irreducible, reduced, and of finite type. By the first case, there is therefore an integer m such that

$$g'^m(Y'^m) = \Gamma_G(g).$$

For every $n \geq 1$, however,*

$$f'^n(X'^n) \subset g'^n(Y'^n) \subset f'^{4n}(X'^{4n}).$$

It follows that

$$\Gamma_G(f) = \Gamma_G(g) \quad \text{and} \quad f'^{4m}(X'^{4m}) = \Gamma_G(f).$$

General case. We argue by induction on the number of irreducible components of X ; this number is finite because X , being of finite type over k , is Noetherian. Suppose the proposition has been proved when X has at most $r-1$ irreducible components, and suppose that X has r such components, say $A_1, \dots, A_r$. The unit e belongs to the image of one of the A_i ; suppose, for example, that

$$e \in f(A_1).$$

Put

$$X_r = A_1 \cup \dots \cup A_{r-1}, \quad Y = \Gamma_G(f_r),$$

*Translator's note: the French source prints $g'^n(Y'^n)$ in the middle term. Since the source of g'^n is Y'^n , the prime is restored here; compare the preceding equality $g'^m(Y'^m) = \Gamma_G(g)$.

where f_r denotes the restriction of f to the reduced subscheme X_r . We choose the numbering of the A_i so that X_r is connected, which is always possible. Then Y is a closed, reduced, irreducible subgroup of G , by Corollary 7.2.1.

Put

$$Z = \overline{f(A_r)}$$

and let $T = Y \cup Z$, endowed with its closed reduced subscheme structure; let $i : T \hookrightarrow G$ be the inclusion. It follows immediately from 7.2(ii) that

$$\Gamma_G(i) = \Gamma_G(f).$$

Moreover, T is connected: since X is connected, X_r and A_r have a common point a , and Y and Z have the common point $f(a)$. Also, $e \in T$, and T has at most two irreducible components, because Y and Z are irreducible.

By the induction hypothesis, there exists an integer m such that

$$f_r'^m(X_r'^m) = \Gamma_G(f_r) = Y.$$

Since $X = X_r \cup A_r$, and since

$$Z = \overline{f(A_r)} \subset \overline{f'^m(X'^m)}$$

because $e \in f(A_1)$, we obtain

$$f(X) \subset Y \cup Z \subset \overline{f'^m(X'^m)}.$$

On the other hand, because T has at most two irreducible components, the second case gives an integer p such that

$$i'^p(T'^p) = \Gamma_G(i) = \Gamma_G(f).$$

Now

$$T \subset \overline{f'^m(X'^m)},$$

and hence

$$\overline{f'^{mp}(X'^{mp})} = \Gamma_G(f).$$

As in the first case, this last equality implies

$$f'^{2mp}(X'^{2mp}) = \Gamma_G(f).$$

This completes the proof.

Lemma 7.5.— *Let S be a scheme, let G be an S -group scheme, let X be an S -prescheme, and let $f : X \rightarrow G$ be an S -morphism. Suppose that X and G are locally of finite presentation over S , and that, for every $s \in S$ and every maximal point g of G_s , there exists a point x of X such that $f(x) = g$ and f is flat at x . Then the morphism*

$$\mu \circ (f \times_S f) : X \times_S X \longrightarrow G$$

is a covering for the (fppf) topology.

⁷⁵ By EGA IV₃, 11.3.1, the set V of points of X at which f is flat is open, and $f|_V$ is an open morphism. Thus

$$U = f(V)$$

⁷⁵Editor's note: the argument in the original has been expanded in what follows.

is an open subset of G . Moreover, by hypothesis,

$$U \cap G_s$$

is dense in G_s for every $s \in S$. Let ϕ denote the restriction of $\mu \circ (f \times_S f)$ to $V \times_S V$.

It is enough to show that ϕ is a covering for the (fppf) topology, and for this it is enough to show that ϕ is faithfully flat and of finite presentation (Exposé IV, 6.3.1(iv)). Now ϕ is the composite

$$V \times_S V \xrightarrow{f|_V \times f|_V} U \times_S U \xrightarrow{\mu} G,$$

where the first morphism is faithfully flat and locally of finite presentation, since $f|_V$ is. It therefore remains only to prove the same assertions for the restriction of μ to $U \times_S U$.

Since $G \rightarrow S$ is locally of finite presentation and flat, so is

$$\mu : G \times_S G \longrightarrow G.$$

Indeed, this morphism is isomorphic to the morphism obtained from $G \rightarrow S$ by the base change $G \rightarrow S$ (VI_A, 0.1). The induced morphism

$$U \times_S U \longrightarrow G$$

therefore has the same properties. To prove that this last morphism is surjective, it is enough to check fiber by fiber. There the result follows from VI_A, 0.5, since $U \cap G_s$ is a dense open subset of G_s for every $s \in S$.

Remark 7.6.0.—⁷⁶ Let S be a scheme, let H be an S -group, and let $f : X \rightarrow H$ be a morphism of S -schemes. The group subsheaf of H generated by the image of f , denoted

$$\langle \text{Im } f \rangle,$$

is the group subfunctor of H which assigns to every S -scheme S' the subgroup of $H(S')$ generated by $f(X(S'))$. Since every element of this subgroup can be written as a finite product

$$f(x_1)^{\varepsilon_1} \cdots f(x_n)^{\varepsilon_n}, \quad n \in \mathbb{N}^*, \quad x_i \in X(S'), \quad \varepsilon_i = \pm 1,$$

it follows that, if one puts

$$X^1 = X \amalg X$$

and defines the morphisms $f^n : X^n \rightarrow H$ as in 7.1, then $\langle \text{Im } f \rangle$ is precisely the image presheaf of the morphism

$$f^\infty : X^\infty \longrightarrow H,$$

where

$$X^\infty = \coprod_{n \geq 1} X^n$$

and $f^\infty : X^\infty \rightarrow H$ is the S -morphism whose restriction to each X^n is f^n .

Proposition 7.6.—⁷⁷ Let S be a scheme, let X be an S -scheme flat and locally of finite presentation over S , let H be an S -group locally of finite presentation over S and with reduced fibers, let $f : X \rightarrow H$ be a morphism of S -schemes, and let $f^\infty : X^\infty \rightarrow H$ be the S -morphism introduced above. The following conditions are equivalent:

⁷⁶Editor's note: this definition has been added; in the original it was contained in the statement of Proposition 7.6.

⁷⁷Editor's note: the original stated this result under the hypotheses of the special case, but the more general form was used implicitly in the proof of 10.12; the statement has been rewritten accordingly.

- (i) H represents the S -sheaf for the (fppf) topology associated with the presheaf $\langle \text{Im } f \rangle$.
- (ii) f^∞ is a covering for the (fppf) topology.
- (iii) f^∞ is surjective, that is,

$$H = \bigcup_{n \geq 1} f^n(X^n).$$

If, in addition, H is quasi-compact, these conditions are also equivalent to the following one:

- (iv) There exists an integer n such that $f^n : X^n \rightarrow H$ is a covering for the (fppf) topology, and hence is surjective.

This applies in particular when X is a geometrically reduced k -scheme locally of finite type, G is a k -group locally of finite type, $\varphi : X \rightarrow G$ is a morphism of k -schemes, $H = \Gamma_G(\varphi)$, and $f : X \rightarrow H$ is the morphism induced by φ .

Proof. The sheaf considered in (i) is the image sheaf of X^∞ under f^∞ . Hence, by IV, 4.4.3, saying that H represents this sheaf is equivalent to saying that f^∞ is a covering; this in turn implies that f^∞ is surjective. Conversely, suppose that f^∞ is surjective. We shall show that it is then a covering for the (fppf) topology.

Let $s \in S$, let η be a maximal point of the fiber H_s , and let $x \in X_s^\infty$ be such that

$$f^\infty(x) = \eta;$$

such an x exists because f^∞ is surjective. Since the fiber H_s is reduced, $\mathcal{O}_{H_s, \eta}$ is a field, and therefore f_s^∞ is flat at x . Since X^∞ and H are locally of finite presentation over S , and X^∞ is flat over S , the fiberwise criterion for flatness (EGA IV₃, 11.3.10) shows that f^∞ is flat at x . Thus, by Lemma 7.5, the morphism

$$\mu \circ (f^\infty \times_S f^\infty) : X^\infty \times_S X^\infty \longrightarrow H$$

is a covering for the (fppf) topology.

Now $X^n \times_S X^m$ is canonically isomorphic to X^{n+m} , and, under this isomorphism,

$$\mu \circ (f^n \times_S f^m)$$

corresponds to f^{n+m} . It is therefore clear that

$$\mu \circ (f^\infty \times_S f^\infty)$$

factors through f^∞ , so f^∞ is a covering for the (fppf) topology. This proves (iii) $\Rightarrow$ (ii), and hence the equivalence of conditions (i), (ii), and (iii).

Furthermore, since the morphisms $X \rightarrow S$ and $H \rightarrow S$ are locally of finite presentation, so is $f : X \rightarrow H$ (EGA IV₁, 1.4.3(v)). The multiplication

$$\mu : H \times_S H \longrightarrow H$$

is also locally of finite presentation (VI_A, 0.1), and it follows that each $f^n : X^n \rightarrow H$ is locally of finite presentation. Consequently, by EGA IV₁, 1.9.5(viii), the sets

$$f^n(X^n)$$

are ind-constructible subsets of H .

Suppose, in addition, that H is quasi-compact; then S is also quasi-compact, since $H \rightarrow S$ is surjective. By EGA IV₁, 1.9.9, there exists an integer p such that

$$f^p(X^p) = H.$$

As above, the fact that the fibers of H are reduced, together with Lemma 7.5, shows that the morphism

$$\mu \circ (f^p \times_S f^p) : X^p \times_S X^p \longrightarrow H$$

is a covering for the (fppf) topology. Since this morphism is

$$f^{2p} : X^{2p} \longrightarrow H,$$

this completes the proof of Proposition 7.6.

Remark 7.6.1.— The equivalent conditions of 7.6 plainly imply that the sheaf F under consideration is representable. The converse is false in general:⁷⁸ for example, if k has characteristic 0, let $G = \mathbb{G}_{a,k}$, and let

$$f : \operatorname{Spec} k \longrightarrow G$$

be the morphism defined by the point 1 of $G(k)$. Then F is represented by the constant k -group $\mathbb{Z}_k$, whereas $\Gamma_G(f) = G$; hence the monomorphism

$$\mathbb{Z}_k \hookrightarrow G$$

is not surjective.

For simplicity, suppose now that the hypotheses of the special case in 7.6 hold, and suppose that F is representable. By EGA IV₃, 8.14.2, the sheaf F is locally of finite presentation over k . The question is therefore whether the dominant monomorphism

$$F \longrightarrow \Gamma_G(f)$$

is an isomorphism, or, equivalently, a closed immersion. By 1.4.2, this is so if F is quasi-compact; and by VI_A, 0.5.1, it is so if F is connected. In particular, by 7.2.1, it is so when X is connected and $f(X)$ contains the unit element of G .

Lemma 7.7.— *Let k be an algebraically closed field, let G be a k -group locally of finite type, let X be a geometrically reduced k -scheme locally of finite type, let $f : X \rightarrow G$ be a k -morphism, and let H be a group subscheme of G such that $H \subset f(X)$. Put*

$$\Gamma' = \bigcup_{n \geq 1} f^n(X^n), \quad \Gamma'_0 = \Gamma' \cap G(k), \quad H_0 = H(k),$$

and suppose that H_0 has finite index in Γ'_0 . Then there exists an integer m such that

$$f^m(X^m) = \Gamma_G(f)$$

(cf. 7.6), and $\Gamma_G(f)$ is the union of finitely many translates of H .

For every $n \geq 1$, the subset $f^n(X^n)$ of G is ind-constructible (EGA IV₁, 1.9.5(viii)). The same is therefore true of Γ' . Since G is a Jacobson scheme, Γ'_0 is dense in Γ' . By hypothesis, there are finitely many points

$$a_1, \dots, a_r \in \Gamma'_0$$

⁷⁸Editor's note: the argument in the original has been expanded in what follows.

such that

$$\Gamma'_0 = a_1 H_0 \cup \cdots \cup a_r H_0.$$

Consequently,

$$\begin{aligned} \Gamma_G(f) &= \overline{\Gamma'} = \overline{\Gamma'_0} = \overline{a_1 H_0 \cup \cdots \cup a_r H_0} \\ &= \overline{a_1 H_0} \cup \cdots \cup \overline{a_r H_0} \\ &= a_1 \overline{H_0} \cup \cdots \cup a_r \overline{H_0}, \end{aligned}$$

where the last equality follows because translation by each a_i is a homeomorphism of G onto itself. Thus

$$\Gamma_G(f) = a_1 H \cup \cdots \cup a_r H.$$

It is also clear that there exists an integer p such that every a_i , for $1 \leq i \leq r$, belongs to $f^p(X^p)$. Finally, since $H \subset f(X)$, one has

$$a_i H \subset f^{p+1}(X^{p+1})$$

for every i . Hence

$$f^{p+1}(X^{p+1}) = \Gamma_G(f).$$

Proposition 7.8.— *Let G be a k -group locally of finite type, and let A and B be two geometrically reduced k -group subschemes of G (hence smooth at the generic points of their irreducible components, and therefore smooth by 1.3). Suppose that one of the following conditions holds:*

- a) *A and B are invariant and of finite type over k ;*
- b) *A is connected and B is of finite type over k .*

Then (A, B) is of finite type over k , and represents the sheaf associated, for the (fppf) topology (or the (fpqc) topology), with the group presheaf of commutators of A and B in G . Moreover,⁷⁹ the k -groups (A, B^0) and (A^0, B) are connected, and

$$(A, B)^0 = (A, B^0) \cdot (A^0, B).$$

Proof. By 7.6, to show that (A, B) is the desired associated sheaf, it is enough to show that there exists an integer n such that

$$\nu^n((A \times_k B)^n) = (A, B),$$

with the notation of 7.2.2. To prove this and the other two assertions, we may suppose that k is algebraically closed.

Indeed, let $\bar{k}$ be an algebraic closure of k . By 7.1(iii) and VI_A, 2.4, one has, with the evident notation,

$$\begin{aligned} (A, B)_{\bar{k}} &= (A_{\bar{k}}, B_{\bar{k}}), \\ ((A, B)^0)_{\bar{k}} &= (A_{\bar{k}}, B_{\bar{k}})^0, \\ (A, B^0)_{\bar{k}} &= (A_{\bar{k}}, B_{\bar{k}}^0), \quad \text{and so on.} \end{aligned}$$

Consequently, suppose that $(A_{\bar{k}}, B_{\bar{k}})$ is of finite type over $\bar{k}$; respectively, suppose that $(A_{\bar{k}}, B_{\bar{k}}^0)$ and $(A_{\bar{k}}^0, B_{\bar{k}})$ are connected and that the morphism*

$$(A_{\bar{k}}, B_{\bar{k}}^0) \times_{\bar{k}} (A_{\bar{k}}^0, B_{\bar{k}}) \longrightarrow (A_{\bar{k}}, B_{\bar{k}})^0$$

⁷⁹Editor's note: the assertions that follow have been made explicit, and the reduction to the algebraically closed case has been expanded in the proof.

* Translator's note: the French source prints $(A_{\bar{k}}, B_{\bar{k}}^0)$ in both factors of this product. The second factor must be $(A_{\bar{k}}^0, B_{\bar{k}})$, as in the asserted identity and throughout the proof; the mixed product is restored here.

is surjective. The analogous assertions then hold over k , by EGA IV₂, 2.7.1 and 2.6.1.

Let

$$B^1, \dots, B^p$$

be the connected components of B other than the identity component B^0 . There are only finitely many, since B is of finite type over k , hence Noetherian. In case a), likewise let

$$A^1, \dots, A^q$$

be the connected components of A other than A^0 . In case b), only $A^0 = A$ is considered. Let ν_{ij} be the restriction of ν to

$$A^i \times_k B^j.$$

Each A^i and B^j is irreducible, by VI_A, 2.4.1; hence so are $A^0 \times_k B^j$ and $A^i \times_k B^0$. Since the unit element of G belongs to A^0 and to B^0 , it belongs to

$$\nu_{0j}(A^0 \times_k B^j) \quad \text{and} \quad \nu_{i0}(A^i \times_k B^0).$$

Thus every $\Gamma_G(\nu_{0j})$ and every $\Gamma_G(\nu_{i0})$ is connected, by 7.2.1.

Let u_{0j} , respectively u_{i0} , denote the inclusion of $\Gamma_G(\nu_{0j})$, respectively $\Gamma_G(\nu_{i0})$, in G . Then

$$(A^0, B) = \Gamma_G((u_{0j})_{j=0}^p), \quad (A, B^0) = \Gamma_G((u_{i0})_{i=0}^q),$$

and these groups are connected. Moreover, it follows from 7.4 and the preceding constructions that there is an integer r such that both (A^0, B) and (A, B^0) are contained in

$$\nu^r((A \times_k B)^r).$$

In case b), one has $(A, B) = (A^0, B)$, and the proof is complete.

We now consider case a).⁸⁰ We already know that (A^0, B) and (A, B^0) are smooth connected k -subgroups of G , hence are of finite type over k (VI_A, 2.4). Since A^0 is a characteristic subgroup of A (VI_A, 2.6.5), it is an invariant subgroup of G ; hence 7.3(v) shows that (A^0, B) is an invariant subgroup of G , and the same holds for (A, B^0) .

By 7.1.1, the subgroup H of G generated by (A, B^0) and (A^0, B) is therefore

$$H = (A, B^0) \cdot (A^0, B).$$

In particular,

$$H \subset \nu^{2r}((A \times_k B)^{2r}).$$

If T is a subset of G stable under the group law (cf. 3.0), write T_0 for the group of k -points of G belonging to T . Put

$$\Gamma' = \bigcup_{q \geq 1} \nu^q((A \times_k B)^q).$$

By Proposition 7.9 below,

$$(A^0, B)_0 = (A_0^0, B_0), \quad (A, B^0)_0 = (A_0, B_0^0), \quad \Gamma'_0 = (A_0, B_0).$$

Therefore

$$H_0 = (A_0^0, B_0) \cdot (A_0, B_0^0)$$

has finite index in Γ'_0 (Bible, Exposé III, Appendix), since A_0 and B_0 are invariant and A_0^0 , respectively B_0^0 , has finite index in A_0 , respectively B_0 .

⁸⁰Editor's note: the argument that follows has been expanded to take account of the added Corollary 7.1.1.

The hypotheses of Lemma 7.7 now hold. Since

$$H \subset \nu^{2r}((A \times_k B)^{2r}),$$

there exists an integer m such that

$$\nu^{2rm}((A \times_k B)^{2rm}) = (A, B),$$

and there are finitely many k -points $a_1, \dots, a_n$ of G such that

$$(A, B) = a_1 H \cup \dots \cup a_n H.$$

Since H is of finite type over k , every $a_i H$ is quasi-compact. Their union (A, B) is therefore quasi-compact, hence of finite type over k . Finally, H is irreducible, so each $a_i H$ is irreducible; and because the unit e belongs to H , it follows that

$$H = (A, B)^0.$$

Proposition 7.9.—*Let k be an algebraically closed field, and let G be a k -group locally of finite type.*

- (i) *Let A and B be two ind-constructible subsets of G . Denote by A_0 the set of rational points of G that belong to A . Then*

$$(A \cdot B)_0 = A_0 \cdot B_0,$$

where the product on the right is taken in the group $G(k)$.

- (ii) *Let X be a geometrically reduced k -scheme locally of finite type, and let $f : X \rightarrow G$ be a k -morphism. Put*

$$\Gamma'_G(f) = \bigcup_{n \geq 1} f^n(X^n).$$

Then $\Gamma'_G(f)_0$ is the subgroup of $G(k)$ generated by $f(X)_0$.

- (iii) *In particular, let A and B be two smooth group subschemes of G , and put*

$$\Gamma' = \bigcup_{n \geq 1} \nu^n((A \times_k B)^n)$$

with the notation of 7.2.2. Then Γ'_0 is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.

Proof. We first prove (i). It is clear that

$$A_0 \cdot B_0 \subset (A \cdot B)_0.$$

Conversely, let $z \in (A \cdot B)_0$. Then $\mu^{-1}(z)$ is a closed subset of $G \times_k G$, while $A \times_k B$ is an ind-constructible subset of $G \times_k G$ (cf. 3.0). Consequently,

$$\mu^{-1}(z) \cap (A \times_k B)$$

is a nonempty ind-constructible subset of $G \times_k G$. By EGA IV₃, 10.4.8, it therefore contains a rational point of $G \times_k G$. Its projections x and y are rational points of G such that $x \in A$, $y \in B$, and $x \cdot y = z$. Hence *

$$(A \cdot B)_0 = A_0 \cdot B_0.$$

*Translator's note: the French proof prints $(A, B)_0 = A_0 \cdot B_0$ here. The statement and the multiplication argument require $(A \cdot B)_0 = A_0 \cdot B_0$, which is restored in the English text.

To prove (ii), observe that, since f^n is locally of finite type, $f^n(X^n)$ is an ind-constructible subset of G (EGA IV₁, 1.9.5(viii))[†]. Assertion (i) then shows by induction that, if

$$A = f(X)_0,$$

one has

$$f^n(X^n)_0 = (A \cup A^{-1})^n.$$

Consequently,

$$\Gamma'_G(f)_0 = \bigcup_{n \geq 1} f^n(X^n)_0 = \bigcup_{n \geq 1} (A \cup A^{-1})^n,$$

which is the subgroup of $G(k)$ generated by $A = f(X)_0$.

Finally, (iii) follows from (ii) and the definitions.

Corollary 7.10.—*Under the hypotheses of 7.8, if k is algebraically closed, then $(A, B)(k)$ is the commutator subgroup of $A(k)$ and $B(k)$ in $G(k)$.*

Indeed, it is enough to apply 7.9(iii), 7.8, and 7.6.

8. Solvable or nilpotent group schemes

8.1.— Let $\mathcal{C}$ be a category equipped with a topology T (cf. Exposé IV, §4). For every presheaf P on $\mathcal{C}$, write P^b for the associated sheaf.

Let G be a presheaf of groups on $\mathcal{C}$, let A and B be two group subsheaves of G , and let $\text{Comm}(A, B)$ be the group presheaf of commutators of A and B in G . Thus, for every $S \in \text{Ob } \mathcal{C}$, $\text{Comm}(A, B)(S)$ is the subgroup of $G(S)$ generated by the commutators

$$aba^{-1}b^{-1}, \quad a \in A(S), \quad b \in B(S).$$

We write

$$\text{Comm}_T(A, B) = \text{Comm}(A, B)^b.$$

In the proof of 8.2, we shall need the following results.⁸¹

Lemma 8.1.1.—*Let $A \subset G$ be sheaves of groups, with A invariant in G .*

- (i) *$\text{Comm}_T(G, A)$ is the smallest invariant group subsheaf C of G such that the sheaf $(A/C)^b$, associated with the quotient presheaf A/C , is central in $(G/C)^b$.*
- (ii) *In particular, $\text{Comm}_T(G, G)$ is the smallest invariant group subsheaf C of G such that the quotient sheaf $(G/C)^b$ is commutative.*

Clearly, (ii) is the special case $A = G$ of (i), so it is enough to prove (i). Let C be a group subsheaf of A , invariant in G , such that the quotient sheaf $(A/C)^b$ is central in $(G/C)^b$. By Lemma IV, 4.4.8.1, the presheaves A/C and G/C are separated. Hence, by IV, 4.3.11, all the morphisms in the following diagram are monomorphisms:

$$\begin{array}{ccc} A/C & \hookrightarrow & G/C \\ \downarrow & & \downarrow \\ (A/C)^b & \hookrightarrow & (G/C)^b \end{array}$$

[†] Translator's note: the French source prints EGA IV₄ here. The cited result 1.9.5(viii) belongs to EGA IV₁, which is restored in the English text.

⁸¹Editor's note: these results were stated without proof in the original. We have highlighted them as Lemmas 8.1.1 and 8.1.2 and supplied the details of the proofs.

Since $(A/C)^b$ is central in $(G/C)^b$, the presheaf A/C is central in G/C . It follows that

$$\mathrm{Comm}(G, A) \subset C,$$

and hence, by IV, 4.3.12, C contains $\mathrm{Comm}_T(G, A)$.

Conversely, $\mathrm{Comm}(G, A)$ is an invariant, separated group subpresheaf of A (cf. IV, 4.3.1, editor's note 24). Therefore, by IV, 4.4.8.2(i) and IV, 4.3.11,

$$C = \mathrm{Comm}_T(G, A)$$

is an invariant group subsheaf of A containing $\mathrm{Comm}(G, A)$. Consequently, the presheaf A/C is central in G/C , and hence, by IV, 4.4.8.2(ii), $(A/C)^b$ is central in $(G/C)^b$. This proves Lemma 8.1.1.

Lemma 8.1.2.— *Let G be a sheaf of groups, and let A, B be two group subpresheaves of G .*

(i) *The morphism*

$$\tau : \mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^b, B^b)$$

is a covering monomorphism.

(ii) *Consequently, there is an isomorphism*

$$\mathrm{Comm}_T(A, B) \xrightarrow{\sim} \mathrm{Comm}_T(A^b, B^b).$$

Proof. (i) Since A (respectively, B) is a subpresheaf of G , the associated sheaf A^b (respectively, B^b) is a subpresheaf of G containing A (respectively, B). It follows that τ is a monomorphism.

We now show that τ is a covering. Let $S \in \mathrm{Ob} \mathcal{C}$ and let

$$g \in \mathrm{Comm}(A^b, B^b)(S).$$

There exist an integer $n \geq 1$ and, for $1 \leq i \leq n$, elements

$$a'_i \in A^b(S), \quad b'_i \in B^b(S), \quad \varepsilon_i \in \{\pm 1\},$$

such that

$$g = (a'_1, b'_1)^{\varepsilon_1} \cdots (a'_n, b'_n)^{\varepsilon_n},$$

where (a, b) denotes the commutator $aba^{-1}b^{-1}$. There is a refinement R of S such that

$$a'_i \in A(R) \quad \text{and} \quad b'_i \in B(R)$$

for every i . Then g is the composite morphism

$$R \xrightarrow{(a'_1, \dots, b'_n)} (A \times B)^n \xrightarrow{\Phi^{\varepsilon_1, \dots, \varepsilon_n}} \mathrm{Comm}(A, B)$$

where $\Phi = \Phi^{\varepsilon_1, \dots, \varepsilon_n}$ is the morphism defined set-theoretically by

$$\Phi(a_1, b_1, \dots, a_n, b_n) = (a_1, b_1)^{\varepsilon_1} \cdots (a_n, b_n)^{\varepsilon_n}$$

for every $T \in \mathrm{Ob} \mathcal{C}$, with $a_i \in A(T)$ and $b_i \in B(T)$. This shows that

$$g \in \mathrm{Comm}(A, B)(R).$$

It follows, as in the proof of IV, 4.3.11(i), that

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^b, B^b)$$

is a covering.

The morphism

$$\mathrm{Comm}(A^b, B^b) \longrightarrow \mathrm{Comm}(A^b, B^b)^b$$

is likewise a covering monomorphism (IV, 4.3.11(iv)). Hence the same is true of

$$\mathrm{Comm}(A, B) \longrightarrow \mathrm{Comm}(A^b, B^b)^b.$$

By IV, 4.3.12, we therefore obtain an isomorphism

$$\mathrm{Comm}(A, B)^b \xrightarrow{\sim} \mathrm{Comm}(A^b, B^b)^b.$$

This proves Lemma 8.1.2.

Proposition 8.2.— *Let $\mathcal{C}$ be a category, let T be a topology on $\mathcal{C}$, let G be a sheaf of groups on $\mathcal{C}$, and let n be an integer with $n \geq 0$. The following conditions are equivalent:*

- (i) *Put $K_0 = G$, and, for $p \geq 1$, put*

$$K_p = \mathrm{Comm}(K_{p-1}, K_{p-1}) \quad (\text{respectively, } K_p = \mathrm{Comm}(G, K_{p-1})).$$

Then K_n is the unit presheaf of groups.

- (ii) *Put $K'_0 = G$, and, for $p \geq 1$, put*

$$K'_p = \mathrm{Comm}_T(K'_{p-1}, K'_{p-1}) \quad (\text{respectively, } K'_p = \mathrm{Comm}_T(G, K'_{p-1})).$$

Then K'_n is the unit sheaf of groups.

- (iii) *There is a sequence*

$$G = H_0 \supset H_1 \supset \cdots \supset H_n$$

of invariant group subsheaves of G such that, for every i , the quotient sheaf H_i/H_{i+1} is commutative (respectively, central in G/H_{i+1}), and H_n is the unit sheaf.

It is clear that $K_n \subset K'_n$; consequently, (ii) implies (i). We show that (i) implies (ii). One has

$$K'_1 = \mathrm{Comm}_T(G, G) = K_1^b,$$

and induction using Lemma 8.1.2 gives

$$K'_p = K_p^b$$

for every p . Consequently, if K_n is the unit presheaf, then $K'_n = K_n^b$ is the unit sheaf.

Finally, conditions (ii) and (iii) are equivalent by Lemma 8.1.1.

Definition 8.2.1.— *When these conditions hold, the sheaf G is said to be solvable of class n (respectively, nilpotent of class n). If there is an integer n for which these conditions hold, G is said to be solvable (respectively, nilpotent).*

Observe that, by condition (i), this does not depend on the topology T .

Proposition 8.3.— *Let k be a field, let S be a nonempty k -scheme, let Ω be an algebraically closed extension of k , and let G be a smooth k -group of finite type. The following conditions are equivalent:*

- (i) G is solvable of class n (respectively, nilpotent of class n).
- (ii) $G \times_k S$ is solvable of class n (respectively, nilpotent of class n).
- (iii) The group $G(\Omega)$ is solvable of class n (respectively, nilpotent of class n).
- (iv) Put $K_0 = G$, and, for $p \geq 1$, consider the k -groups

$$K_p = (K_{p-1}, K_{p-1}) \quad (\text{respectively, } K_p = (G, K_{p-1}))$$

(cf. 7.2.2). Then K_n is the unit k -group.

The equivalence of conditions (i) and (ii) follows from Proposition 8.2, since the formation of the group presheaf of commutators commutes with base change (IV, 4.1.3).

The equivalence of (i) and (iv) follows from the fact that, by 7.8, the k -group

$$K_p = (K_{p-1}, K_{p-1}) \quad (\text{respectively, } K_p = (G, K_{p-1}))$$

represents the sheaf

$$\text{Comm}_T(K_{p-1}, K_{p-1}) \quad (\text{respectively, } \text{Comm}_T(G, K_{p-1})),$$

where T is the (fppf) or (fpqc) topology.

To prove the equivalence of conditions (iii) and (iv), we may assume that $k = \Omega$; the result then follows from 7.10.

Proposition 8.4.— *Let G be an S -group of finite presentation such that, for every $s \in S$, the fiber G_s is smooth over $\kappa(s)$. Let T be the set of points $s \in S$ for which G_s is solvable (respectively, nilpotent).*

- (i) *Then T is locally constructible in S .*
- (ii) *If, in addition, G is assumed flat and separated over S (that is, when G is smooth, quasi-compact, and separated over S), then T is closed in S .*

Proof. We may clearly suppose that S is affine, with ring A . By 10.1 and 10.10b),* there exist a Noetherian subring A' of A and an A' -group G' of finite type such that

$$G' \otimes_{A'} A \simeq G.$$

By EGA IV₃, 11.2.6 and 8.10.5,⁸² if G is flat and separated over S , we may suppose that G' is flat and separated over $S' = \text{Spec } A'$.⁸³ Since G is of finite presentation over S , the set of points $s \in S$ for which G_s is geometrically reduced—or, equivalently, smooth over $\kappa(s)$ —is locally constructible (EGA IV₃, 9.7.7). Hence, by EGA IV₃, 9.3.3, we may suppose that $G'_{s'}$ is smooth over $\kappa(s')$ for every $s' \in S'$. On the other hand, if s' denotes the image of s in S' , then

$$G'_{s'} \otimes_{\kappa(s')} \kappa(s) \simeq G_s.$$

It follows from 8.3 that G_s is solvable (respectively, nilpotent) if and only if the same is true of $G'_{s'}$. We are therefore reduced to the case where S is a Noetherian affine scheme.

* In the course of this proof we shall use results established in Section 10; like Section 9, they do not depend on the present Section 8.

⁸²Editor's note: the reference 10.8.5 has been corrected to 8.10.5.

⁸³Editor's note: what follows has been expanded.

We first show that T is constructible. Applying the criterion of EGA 0_{III}, 9.2.3, and arguing as above, we are reduced to showing that, when S is Noetherian and integral, either T or $S - T$ contains a nonempty open subset of S .

Suppose, then, that S is integral and Noetherian, with generic point η . For every $s \in S$, put $K_s^0 = G_s$, and put, for $p \geq 1$,

$$K_s^p = (K_s^{p-1}, K_s^{p-1}) \quad (\text{respectively, } K_s^p = (G_s, K_s^{p-1})). \quad \dagger$$

We first show that the sequence of closed subschemes K_η^p is stationary. By 7.3(v), each K_η^p is invariant, and hence the sequence $(K_\eta^p)_p$ is decreasing. It is therefore stationary, since G_η is Noetherian. Thus there is an integer n such that

$$K_\eta^p = K_\eta^n \quad \text{for every } p \geq n.$$

On the other hand, by 10.12.1 and 10.13, there exist a nonempty open subset S' of S and an S' -group D of finite presentation such that, for every $s \in S'$,

$$D_s = K_s^n, \quad (D_s, D_s) = D_s \quad (\text{respectively, } (G_s, D_s) = D_s).$$

We may suppose that $S' = S$. Then, for every $s \in S$ and every $p \geq n$, one has $D_s = K_s^p$. Consequently, G_s is solvable (respectively, nilpotent) if and only if D_s is isomorphic to the unit $\kappa(s)$ -group.

By EGA IV₃, 9.6.1(xi), the set of points $s \in S$ for which the structural morphism

$$D_s \longrightarrow \text{Spec } \kappa(s)$$

is an isomorphism is constructible.⁸⁴ It is therefore either nowhere dense or contains a nonempty open subset of S . We have thus proved that T is locally constructible.

We now show that, if G is also flat and separated over S , then T is closed. Since T is locally constructible, it is closed if and only if it is stable under specialization (EGA IV₁, 1.10.1).

Let $s \in S$, and let s' be a specialization of s in S . Since we have reduced to the case where S is Noetherian, EGA II, 7.1.9 provides a discrete valuation ring A and a morphism

$$\text{Spec } A \longrightarrow S$$

such that s , respectively s' , is the image of the generic point α , respectively the closed point a , of $\text{Spec } A$. It is enough to show that, if

$$G' = G \times_S \text{Spec } A$$

and G'_α is solvable (respectively, nilpotent), then the same is true of G'_a . Since G is flat and separated over S , the A -group G' is flat and separated.[‡] We are thus reduced to the case where S is the spectrum of a discrete valuation ring A .

Since G_α is assumed solvable (respectively, nilpotent), there is an integer q such that K_α^q , with the notation introduced above, is isomorphic to the unit $\kappa(\alpha)$ -group. For every n ,

[†] Translator's note: in the nilpotent branch of this display, the French source prints G . Since K_s^{p-1} is a $\kappa(s)$ -group and the subsequent formulas use G_s , the fiber subscript is restored here.

⁸⁴Editor's note: here one uses the fact that S is assumed affine, and is therefore quasi-compact and quasi-separated (cf. EGA 0_{III}, 9.1.12).

[‡] Translator's note: the French source prints G'_α as being flat and separated over A . The intended object is the base-changed A -group G' ; its generic fiber G'_α is a group over $\kappa(\alpha)$.

let $\overline{K_\alpha^n}$ denote the schematic closure of K_α^n in G , in the sense of EGA IV₂, 2.8.5. We show by induction on p that

$$(\overline{K_\alpha^p})_a \supset K_a^p.$$

First, since G is flat over A , one has $\overline{G_\alpha} = G$ by EGA IV₂, 2.8.5, and hence

$$(\overline{K_\alpha^0})_a = K_a^0.$$

Let $p \geq 0$, and suppose that

$$K_a^p \subset (\overline{K_\alpha^p})_a.$$

Following 7.2.2, denote by $\nu_a, \nu_\alpha, \bar{\nu}, \bar{\nu}_a$ the following morphisms. In the solvable case they are

$$\begin{aligned} \nu_a &: K_a^p \times_{\kappa(a)} K_a^p \longrightarrow G_a, \\ \nu_\alpha &: K_\alpha^p \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha, \\ \bar{\nu} &: \overline{K_\alpha^p} \times_A \overline{K_\alpha^p} \longrightarrow G, \\ \bar{\nu}_a &: (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a; \end{aligned}$$

respectively, in the nilpotent case they are

$$\begin{aligned} \nu_a &: G_a \times_{\kappa(a)} K_a^p \longrightarrow G_a, \\ \nu_\alpha &: G_\alpha \times_{\kappa(\alpha)} K_\alpha^p \longrightarrow G_\alpha, \\ \bar{\nu} &: G \times_A \overline{K_\alpha^p} \longrightarrow G, \\ \bar{\nu}_a &: G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a \longrightarrow G_a. \end{aligned}$$

Since ν_α factors through K_α^{p+1} , $\bar{\nu}$ factors through $\overline{K_\alpha^{p+1}}$. The latter is clearly a group subscheme of G , and therefore contains $\Gamma_G(\bar{\nu})$. By 7.1(iii),

$$\Gamma_G(\bar{\nu})_a = \Gamma_{G_a}(\bar{\nu}_a).$$

Moreover, the induction hypothesis gives

$$\begin{aligned} K_a^p \times_{\kappa(a)} K_a^p &\subset (\overline{K_\alpha^p})_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a && \text{in the solvable case,} \\ G_a \times_{\kappa(a)} K_a^p &\subset G_a \times_{\kappa(a)} (\overline{K_\alpha^p})_a && \text{in the nilpotent case.} \end{aligned}$$

Consequently,

$$K_a^{p+1} = \Gamma_{G_a}(\nu_a) \subset \Gamma_{G_a}(\bar{\nu}_a) = \Gamma_G(\bar{\nu})_a \subset (\overline{K_\alpha^{p+1}})_a.$$

Finally, K_α^q is isomorphic to the unit $\kappa(\alpha)$ -group. The unit A -group is flat over A and is isomorphic to a closed subscheme of G , since G is separated over A (cf. 5.1). It follows from EGA IV₂, 2.8.5 that $\overline{K_\alpha^q}$ is isomorphic to the unit A -group. Since we have proved that

$$K_a^q \subset (\overline{K_\alpha^q})_a,$$

the group K_a^q is isomorphic to the unit $\kappa(a)$ -group. Thus G_a is solvable (respectively, nilpotent).

9. Quotient sheaves

The present section is essentially limited to recalling, in the particular case of homogeneous spaces of groups, well-known general facts concerning passage to the quotient by flat equivalence relations (cf. Exposé IV).

Definition 9.1.— *Given a monomorphism $u : G' \rightarrow G$ of S -groups, we write G/G' (respectively, $G' \backslash G$) and call it the right (respectively, left) quotient sheaf of G by G' , meaning the quotient sheaf, for the (fpqc) topology, of G by the equivalence relation defined by the monomorphism*

$$\begin{aligned} G \times_S G' &\xrightarrow{\delta \circ (\text{id}_G \times u)} G \times_S G, \\ &\text{(respectively,} \\ G' \times_S G &\xrightarrow{\gamma \circ (u \times \text{id}_G)} G \times_S G), \end{aligned}$$

where δ (respectively, γ) denotes the automorphism of $G \times_S G$ defined, for $g, h \in G(T)$, by

$$(g, h) \mapsto (g, gh) \quad \text{(respectively, } (h, g) \mapsto (hg, g)).$$

Proposition 9.2.— *Let $u : G' \rightarrow G$ be a monomorphism of S -groups. Suppose that G/G' is representable by an S -scheme G'' . Then:*

- (i) *The canonical morphism $p : G \rightarrow G''$ is a covering for the (fpqc) topology.*
- (ii) *If $\varepsilon'' = p \circ \varepsilon$ (this morphism is called the unit section of G''), the following diagrams are cartesian:*

$$\begin{array}{ccc} G \times_S G' & \xrightarrow{\mu \circ (\text{id}_G \times u)} & G \\ \text{pr}_1 \downarrow & & \downarrow p \\ G & \xrightarrow{p} & G'' \end{array} \quad , \quad \begin{array}{ccc} G' & \xrightarrow{u} & G \\ \pi' \downarrow & & \downarrow p \\ S & \xrightarrow{\varepsilon''} & G'' \end{array} .$$

In particular, u is an immersion.

- (iii) *There exists on G'' a unique structure of an S -scheme with left operator group G such that p is a morphism of S -schemes with operator group G .*
- (iv) *If, in addition, G' is assumed invariant in G , there exists on G'' a unique S -group structure such that p is a morphism of S -groups.*
- (v) *Let S_0 be an S -scheme, and put $G_0 = G \times_S S_0$ and $G'_0 = G' \times_S S_0$. Then G_0/G'_0 is representable by $G''_0 = G'' \times_S S_0$.⁸⁵*
- (vi) *Let $\mathcal{P}$ be a property of an S -morphism. Suppose that $\mathcal{P}$ is stable under base change. If $p : G \rightarrow G''$ has property $\mathcal{P}$, then the structural morphism $\pi' : G' \rightarrow S$ has property $\mathcal{P}$ as well.*
- (vii) *Let $\mathcal{P}$ be a property of an S -morphism. Suppose that $\mathcal{P}$ is local for the (fpqc) topology (cf. 2.0 and 2.1.2). Then $p : G \rightarrow G''$ has property $\mathcal{P}$ if and only if $\pi' : G' \rightarrow S$ has property $\mathcal{P}$.^{*}*

⁸⁵Editor's note: that is, the quotient is universal; cf. Exposé IV, §3.

^{*} Translator's note: the French source prints S' as the target of π' in this clause. No S' is defined here, and clause (vi) identifies π' as the structural morphism $G' \rightarrow S$; that target is used here.

(viii) Let $\mathcal{P}$ be a property of an S -morphism. Suppose that $\mathcal{P}$ is local for the (fpqc) topology and stable under composition. If the structural morphisms $G'' \rightarrow S$ and $G' \rightarrow S$ have property $\mathcal{P}$, then the structural morphism $G \rightarrow S$ has property $\mathcal{P}$ as well.

(ix) If G is reduced, then G'' is reduced.

(x) For G'' to be separated over S , it is necessary and sufficient that u (or, equivalently, ε'') be a closed immersion.

(xi) For G' to be flat over S , it is necessary and sufficient that p be flat (or, equivalently, faithfully flat).

In this case, for G'' to be flat over S , it is necessary and sufficient that G be flat over S .

(xii) For G' to be flat and locally of finite presentation over S , it is necessary and sufficient that $p : G \rightarrow G''$ be faithfully flat and locally of finite presentation.

In this case, for G'' to be locally of finite presentation (respectively, locally of finite type, of finite type, smooth, étale, unramified, locally quasi-finite, quasi-finite) over S , it is enough that G have the same property over S (and the condition is also necessary in the first two cases; cf. (viii)).

(xiii) Suppose that G' is flat and of finite presentation over S .

a) Then p is of finite presentation and faithfully flat.

b) Moreover, for G to be of finite presentation over S , it is necessary and sufficient that G'' be so.

Proof. Recall that the equivalence relation under consideration is universally effective (IV, 4.4.9). Assertions (i), (iii), (iv), (v), and the first assertion of (ii) then follow, respectively, from IV, 4.4.3, 5.2.2, 5.2.4, 3.4.5, and 3.3.2(iii). The second assertion of (ii) follows from the first, as shown by the following cartesian diagram, since

$$(G \times_S G') \times_G S$$

is isomorphic to G' :

$$\begin{array}{ccccc} G' & \xrightarrow{(\varepsilon \circ \pi'), \text{id}_{G'}} & G \times_S G' & \xrightarrow{\mu \circ (\text{id}_G \times u)} & G \\ \pi' \downarrow & & \downarrow \text{pr}_1 & & \downarrow p \\ S & \xrightarrow{\varepsilon} & G & \xrightarrow{p} & G'' \end{array}$$

Indeed, ε'' is an S -section of G'' , hence an immersion (EGA I, 5.3.13); by the preceding cartesian diagram the same is true of u , which completes the proof of (ii). Moreover, (vi) is an immediate consequence of the second cartesian diagram in (ii).

We prove (vii). By (i), p is a covering for the (fpqc) topology. Hence, by (ii), in order to show that p has property $\mathcal{P}$, it is enough to show that the first projection

$$\text{pr}_1 : G \times_S G' \longrightarrow G$$

has property $\mathcal{P}$. This follows from the stability of $\mathcal{P}$ under base change, since pr_1 is obtained from π' by base change.

It is clear that (viii) follows from (vii), since

$$\pi = \pi'' \circ p,$$

where $\pi'' : G'' \rightarrow S$ denotes the structural morphism.

We prove (ix). By (i), p is an epimorphism. Since G is reduced, p factors through the immersion

$$G''_{\text{red}} \longrightarrow G'',$$

which is therefore also an epimorphism, and hence an isomorphism (IV, 4.4.4).

We prove (x). If G'' is separated over S , then ε'' is a closed immersion by EGA I, 5.4.6. On the other hand, we saw in (ii) that ε'' is a closed immersion if and only if u is. Finally, if u is a closed immersion, then so is

$$\delta \circ (\text{id}_G \times u) : G \times_S G' \longrightarrow G \times_S G.$$

It follows from Lemma 9.2.1 below that G'' is separated over S .

Assertion (xi) follows from (vii) and EGA IV₂, 2.2.13.

Assertion (xii) follows from (vii), EGA IV₄, 17.7.5 and 17.7.7, and the fact that p is universally open: for every $s \in S$, if the underlying space of G_s is discrete, then the same is true of the underlying space of G''_s .

Finally, assertion (xiii) follows from (vii), (viii), and EGA IV₄, 17.7.5.

Lemma 9.2.1.— *Let X be an S -scheme, and let R be an equivalence relation on X defined by the monomorphism $v : R \rightarrow X \times_S X$. Suppose that R is effective. Then:*

- (i) *The morphism v is an immersion; it is a closed immersion if $Y = X/R$ is separated.*
- (ii) *Suppose, in addition, that Y represents the (fpqc) quotient sheaf of X by R ,⁸⁶ and that v is a closed immersion. Then $Y = X/R$ is separated over S .*

Recall (IV, Definition 3.3.2) that the hypothesis “ R is effective” means that there is a morphism of S -schemes

$$p : X \longrightarrow Y$$

such that the natural morphism

$$R \longrightarrow X \times_Y X$$

is an isomorphism. It follows (EGA I, 5.3.5) that the following diagram is cartesian:

$$\begin{array}{ccc} R & \xrightarrow{v} & X \times_S X \\ \downarrow & & \downarrow p \times p \\ Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y \end{array}$$

Since $\Delta_{Y/S}$ is an immersion (EGA I, 5.3.9), the same is true of v . Likewise, if Y is separated over S , then $\Delta_{Y/S}$ is a closed immersion, and hence so is v .

Conversely, suppose that v is a closed immersion and that Y represents the (fpqc) quotient sheaf of X by R . Then p is a covering for the (fpqc) topology (IV, 4.4.3), and therefore $p \times p$ is also a covering: by base change, $p \times \text{id}_X$ and $\text{id}_Y \times p$ are coverings, and so is their composite $p \times p$. Thus, by (fpqc) descent (cf. EGA IV₂, 2.7.1), $\Delta_{Y/S}$ is a closed immersion; in other words, Y is separated over S .

Remark 9.2.2.— *Under the general hypotheses of 9.2, if G' is flat and locally of finite presentation over S , then p is a covering for the (fppf) topology,⁸⁷ by 9.2(vii). Consequently, assertions 9.2(vii) and 9.2(viii) extend to properties $\mathcal{P}$ that are local for the (fppf) topology.*

⁸⁶Editor's note: as O. Gabber pointed out, this condition is used in the proof and must be included among the hypotheses.

⁸⁷Editor's note: (fpqc) has been corrected to (fppf).

Remark 9.3.— a) The question whether a quotient G/G' is representable is often delicate; in this seminar we prove the representability of certain particular quotients.

In general, in order to conclude that G/G' is representable, it is not enough to suppose that G and G' are of finite presentation over S and that G' is flat over S . Indeed, suppose in addition that G is smooth with connected fibers. In that case, if G/G' is a scheme, it is separated by Corollary 5.4; therefore $G' \hookrightarrow G$ is a closed immersion by 9.2(x). Consequently, if G' is not closed in G , then G/G' is not representable.

To obtain such a counterexample, take S to be the spectrum of a discrete valuation ring and put $G = \mathbb{G}_{m,S}$. Let $n > 1$ be an integer invertible on S . Then

$$\mu_n = \ker(G \xrightarrow{n} G)$$

is a closed subgroup of G , étale over S (cf. Exposé VII_A⁸⁸). Let G' be the open subgroup of μ_n obtained by removing from μ_n the closed subset of its closed fiber that is complementary to the origin. Then G' is not closed in G , and hence G/G' is not representable. One can also construct such examples in which G' is smooth with connected fibers.

b) It remains possible, however, that G/G' is representable when G and G' are of finite presentation over S , and G' is flat over S and closed in G^* ⁸⁹. Under these hypotheses, G/G' is known to be representable in the following particular cases:

- 1°— S is the spectrum of an Artinian ring (cf. VI_A, 3.2 and 3.3.2).
- 2°— G' is proper over S , and G is quasi-projective over S (cf. V, 7.1).
- 3°— S is locally Noetherian of dimension 1 (cf. [An73], Theorem 4.C).

10. Passage to the projective limit in group schemes and schemes with an operator group

10.0.— Recall the essential result of EGA IV₃, §8.8. Suppose that the following data are given: a quasi-compact and quasi-separated scheme S_0 , an upward filtered preordered set I , a direct system $(\mathcal{A}_i)_{i \in I}$ of quasi-coherent commutative $\mathcal{O}_{S_0}$ -algebras,

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i, \quad S_i = \operatorname{Spec} \mathcal{A}_i \quad (i \in I), \quad S = \operatorname{Spec} \mathcal{A}^{90}.$$

Then the category of S -schemes of finite presentation is determined, up to equivalence, by the categories of S_i -schemes of finite presentation, the functors between these categories

$$\rho_{ji} : X_i \longmapsto X_i \times_{S_i} S_j \quad (i \leq j),$$

and the transitivity isomorphisms

$$\rho_{kj} \circ \rho_{ji} \xrightarrow{\sim} \rho_{ki}.$$

More precisely, given $j \in I$ and an S_j -scheme X_j of finite presentation, put, for every $i \in I$ with $i \geq j$,

$$X_i = X_j \times_{S_j} S_i, \quad X = X_j \times_{S_j} S.$$

Then EGA IV₃, 8.8.2 gives:

⁸⁸Editor's note: see Exposé VII_A, 8.4, or Exposé VIII, 2.1.

^{*}This is too optimistic, as M. Raynaud shows in his thesis (loc. cit., X, 14).

⁸⁹Editor's note: the starred remark refers to counterexample X, 14 in [Ray70a]. The base there is a regular local scheme of dimension 2.

⁹⁰Editor's note: since S is affine over S_0 , it is quasi-compact and quasi-separated; cf. Editor's note 92 below.

- (i) Given $j \in I$ and two S_j -schemes X_j and Y_j of finite presentation, the canonical map

$$\varinjlim_{i \geq j} \operatorname{Hom}_{S_i}(X_i, Y_i) \longrightarrow \operatorname{Hom}_S(X, Y)$$

is bijective.

- (ii) For every S -scheme X of finite presentation, there are an index $j \in I$, an S_j -scheme X_j of finite presentation, and an S -isomorphism

$$X \xrightarrow{\sim} X_j \times_{S_j} S.$$

It follows from EGA IV₃, 8.8.3 that whenever a diagram $\mathcal{D}$ involves finitely many objects and arrows in the category of S -schemes of finite presentation, one can find an index $i \in I$ and a diagram $\mathcal{D}_i$ in the category of S_i -schemes of finite presentation such that $\mathcal{D}$, up to isomorphism, is obtained from $\mathcal{D}_i$ by the base change $S \rightarrow S_i$. One can even find i and $\mathcal{D}_i$ such that every cartesian square in $\mathcal{D}$ is obtained from a cartesian square in $\mathcal{D}_i$.

10.1.— Moreover, many familiar properties of a morphism that are stable under base change have the following property:

Let $u : X \rightarrow Y$ be an S -morphism between S -schemes of finite presentation. By 10.0, it is obtained by base change from an S_j -morphism $u_j : X_j \rightarrow Y_j$ between S_j -schemes of finite presentation. Then u has the property $\mathcal{P}$ if and only if there is an $i \geq j$ such that

$$u_i = u_j \times_{S_j} S_i$$

has the property $\mathcal{P}$.

This applies when $\mathcal{P}$ is any of the following properties of a morphism: separated, surjective, radicial, affine, quasi-affine, finite, quasi-finite, proper, projective, quasi-projective, an isomorphism, a monomorphism, an immersion, an open immersion, a closed immersion (EGA IV₃, 8.10.5), flat (EGA IV₃, 11.2.6), smooth, unramified, or étale (EGA IV₄, 17.7.8).⁹¹

This also applies when $\mathcal{P}$ is the property of being a covering for the (fppf) topology. Indeed, given two S -schemes X and Y of finite presentation and an S -morphism $u : X \rightarrow Y$, it follows from Exposé IV, 6.3.1(i)⁹² that u is a covering for the (fppf) topology if and only if there are an S -scheme Z and a faithfully flat S -morphism $v : Z \rightarrow Y$ of finite presentation that factors through u .

The purpose of this Section 10 is to give variants of results of this kind for the category of S -groups of finite presentation, for the category of S -schemes with an operator group, and for certain properties of group monomorphisms (being invariant, being central with representable quotient sheaf, and so on).

The first two preliminary results of this kind follow. In 10.2 through 10.9, we retain the notation introduced in 10.0.

Lemma 10.2.— *Let G_j and H_j be two S_j -groups of finite presentation. For every $i \geq j$, put*

$$G_i = G_j \times_{S_j} S_i, \quad G = G_j \times_{S_j} S,$$

and define H_i and H similarly. Then the canonical map

$$\varinjlim_{i \geq j} \operatorname{Hom}_{S_i\text{-gr}}(G_i, H_i) \longrightarrow \operatorname{Hom}_{S\text{-gr}}(G, H)$$

⁹¹Editor's note: the word "flat" has been added, and the reference 17.7.6 has been corrected to 17.7.8.

⁹²Editor's note: one also uses the fact that Y , being of finite presentation over S , is quasi-compact.

is bijective.

Lemma 10.3.— *Let G be an S -group of finite presentation. Then there are an index $j \in I$, an S_j -group G_j of finite presentation, and an isomorphism of S -groups*

$$G \xrightarrow{\sim} G_j \times_{S_j} S.$$

Assertions 10.2 and 10.3 are easy consequences of 10.0 and 10.1, taking into account the interpretation⁹³ of the structure of an S -group given in EGA 0_{III}, 8.2.5 and 8.2.6.

Lemma 10.4.— *Let $u : G \rightarrow H$ be a morphism of S -groups between S -groups of finite presentation. By 10.3 and 10.2, u is obtained by base change from a morphism $u_j : G_j \rightarrow H_j$ between S_j -groups of finite presentation. Then u is a central monomorphism (respectively, an invariant monomorphism) if and only if there is an $i \geq j$ such that*

$$u_i = u_j \times_{S_j} S_i$$

has the same property.

This follows immediately from 10.0 and 10.1, in view of the characterization in 6.7 of central or invariant group monomorphisms.

Corollary 10.5.— *Let G_j be an S_j -group of finite presentation. Then $G_j \times_{S_j} S$ is commutative if and only if there is an $i \geq j$ such that $G_j \times_{S_j} S_i$ is commutative.*

Indeed, to say that an S -group is commutative is the same as saying that it is central when regarded as a group subscheme of itself.

Proposition 10.6.— *Let G_j be an S_j -group of finite presentation, and let G'_j be a group subscheme of G_j , flat and of finite presentation over S_j . Then the quotient*

$$(G_j \times_{S_j} S)/(G'_j \times_{S_j} S)$$

is representable for the (fpqc) topology if and only if there is an $i \geq j$ such that

$$(G_j \times_{S_j} S_i)/(G'_j \times_{S_j} S_i)$$

is representable.

This is a consequence of the following more general lemma.

Lemma 10.7.— *Let X_j be an S_j -scheme of finite presentation, and let R_j be an equivalence relation on X_j , flat and of finite presentation.⁹⁴ Then the quotient sheaf*

$$(X_j \times_{S_j} S)/(R_j \times_{S_j} S)$$

for the topology $\mathcal{T} = (\text{fppf})$ or (fpqc) is representable if and only if there is an $i \geq j$ such that the quotient sheaf

$$(X_j \times_{S_j} S_i)/(R_j \times_{S_j} S_i)$$

for the topology $\mathcal{T}$ is representable.

In view of the statements EGA IV₂, 8.8.2, 8.8.3, 8.10.5, and 11.2.6 recalled in 10.0, this lemma follows from the next result.

⁹³Editor's note: in terms of commutative diagrams of S -morphisms.

⁹⁴Editor's note: that is, such that the composite

$$R_j \hookrightarrow X_j \times_{S_j} X_j \xrightarrow{\text{pr}_1} X_j$$

is flat and of finite presentation.

Lemma 10.8.— *Let $\mathcal{T}$ be the (fppf) or (fpqc) topology. Let X be an S -scheme of finite presentation (respectively, locally of finite presentation), and let R be an equivalence relation on X defined by a monomorphism*

$$v : R \longrightarrow X \times_S X$$

such that $\mathrm{pr}_1 \circ v$ is flat and of finite presentation (respectively, flat and locally of finite presentation). Then the following conditions are equivalent:

- (i) *The quotient sheaf X/R for the topology $\mathcal{T}$ is representable.*
- (ii) *There are an S -scheme Y of finite presentation (respectively, locally of finite presentation) and a faithfully flat morphism $p : X \rightarrow Y$ such that the diagram*

$$(D) \quad \begin{array}{ccc} R & \xrightarrow{\mathrm{pr}_1 \circ v} & X \\ \mathrm{pr}_2 \circ v \downarrow & & \downarrow p \\ X & \xrightarrow{p} & Y \end{array}$$

is cartesian.

First note that, by IV, 3.3.2 and 4.4.3, the sheaf X/R for the topology $\mathcal{T}$ is representable by Y if and only if diagram (D) is cartesian and p is a covering for the topology $\mathcal{T}$.

Let us prove that (i) implies (ii). Hypothesis (i) implies that diagram (D) is cartesian. Thus $\mathrm{pr}_1 \circ v$ is obtained from p by base change along p , and p is a covering for the (fpqc) topology. Consequently, by (fpqc) descent (EGA IV₂, 2.7.1), since $\mathrm{pr}_1 \circ v$ is faithfully flat and (locally) of finite presentation, the same is true of p . Then EGA IV₄, 17.7.5 shows that, since X is (locally) of finite presentation over S , so is Y .

Let us prove that (ii) implies (i). It is enough to show that p is a covering for the (fppf) topology. But p is faithfully flat by hypothesis, and it is locally of finite presentation because X and Y are locally of finite presentation over S (EGA IV₁, 1.4.3(v)).

Lemma 10.9.— *Let G_j be an S_j -group of finite presentation, and put*

$$G = G_j \times_{S_j} S.$$

Then G^0 is representable if and only if there is an $i \geq j$ such that

$$(G_i)^0 = (G_j \times_{S_j} S_i)^0$$

is representable.

The condition is sufficient, since the functor $G \mapsto G^0$ commutes with base change, by 3.3.

Conversely, suppose that G^0 is representable. Then, by 3.9, G^0 is open in G and quasi-compact over S , and hence is of finite presentation over S , since G is. By 10.3 and 10.1, there are an $i \geq j$ and an open group subscheme H_i of G_i such that

$$H_i \times_{S_i} S = G^0.$$

The structural morphism $G^0 \rightarrow S$ has geometrically connected fibers (VI_A, 2.1.1). Consequently, by EGA IV₃, 9.3.3 and 9.7.7, after increasing i we may suppose that the structural morphism $H_i \rightarrow S_i$ has geometrically connected fibers.* It follows from 3.10.1 that the underlying space of H_i is precisely $(G_i)^0$; hence H_i represents $(G_i)^0$.

* Translator's note: the French source prints $H_i \rightarrow S$. Since H_i is an S_i -group and the argument spreads out the connectedness of $G^0 \rightarrow S$, the base here is S_i .

10.10.— Recall two very useful special cases of the situation stated in 10.0 (cf. EGA IV₃, 8.1.2(a) and (c)):

a) Given a point x of a scheme X , put $S_0 = \operatorname{Spec} \mathbb{Z}$, and consider the decreasing filtered inverse system $(S_i)_{i \in I}$ of affine open neighborhoods of x . Then

$$S = \operatorname{Spec} \mathcal{O}_{X,x}.$$

In particular, if x is the generic point of an integral scheme X , then

$$S = \operatorname{Spec} \kappa(x).$$

b) Put $S_0 = \operatorname{Spec} \mathbb{Z}$, and consider the family $(\mathcal{A}_i)_{i \in I}$, preordered by inclusion, of the finitely generated $\mathbb{Z}$ -subalgebras of the ring of an affine scheme S . Since the $\mathcal{A}_i$ are Noetherian rings, this makes it possible in many cases to pass from the Noetherian case to the general case.

We shall now give two results concerning the special case considered in (a).⁹⁵

Proposition 10.11.—⁹⁶ *Let S be an integral scheme with generic point η . Let G be an S -group and let Y, Z be S -schemes, all of finite presentation. Let*

$$u, v, i : Y \longrightarrow G, \quad j : Z \longrightarrow G$$

be morphisms of S -schemes. Suppose that

$$i_\eta : Y_\eta \longrightarrow G_\eta, \quad j_\eta : Z_\eta \longrightarrow G_\eta$$

are closed immersions.

Then there is a nonempty open subset S' of S such that the morphisms

$$i' : Y' \longrightarrow G', \quad j' : Z' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\operatorname{Transp}}(u', v'), \quad \underline{\operatorname{Transp}}_{G'}(i'(Y'), j'(Z')), \\ \underline{\operatorname{Transp}}_{G'}^{\operatorname{str}}(i'(Y'), j'(Z'))$$

respectively, the functors

$$\underline{\operatorname{Centr}}(u'), \quad \underline{\operatorname{Norm}}_{G'} i'(Y'),$$

are representable by closed S' -subschemes (respectively, closed S' -subgroups) of G' , of finite presentation over S' .

Proof. We shall apply the results of 10.1, first in the situation of 10.10(a), and then in that of 10.10(b). The schemes G_η, Y_η, Z_η are flat over the field $\kappa(\eta)$; the group G_η is separated over $\kappa(\eta)$ (VI_A, 0.3); and i_η, j_η are closed immersions. Therefore, by 10.1, there are an affine open subset $S' = \operatorname{Spec}(A')$ of S , a Noetherian subring A of A' , A -schemes G_A, Y_A, Z_A , flat and of finite presentation over A , and morphisms

$$u_A, v_A, i_A : Y_A \longrightarrow G_A, \quad j_A : Z_A \longrightarrow G_A,$$

such that G_A is an A -group separated over A , i_A and j_A are closed immersions, and

$$G \times_S S' = G_A \otimes_A A',$$

⁹⁵Editor's note: observe that the proof also uses case (b).

⁹⁶Editor's note: the statement has been simplified, and the case of group subschemes has been treated separately in Corollary 10.11.1.

with analogous equalities for the remaining schemes and morphisms. Since the functors under consideration over S' are obtained by base change from the analogous functors over $\mathrm{Spec}(A)$, it is enough to prove the result for the latter.

By EGA IV₂, 6.9.2, after replacing A by a localization A_f (and hence S' by the affine open subset S'_f), we may suppose that G_A, Y_A, Z_A are essentially free over A , in the sense of 6.2.1.⁹⁷ It then follows from 6.2.4(b) and 6.2.4(e) that, under the hypotheses of the proposition, the functors under consideration are representable by closed A -subschemes of G_A . These subschemes are of finite presentation over A , since A is Noetherian and G_A is of finite presentation over A ; and in the cases of $\underline{\mathrm{Centr}}(u_A)$ and $\underline{\mathrm{Norm}}_{G_A} i_A(Y_A)$, they are closed A -subgroups of G_A .

Corollary 10.11.1.— *Let S be an integral scheme with generic point η . Let G, H, K be S -groups of finite presentation, and let*

$$i : H \longrightarrow G, \quad j : K \longrightarrow G$$

be two quasi-compact monomorphisms of S -groups. Then there is a nonempty open subset S' of S such that the morphisms

$$i' : H' \longrightarrow G', \quad j' : K' \longrightarrow G',$$

obtained by base change, are closed immersions, and such that the functors

$$\underline{\mathrm{Transp}}_{G'}(H', K'), \quad \underline{\mathrm{Transp}}_{G'}^{\mathrm{str}}(H', K'),$$

respectively, the functors

$$\underline{\mathrm{Centr}}_{G'} H', \quad \underline{\mathrm{Norm}}_{G'} H',$$

are representable by closed S' -subschemes (respectively, closed S' -subgroups) of G' , of finite presentation over S' .

This follows from the preceding proposition, since the hypotheses imply, by 1.4.2, that i_η and j_η are closed immersions.

Proposition 10.12.— *Let S be an integral scheme, let G be an S -group of finite presentation, and let A and B be two group subschemes of G , of finite presentation over S and with smooth generic fibers. Suppose moreover that one of the following conditions holds:*

- a) *A has connected generic fiber;*
- b) *A and B are invariant in G .*

Then there are a nonempty open subset S' of S and a closed group subscheme D' of

$$G' = G|_{S'},$$

of finite presentation over S' and with smooth fibers, that represents the (fppf) sheaf associated with the presheaf of groups of commutators of

$$A' = A|_{S'} \quad \text{and} \quad B' = B|_{S'}$$

in G' . Moreover, D' has connected fibers in case (a), and is invariant in G' in case (b).

⁹⁷Editor's note: indeed, since G_A is flat and of finite presentation over A , it is covered by affine open subsets $G_1, \dots, G_n$ such that each $\mathcal{O}(G_i)$ is a flat A -algebra of finite presentation. By EGA IV₂, 6.9.2, there is an $f_i \in A$ such that $\mathcal{O}(G_i)_{f_i}$ is a free A_{f_i} -module. One may therefore replace $\mathrm{Spec}(A)$ by the affine open subset $D(f)$, where $f = f_1 \cdots f_n$, and proceed in the same way for Y_A and Z_A .

In particular, for every $s \in S'$,

$$D'_s = (A_s, B_s),$$

with the notation of 7.2.2.

Proof. Let η be the generic point of S , and put

$$H_\eta = (A_\eta, B_\eta).$$

Since A_η and B_η are smooth, 7.8 in case (a) and 7.3(v) in case (b) show that H_η is connected, respectively invariant in G_η .

We are in the situation of 10.0 corresponding to 10.10(a). Consequently, by 10.3 and 10.1, there are a nonempty open subset S' of S and an S' -group subscheme D' , of finite presentation and closed in G' , such that

$$D'_\eta = D' \otimes_{S'} \kappa(\eta) = H_\eta.$$

Moreover, by EGA IV₃, 9.7.7 and 9.3.3, we may suppose that D' has geometrically reduced fibers. In case (a), another application of EGA IV₃, 9.7.7 and 9.3.3 allows us to suppose that D' has connected fibers, hence geometrically connected fibers (cf. VI_A, 2.1.1). In case (b), we may suppose, by 10.4, that D' is invariant in G' .

We also saw, in the course of the proof of 7.8, that there is an integer n such that

$$\nu_\eta^n ((A_\eta \times_{\kappa(\eta)} B_\eta)^n) = D'_\eta,$$

where ν_η and ν_η^n are defined as in 7.2.2(a) and 7.1(ii). The same formulas define morphisms

$$\nu' : A' \times_{S'} B' \longrightarrow G', \quad \nu'^n : (A' \times_{S'} B')^n \longrightarrow G',$$

and

$$\nu'^n \otimes_{S'} \kappa(\eta) = \nu_\eta^n.$$

It follows from 10.1 that we may choose S' so that the morphism ν'^n is flat and factors through D' , and so that the resulting morphism

$$\nu'^n : (A' \times_{S'} B')^n \longrightarrow D'$$

is surjective. By 7.5, the morphism⁹⁸

$$\nu'^{2n} = \mu \circ (\nu'^n \times_{S'} \nu'^n) : (A' \times_{S'} B')^{2n} \longrightarrow D'$$

is a covering for the (fppf) topology. Therefore, by 7.6, D' represents the (fppf) sheaf associated with the presheaf of commutators of A' and B' in G' .

Furthermore, for every $s \in S'$, ν'^n induces a surjective morphism

$$\nu_s^n : (A_s \times_{\kappa(s)} B_s)^n \longrightarrow D'_s.$$

⁹⁹ Thus D'_s is a closed subgroup of G_s containing

$$\nu_s(A_s \times_{\kappa(s)} B_s),$$

and hence containing (A_s, B_s) . Since ν_s^n is surjective, $D'_s = (A_s, B_s)$; and by 7.6, it represents the (fppf) sheaf of commutators of A_s and B_s in G_s .

⁹⁸Editor's note: $n + 1$ has been corrected below to $2n$.

⁹⁹Editor's note: the argument that follows has been expanded from the original.

Corollary 10.12.1.—¹⁰⁰ *Let S be an integral scheme with generic point η , and let G be an S -group of finite presentation with smooth fibers. Put*

$$K_\eta^0 = G_\eta, \quad K_\eta^p = (K_\eta^{p-1}, K_\eta^{p-1}) \quad (\text{respectively, } K_\eta^p = (G_\eta, K_\eta^{p-1}))$$

for every $p \in \mathbb{N}^$. *Fix $n \in \mathbb{N}^*$. Then there are a nonempty open subset S' of S and a group subscheme D , invariant in $G|_{S'}$, of finite presentation and with smooth fibers, such that*

$$D_s = K_s^n$$

for every $s \in S'$.

This follows from 10.12, by induction on n .

Corollary 10.13.— *Let S be an integral scheme with generic point η . Let G be an S -group and let H be an invariant group subscheme of G . Suppose that G and H are of finite presentation over S and have smooth generic fibers.¹⁰¹*

If

$$(H_\eta, H_\eta) = H_\eta \quad (\text{respectively, } (G_\eta, H_\eta) = H_\eta),$$

then there is a nonempty open subset S' of S such that, for every $s \in S'$,

$$(H_s, H_s) = H_s \quad (\text{respectively, } (G_s, H_s) = H_s).$$

Indeed, the proof of 10.12 gives a nonempty open subset S' of S and an S' -group subscheme D of $G|_{S'}$, of finite presentation and with smooth fibers, such that, for every $s \in S'$,

$$D_s = (H_s, H_s) \quad (\text{respectively, } D_s = (G_s, H_s)).$$

On the other hand, since $D_\eta = H_\eta$ and D and H are of finite presentation over S' , EGA IV₃, 8.8.2.5 gives a nonempty open subset S'' of S' such that

$$D|_{S''} = H|_{S''}.$$

Consequently, for every $s \in S''$,

$$H_s = D_s = (H_s, H_s) \quad (\text{respectively, } H_s = D_s = (G_s, H_s)).$$

10.14.— Statements 10.2 and 10.3, concerning the category of S -groups of finite presentation, extend to the category of pairs consisting of an S -group of finite presentation and an S -scheme of finite presentation with operator group G . More precisely, in the situation recalled at the beginning of 10.0:

- (i) let $j \in I$, and let G_j and G'_j be two S_j -groups of finite presentation; let H_j (respectively, H'_j) be an S_j -scheme of finite presentation with operator group G_j (respectively, G'_j). For $i \in I$, $i \geq j$, put

$$G_i = G_j \times_{S_j} S_i, \quad G = G_j \times_{S_j} S,$$

and define G'_i , G' , H_i , H , H'_i , and H' similarly. Denote by

$$\text{Dihom}_{S\text{-gr.}}((G, H), (G', H'))$$

¹⁰⁰Editor's note: this corollary, which is used in the proof of 8.4, has been added.

* Translator's note: the French re-edition prints G in the second recursion. Since K_η^{p-1} is a $\kappa(\eta)$ -group subscheme of G_η , the commutator must be formed with G_η .

¹⁰¹Editor's note: the original assumed that G and H were of finite presentation and that H had smooth fibers; the hypothesis has been modified in order to apply 10.12. The proof has also been expanded.

the set of di-morphisms of S -groups and S -schemes with operator groups from the pair (G, H) to the pair (G', H') . Then the canonical map

$$\varinjlim_{i \geq j} \text{Dihom}_{S\text{-gr.}}((G_i, H_i), (G'_i, H'_i)) \longrightarrow \text{Dihom}_{S\text{-gr.}}((G, H), (G', H'))$$

is bijective.

- (ii) let G be an S -group of finite presentation and H an S -scheme of finite presentation with operator group G . Then there are an index $j \in I$, an S_j -group G_j of finite presentation, an S_j -scheme H_j of finite presentation with operator group G_j , and a di-isomorphism of S -groups and S -schemes with operator groups from

$$(G_j \times_{S_j} S, H_j \times_{S_j} S)$$

onto (G, H) .*

Definition 10.15.—¹⁰² Let $\mathcal{T}$ be a topology on $(\mathbf{Sch}/S)$ coarser than the canonical topology. Given an S -group scheme G and an S -scheme X with operator group G , one says that X is a formally homogeneous space under G (relative to the topology $\mathcal{T}$) if the morphism

$$\Phi : G \times_S X \longrightarrow X \times_S X, \quad (g, x) \longmapsto (gx, x),$$

defined for every $S' \rightarrow S$, $g \in G(S')$, and $x \in X(S')$, is an epimorphism in the category of sheaves for the topology $\mathcal{T}$; equivalently, Φ is a covering for the topology $\mathcal{T}$ (cf. IV, 4.4.3).

One says that X is a homogeneous space if it is formally homogeneous and, in addition, the morphism $X \rightarrow S$ is also a covering for the topology $\mathcal{T}$.

In particular, one says that X is a formally principal homogeneous space under G if Φ is an isomorphism, and that X is a principal homogeneous bundle (or a G -torsor) if Φ is an isomorphism and, in addition, the morphism $X \rightarrow S$ is a covering for the topology $\mathcal{T}$ (cf. IV, 5.1.5 and 5.1.6(ii)).

Proposition 10.16.— Retain the situation considered at the beginning of 10.0. Let $j \in I$, let G_j be an S_j -group, and let X_j be an S_j -scheme with operator group G_j . Suppose that G_j and X_j are of finite presentation over S_j .

For

$$X = X_j \times_{S_j} S$$

to be a homogeneous space (respectively, a principal homogeneous bundle) under

$$G = G_j \times_{S_j} S$$

for the (fppf) topology, it is necessary and sufficient that there be an index $i \geq j$ such that

$$X_i = X_j \times_{S_j} S_i$$

is a homogeneous space (respectively, a principal homogeneous bundle) under

$$G_i = G_j \times_{S_j} S_i.$$

Taking into account 10.14 and EGA IV₃, 8.8.2, 8.8.3, and 8.10.5, the assertion follows from the property of covering morphisms for the (fppf) topology recalled in 10.1.¹⁰³

* Translator's note: the French re-edition prints (G, X) here. Since this assertion begins with the pair (G, H) and X is not defined in 10.14, H is restored.

¹⁰²Editor's note: the original has been corrected by distinguishing the notions of a "formally homogeneous" object and a "homogeneous" object; see IV, §§5.1 and 6.7.

¹⁰³Editor's note: for limit-passage properties of torsors expressed in terms of Čech cohomology, see also SGA 4, VII, 5.7 and its corollaries. This is developed in detail in [Ma07].

11. Affine group schemes

11.0. Reminders.—¹⁰⁴ Let $q : X \rightarrow S$ be a quasi-compact and quasi-separated morphism of schemes (cf. EGA IV₁, 1.1 and 1.2), and let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -module. Recall that $q_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_S$ -module (EGA I, 9.2.1). Moreover, by EGA III, 1.4.15 (completed by EGA IV₁, 1.7.21), one has (c) below, and the proof in *loc. cit.* also gives (a) and (b):

(a) If $\mathcal{F}$ is a filtered direct limit of quasi-coherent submodules $\mathcal{F}_\alpha$, then

$$q_*(\mathcal{F}) = \varinjlim_{\alpha} q_*(\mathcal{F}_\alpha).$$

(b) If $\mathcal{E}$ is a flat $\mathcal{O}_S$ -module, the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} q_*(\mathcal{O}_X) \longrightarrow q_*q^*(\mathcal{E})$$

is an isomorphism.

(c) Let $p : S' \rightarrow S$ be a flat morphism, let $q' : X' \rightarrow S'$ be obtained from q by base change, and let $\mathcal{F}'$ be the inverse image of $\mathcal{F}$ on X' . Then the canonical morphism

$$p^*q_*(\mathcal{F}) \longrightarrow q'_*(\mathcal{F}')$$

is an isomorphism.

Indeed, let $U = \operatorname{Spec}(A)$ be an arbitrary affine open subset of S . By hypothesis, $q^{-1}(U)$ is the union of affine open subsets

$$V_i = \operatorname{Spec}(B_i), \quad i = 1, \dots, n,$$

and every intersection $V_i \cap V_j$ is the union of finitely many affine open subsets

$$W_{ijk} = \operatorname{Spec}(C_{ijk}).$$

Then $\Gamma(U, q_*(\mathcal{F})) = \Gamma(q^{-1}(U), \mathcal{F})$ is the kernel of the morphism

$$\bigoplus_{i=1}^n \Gamma(V_i, \mathcal{F}) \longrightarrow \bigoplus_{i,j,k} \Gamma(W_{ijk}, \mathcal{F}).$$

Assertion (a) follows, since every term above commutes with filtered direct limits (the V_i and W_{ijk} being affine, hence quasi-compact).

Let us prove (b). The module $E = \Gamma(U, \mathcal{E})$ is flat over A , and $\Gamma(U, q_*q^*(\mathcal{E}))$ is the kernel $K(E)$ of the morphism

$$\bigoplus_{i=1}^n B_i \otimes_A E \longrightarrow \bigoplus_{i,j,k} C_{ijk} \otimes_A E.$$

Since E is flat over A , this kernel is identified with

$$K(A) \otimes_A E = \mathcal{O}_X(q^{-1}(U)) \otimes_A E.$$

¹⁰⁴Editor's note: this paragraph of reminders has been added.

Finally, if U' is an arbitrary affine open subset of S' lying over U , then $A' = \mathcal{O}(U')$ is a flat A -algebra, and the same argument gives*

$$\mathcal{F}(q^{-1}(U)) \otimes_A A' \xrightarrow{\sim} \mathcal{F}'(q'^{-1}(U')).$$

Notation.— Let S be a scheme, let X be an S -scheme, and let $f : X \rightarrow S$ be the structural morphism. We shall put

$$\mathcal{A}(X) = f_*(\mathcal{O}_X).$$

Lemma 11.1.— *Let X and Y be two quasi-compact and quasi-separated S -schemes, and let $f : X \rightarrow S$ and $g : Y \rightarrow S$ be their structural morphisms. Then the canonical homomorphism*

$$\varphi : \mathcal{A}(X) \otimes_{\mathcal{O}_S} \mathcal{A}(Y) \longrightarrow \mathcal{A}(X \times_S Y)$$

*is an isomorphism in each of the following cases:*¹⁰⁵

- (a) f and g are affine;
- (b) g (or f) is flat and affine;
- (c) g is flat and $f_*(\mathcal{O}_X)$ is a flat $\mathcal{O}_S$ -module.

In case (b), we shall suppose that it is g that is flat and affine. Put

$$S' = \operatorname{Spec} \mathcal{A}(X), \quad Y' = Y \times_S S', \quad g' = g \times_S S',$$

and write $v : S' \rightarrow S$ for the structural morphism:

$$\begin{array}{ccc} Y' & \xrightarrow{\quad} & Y \\ g' \downarrow & & \downarrow g \\ \operatorname{Spec} \mathcal{A}(X) = S' & \xrightarrow{\quad v \quad} & S \end{array}$$

In cases (a) and (b), the morphism g is affine, and hence EGA II, 1.5.2 gives

$$g'_*(\mathcal{O}_{Y'}) = v^* g_*(\mathcal{O}_Y) = \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}. \quad (1)$$

The same equality holds in case (c), by 11.0(c), since S' is flat over S and g is quasi-compact and quasi-separated.

On the other hand, by EGA II, 1.2.7, the morphism $f : X \rightarrow S$ factors through v by means of a morphism $p : X \rightarrow S'$, and

$$p_*(\mathcal{O}_X) = \mathcal{O}_{S'}, \quad X \times_S Y = X \times_{S'} Y'.$$

Since f is quasi-separated, so is p (EGA IV₁, 1.2.2); and since f is quasi-compact and v is quasi-separated, p is also quasi-compact (EGA IV₁, 1.2.4). Consider the cartesian square

* Translator's note: the French re-edition prints $q'^{-1}(U)$ in the right-hand side below. Since $q' : X' \rightarrow S'$ and the affine open subset just introduced is $U' \subset S'$, the inverse image is $q'^{-1}(U')$.

¹⁰⁵Editor's note: if k is a field and X is an infinite disjoint union of copies of $S = \operatorname{Spec} k$ (so that X is not quasi-compact), then $\mathcal{A}(X) = k^X$, and the canonical morphism

$$k^X \otimes_k k^X \longrightarrow k^{X \times X}$$

is not surjective.

$$\begin{array}{ccc}
X \times_S Y & \xrightarrow{p'} & Y' \\
\downarrow & & \downarrow g' \\
X & \xrightarrow{p} & S'
\end{array}$$

In cases (b) and (c), the scheme Y is flat over S , and hence Y' is flat over S' . Applying 11.0(c) once again, we obtain

$$p'_*(\mathcal{O}_{X \times_S Y}) = g'^*p_*(\mathcal{O}_X) = g'^*(\mathcal{O}_{S'}) = \mathcal{O}_{Y'}. \quad (2)$$

The same equality holds in case (a), because p and p' are then isomorphisms.

Finally, since v is affine, EGA II, 1.4.7 gives

$$v_*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(X)$$

for every quasi-coherent $\mathcal{O}_S$ -module $\mathcal{F}$. Combined with (2) and (1), this gives

$$\begin{aligned}
\mathcal{A}(X \times_S Y) &= v_* g'_* p'_*(\mathcal{O}_{X \times_S Y}) \\
&\stackrel{(2)}{=} v_* g'_*(\mathcal{O}_{Y'}) \stackrel{(1)}{=} v_*(\mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}) \\
&= \mathcal{A}(Y) \otimes_{\mathcal{O}_S} \mathcal{A}(X).
\end{aligned}$$

Corollary 11.2.— *The functor*

$$X \longmapsto \operatorname{Spec} \mathcal{A}(X),$$

from the full subcategory of $(\mathbf{Sch}/S)$ formed by the S -schemes X that are flat, quasi-compact, and quasi-separated over S , and for which $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module, to the category of flat affine S -schemes, commutes with finite products and therefore takes S -groups to S -groups.

Definition 11.3.— *Given a flat S -group G , quasi-compact and quasi-separated over S , such that $\mathcal{A}(G)$ is flat over $\mathcal{O}_S$,¹⁰⁶ we write G_{af} , and call the S -group*

$$G_{\text{af}} = \operatorname{Spec} \mathcal{A}(G)$$

the affine envelope of G .

Proposition 11.3.1.—¹⁰⁷ *The canonical morphism*

$$\tau_G : G \longrightarrow G_{\text{af}}$$

is a morphism of S -groups. Moreover, it has the following universal property:

- (i) *For every morphism of S -schemes $\varphi : G \rightarrow H$, where H is affine over S , there is a unique morphism of S -schemes $\varphi' : G_{\text{af}} \rightarrow H$ such that*

$$\varphi = \varphi' \circ \tau_G.$$

- (ii) *If, moreover, H is an S -group and φ is a morphism of S -groups, then the same is true of φ' .*

¹⁰⁶Editor's note: if S is a regular locally Noetherian scheme of dimension at most 2, and X is an S -scheme that is flat, quasi-compact, and quasi-separated, then $\mathcal{A}(X)$ is a flat $\mathcal{O}_S$ -module; cf. [Ray70a], VII, 3.2.

¹⁰⁷Editor's note: this proposition has been added; see also the additional Section 12 below for a study of the morphism $G \rightarrow G_{\text{af}}$ and of its kernel.

11.4.—¹⁰⁸ Let $\mathcal{E}$ and $\mathcal{F}$ be two quasi-coherent $\mathcal{O}_S$ -modules. Consider the S -functor

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$$

(cf. I, 3.1.4), that is, for every S -scheme $f : X \rightarrow S$,

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(\mathcal{E})_X, \mathrm{W}(\mathcal{F})_X).$$

Moreover, by I, 4.6.2, one has $\mathrm{W}(\mathcal{E})_X = \mathrm{W}(f^*(\mathcal{E}))$ (and similarly for $\mathcal{F}$), and

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathrm{W}(f^*(\mathcal{E})), \mathrm{W}(f^*(\mathcal{F}))) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})).$$

We therefore obtain (using the adjunction formula for the last equality)

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))(X) = \mathrm{Hom}_{\mathcal{O}_X}(f^*(\mathcal{E}), f^*(\mathcal{F})) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_*f^*(\mathcal{F})). \quad (\dagger)$$

Proposition 11.5.— *Let X be an S -scheme that is quasi-compact and quasi-separated over S , let $f : X \rightarrow S$ be the structural morphism, and let $\mathcal{E}$ and $\mathcal{E}'$ be two quasi-coherent $\mathcal{O}_S$ -modules. Suppose that one of the following two conditions holds:*

- a) f is affine;
- b) $\mathcal{E}$ is flat over $\mathcal{O}_S$.

Then the canonical morphism

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X) \longrightarrow f_*f^*(\mathcal{E})$$

is an isomorphism, and hence

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E}))(X) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(X)).$$

Indeed, the second assertion follows from 11.4 and the first; the latter follows from EGA II, 1.4.7 in case (a), and from 11.0(b) in case (b).¹⁰⁹

11.6.—¹¹⁰ Let G be an S -group and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module. Giving $\mathcal{E}$ a G - $\mathcal{O}_S$ -module structure (that is, an $\mathcal{O}_S$ -linear action of G on $\mathrm{W}(\mathcal{E})$, cf. I, 4.7.1) is equivalent to giving a morphism of S -functors in monoids

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}));$$

indeed, such a ρ necessarily takes G into $\underline{\mathrm{Aut}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$.

By 11.4, giving a morphism of S -functors

$$\rho : G \longrightarrow \underline{\mathrm{End}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}))$$

¹⁰⁸Editor's note: in 11.4–11.6, we have considered $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}), \mathrm{W}(\mathcal{F}))$ instead of $\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{F}), \mathrm{V}(\mathcal{E}))$ (cf. I, 4.6.3), and simplified the original by taking account of the addition 11.0(b).

¹⁰⁹Editor's note: we have simplified the original, which used the isomorphism

$$\underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{W}(\mathcal{E}'), \mathrm{W}(\mathcal{E})) \simeq \underline{\mathrm{Hom}}_{\mathcal{O}_S}(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')),$$

then the inclusion of the right-hand term in

$$\underline{\mathrm{Hom}}_S(\mathrm{V}(\mathcal{E}), \mathrm{V}(\mathcal{E}')) = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}', f_*f^*(\mathrm{Sym}(\mathcal{E}))),$$

and applied EGA III, 4.1.15 to $\mathrm{V}(\mathcal{E}) = \mathrm{Spec}(\mathrm{Sym}(\mathcal{E}))$ to deduce 11.0(b).

¹¹⁰Editor's note: we have expanded 11.6 and displayed the results obtained in the form of Proposition 11.6.1.

is equivalent to giving an element θ of

$$\mathrm{Hom}_{\mathcal{O}_G}(f^*(\mathcal{E}), f^*(\mathcal{E})),$$

which corresponds by adjunction to an element δ of

$$\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{E}, f_* f^*(\mathcal{E})),$$

where f denotes the projection $G \rightarrow S$.

Let $m : G \times_S G \rightarrow G$ be multiplication, let δ_G be the morphism

$$\mathcal{O}_G \longrightarrow m_*(\mathcal{O}_{G \times_S G}),$$

and let $\varphi : G \times_S G \rightarrow S$ be the projection (which equals $f \circ m$). It is convenient to write $\boxtimes$ for the “external” tensor product. We thus obtain a morphism

$$\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow \mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}).$$

By abuse of notation, we again write $\mathrm{id}_{\mathcal{E}} \boxtimes \delta_G$ for the composite of this morphism with the canonical morphism

$$\mathcal{E} \boxtimes_{\mathcal{O}_S} m_*(\mathcal{O}_{G \times_S G}) \longrightarrow m_* m^* f^*(\mathcal{E}) = m_* \varphi^*(\mathcal{E}).$$

On the other hand, let $h : G \rightarrow S$ denote a second copy of $f : G \rightarrow S$, and consider the following commutative diagram, in which p, q denote the two projections:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ p \downarrow & \searrow \varphi & \downarrow h \\ G & \xrightarrow{f} & S \end{array}$$

Writing δ again for the morphism $\mathcal{E} \rightarrow h_* h^*(\mathcal{E})$, we obtain the morphism

$$\delta \boxtimes \mathrm{id}_{\mathcal{O}_G} : f^*(\mathcal{E}) = \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \longrightarrow h_* h^*(\mathcal{E}) \boxtimes_{\mathcal{O}_S} \mathcal{O}_G = f^* h_* h^*(\mathcal{E}).$$

By abuse of notation, we again write $\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}$ for the composite of this morphism with the canonical morphism

$$f^* h_* h^*(\mathcal{E}) \longrightarrow p_* \varphi^*(\mathcal{E}).$$

The condition that ρ be compatible with multiplication is then equivalent to saying that, for every open subset U of S , the following diagram is commutative:

$$(1) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ \delta \downarrow & & \downarrow \mathrm{id}_{\mathcal{E}} \boxtimes \delta_G \\ \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) & \xrightarrow{\delta \boxtimes \mathrm{id}_{\mathcal{O}_G}} & \Gamma(U \times_S G \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \end{array}$$

Furthermore, the unit section $\varepsilon : S \rightarrow G$ induces a morphism u from $\mathcal{O}_G$ to

$$\varepsilon_* \varepsilon^*(\mathcal{O}_G) = \varepsilon_*(\mathcal{O}_S),$$

and the condition that ρ preserve unit elements is equivalent to the commutativity of the diagram

$$(2) \quad \begin{array}{ccc} \Gamma(U, \mathcal{E}) & \xrightarrow{\delta} & \Gamma(U \times_S G, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_G) \\ & \searrow \sim & \swarrow \text{id}_{\mathcal{E}} \boxtimes u \\ & \Gamma(U \times_S S, \mathcal{E} \boxtimes_{\mathcal{O}_S} \mathcal{O}_S) & \end{array}$$

Thus, giving $\mathcal{E}$ a G - $\mathcal{O}_S$ -module structure is equivalent to giving a morphism of $\mathcal{O}_S$ -modules

$$\delta : \mathcal{E} \longrightarrow f_* f^*(\mathcal{E})$$

satisfying conditions (1) and (2) above. In that case, the morphism

$$\theta : f^*(\mathcal{E}) \longrightarrow f^*(\mathcal{E})$$

deduced from δ by adjunction is an isomorphism, because it corresponds to the isomorphism

$$G \times_S W(\mathcal{E}) \xrightarrow{\sim} G \times_S W(\mathcal{E})$$

defined set-theoretically by $(g, x) \mapsto (g, gx)$; see also I, 6.5.4.

Now suppose that G is flat, quasi-compact, and quasi-separated over S , and that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module. Then, by 11.1(c), the canonical morphism

$$\mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow \mathcal{A}(G \times_S G)$$

is an isomorphism, and we write Δ for the morphism

$$\delta_G : \mathcal{A}(G) \longrightarrow \mathcal{A}(G) \otimes_{\mathcal{O}_S} \mathcal{A}(G).$$

If, moreover,

$$\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G) = f_* f^*(\mathcal{E}),$$

as is the case by 11.5 if $G \rightarrow S$ is affine or if $\mathcal{E}$ is flat over $\mathcal{O}_S$, then conditions (1) and (2) are equivalent to the following conditions, which express that

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

makes $\mathcal{E}$ a right $\mathcal{A}(G)$ -comodule (cf. I, 4.7.2):

(CM 1) Put $\mathcal{A} = \mathcal{A}(G)$. The following diagram is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ \delta \downarrow & & \downarrow \text{id}_{\mathcal{E}} \otimes \Delta \\ \mathcal{E} \otimes \mathcal{A} & \xrightarrow{\delta \otimes \text{id}_{\mathcal{A}}} & \mathcal{E} \otimes \mathcal{A} \otimes \mathcal{A} \end{array}$$

(CM 2) Write $\eta : \mathcal{A} \rightarrow \mathcal{O}_S$ for the morphism $\mathcal{A}(\varepsilon)$. The following diagram is commutative:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{\delta} & \mathcal{E} \otimes \mathcal{A} \\ \sim \searrow & & \swarrow \text{id}_{\mathcal{E}} \otimes \eta \\ & \mathcal{E} \otimes \mathcal{O}_S & \end{array}$$

Remark 11.6.A.— Recall that $V(\mathcal{E})$ denotes the vector fibration over S representing the functor $V(\mathcal{E})$; that is, for every $S' \rightarrow S$,

$$V(\mathcal{E})(S') = \operatorname{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'}).$$

Since I, 4.6.2 gives an anti-isomorphism of S -functors in monoids

$$\underline{\operatorname{End}}_{\mathcal{O}_S}(W(\mathcal{E})) \simeq \underline{\operatorname{End}}_{\mathcal{O}_S}(V(\mathcal{E})),$$

if $\mathcal{E}$ is a left G - $\mathcal{O}_S$ -module, then there is a right action

$$\mu : V(\mathcal{E}) \times_S G \longrightarrow V(\mathcal{E})$$

of G on $V(\mathcal{E})$, defined set-theoretically by

$$(\phi g)(x) = \phi(gx)$$

for every $g \in G(S')$, $x \in \Gamma(S', \mathcal{E}_{S'})$, and $\phi \in \operatorname{Hom}_{\mathcal{O}_{S'}}(\mathcal{E}_{S'}, \mathcal{O}_{S'})$. We therefore have the commutative diagrams

$$\begin{array}{ccc} V(\mathcal{E}) \times_S G \times_S G & \xrightarrow{\mu \times \operatorname{id}_G} & V(\mathcal{E}) \times_S G \\ \operatorname{id}_{V(\mathcal{E})} \times m \downarrow & & \downarrow \mu \\ V(\mathcal{E}) \times_S G & \xrightarrow{\mu} & V(\mathcal{E}) \end{array} \quad \begin{array}{ccc} V(\mathcal{E}) & \xleftarrow{\mu} & V(\mathcal{E}) \times_S G \\ & \nwarrow \tilde{\operatorname{id}}_{V(\mathcal{E})} \times \varepsilon & \uparrow \\ & & V(\mathcal{E}) \times_S S \end{array}$$

When G is flat, quasi-compact, and quasi-separated over S , when $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and when one of the conditions of 11.5 is satisfied, one similarly recovers conditions (CM 1) and (CM 2).

Consequently, we have proved:

Proposition 11.6.1.— *Let G be an S -group that is flat, quasi-compact, and quasi-separated over S , such that $\mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module, and let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module.*

(i) *Giving an $\mathcal{A}(G)$ -comodule structure*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$$

is equivalent to giving $\mathcal{E}$ a G_{af} - $\mathcal{O}_S$ -module structure (that is, an $\mathcal{O}_S$ -linear action of G_{af} on $\mathcal{E}$). By composition with the morphism of S -groups $G \rightarrow G_{\text{af}}$, this defines a G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$.

(ii) *If, moreover, $\mathcal{E}$ is flat, every $\mathcal{O}_S$ -linear action of G on $\mathcal{E}$ factors through G_{af} and corresponds to a unique $\mathcal{A}(G)$ -comodule structure on $\mathcal{E}$.*

Lemma 11.7.— *Let G be an S -group that is flat, quasi-compact, and quasi-separated over S , such that $\mathcal{A} = \mathcal{A}(G)$ is a flat $\mathcal{O}_S$ -module. Let $\mathcal{E}$ be a quasi-coherent $\mathcal{O}_S$ -module, let*

$$\delta : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{A}$$

be an $\mathcal{A}$ -comodule structure, and let

$$\rho : G \longrightarrow \underline{\operatorname{Aut}}_{\mathcal{O}_S}(W(\mathcal{E}))$$

be the associated action of G on $\mathcal{E}$. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$ -submodule of $\mathcal{E}$ such that the restriction δ_0 of δ to $\mathcal{E}_0^$ factors through*

$$\mathcal{E}_0 \longrightarrow \mathcal{E}_0 \otimes \mathcal{A};$$

that is, suppose that the following diagram is commutative:

* Translator's note: the French source writes “the restriction δ_0 of μ ” here. The surrounding notation and the displayed square show that δ is intended.

$$\begin{array}{ccc}
\mathcal{E}_0 & \hookrightarrow & \mathcal{E} \\
\delta_0 \downarrow & & \downarrow \delta \\
\mathcal{E}_0 \otimes \mathcal{A} & \hookrightarrow & \mathcal{E} \otimes \mathcal{A}
\end{array}$$

(N.B. The morphism $\mathcal{E}_0 \otimes \mathcal{A} \rightarrow \mathcal{E} \otimes \mathcal{A}$ is injective because $\mathcal{A}$ is flat over $\mathcal{O}_S$.) Then δ_0 makes $\mathcal{E}_0$ an $\mathcal{A}(G)$ -comodule, and hence defines an action ρ_0 of G on $\mathcal{E}_0$. This is called the action on $\mathcal{E}_0$ induced by ρ , and we say that $\mathcal{E}_0$ is stable under ρ .

This follows immediately from the definitions and from 11.6. Notice, however, that in general the canonical map

$$W(\mathcal{E}_0) \longrightarrow W(\mathcal{E})$$

is not a monomorphism.

Remark 11.7.bis.—¹¹¹ Let G be a flat S -group and let $\mathcal{E}$ be a quasi-coherent G - $\mathcal{O}_S$ -module. Write f for the morphism $G \rightarrow S$ and δ for the morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E})$$

defined in 11.6. Let $\mathcal{E}_0$ be a quasi-coherent $\mathcal{O}_S$ -submodule of $\mathcal{E}$. Since f is flat, $f_* f^*(\mathcal{E}_0)$ is an $\mathcal{O}_S$ -submodule of $f_* f^*(\mathcal{E})$, and the same holds for $\varphi = f \times f$. Consequently, if the restriction δ_0 of δ to $\mathcal{E}_0$ factors through $f_* f^*(\mathcal{E}_0)$, then it makes $\mathcal{E}_0$ a G - $\mathcal{O}_S$ -module. In this case, we say that $\mathcal{E}_0$ is a G -stable submodule of $\mathcal{E}$.

Definition 11.8.0.—¹¹² Let S be a scheme. An $\mathcal{O}_S$ -coalgebra is an $\mathcal{O}_S$ -module $\mathcal{C}$, together with two morphisms of $\mathcal{O}_S$ -modules

$$\Delta : \mathcal{C} \longrightarrow \mathcal{C} \otimes \mathcal{C}, \quad \varepsilon : \mathcal{C} \longrightarrow \mathcal{O}_S,$$

satisfying the following two axioms (cf. I, 4.2):

(CO 1) Δ is coassociative: the following diagram is commutative:

$$\begin{array}{ccccc}
& & \mathcal{C} \otimes \mathcal{C} & & \\
& \nearrow \Delta & & \searrow \text{id} \otimes \Delta & \\
\mathcal{C} & & & & \mathcal{C} \otimes \mathcal{C} \otimes \mathcal{C} \\
& \searrow \Delta & & \nearrow \Delta \otimes \text{id} & \\
& & \mathcal{C} \otimes \mathcal{C} & &
\end{array}$$

(CO 2) ε is a counit; that is, the following two composites are the identity:

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\text{id} \otimes \varepsilon} \mathcal{C} \otimes \mathcal{O}_S \xrightarrow{\sim} \mathcal{C},$$

$$\mathcal{C} \xrightarrow{\Delta} \mathcal{C} \otimes \mathcal{C} \xrightarrow{\varepsilon \otimes \text{id}} \mathcal{O}_S \otimes \mathcal{C} \xrightarrow{\sim} \mathcal{C}.$$

¹¹¹Editor's note: we have added this remark, which generalizes 11.7 and will be useful in 11.10.bis.

¹¹²Editor's note: the statements 11.8 and 11.9 concern only the notion of a comodule over a coalgebra. We have therefore introduced Definition 11.8.0 and reformulated 11.8 and 11.9 accordingly.

A $\mathcal{C}$ -comodule (on the right) is an $\mathcal{O}_S$ -module $\mathcal{E}$, together with a morphism of $\mathcal{O}_S$ -modules

$$\mu : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathcal{C}$$

satisfying axioms (CM 1) and (CM 2) of 11.6.

We say that $\mathcal{C}$ (respectively, $\mathcal{E}$) is a *quasi-coherent coalgebra* (respectively, a *quasi-coherent comodule*) if it is a quasi-coherent $\mathcal{O}_S$ -module.

Let A be a commutative ring and let C be an A -coalgebra. Then

$$C^\vee = \operatorname{Hom}_A(C, A)$$

is an A -algebra. We denote by

$$\operatorname{ev} : C \otimes_A C^\vee \longrightarrow A$$

the natural evaluation map.

Lemma 11.8.— *Let C be an A -coalgebra, V a C -comodule, and M an A -submodule of V . Suppose that C is a projective A -module.¹¹³ Let $c(M)$ be the image of the morphism of A -modules*

$$\theta : M \otimes_A C^\vee \xrightarrow{\mu \otimes \operatorname{id}} V \otimes_A C \otimes_A C^\vee \xrightarrow{\operatorname{id} \otimes \operatorname{ev}} V.$$

Then $c(M)$ is the smallest subcomodule of V containing M , and it is a finitely generated A -module if M is. We call $c(M)$ the subcomodule generated by M .

Moreover, for every homomorphism of rings $A \rightarrow A'$, if M' denotes the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$, then $c(M')$ is the image of $c(M) \otimes_A A'$ in V' . Thus the formation of $c(M)$ commutes with base change.

First, $M \subset c(M)$ by (CM 2). If N is a subcomodule of V containing M , then $\mu(M) \subset N \otimes C$, and hence $c(M) \subset N$.

By hypothesis, C is a direct summand of a free A -module L with basis $(e_i)_{i \in I}$. Let φ_i denote the restriction to C of the linear form e_i^* , defined by

$$e_i^*(e_j) = \delta_{ij}.$$

Let $x \in M$. We may write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i, \tag{1}$$

where $x_i \in V$ and J is a finite subset of I . Then

$$x_i = \theta(x \otimes \varphi_i) \in c(Ax),$$

and consequently

$$c(Ax) = \sum_{i \in J} Ax_i.$$

Since C is a direct summand of L , say $L = C \oplus R$, whence

$$V \otimes L = (V \otimes C) \oplus (V \otimes R),$$

we obtain

$$(c(Ax) \otimes L) \cap (V \otimes C) = c(Ax) \otimes C.$$

¹¹³Editor's note: in the original, C is assumed to be a free A -module. The generalization to the case where C is a projective A -module, pointed out by J.-P. Serre, is mentioned in Remark 11.10.1. We have incorporated this generalization here and in 11.9, and have expanded the proofs accordingly.

Consequently, $\mu(x)$ may also be written in the form

$$\mu(x) = \sum_{j \in J} x_j \otimes b_j, \quad (2)$$

with $b_j \in C$. Write

$$\Delta(b_j) = \sum_{i \in I} b_{ij} \otimes e_i, \quad b_{ij} \in C.$$

Applying $\mu \otimes \text{id}$ to (1) and $\text{id} \otimes \Delta$ to (2), and using axiom (CM 1), we obtain, for every $i \in J$,

$$\mu(x_i) = \sum_{j \in J} x_j \otimes b_{ij} \in c(Ax) \otimes C.$$

This proves that $c(M)$ is a subcomodule of V , and therefore that it is the smallest subcomodule of V containing M .

It is clear that $c(M)$ is a finitely generated A -module whenever M is: if

$$M = Ax_1 + \cdots + Ax_n, \quad \mu(x_k) = \sum_i x_{ik} \otimes e_i,$$

then the x_{ik} , for $k = 1, \dots, n$ and i ranging over a finite subset of I , generate $c(M)$.

Finally, let $A \rightarrow A'$ be a homomorphism of rings, and let M' be the image of $M \otimes_A A'$ in $V' = V \otimes_A A'$. Then $c(M')$, respectively the image of $c(M) \otimes_A A'$ in V' , is the image of the morphism θ' below, respectively of the composite $\theta' \circ \tau$:

$$M \otimes_A A' \otimes_A C^\vee \xrightarrow{\tau} M \otimes_A \text{Hom}_A(C, A') \xrightarrow{\theta'} V'.$$

These two morphisms have the same image. Indeed, let $\psi \in \text{Hom}_A(C, A')$ and $x \in M$, and write

$$\mu(x) = \sum_{i \in J} x_i \otimes e_i.$$

Then

$$\theta'(x \otimes \psi) = \sum_{i \in J} \psi(e_i) x_i$$

is the image under $\theta' \circ \tau$ of the element

$$\sum_{i \in J} x \otimes \psi(e_i) \otimes \varphi_i \in M \otimes_A A' \otimes_A C^\vee.$$

This proves the lemma.

We also have the following proposition.

Proposition 11.8.bis.—¹¹⁴ *Let A be a Noetherian ring, let C be an A -coalgebra flat over A , let V be a C -comodule, and let M be a finitely generated A -submodule of V . Then there is a subcomodule W of V , finitely generated over A , that contains M .*

Indeed, since M is finitely generated, so is $\Delta_V(M)$. Hence there is a finitely generated A -submodule M' of V such that

$$\Delta_V(M) \subset M' \otimes_A C.$$

Let $\pi : V \rightarrow V/M'$ be the projection, put

$$\overline{\Delta}_V = (\pi \otimes \text{id}_C) \Delta_V,$$

¹¹⁴Editor's note: this proposition, taken from [Se68], Section 1.5, Proposition 2, has been added.

and set

$$W = \{x \in V \mid \Delta_V(x) \in M' \otimes_A C\} = \ker \bar{\Delta}_V.$$

This is an A -submodule of V that contains M and is contained in M' , because

$$x = (\mathrm{id}_V \otimes \varepsilon)\Delta_V(x).$$

It is therefore finitely generated over A . Moreover,

$$(\bar{\Delta}_V \otimes \mathrm{id}_C)\Delta_V = (\pi \otimes \Delta_C)\Delta_V$$

vanishes on W ; in other words, $\Delta_V(W)$ is contained in the kernel K of $\bar{\Delta}_V \otimes \mathrm{id}_C$. Since C is flat over A , one has

$$K = W \otimes_A C.$$

Thus W is a subcomodule of V .

Lemma 11.8.1.—¹¹⁵ *Let C be an A -coalgebra, let V be a C -comodule, let M be an A -submodule of V , and let $f : A \rightarrow A'$ be a faithfully flat homomorphism of rings. Suppose that $C' = C \otimes_A A'$ is a projective A' -module.*

- (i) *There is a smallest subcomodule $t(M)$ of V containing M , and $t(M)$ is a finitely generated A -module if M is. Moreover, the formation of $t(M)$ commutes with base change.*
- (ii) *More precisely, C is a projective A -module, and $t(M) = c(M)$.*

Proof. For (ii), [RG71] (see Proposition 11.8.2 below) shows that C is a projective A -module. We may therefore apply Lemma 11.8: $c(M)$ is the smallest subcomodule of V containing M , it is a finitely generated A -module if M is, and its formation commutes with base change.

To avoid an anachronism ([RG71] postdates SGA 3), let us sketch a direct proof of (i). Since $A \rightarrow A'$ is flat,

$$M' = M \otimes_A A'$$

is an A' -submodule of $V' = V \otimes_A A'$. Since C' is a projective A' -module, $c(M')$ is the smallest subcomodule of V' containing M' . Denote by $V' \otimes_A A'$ and $A' \otimes_A V'$ the two $A' \otimes_A A'$ -comodule structures on

$$V'' = V' \otimes_{A'} (A' \otimes_A A')$$

obtained by the two base changes

$$A' \rightrightarrows A' \otimes_A A', \quad a' \mapsto a' \otimes 1, \quad a' \mapsto 1 \otimes a'.$$

The A' -comodule V' carries an isomorphism of $A' \otimes_A A'$ -comodules

$$\varphi : V' \otimes_A A' \xrightarrow{\sim} A' \otimes_A V', \quad (x \otimes a') \otimes b' \mapsto b' \otimes (x \otimes a'),$$

which is a descent datum, that is,

$$\varphi_{31} = \varphi_{32} \circ \varphi_{21}.$$

¹¹⁵Editor's note: this lemma, which is used in the proof of 11.9, has been added.

Since $M' = M \otimes_A A'$, the isomorphism φ takes $M' \otimes_A A'$ onto $A' \otimes_A M'$, and hence takes $c(M' \otimes_A A')$ onto $c(A' \otimes_A M')$. Because the formation of $c(M')$ commutes with base change,

$$c(M' \otimes_A A') = c(M') \otimes_A A', \quad c(A' \otimes_A M') = A' \otimes_A c(M').$$

Thus

$$\varphi(c(M') \otimes_A A') = A' \otimes_A c(M'),$$

and φ equips $c(M')$ with a descent datum. By fpqc descent, there is a unique subcomodule $t(M)$ of V such that

$$c(M') = t(M) \otimes_A A'.$$

It contains M , since $t(M) \otimes_A A'$ contains M' . Moreover, if N is a subcomodule of V containing M , then N contains $t(M)$, because $N \otimes_A A'$ contains

$$c(M') = t(M) \otimes_A A'.$$

Hence $t(M)$ is the smallest subcomodule of V containing M .

Finally, let $A \rightarrow B$ be a homomorphism of rings. Put

$$B' = B \otimes_A A',$$

and let M_B , respectively $M'_{B'}$, be the image of $M \otimes_A B$ in $V_B = V \otimes_A B$, respectively of $M' \otimes_{A'} B'$ in $V' \otimes_{A'} B' = V \otimes_A B'$. Then

$$M_B \otimes_B B' = M'_{B'}.$$

On the one hand, the preceding construction, applied to C_B and $B \rightarrow B'$, gives

$$c(M_B \otimes_B B') = t(M_B) \otimes_B B' = t(M_B) \otimes_A A'.$$

On the other hand, because the formation of $c(M')$ commutes with base change, $c(M'_{B'})$ is the image in

$$V' \otimes_{A'} B' = V \otimes_A B \otimes_A A'$$

of

$$c(M') \otimes_{A'} B' = t(M) \otimes_A B \otimes_A A'.$$

It follows that $t(M_B)$ is the image of $t(M) \otimes_A B$ in V_B .

Proposition 11.8.2 (Gruson–Raynaud).— *Let $f : A \rightarrow A'$ be a faithfully flat homomorphism. Then f “descends projectivity”: if M is an A -module and $M \otimes_A A'$ is a projective A' -module, then M is a projective A -module.*

Indeed, by [RG71], II, 2.5.1, f descends the Mittag–Leffler condition, and therefore, by *loc. cit.*, II, 3.1.3, f descends projectivity.¹¹⁶

Proposition 11.9.—¹¹⁷ *Let $\mathcal{C}$ be an $\mathcal{O}_S$ -coalgebra, let $\mathcal{E}$ be a $\mathcal{C}$ -comodule, and let $\mathcal{F}$ be an $\mathcal{O}_S$ -submodule of $\mathcal{E}$, all of them quasi-coherent. Suppose that S is covered by affine open subsets $U_\alpha = \text{Spec } A_\alpha$, and that for each α there is a faithfully flat homomorphism of rings $A_\alpha \rightarrow A'_\alpha$ such that*

$$\Gamma(U_\alpha, \mathcal{C}) \otimes_{A_\alpha} A'_\alpha$$

¹¹⁶Editor’s note: let us also point out that assertion II, 2.5.2 of *loc. cit.*, which is more general than II, 2.5.1, is corrected in [Gr73]; this does not affect the case of faithfully flat homomorphisms.

¹¹⁷Editor’s note: as indicated in editor’s note 112, we have rewritten the statement for an $\mathcal{O}_S$ -coalgebra $\mathcal{C}$ (rather than for an S -group G satisfying the hypotheses in Corollary 11.10). We have also expanded the proof (the original merely said: “... the proposition follows from Lemma 11.8”).

is a projective A'_α -module.^(*)

Then there is a smallest quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$, and $t(\mathcal{F})$ is a finitely generated $\mathcal{O}_S$ -module if $\mathcal{F}$ is. Moreover, for every base change $S' \rightarrow S$, if $\mathcal{F}'$ denotes the image of $\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in

$$\mathcal{E}' = \mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'},$$

then $t(\mathcal{F}')$ is the image of $t(\mathcal{F}) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ in $\mathcal{E}'$. In other words, the formation of $t(\mathcal{F})$ commutes with base change.

Proof. For each α , let T_α be the A_α -module

$$T_\alpha = t(\Gamma(U_\alpha, \mathcal{F})).$$

By 11.8.1(i), the associated $\mathcal{O}_{U_\alpha}$ -module $\tilde{T}_\alpha$ is the smallest quasi-coherent subcomodule of $\mathcal{E}|_{U_\alpha}$ containing $\mathcal{F}|_{U_\alpha}$; and it is a finitely generated $\mathcal{O}_{U_\alpha}$ -module if $\mathcal{F}|_{U_\alpha}$ is.

For every α, β , put

$$U_{\alpha\beta} = U_\alpha \cap U_\beta.$$

Since the construction of $t(M)$ commutes with base change, there are canonical isomorphisms of $\mathcal{O}_{U_{\alpha\beta}}$ -modules

$$\varphi_{\alpha\beta} : \tilde{T}_\beta \otimes_{A_\beta} \mathcal{O}_{U_{\alpha\beta}} \xrightarrow{\sim} \tilde{T}_\alpha \otimes_{A_\alpha} \mathcal{O}_{U_{\alpha\beta}}.$$

They satisfy the cocycle condition

$$\varphi'_{\alpha\gamma} = \varphi'_{\alpha\beta} \circ \varphi'_{\beta\gamma},$$

where $\varphi'_{\alpha\gamma}$, respectively the other maps, denotes the restriction of $\varphi_{\alpha\gamma}$, respectively the other maps, to $U_\alpha \cap U_\beta \cap U_\gamma$.

Consequently, the $\tilde{T}_\alpha$ glue to a quasi-coherent subcomodule $t(\mathcal{F})$ of $\mathcal{E}$ containing $\mathcal{F}$. We leave it to the reader to verify that $t(\mathcal{F})$ is the smallest quasi-coherent subcomodule of $\mathcal{E}$ containing $\mathcal{F}$, and that its formation commutes with base change.

Definition 11.9.1.—¹¹⁸ Let S be a scheme and let $\mathcal{P}$ be a quasi-coherent $\mathcal{O}_S$ -module. The following conditions are equivalent:

- (i) For every affine open subset U of S , the module $\Gamma(U, \mathcal{P})$ is a projective $\mathcal{O}_S(U)$ -module.
- (ii) There is a covering (U_α) of S by affine open subsets such that every $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module.
- (iii) There is a covering (U_α) of S by affine open subsets, and faithfully flat homomorphisms of rings

$$A_\alpha = \mathcal{O}_S(U_\alpha) \longrightarrow A'_\alpha,$$

such that, for every α ,

$$\Gamma(U_\alpha, \mathcal{P}) \otimes_{A_\alpha} A'_\alpha$$

is a projective A'_α -module.

^{*}This holds, for example, when $\mathcal{C} = \mathcal{A}(G)$, where G is a reductive S -group, as we shall see in Exposé XXII, 5.7.8.

¹¹⁸Editor's note: this definition, taken from the bottom of page 82 of [RG71], has been added. Thus the hypothesis in Proposition 11.9 is that the coalgebra $\mathcal{C}$ be a locally projective $\mathcal{O}_S$ -module, and we have used this terminology in Corollary 11.10.

It is clear that

$$(i) \implies (ii) \implies (iii).$$

Conversely, if (iii) holds, 11.8.2 implies that every $\Gamma(U_\alpha, \mathcal{P})$ is a projective $\mathcal{O}_S(U_\alpha)$ -module, and hence (ii) holds.

Finally, suppose that (ii) holds, and let

$$V = \operatorname{Spec} A$$

be an arbitrary affine open subset. It is covered by finitely many affine open subsets

$$V_1, \dots, V_n, \quad V_i = \operatorname{Spec} A_i,$$

each contained in at least one $V \cap U_\alpha$, so that $\Gamma(V_i, \mathcal{P})$ is a projective A_i -module. Put

$$A' = A_1 \times \dots \times A_n.$$

Then $A \rightarrow A'$ is faithfully flat and

$$\Gamma(V, \mathcal{P}) \otimes_A A'$$

is a projective A' -module. Therefore 11.8.2 implies that $\Gamma(V, \mathcal{P})$ is a projective A -module.

When these equivalent conditions hold, we say that $\mathcal{P}$ is a *locally projective* $\mathcal{O}_S$ -module.

Corollary 11.10.— *Let S be a quasi-compact and quasi-separated scheme, let G be an S -group, and let ρ be a linear action of G on a quasi-coherent $\mathcal{O}_S$ -module $\mathcal{E}$. Suppose that:*

(i) *G satisfies one of the following conditions:*

(a) *G is affine and flat over S ;*

(b) *G is flat, quasi-compact, and quasi-separated over S , and $\mathcal{E}$ is flat;*

(ii) *$\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module.*

Then $\mathcal{E}$ is the direct limit of an increasing filtered family of finitely generated quasi-coherent $\mathcal{O}_S$ -submodules of $\mathcal{E}$ that are stable under G .

By hypothesis (i) and 11.6.1, $\mathcal{E}$ is endowed with an $\mathcal{A}(G)$ -comodule structure. On the other hand, since S is quasi-compact and quasi-separated, $\mathcal{E}$ is the direct limit of its finitely generated quasi-coherent submodules (EGA I, 9.4.9 and EGA IV₁, 1.7.7). Consequently, the corollary follows from Proposition 11.9, applied to the coalgebra $\mathcal{A}(G)$.

We also have the following proposition.

Proposition 11.10.bis.¹¹⁹ *Let S be a Noetherian scheme, let G be an S -group that is flat, quasi-compact, and quasi-separated over S , let $\mathcal{E}$ be a quasi-coherent G - $\mathcal{O}_S$ -module, and let $\mathcal{M}$ be a coherent $\mathcal{O}_S$ -submodule of $\mathcal{E}$. Then $\mathcal{M}$ is contained in a coherent $\mathcal{O}_S$ -submodule stable under G .*

Proof. Write f for the morphism $G \rightarrow S$, and τ for the adjunction morphism

$$\mathcal{E} \longrightarrow f_* f^*(\mathcal{E}).$$

By 11.6, the G - $\mathcal{O}_S$ -module structure on $\mathcal{E}$ is given by an automorphism θ of the $\mathcal{O}_G$ -module $f^*(\mathcal{E})$, such that the morphism

$$\delta = \theta \circ \tau$$

¹¹⁹Editor's note: this proposition has been added; it is a special case of [Th87], 1.4–1.5. The author there refers to an argument of Deligne (cf. [Kn71], III, Th. 1.1); one may also note its similarity with Serre's argument ([Se68], Prop. 2) recalled in 11.8.bis.

satisfies conditions (1) and (2) of 11.6. (The situation considered in [Th87] is more general: the author considers a G -scheme X and a G -equivariant $\mathcal{O}_X$ -module $\mathcal{E}$ (cf. Exposé I, Section 6); here $X = S$, endowed with the trivial action of G .)

Since S is Noetherian, $\mathcal{E}$ is the filtered direct limit of its coherent submodules $\mathcal{F}_\alpha$ (cf. EGA I, 9.4.9). Then $f^*(\mathcal{E})$ is the filtered direct limit of the $f^*(\mathcal{F}_\alpha)$, which are submodules of $f^*(\mathcal{E})$ because f is flat. Moreover, since f is quasi-compact and quasi-separated, by 11.0 the $\mathcal{O}_S$ -module $f_*f^*(\mathcal{E})$ is quasi-coherent and is the filtered direct limit of the quasi-coherent submodules $f_*f^*(\mathcal{F}_\alpha)$. Consequently, $\mathcal{E}$ is the filtered direct limit of the quasi-coherent $\mathcal{O}_S$ -submodules $\mathcal{E}_\alpha$, where $\mathcal{E}_\alpha$ denotes the inverse image under δ of $f_*f^*(\mathcal{F}_\alpha)$; equivalently, it is the kernel of the composite

$$\bar{\delta}_\alpha : \mathcal{E} \xrightarrow{\delta} f_*f^*(\mathcal{E}) \longrightarrow f_*f^*(\mathcal{E}/\mathcal{F}_\alpha).$$

Since $\mathcal{M}$ is coherent and S is Noetherian, every increasing sequence of submodules of $\mathcal{M}$ is stationary; hence $\mathcal{M}$ is contained in some $\mathcal{E}_\alpha$. We shall show that every $\mathcal{E}_\alpha$ is coherent and G -stable.

Let

$$u : f^*(\mathcal{E}) \longrightarrow \varepsilon_*(\mathcal{E})$$

be the morphism corresponding by adjunction to the identity morphism from $\mathcal{E}$ to $f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$. Then the composite

$$\mathcal{E} \xrightarrow{\tau} f_*f^*(\mathcal{E}) \xrightarrow{f_*(u)} f_*\varepsilon_*(\mathcal{E}) = \mathcal{E}$$

is the identity (cf. 11.6(2)). We have the commutative diagram

$$\begin{array}{ccccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_*f^*(\mathcal{F}_\alpha) & \xrightarrow{f_*(u)} & \mathcal{F}_\alpha \\ i_\alpha \downarrow & & \downarrow & & \downarrow j_\alpha \\ \mathcal{E} & \xrightarrow{\tau} & f_*f^*(\mathcal{E}) & \xrightarrow{f_*(u)} & \mathcal{E} \end{array}$$

where i_α (respectively j_α) denotes the inclusion of $\mathcal{E}_\alpha$ (respectively $\mathcal{F}_\alpha$) in $\mathcal{E}$. It follows that i_α factors through j_α ; thus $\mathcal{E}_\alpha$ is a submodule of $\mathcal{F}_\alpha$, and is therefore coherent.

Finally, let us show that $\mathcal{E}_\alpha$ is G -stable (cf. 11.7.bis). Let $h : G \rightarrow S$ denote a second copy of $f : G \rightarrow S$, and consider the following commutative diagram:

$$\begin{array}{ccc} G \times_S G & \xrightarrow{q} & G \\ p \downarrow & \searrow \varphi & \downarrow h \\ G & \xrightarrow{f} & S \end{array}$$

The exact sequence

$$0 \longrightarrow \mathcal{E}_\alpha \longrightarrow \mathcal{E} \xrightarrow{\bar{\delta}_\alpha} h_*h^*(\mathcal{E}/\mathcal{F}_\alpha)$$

therefore gives, since f is flat, the exact sequence

$$0 \longrightarrow f_*f^*(\mathcal{E}_\alpha) \longrightarrow f_*f^*(\mathcal{E}) \xrightarrow{f_*f^*(\bar{\delta}_\alpha)} f_*f^*h_*h^*(\mathcal{E}/\mathcal{F}_\alpha). \quad (\dagger)$$

Moreover, since h is quasi-compact and quasi-separated and f is flat, the canonical morphism

$$f^*h_*h^*(\mathcal{E}/\mathcal{F}_\alpha) \longrightarrow p_*q^*h^*(\mathcal{E}/\mathcal{F}_\alpha) = p_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$$

is an isomorphism. Thus the right-hand term in (†) is $\varphi_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha)$. Using the notation of 11.6 and writing π_α for the projection $\mathcal{E} \rightarrow \mathcal{E}/\mathcal{F}_\alpha$, we obtain the following commutative diagram, whose bottom row is exact:

$$\begin{array}{ccccc} \mathcal{E}_\alpha & \xrightarrow{\delta} & f_*f^*(\mathcal{E}) & & \\ \delta \downarrow & & \downarrow f_*(f^*(\pi_\alpha) \boxtimes \delta_G) & & \\ f_*f^*(\mathcal{E}_\alpha) & \longrightarrow & f_*f^*(\mathcal{E}) & \xrightarrow{f_*f^*(\bar{\delta}_\alpha)} & \varphi_*\varphi^*(\mathcal{E}/\mathcal{F}_\alpha) \end{array}$$

Since

$$f_*(f^*(\pi_\alpha) \boxtimes \delta_G) \circ \delta$$

vanishes on $\mathcal{E}_\alpha$, it follows that δ carries $\mathcal{E}_\alpha$ into $f_*f^*(\mathcal{E}_\alpha)$; in other words, $\mathcal{E}_\alpha$ is G -stable. This proves the proposition.

Remark 11.10.1.—¹²⁰ In 11.8, it is not enough to assume that C is a flat A -module, even when A is a principal ideal domain. Indeed, we have the following counterexamples, pointed out to us independently by O. Gabber and J.-P. Serre.

(a) Let $(A, \mathfrak{m})$ be a discrete valuation ring, let K be its field of fractions, and let G be the A -group obtained by “extension by zero” from the K -group $(\mathbb{Z}/2\mathbb{Z})_K$. The constant group $(\mathbb{Z}/2\mathbb{Z})_A$, and hence also its subgroup G , acts on the free A -module V with basis v_1, v_2 by interchanging v_1 and v_2 . Then Av_1 is not a G -submodule of V , but it is the intersection of the G -submodules

$$Av_1 + \mathfrak{m}^n v_2, \quad n \in \mathbb{N}^*.$$

Thus there is no smallest G -submodule of V containing v_1 .

(b) Let A be an integral domain distinct from its field of fractions K , and let G be the flat affine A -group corresponding to the Hopf algebra

$$\mathcal{A}(G) = \{P \in K[T] \mid P(0) \in A\}.$$

Its comultiplication, counit, and antipode are respectively defined by

$$\Delta(T) = T \otimes 1 + 1 \otimes T, \quad \varepsilon(T) = 0, \quad \tau(T) = -T.$$

(Thus $G \otimes_A K = \mathbb{G}_{a,K}$.)

Let $V = Av_1 \oplus Av_2$ be free over A , and let u be the endomorphism of V defined by

$$u(v_1) = v_2, \quad u(v_2) = 0,$$

so that $u^2 = 0$. Then V has the $\mathcal{A}(G)$ -comodule structure

$$\mu(m) = 1 \otimes m + T \otimes u(m).$$

The G -submodules of V containing v_1 are precisely the A -submodules $Av_1 \oplus Iv_2$, where I is a nonzero ideal of A . Their intersection is Av_1 , which is not a G -submodule. Hence there is no smallest G -submodule of V containing v_1 . Moreover,

$$C = A \oplus K \cdot T$$

is a subcoalgebra of $\mathcal{A}(G)$, flat over A , and the coaction

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

¹²⁰Editor’s note: the first part of the original Remark 11.10.1 has been incorporated into 11.8 and 11.9 (with “free” replaced by “projective”); the counterexamples below correct its second part.

factors through $V \otimes C$. Thus the “very simple” coalgebra C also gives a counterexample.

Finally, the preceding two examples are special cases of the following construction. Let A be an integral domain distinct from its field of fractions K , and let B be a Hopf A -algebra, free over A . Write

$$\varepsilon : B \longrightarrow A, \quad I = \ker(\varepsilon)$$

for the augmentation and the augmentation ideal. Since $B = A \cdot 1 \oplus I$, one readily sees that

$$B' = \{b \in B \otimes_A K \mid \varepsilon(b) \in A\}$$

is a Hopf subalgebra of B_K . Suppose that V is a B -comodule, free with basis $(v_1, \dots, v_n)$ as an A -module, and that $\mu_V(v_1) \neq v_1 \otimes 1$. Then Av_1 is not a subcomodule of V , but it is the intersection of the subcomodules

$$Av_1 + J \cdot V$$

as J ranges over the nonzero ideals of A .^{*} Thus there is no smallest subcomodule of V containing v_1 .

Proposition 11.11.— *Let G be an algebraic group over a field k . The following conditions are equivalent:*

- (i) G is affine.
- (ii) G is quasi-affine.
- (iii) G acts faithfully on a quasi-affine k -scheme X .
- (iv) G acts linearly and faithfully on a k -vector space (not necessarily finite-dimensional).
- (v) G is isomorphic to a closed subgroup of $\mathrm{GL}(n)_k$.

Proof. The implication (i) $\Rightarrow$ (ii) is immediate, and (ii) $\Rightarrow$ (iii) because G acts faithfully on itself by translations.

Suppose that G acts faithfully on the right on a quasi-affine k -scheme X .¹²¹ Since X is quasi-affine, it is separated and quasi-compact.¹²² Similarly, G is separated (VI_A, 0.3) and quasi-compact (since it is of finite type over k). Hence, by 11.1(c), there are canonical isomorphisms

$$\mathcal{O}(X \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G), \quad \mathcal{O}(X \times G \times G) = \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G).$$

It follows that the morphism

$$\mu : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

induced by $X \times G \rightarrow X$ gives $\mathcal{O}(X)$ the structure of a right $\mathcal{O}(G)$ -comodule; that is, G acts linearly on the left on the k -algebra $\mathcal{O}(X)$. Consequently, G also acts on the right on the affine envelope

$$X_{\mathrm{af}} = \mathrm{Spec} \mathcal{O}(X)$$

^{*} Translator’s note: the French re-edition uses I both for the fixed augmentation ideal above and for the ideals ranging in this final intersection. We use J for the latter to keep the two roles distinct.

¹²¹Editor’s note: the original has been expanded in what follows. We have also chosen a right action of G on X , so as to obtain a left linear action of G on $\mathcal{O}(X)$.

¹²²Editor’s note: recall that a k -scheme X is called quasi-affine if it is isomorphic to a quasi-compact open subscheme of an affine k -scheme (EGA II, 5.1.1).

of X , and the canonical morphism $X \rightarrow X_{\text{af}}$ is G -equivariant.

Since X is quasi-affine, the morphism $X \rightarrow X_{\text{af}}$ is an open immersion (EGA II, 5.1.2). Hence, a fortiori, G acts faithfully on X_{af} , and therefore the left linear action of G on the k -algebra $\mathcal{O}(X) = \mathcal{O}(X_{\text{af}})$ is faithful. This proves (iii) $\Rightarrow$ (iv).

Now suppose that G acts faithfully on a k -vector space V . By 11.10, V is the direct limit of finite-dimensional vector subspaces V_i stable under G . Let K_i be the kernel of the induced action of G on V_i ; that is, the kernel of $G \rightarrow \text{Aut}(V_i)$. Then K_i is a closed subscheme of G , and the faithfulness of the action of G means that the intersection of the K_i is the identity subgroup of G . Since G is Noetherian, one of the K_i is already the identity subgroup. Thus $G \rightarrow \text{Aut}(V_i)$ is a monomorphism, and hence, by 1.4.2, a closed immersion. This proves (iv) $\Rightarrow$ (v). Finally, (v) $\Rightarrow$ (i) is immediate, and the proposition follows.

Remark 11.11.1.— Proposition 11.11 admits the following generalization. Let S be a regular, locally Noetherian scheme of dimension at most 1, and let G be a group scheme that is flat, quasi-compact, and quasi-separated over S . (In this case, $\mathcal{A}(G)$ is a torsion-free $\mathcal{O}_S$ -module, and hence is flat.)

(a) The following conditions are equivalent:¹²³

- (i) G is affine over S .
- (ii) G is quasi-affine over S .
- (iii) G acts faithfully on a quasi-affine flat S -scheme.
- (iv) G acts linearly and faithfully on a flat quasi-coherent $\mathcal{O}_S$ -module.

(b) If, moreover, G is of finite type over S and S is Noetherian, these conditions imply that G is isomorphic to a closed subgroup of $\text{Aut}(\mathcal{E})$, where $\mathcal{E}$ is a finite-rank locally free $\mathcal{O}_S$ -module.¹²⁴

Lemma 11.12.— Let k be a field and let G be an affine k -group. Put $A = \mathcal{A}(G)$. Given $x \in A$, there is a finitely generated k -subalgebra B of A such that

$$x \in B, \quad \Delta(B) \subset B \otimes_k B, \quad u(B) \subset B,$$

where u denotes the involution of A corresponding to the inversion morphism of G .¹²⁵

We may suppose that $x \neq 0$. Then $\Delta(x) \neq 0$, since

$$(\varepsilon \otimes \text{id})\Delta(x) = x,$$

where ε denotes the augmentation (counit) of A . Write

$$\Delta(x) = \sum_{j=1}^n e_j \otimes a_j, \quad n \text{ minimal.} \tag{1}$$

The e_j (respectively the a_j) are then linearly independent. Complete $(e_1, \dots, e_n)$ to a basis $(e_i)_{i \in I}$ of A , and, for

$$j \in J = \{1, \dots, n\},$$

¹²³Editor's note: the equivalence of these conditions is proved in the added Section 12.

¹²⁴Editor's note: this is proved, with various generalizations, in the added Section 13.

¹²⁵Editor's note: on the one hand, this is generalized in Section 13 to the case where G is affine and flat over a regular base of dimension at most 2. On the other hand, the original has been expanded and corrected in what follows.

write

$$\Delta(e_j) = \sum_{i \in I} e_i \otimes b_{ij}.$$

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (1), the axiom (CO 1) of 11.8.0 (see also (HA 1) in I, 4.2) gives

$$\sum_{j \in J} e_j \otimes \Delta(a_j) = \sum_{\ell \in J} \Delta(e_\ell) \otimes a_\ell = \sum_{i \in I} e_i \otimes \left(\sum_{\ell \in J} b_{i\ell} \otimes a_\ell \right).$$

Since the e_i are linearly independent, it follows that

$$\Delta(a_j) = \sum_{\ell \in J} b_{j\ell} \otimes a_\ell \quad (j \in J). \quad (2)$$

Let B be the finitely generated k -subalgebra of A generated by the b_{ij} and the $u(b_{ij})$, for $i, j \in J$. It is clear that $u(B) = B$.

Applying $\Delta \otimes \text{id}$ and $\text{id} \otimes \Delta$ to (2), we obtain once again from (CO 1) the equalities

$$\sum_{\ell \in J} \Delta(b_{j\ell}) \otimes a_\ell = \sum_{i \in J} b_{ji} \otimes \Delta(a_i) = \sum_{i, \ell \in J} b_{ji} \otimes b_{i\ell} \otimes a_\ell.$$

Since the a_ℓ are linearly independent, this yields

$$\Delta(b_{j\ell}) = \sum_{i \in J} b_{ji} \otimes b_{i\ell} \quad (j, \ell \in J). \quad (3)$$

Moreover,

$$\Delta \circ u = (u \otimes u) \circ v \circ \Delta, \quad v(a \otimes b) = b \otimes a,$$

and hence

$$\Delta(u(b_{j\ell})) = \sum_{i \in J} u(b_{i\ell}) \otimes u(b_{ji}) \quad (j, \ell \in J). \quad (4)$$

Since Δ is an algebra homomorphism, (3) and (4) imply

$$\Delta(B) \subset B \otimes_k B.$$

Finally, axiom (CO 2) of 11.8.0 (see also (HA 2) in I, 4.2), applied to (2) and (1), gives*

$$a_j = \sum_{i \in J} b_{ji} \varepsilon(a_i), \quad x = \sum_{j \in J} \varepsilon(e_j) a_j.$$

Thus $x \in B$, proving the lemma.

Proposition 11.13.— *Let k be a field and let G be an affine k -group with algebra A . Then G is the projective limit of an increasing filtered system of affine k -groups of finite type whose transition morphisms are faithfully flat.*

If B and B' are two finitely generated subalgebras of A stable under Δ and u , the subalgebra generated by B and B' has the same properties. Hence, by Lemma 11.12, A is the direct limit of an increasing filtered family $(B_i)_{i \in I}$ of finitely generated subalgebras stable under Δ and u . Each B_i , equipped with the restriction of u and with the morphism

$$B_i \longrightarrow B_i \otimes_k B_i$$

* Source-reading note: the French re-edition prints $a_j = \sum_{i \in I} \varepsilon(a_i) b_{ij}$. Applying $\text{id} \otimes \varepsilon$ to (2) gives the first identity displayed here. The Reinhold comparison lineage repeats the printed formula.

induced by Δ , is a Hopf algebra. Thus, by I, 4.2, it is the algebra of an affine k -group G_i of finite type over k . Finally, since

$$A = \varinjlim_i B_i,$$

we have

$$G = \varprojlim_i G_i$$

(cf. EGA IV₃, 8.2.3). The transition morphisms are faithfully flat by the following lemma.

Lemma 11.14.— *Let k be a field, let $u : G \rightarrow H$ be a morphism between affine k -groups of finite type, and let*

$$u^\natural : B \longrightarrow A$$

be the corresponding morphism of k -algebras.¹²⁶ Then u is faithfully flat if and only if $u^\natural$ is injective.

The condition is plainly necessary (cf. EGA 0_I, 6.6.1). Conversely, suppose that $u^\natural$ is injective, and put

$$N = \text{Ker}(u).$$

By VI_A, 3.3.2 and VI_A, 5.4.1, G/N is a k -group of finite type and u factors as

$$G \xrightarrow{p} G/N \xrightarrow{v} H,$$

where p is faithfully flat and v is a closed immersion. Since H is affine, G/N is affine, and

$$v^\natural : B \longrightarrow \mathcal{O}(G/N)$$

is surjective (cf. EGA I, 4.2.3). But

$$u^\natural = p^\natural \circ v^\natural$$

is injective, so $v^\natural$ is also injective. Hence $v^\natural$, and therefore v , is an isomorphism. Since p is faithfully flat, so is u .

Definition 11.15.—¹²⁷ *Let k be a field, let G be a quasi-compact k -group, and let V be a k -vector space endowed with a k -linear action of G , hence, by 11.6.1, with an $\mathcal{A}(G)$ -comodule structure*

$$\delta : V \longrightarrow V \otimes \mathcal{A}(G).$$

Let $v \in V$ be nonzero. The following conditions are equivalent:

- (i) *There is a $\lambda \in \mathcal{A}(G)$, necessarily unique, such that*

$$\delta(v) = v \otimes \lambda.$$

- (ii) *For every k -algebra R and every $g \in G(R)$, one has $g \cdot v \in Rv$; that is, there is a necessarily unique $f \in R$ such that, in $V \otimes R$,*

$$g \cdot v = v \otimes f.$$

¹²⁶Editor's note: we write $u^\natural$ (instead of u°) for the morphism $B \rightarrow A$ corresponding to $u : \text{Spec}(A) \rightarrow \text{Spec}(B)$.

¹²⁷Editor's note: we have added the hypothesis that G be quasi-compact and have spelled out the equivalence of conditions (i) and (ii).

It is clear that (i) implies (ii). Conversely, if (ii) holds, apply it with

$$R = \mathcal{A}(G), \quad g = \text{id}_{\mathcal{A}(G)}.$$

There is then a unique $\lambda \in \mathcal{A}(G)$ such that $\delta(v) = v \otimes \lambda$.

When v satisfies these conditions, we call v a *semi-invariant vector under G* , and λ the *weight* of v . We also say that v is a *semi-invariant of weight λ* .

Let Δ denote the comultiplication of $\mathcal{A}(G)$. The equality

$$v \otimes \lambda \otimes \lambda = (\delta \otimes \text{id})(\delta(v)) = (\text{id} \otimes \Delta)(\delta(v)) = v \otimes \Delta(\lambda)$$

implies

$$\Delta(\lambda) = \lambda \otimes \lambda.$$

Consequently, λ defines a morphism of Hopf algebras

$$\mathcal{A}(\mathbb{G}_{m,k}) = k[T, T^{-1}] \longrightarrow \mathcal{A}(G), \quad T \longmapsto \lambda,$$

and hence a morphism of k -groups

$$\lambda : G \longrightarrow \mathbb{G}_{m,k}.$$

Thus λ is a *character* of G , called the *character associated with the semi-invariant vector v* .

Lemma 11.16.0.—¹²⁸ *Let k be a field, let H be an affine k -group, let V be an H -module of dimension n , and let U be a d -dimensional vector subspace of V . Consider the line*

$$D = \bigwedge^d U \subset \bigwedge^d V.$$

Then U is stable under H if and only if D is stable under H .

Necessity is clear. To prove sufficiency, we may assume $d < n$. Let $(e_1, \dots, e_d)$ be a basis of U , and complete it to a basis $(e_1, \dots, e_n)$ of V . For every k -algebra R , the module

$$V_R = V \otimes R$$

is free over R , and

$$U_R = \{v \in V_R \mid v \wedge (e_1 \wedge \cdots \wedge e_d) = 0\}.$$

Indeed, for $i > d$, the elements

$$e_i \wedge e_1 \wedge \cdots \wedge e_d$$

are linearly independent in $\bigwedge_R^{d+1} V_R$. Since $H(R)$ acts on $\bigwedge_R^\bullet V_R$ by

$$h(x_1 \wedge \cdots \wedge x_s) = h(x_1) \wedge \cdots \wedge h(x_s),$$

it follows that, if $H(R)$ stabilizes

$$D_R = R(e_1 \wedge \cdots \wedge e_d),$$

then it also stabilizes U_R .

Theorem 11.16 (Chevalley).— *Let k be a field, let G be an affine algebraic k -group, and let H be a closed group subscheme of G .¹²⁹ Then there are a finite-dimensional G -module V and a line D in V such that*

$$H = \text{Norm}_G(D);$$

¹²⁸Editor's note: this lemma, taken from [DG70], Section II.2, 3.5, has been added.

¹²⁹Editor's note: first, recall that every group subscheme of G is closed (1.4.2). Second, we have stated the result in its usual form, namely that “ H is the stabilizer of a line in a representation of G ,” while retaining the original formulation in terms of a sequence $a_1, \dots, a_n$ of semi-invariants in $\mathcal{A}(G)$.

that is, for every k -algebra R ,

$$H(R) = \{ g \in G(R) \mid g(D_R) = D_R \}.$$

Equivalently, there are finitely many elements $a_1, \dots, a_n \in \mathcal{A}(G)$, all semi-invariant of the same weight λ for the “right” action of H ; that is,

$$a_i(gh) = \lambda(h)a_i(g)$$

for every k -algebra R , $g \in G(R)$, and $h \in H(R)$, such that H is the largest closed group subscheme of G under which the a_i are semi-invariant.

Proof. Let Δ and ε denote, respectively, the comultiplication and the augmentation of

$$A = \mathcal{A}(G).$$

There is an ideal I of A , contained in $\text{Ker } \varepsilon$, such that

$$H = \text{Spec}(A/I), \quad \Delta(I) \subset I \otimes A + A \otimes I.$$

Put $B = A/I$, and let $\pi : A \rightarrow B$ be the projection. Consider the left action of H on A given by

$$(h\phi)(g) = \phi(gh).$$

The corresponding B -comodule structure is

$$\overline{\Delta} : A \xrightarrow{\Delta} A \otimes A \xrightarrow{\text{id}_A \otimes \pi} A \otimes B.$$

Then I is an H -submodule of A , since

$$\overline{\Delta}(I) \subset I \otimes B.$$

The k -algebra A is finitely generated, and hence Noetherian, so I has a finite system of generators $(x_1, \dots, x_r)$. By 11.8, the x_i are contained in a finite-dimensional G -submodule V of A . Then

$$W = V \cap I$$

is a finite-dimensional H -module; write

$$d = \dim_k W.$$

Since V contains every x_i , W generates the ideal I .

Put

$$E = \bigwedge^d V,$$

choose a basis $(w_1, \dots, w_d)$ of W , and choose a basis $(e_0, \dots, e_n)$ of E containing

$$e_0 = w_1 \wedge \dots \wedge w_d.$$

The action of G on V canonically induces an action of G on E , hence an A -comodule structure

$$\rho : E \longrightarrow E \otimes A.$$

For $j = 0, \dots, n$, write[†]

$$\rho(e_j) = \sum_{i=0}^n e_i \otimes b_{ij}.$$

[†] Source-reading note: the French re-edition and the Reinhold comparison lineage print the lower summation bound as $i = 1$, although the chosen basis begins with e_0 and the ensuing identity sums from 0. The lower bound is therefore $i = 0$ here.

As in the proof of 11.12, one has

$$\Delta(b_{ij}) = \sum_{\ell=0}^n b_{i\ell} \otimes b_{\ell j}. \quad (*)$$

Put $a_i = b_{i0}$. Equivalently, if $(e_0^*, \dots, e_n^*)$ is the dual basis, then the a_i are the matrix coefficients

$$g \mapsto e_i^*(ge_0).$$

The action of H on E corresponds to

$$\bar{\rho}: E \xrightarrow{\rho} E \otimes A \xrightarrow{\text{id}_E \otimes \pi} E \otimes B.$$

Since W is stable under H , the vector e_0 is semi-invariant under H . Hence

$$\bar{\rho}(e_0) = e_0 \otimes \pi(a_0), \quad a_i = b_{i0} \in I \quad (1 \leq i \leq n).$$

Substitution in $(*)$ gives

$$\bar{\Delta}(a_i) = a_i \otimes \pi(a_0) \quad (0 \leq i \leq n).$$

Thus the a_i are semi-invariant under H , all of weight $\pi(a_0)$. Moreover, after replacing E , if necessary, by the G -submodule generated by e_0 (cf. 11.8), we may suppose that the a_i are linearly independent.

Conversely, let

$$H' = \text{Spec } B', \quad B' = A/I',$$

be a closed group subscheme of G under which every a_i is semi-invariant, of weight $\lambda_i \in B'$. This includes, in particular, the case in which e_0 is semi-invariant under H' of weight λ' . We shall prove that $H' \subset H$.[‡] Let $\pi': A \rightarrow B'$ be the projection. The hypothesis gives

$$a_i \otimes \lambda_i = (\text{id}_A \otimes \pi')\Delta(b_{i0}) = \sum_{\ell=0}^n b_{i\ell} \otimes \pi'(a_\ell).$$

It follows that

$$\lambda_i = \pi'(a_0), \quad a_\ell \in I' \quad (1 \leq \ell \leq n),$$

and hence e_0 is semi-invariant under H' . By Lemma 11.16.0, W is stable under H' . Since W generates I , the ideal I is also stable under H' , and therefore

$$\Delta(I) \subset I \otimes A + A \otimes I'.$$

Finally,

$$I \subset \text{Ker } \varepsilon, \quad (\varepsilon \otimes \text{id}_A) \circ \Delta = \text{id}_A,$$

so $I \subset I'$, whence $H' \subset H$. This completes the proof.

Lemma 11.17.0.¹³⁰ *Let k be an algebraically closed field, let G be a reduced affine algebraic k -group, let V be a finite-dimensional G -module over k , and let $v \in V$. Let E be*

[‡] Source-reading note: the French re-edition says at this point “let us show that $H' = H$,” as does the Reinhold comparison lineage. The maximality assertion requires $H' \subset H$, and that is exactly the conclusion proved below.

¹³⁰Editor’s note: we have added this lemma, drawn from the proof of Theorem 5.6 of [DG70], Section III.3. By an abuse of notation, the same letter E denotes a k -vector space and the k -module scheme $W(E) = \text{Spec } S(E^*)$.

the vector subspace of V spanned by the vectors gv , for $g \in G(k)$. Then E is the smallest G -submodule of V containing v . Consequently, the morphism

$$G \times E \longrightarrow V, \quad (g, x) \longmapsto gx,$$

factors through E .

Proof. By 11.8, there is a smallest G -submodule U of V containing v . Indeed, if

$$\mu : V \longrightarrow V \otimes \mathcal{A}(G)$$

denotes the comodule structure and

$$\mu(v) = \sum_{i=1}^n v_i \otimes a_i$$

with the a_i linearly independent, then

$$U = \text{Span}_k(v_1, \dots, v_n).$$

It is clear that U contains E , and that the morphism $G \times U \rightarrow V$ factors through U .

Conversely, the inverse image of E under the morphism

$$\mu_v : G \longrightarrow V, \quad g \longmapsto gv,$$

is a closed subset of G containing all the rational points. Those points are dense in G , since G is of finite type over k (cf. EGA IV₃, 10.4.8). Hence

$$\mu_v^{-1}(E) = G.$$

Since G is reduced, μ_v therefore factors through E , whence $E = U$.

Theorem 11.17 (Chevalley).— *Let k be a field, let G be an affine k -group (not necessarily of finite type), and let N be a closed group subscheme of G , invariant in G . Then the (fpqc) sheaf quotient G/N is representable by an affine k -group.*¹³¹

Proof. Suppose first that G is of finite type. By VI_A, 3.2 and 5.2, the (fpqc) sheaf quotient G/N is representable by a k -group Q ; it remains to prove that Q is affine. We proceed in several steps, first supposing that k is algebraically closed.¹³²

(a) Suppose, in addition, that G is reduced and connected, and that N is reduced. By 11.16, there are a finite-dimensional G -module V over k and a line $D = ke_0$ such that

$$\text{Norm}_G(D) = N.$$

In particular, N acts on D through a character

$$\chi : N \longrightarrow \mathbb{G}_{m,k}.$$

Fix $h \in N(k)$. For every $g \in G(k)$, one has

$$hge_0 = g(g^{-1}hg)e_0 = \chi(g^{-1}hg)ge_0.$$

¹³¹Editor's note: for another proof of this theorem that does not use the results of Exposé VI_A, see [Ta72], Theorem 5.2 (see also Remark 11.18.5).

¹³²Editor's note: in what follows, we have expanded the original and corrected the erroneous assertion $(G/N)_{\text{red}} = G_{\text{red}}/N_{\text{red}}$, drawing on [DG70], Section III.3, 5.6.

Thus $\chi(g^{-1}hg)$ is an eigenvalue of h . The continuous map

$$\varphi : G(k) \longrightarrow k, \quad g \longmapsto \chi(g^{-1}hg),$$

therefore takes only finitely many values. Since $G(k)$ is irreducible (it is dense in G), it follows that

$$\varphi(g) = \varphi(e) = \chi(h) \quad \text{for every } g \in G(k).$$

Consequently,

$$\chi(g^{-1}hg) = \chi(h) \quad \text{for all } g \in G(k), h \in N(k).$$

Let E be the vector subspace of V spanned by the vectors ge_0 , for $g \in G(k)$. By Lemma 11.17.0, this is the G -submodule of V generated by e_0 .

By the preceding identity, the two morphisms

$$N \times E \longrightarrow E, \quad (h, x) \longmapsto hx \quad \text{and} \quad (h, x) \longmapsto \chi(h)x,$$

agree on

$$(N \times E)(k) = N(k) \times E(k),$$

which is dense in $N \times E$. Since $N \times E$ is reduced (and E is separated), the two morphisms are equal. Thus N acts on E by scalar multiplications.

It follows that N is contained in the kernel K of the morphism

$$\rho : G \longrightarrow \mathrm{GL}(\mathrm{End}_k(E))$$

defined, for every k -algebra R , by

$$\rho(g)(u) = gug^{-1} \quad (g \in G(R), u \in \mathrm{End}_R(E \otimes_k R)).$$

Conversely, if $g \in K(R)$, then

$$g(Re_0) = Re_0,$$

and hence $g \in N(R)$. Thus $N = K$. By VI_A, 5.4.1, the morphism

$$G/N \longrightarrow \mathrm{GL}(\mathrm{End}_k(E))$$

is a closed immersion. Therefore G/N is affine.

(b) Suppose now that G and N are reduced, without assuming that G is connected. Put

$$N' = N \cap G^0.$$

Then G^0/N' is affine by (a). Moreover, NG^0 is an invariant subgroup of G , and G/NG^0 , being a quotient of the finite constant group G/G^0 (cf. VI_A, 5.5.1), is itself a finite constant group. The scheme G/N is therefore the disjoint union of the fibers of

$$G/N \longrightarrow G/NG^0.$$

By VI_A, 5.3.3, all these fibers are isomorphic to NG^0/N , hence to G^0/N' . Thus G/N is affine.

(c) Suppose that G is reduced, with N arbitrary. The morphism

$$G \times N \longrightarrow N, \quad (g, h) \longmapsto ghg^{-1},$$

induces a morphism

$$(G \times N)_{\mathrm{red}} \longrightarrow N_{\mathrm{red}}.$$

Since G is reduced and k is algebraically closed,

$$(G \times N)_{\text{red}} = G \times N_{\text{red}},$$

so N_{red} is an invariant subgroup of G . (This fails when G is not reduced; see VI_A, 0.2.)

Hence, by (b),*

$$G' = G/N_{\text{red}}$$

is affine. By VI_A, 5.6.1,

$$N' = N/N_{\text{red}}$$

is a finite k -group. Hence, by Theorem 4.1 of Exposé V,

$$G/N = G'/N'$$

is affine.

(d) For arbitrary G and N , base change of the equivalence relation

$$G \times N \rightrightarrows G$$

along $G_{\text{red}} \rightarrow G$ gives

$$G_{\text{red}} \times N' \rightrightarrows G_{\text{red}}, \quad N' = N \cap G_{\text{red}}.$$

Since the underlying spaces are the same (and the quotient is the quotient ringed space), the morphism

$$G_{\text{red}}/N' \longrightarrow G/N$$

is a homeomorphism. The quotient G_{red}/N' is reduced, because

$$p : G_{\text{red}} \longrightarrow G_{\text{red}}/N'$$

is faithfully flat. It follows that

$$(G/N)_{\text{red}} \simeq G_{\text{red}}/N'.$$

The scheme on the right is affine by (c). Since G/N is of finite type over k (cf. VI_A, 3.3.2), EGA I, 5.1.10 implies that G/N is affine.

Finally, let k be arbitrary and let $\bar{k}$ be an algebraic closure. By 9.2(v), one has

$$(G \otimes_k \bar{k})/(N \otimes_k \bar{k}) \simeq (G/N) \otimes_k \bar{k}.$$

The first scheme is affine, hence so is the second. Therefore G/N is affine by (fpqc) descent (cf. EGA IV₂, 2.7.1). This proves Theorem 11.17 when G is of finite type. To pass to the general case, we need the following lemma.¹³³

Lemma 11.17.1.— *Let $(C_i \rightarrow A_i)_{i \in I}$ be a filtered direct system of ring homomorphisms, all faithfully flat. Then*

$$A = \varinjlim_i A_i$$

is faithfully flat over

$$C = \varinjlim_i C_i.$$

* Source-reading note: the French re-edition prints “by (a)” here, and the Reinhold comparison lineage repeats that reference. At this stage G need not be connected; the applicable preceding case is (b).

¹³³Editor’s note: we have added this lemma, taken from [DG70], Section III.3, 7.1.

Proof. By [BAC], Section I.3, Proposition 9, a ring homomorphism $B \rightarrow B'$ is faithfully flat if and only if it is injective and B'/B is a flat B -module. Since each $C_i \rightarrow A_i$ is faithfully flat, there are exact sequences

$$0 \longrightarrow C_i \longrightarrow A_i \longrightarrow A_i/C_i \longrightarrow 0,$$

and A_i/C_i is a flat C_i -module. Thus

$$(A_i/C_i) \otimes_{C_i} C$$

is a flat C -module. Since direct limits are exact and commute with tensor products, we obtain an exact sequence

$$0 \longrightarrow C \longrightarrow A \longrightarrow A/C \longrightarrow 0$$

and an isomorphism

$$\varinjlim_i ((A_i/C_i) \otimes_{C_i} C) = (A/C) \otimes_C C = A/C.$$

It follows that A/C is a flat C -module. Hence A is faithfully flat over C .

We now return to the proof of Theorem 11.17 in the general case, where G is not assumed to be of finite type. Put

$$\mathcal{A}(G) = A, \quad \mathcal{A}(N) = B = A/J.$$

By 11.13, A is the direct limit of an increasing filtered family $(A_i)_{i \in I}$ of finitely generated sub-Hopf algebras. Thus

$$G = \varprojlim_i G_i, \quad G_i = \text{Spec}(A_i),$$

where the G_i are affine algebraic k -groups. Let Δ and τ denote, respectively, the comultiplication and antipode of A , and put

$$\Delta^2 = (\Delta \otimes \text{id}_A) \circ \Delta.$$

For each i ,

$$B_i = A_i/(J \cap A_i)$$

is a quotient Hopf algebra of A_i , so

$$N_i = \text{Spec}(B_i)$$

is a closed subgroup of G_i . Since N is invariant in G , the morphism

$$G \times N \longrightarrow G, \quad (g, n) \longmapsto gng^{-1},$$

factors through N . Equivalently, the pair (A, J) satisfies

$$(m_{13} \circ (\Delta \otimes \tau) \circ \Delta)(J) \subset A \otimes_k J,$$

where

$$m_{13}(a_1 \otimes a_2 \otimes a_3) = a_1 a_3 \otimes a_2.$$

It follows that $(A_i, A_i \cap J)$ satisfies the analogous property, so N_i is invariant in G_i . Moreover,

$$\varinjlim_i B_i = B, \quad \varprojlim_i N_i = N.$$

By the finite-type case already proved, each (fpqc) sheaf quotient G_i/N_i is represented by an affine k -group

$$Q_i = \operatorname{Spec}(C_i).$$

Put

$$C = \varinjlim_i C_i.$$

The Q_i form a filtered projective system of affine k -groups, and its projective limit

$$Q = \varprojlim_i Q_i$$

is the k -group $\operatorname{Spec}(C)$ (cf. EGA IV₃, 8.2.3). We have an exact sequence of k -groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow Q.$$

We claim that Q represents the (fpqc) sheaf quotient of G by N . It is enough to verify that

$$G \longrightarrow Q$$

is a covering for the (fpqc) topology (cf. IV, 3.3.2.1 and 5.1.7.1). Every morphism

$$G_i \longrightarrow Q_i$$

is faithfully flat (cf. 9.2(xi)); equivalently, A_i is faithfully flat over C_i . Since

$$A = \varinjlim_i A_i, \quad C = \varinjlim_i C_i,$$

Lemma 11.17.1 shows that A is faithfully flat over C . Thus $G \rightarrow Q$ is faithfully flat. It is also affine, hence quasi-compact, and is therefore an (fpqc) covering. This completes the proof of Theorem 11.17.

11.18. Complements.—¹³⁴ Theorem 11.17 and its proof also give the following results, taken from [DG70], Section III.3.7. Let k be a field. We begin with a lemma (cf. [An73], 2.3.3.2) that will be used later (see 12.10).

Lemma 11.18.1.— *Let $u : G \rightarrow G'$ be a morphism of k -groups and put $N = \operatorname{Ker}(u)$. Suppose that G' is affine and that G is of finite type.*

- (i) *The morphism $\tau : G/N \rightarrow G'$ is a closed immersion. In particular, G/N is affine.*
- (ii) *If, moreover, the morphism $u^\sharp : \mathcal{O}(G') \rightarrow \mathcal{O}(G)$ is injective, then τ is an isomorphism. Thus G' is of finite type and u is faithfully flat.*

Indeed, by 11.13, G' is the projective limit of a filtered system of affine algebraic k -groups G'_i . Let

$$u_i : G \longrightarrow G' \longrightarrow G'_i$$

be the composite, and let N_i be its kernel. The N_i form a filtered decreasing system of closed subgroups of G , whose intersection is N . Since G is Noetherian, there is an index i such that

$$N = N_i.$$

¹³⁴Editor's note: we have added this subsection.

The groups G and G'_i are of finite type. Therefore, by VI_A, 3.2 and 5.4.1, the quotient G/N is a k -group of finite type, and u_i is the composite of the faithfully flat projection

$$p : G \longrightarrow G/N$$

and a closed immersion

$$\tau_i : G/N \hookrightarrow G'_i.$$

Consider the commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{u} & G' \\ p \downarrow & \nearrow \tau & \downarrow q_i \\ G/N & \xrightarrow{\tau_i} & G'_i \end{array}$$

Since

$$q_i \circ \tau = \tau_i$$

is a closed immersion and q_i is separated (G' is separated; cf. VI_A, 0.3), τ is a closed immersion (cf. EGA I, 5.4.4). Hence G/N is affine and

$$\tau^\sharp : \mathcal{O}(G') \longrightarrow \mathcal{O}(G/N)$$

is surjective, proving (i).

If $u^\sharp : \mathcal{O}(G') \rightarrow \mathcal{O}(G)$ is also injective, then so is $\tau^\sharp$. It follows that $\tau^\sharp$ is an isomorphism, and hence so is τ , since G' and G/N are affine. This proves (ii).

Theorem 11.18.2.— *Let G and G' be affine k -groups, with algebras A and A' , and let $u : G \rightarrow G'$ be a morphism of k -groups. Put $N = \text{Ker}(u)$, and let $\varphi : A' \rightarrow A$ be the morphism induced by u .*

(i) *If φ is injective, then u is faithfully flat and identifies G' with G/N .*

(ii) *One has $G/N = \text{Spec}(B)$, where $B = \varphi(A')$. Thus u is the composite of the faithfully flat morphism $G \rightarrow G/N$, corresponding to the inclusion $B \hookrightarrow A$, and the closed immersion $G/N \hookrightarrow G'$, corresponding to the surjection $A' \rightarrow B$. Moreover, N is defined in G by the ideal AB_+ , where B_+ denotes the augmentation ideal of B .*

(iii) *In particular, if u is a monomorphism, it is a closed immersion.*

Proof. (i) Suppose that φ is injective and identify A' with a sub-Hopf algebra of A . By 11.13, A is the filtered union of finitely generated sub-Hopf algebras

$$A_i = \mathcal{O}(G_i).$$

Put

$$A'_i = A' \cap A_i, \quad G'_i = \text{Spec}(A'_i),$$

and let N_i be the kernel of the morphism

$$G_i \longrightarrow G'_i$$

induced by $A'_i \hookrightarrow A_i$. By Lemma 11.18.1,

$$G_i/N_i \simeq G'_i.$$

Thus, for each i , there is an exact sequence

$$1 \longrightarrow N_i \longrightarrow G_i \xrightarrow{p_i} G'_i \longrightarrow 1,$$

where p_i is faithfully flat. Since

$$G = \varprojlim_i G_i, \quad G' = \varprojlim_i G'_i,$$

we obtain an exact sequence [†]

$$1 \longrightarrow N \longrightarrow G \xrightarrow{p} G',$$

where $N = \varprojlim_i N_i$. By Lemma 11.17.1, p is faithfully flat (and affine). Therefore G' represents the (fpqc) sheaf quotient of G by N , proving (i).

In general, $B = \varphi(A')$ is a sub-Hopf algebra of A . Let

$$H = \operatorname{Spec}(B),$$

and let N' be the kernel of the morphism $G \rightarrow H$ induced by $B \hookrightarrow A$. By (i), H is identified with G/N' , and u is the composite of the projection $G \rightarrow G/N'$ and the closed immersion

$$G/N' \hookrightarrow G'$$

induced by the surjection $A' \rightarrow B$. Hence $N' = N$. Moreover, by 9.2(ii), the square

$$\begin{array}{ccc} N & \longrightarrow & G \\ \downarrow & & \downarrow p \\ \operatorname{Spec}(k) & \xrightarrow{\varepsilon} & G' \end{array}$$

is cartesian. Here ε is the unit section of G' , corresponding to the augmentation $B \rightarrow k$. It follows that N is defined in G by the ideal AB_+ . This proves (ii), and (iii) follows.

Remark 11.18.3.— Let G be an affine k -group and N an invariant k -subgroup. Since

$$p : G \longrightarrow G/N$$

is faithfully flat and quasi-compact, $\mathcal{O}(G/N)$ is, by IV, 3.3.3.2, the subalgebra of $\mathcal{O}(G)$ formed by the functions ϕ that are right N -invariant; that is,

$$\phi(gh) = \phi(g)$$

for every k -scheme S , every $g \in G(S)$, and every $h \in N(S)$. Let J be the ideal of

$$A = \mathcal{O}(G)$$

defining N . The preceding condition is equivalent to

$$\Delta(\phi) - \phi \otimes 1 \in \mathcal{O}(G) \otimes J,$$

where Δ is the comultiplication of A .

The preceding theorem may therefore be reformulated in terms of Hopf algebras as follows.

Corollary 11.18.4.— *Let k be a field, let A be a commutative k -Hopf algebra, and put $G = \operatorname{Spec}(A)$.*

[†] Source-reading note: the French re-edition prints G/N as the final term of this limiting exact sequence, and the Reinhold comparison lineage repeats it. At this point the argument is proving that G' represents G/N ; taking the projective limits of the preceding sequences gives G' as the final term.

- (i) If B is a sub-Hopf algebra of A , then A is faithfully flat over B .
- (ii) The map

$$N \longmapsto \mathcal{O}(G/N)$$

is a bijection from the set of invariant subgroups of G to the set of sub-Hopf algebras of A . Its inverse is

$$B \longmapsto \operatorname{Spec}(A/AB_+).$$

Moreover, if J is the ideal of A defining N , then

$$\mathcal{O}(G/N) = \{x \in A \mid \Delta(x) - x \otimes 1 \in A \otimes J\}.$$

Remarks 11.18.5.—

- (a) One consequence of the preceding theorem is that the category of commutative affine k -groups is abelian. For this, as well as for other results on affine k -groups, we refer to [DG70], § III.3, 7.4–7.8.
- (b) Finally, let us point out that M. Takeuchi gave another proof of the results of 11.17–11.18.4; see [Ta72], § 5. He moreover strengthened 11.18.4(i) above by showing that A is even a projective B -module; see [Ta79], Theorem 5 (see also [MW94], Theorem 3.6).

12. Complements on G_{af} and the “anti-affine” groups

¹³⁵ We begin with the following lemma, which extends 11.18.1 to the case where G is not assumed to be of finite type.¹³⁶

Lemma 12.1.— *Let k be a field and let $u : G \rightarrow H$ be a monomorphism of k -groups, with H affine. Suppose that u is quasi-compact. Then u is a closed immersion.*

By (fpqc) descent, we may suppose that k is algebraically closed. By VI_A, 6.4, the closed image I of u is a closed group subscheme of H , hence is still affine. Replacing H by I , we may therefore suppose that u is schematically dominant. Since k is algebraically closed, $H' = H_{\text{red}}$ is a group subscheme of H . Put $G' = G \times_H H'$. The morphism

$$u' : G' \longrightarrow H'$$

deduced from u by base change is a quasi-compact monomorphism and is dominant (the underlying continuous map being the same for u and u'). Hence, by VI_A, 6.2, u' is faithfully flat. It is therefore a quasi-compact faithfully flat monomorphism, and thus an isomorphism (cf. IV, 1.14).

Consequently $u : G \rightarrow H$ is a homeomorphism, and hence is affine by 2.9.1. Thus G is affine, and u is a closed immersion by 11.18.2(iii).

Theorem 12.2.— *Let G be an algebraic k -group. Denote by $\rho : G \rightarrow G_{\text{af}}$ the canonical morphism and by N its kernel.*

- (i) *The canonical morphism $G/N \rightarrow G_{\text{af}}$ is an isomorphism. Consequently G_{af} is an affine algebraic group and ρ is faithfully flat.*
- (ii) *There is a canonical isomorphism $(G/N)_{\text{af}} = G_{\text{af}}$.*

¹³⁵Editor’s note: this section has been added.

¹³⁶Editor’s note: this lemma was communicated to us by M. Raynaud; it will be used in the proof of Proposition 12.9.

- (iii) N is a characteristic subgroup of G .
- (iv) $\mathcal{O}(N) = k$.
- (v) N is smooth, connected, and commutative.

Assertion (i) is a special case of 11.18.1, and (ii) follows from the universal properties of G_{af} and $(G/N)_{\text{af}}$.

Let us prove (iii). For an arbitrary k -scheme $\pi : S \rightarrow \text{Spec } k$, consider the cartesian square

$$\begin{array}{ccc} G_S & \longrightarrow & G \\ q \downarrow & & \downarrow p \\ S & \xrightarrow{\pi} & \text{Spec } k \end{array}$$

Since p is quasi-compact and separated and π is flat, EGA III, 1.4.15 and EGA IV₁, 1.7.21 give

$$q_*(\mathcal{O}_{G_S}) = \pi^*(\mathcal{O}(G)) = \pi^*(\mathcal{O}(G_{\text{af}})).$$

Hence EGA II, 1.5.2 gives

$$(G_S)_{\text{af}} = (G_{\text{af}})_S.$$

Thus N_S , being the kernel of the canonical morphism $G_S \rightarrow (G_S)_{\text{af}}$, is invariant under every automorphism of G_S ; in other words, N is a characteristic subgroup of G .

To prove (iv), put

$$N' = \text{Ker}(N \rightarrow N_{\text{af}}).$$

By (ii), this is an invariant subgroup of G . Since N is algebraic (being a closed subgroup of G), (i) gives $N/N' \simeq N_{\text{af}}$. Moreover, by VI_A, 3.2 and 5.3.2, there is an isomorphism of k -groups

$$(G/N')/N_{\text{af}} \simeq (G/N')/(N/N') \simeq G/N.$$

Since N_{af} is affine, the projection $G/N' \rightarrow G/N$ is affine by 9.2(vii); and since $G/N = G_{\text{af}}$ is affine, G/N' is affine as well. The universal property of G_{af} therefore makes the projection

$$p' : G \rightarrow G/N'$$

factor through $G_{\text{af}} = G/N$. Hence $N \subset N'$, and consequently $N = N'$. Thus N_{af} is the trivial group, and $\mathcal{O}(N) = k$.

Finally, (v) follows from the next lemma.

Lemma 12.3.— *Let k be a field and let N be an algebraic k -group such that $\mathcal{O}(N) = k$. Then N is smooth, connected, and commutative.*

Indeed, we may suppose that k is algebraically closed. Then

$$H = N_{\text{red}}^0$$

is a k -subgroup of N , and the quotient k -scheme $X = N/H$ is finite (hence affine) over k , by VI_A, 5.5.1 and 5.6.1. Since $p : N \rightarrow X$ is faithfully flat,

$$\mathcal{O}(X) \subset \mathcal{O}(N) = k.$$

It follows that $N = N_{\text{red}}^0$, and hence that N is smooth (VI_A, 1.3.1) and connected.

Let Z be the center of N . By 6.2.6, N/Z is affine, and the preceding argument gives $\mathcal{O}(N/Z) = k$. Hence $N = Z$. This proves the lemma and completes the proof of 12.2.

We also record, without proof, the following theorem. Recall that an *abelian variety* over a field k is a proper, smooth, connected k -group scheme.

Theorem 12.4 (Chevalley).— *Let k be a perfect field and let G be a smooth, connected algebraic k -group. Then there exists a smooth, connected affine k -subgroup L , invariant in G , such that G/L is an abelian variety. Moreover, L is unique and its formation commutes with extension of the base field.*

Remarks 12.5.—

- (1) This theorem was announced in 1953 by C. Chevalley, who published his proof in 1960 ([Ch60]). Meanwhile, other proofs were obtained independently by I. Barsotti and M. Rosenlicht ([Ba55, Ro56]); see [Se99] for historical comments.
- (2) A modern version (that is, in the language of schemes) of Chevalley’s proof was given by B. Conrad ([Co02]). (In *loc. cit.*, “algebraic group” means a smooth connected k -group scheme.)
- (3) A modern version of Rosenlicht’s proof, on the other hand, was given by Ngô B.-C. in a course at Orsay in 2005–2006.
- (4) If the hypothesis that k be perfect is omitted, there still exists a smallest invariant connected affine subgroup L , not necessarily smooth, such that G/L is an abelian variety ([BLR], § 9.2, Theorem 1).
- (5) One may also omit the hypothesis that G be smooth over k . Indeed, by VII_A, 8.3, there is an integer $n \geq 1$ such that the quotient

$$G' = G/(\mathrm{Fr}^n G)$$

is smooth. Then G' contains a subgroup L' as in (4) above, and the inverse image of L' in G again has the same properties. Thus, for every connected algebraic group G over a field k , there is an exact sequence

$$1 \longrightarrow H \longrightarrow G \longrightarrow A \longrightarrow 1,$$

where H is an affine k -group and A is an abelian k -variety. Moreover, by [Per76], Corollary 4.2.9, such an exact sequence exists for every connected k -group G , not necessarily algebraic.

- (6) Let k be an algebraically closed field and let G be the semidirect product of an elliptic curve E by the constant k -group $\{\pm 1\}_k$, for the action defined by $(-1) \cdot x = -x$. In this case, if L is a closed invariant subgroup of G such that G/L is connected, then $L = G$.

Remark 12.6.— Following [Br09], a k -group N will be called *anti-affine* if $\mathcal{O}(N) = k$. By 12.3 and 12.4, if k is perfect, every anti-affine algebraic k -group is an extension of an abelian variety by a smooth, connected, commutative affine algebraic k -group. For the precise structure of anti-affine algebraic groups over a perfect field, and various consequences, see the recent articles of M. Brion and C. & F. Sancho de Salas ([Br09, SS09]).

To conclude this section, we shall prove two results due to M. Raynaud, the first being Remark 11.11.1, and the second Proposition 2.1 of Exposé XVII, Appendix III. We shall need the following lemma,¹³⁷ which improves, for a complete discrete valuation ring R , the flatness criteria given in [BAC], § III.5.

¹³⁷Editor’s note: this is a version improved by O. Gabber of a statement communicated by M. Raynaud.

Lemma 12.7.— *Let R be a discrete valuation ring, K its fraction field, and π a uniformizer. Let A be a flat R -algebra and let M be an A -module flat over R . Suppose that*

- (i) $M/\pi M$ is flat over $\overline{A} = A/\pi A$,
- (ii) $M \otimes_R K$ is flat over $A \otimes_R K$.

Then M is flat over A .

By the flatness criterion in the nilpotent case (cf. [BAC], III § 5.2, Theorem 1),

$$M/\pi^n M$$

is flat over $A/\pi^n A$ for every $n \in \mathbb{N}^*$. It then follows from [RG71], II, Lemma 1.4.2.1, that M is flat over A . For the reader's convenience, let us briefly indicate the proof. Put

$$s = \pi^n, \quad P = M/sM.$$

Since P is flat over A/sA , and A/sA has projective dimension 1 over A , the spectral sequence of composite functors shows that P has Tor-dimension at most 1 over A . Since M is flat over R , and thus has no π -torsion, M_K/M is the direct limit of the A -modules

$$\pi^{-n}M/M \simeq M/\pi^n M.$$

Consequently M_K/M also has Tor-dimension at most 1. From the exact sequence

$$0 \longrightarrow M \longrightarrow M_K \longrightarrow M_K/M \longrightarrow 0$$

and the hypothesis that M_K is flat over A_K , hence over A , it follows that M is flat.

For completeness, let us also give the following simpler proof, pointed out by O. Gabber. Let I be a finitely generated ideal of A . We must show that

$$u : M \otimes_A I \longrightarrow M$$

is injective. By (ii), $u \otimes_R K$ is injective, so $\text{Ker}(u)$ is an R -module of π -torsion. It therefore suffices to show that the π -torsion submodule of $M \otimes_A I$ is zero. This submodule is a quotient of $\text{Tor}_1^A(M, I/\pi I)$, as is seen by tensoring with M the exact sequence

$$0 \longrightarrow I \xrightarrow{\pi} I \longrightarrow I/\pi I \longrightarrow 0.$$

On the other hand, since M has no π -torsion (being flat over R), one has

$$\text{Tor}_i^A(M, \overline{A}) = 0 \quad (i \geq 1).$$

Consequently, if $P_\bullet$ is a projective resolution of the A -module M , then

$$P_\bullet \otimes_A \overline{A}$$

is a projective resolution of the $\overline{A}$ -module $\overline{M} = M/\pi M$. Thus, for every $\overline{A}$ -module Q ,

$$\text{Tor}_i^A(M, Q) = \text{Tor}_i^{\overline{A}}(\overline{M}, Q),$$

which vanishes for $i \geq 1$, since $\overline{M}$ is flat over $\overline{A}$. Hence $\text{Tor}_1^A(M, I/\pi I) = 0$, proving the lemma.

Remark 12.8.— Let S be a regular locally noetherian scheme of dimension 1, and let X be an S -scheme that is flat, quasi-separated, and quasi-compact over S . Then $\mathcal{A}(X)$ is a

flat $\mathcal{O}_S$ -module. Indeed, we may suppose that S is local. Let s be its closed point, let $i : X_s \hookrightarrow X$ be the inclusion, and let π be a uniformizer of $R = \mathcal{O}(S)$. Since X is flat over S , there is an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\pi} \mathcal{O}_X \longrightarrow i_*(\mathcal{O}_{X_s}) \longrightarrow 0.$$

Taking global sections shows that $\mathcal{A}(X)$ has no π -torsion as an R -module, and hence is flat.¹³⁸ Moreover, the morphism

$$\mathcal{A}((X_{\text{af}})_s) = \mathcal{A}(X)/\pi\mathcal{A}(X) \longrightarrow \mathcal{A}(X_s),$$

induced by $X \rightarrow X_{\text{af}}$, is injective.

We can now prove the following proposition (cf. Remark 11.11.1).

Proposition 12.9.— *Let S be a regular locally noetherian scheme of dimension at most 1, and let G be an S -group scheme that is flat, quasi-separated, and quasi-compact over S . Then the following conditions are equivalent:*

- (i) G is affine over S .
- (ii) G is quasi-affine over S .
- (iii) G acts faithfully on a quasi-affine and flat S -scheme X .
- (iv) G acts linearly and faithfully on a quasi-coherent module $\mathcal{E}$ that is flat over S .
- (v) The morphism $\rho : G \rightarrow G_{\text{af}}$ is a monomorphism.

The implication (i) $\Rightarrow$ (ii) is clear, as is (ii) $\Rightarrow$ (iii) (take $X = G$).

Suppose (iii) holds. Since $\mathcal{A}(G)$ and $\mathcal{A}(X)$ are flat $\mathcal{O}_S$ -modules, the argument of 11.11 shows that G acts faithfully on the right on X_{af} , and hence acts linearly and faithfully on the left on the flat quasi-coherent $\mathcal{O}_S$ -module $\mathcal{A}(X)$.

If (iv) holds, 11.6.1(ii) shows that the monomorphism

$$G \longrightarrow \text{Aut}_{\mathcal{O}_S}(\mathcal{E})$$

factors through G_{af} . Hence $G \rightarrow G_{\text{af}}$ is a monomorphism.

Finally, suppose (v) holds, and let us prove that $\rho : G \rightarrow G_{\text{af}}$ is an isomorphism. Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η . Since the formation of G_{af} commutes with flat base change,

$$(G_{\text{af}})_{\eta} = (G_{\eta})_{\text{af}}.$$

Therefore $G_{\eta} \rightarrow (G_{\eta})_{\text{af}}$ is a monomorphism, hence a closed immersion by 12.1, and consequently an isomorphism because

$$\mathcal{O}((G_{\eta})_{\text{af}}) = \mathcal{O}(G_{\eta}).$$

If $S = \text{Spec}(\kappa(\eta))$, the proof is complete; we may therefore suppose that $\dim S = 1$.

Let s be a closed point of S . We shall show that

$$G_s \longrightarrow (G_{\text{af}})_s$$

¹³⁸Editor's note: this is true more generally when S is regular, locally noetherian, and of dimension at most 2; see [Ray70a], VII 3.2.

is an isomorphism and that ρ is flat at every point of G_s . We may suppose that S is local, with closed point s . The base-change morphism

$$\rho_s : G_s \longrightarrow (G_{\text{af}})_s$$

is a monomorphism, hence a closed immersion by 12.1. Thus the morphism

$$\mathcal{O}((G_{\text{af}})_s) \longrightarrow \mathcal{O}(G_s)$$

induced by ρ_s is surjective. By the preceding remark it is also injective, and hence is an isomorphism. In particular, ρ is surjective.

It follows from Lemma 12.7 that $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat. Since G is quasi-compact over S and G_{af} is separated over S , ρ is also quasi-compact (cf. EGA I, 6.6.4). Therefore ρ is a quasi-compact faithfully flat monomorphism, and is thus an isomorphism. This proves the proposition.

Finally, we prove Proposition 2.1 of Exposé XVII, Appendix III. Taking [Per76], Corollary 4.2.5 into account, we have replaced “of finite type” in the hypotheses by “quasi-compact and quasi-separated”. (If G is assumed to be of finite type, 12.1 may be used in place of *loc. cit.*)

Proposition 12.10.— *Let S be a regular locally noetherian scheme of dimension at most 1, and let G be a flat, quasi-compact, and quasi-separated S -group scheme.*

- (i) *The canonical morphism $\rho : G \rightarrow G_{\text{af}}$ is faithfully flat and quasi-compact. Consequently, G_{af} represents the (fpqc) sheaf quotient of G by $N = \text{Ker}(\rho)$.*
- (ii) *If moreover G is of finite type over S , then ρ is of finite presentation, G_{af} represents the (fppf) sheaf quotient of G by N , and G_{af} is of finite type over S .*
- (iii) *Suppose moreover that G_η is affine for every maximal point η of S . Then N is an étale S -group and is the unit group if G is separated over S .*

Since G_{af} is affine, and therefore separated over S , the morphism ρ is quasi-compact (cf. EGA I, 6.6.4), and its kernel

$$N = \text{Ker}(\rho)$$

is a closed subgroup of G . Replacing S by one of its connected components, we may suppose that S is irreducible, with generic point η .

To prove (i) and (ii), it is enough to show that ρ is faithfully flat. Indeed, Exposé IV, 5.1.7.1 then shows that G_{af} represents the (fpqc) sheaf quotient of G by N . If G is moreover of finite type over S , it is of finite presentation because S is locally noetherian; by 9.2(xiii), ρ is then of finite presentation, as is $G_{\text{af}} \rightarrow S$, and ρ is therefore a covering for the (fppf) topology.

We may thus suppose that

$$S = \text{Spec}(R),$$

where R is a discrete valuation ring if $\dim S = 1$, and is the field $\kappa(\eta)$ if $\dim S = 0$. Let s be the closed point of S . Since formation of G_{af} commutes with flat base change, the canonical morphism $G_\eta \rightarrow (G_\eta)_{\text{af}}$ is identified with

$$\rho_\eta : G_\eta \longrightarrow (G_{\text{af}})_\eta.$$

Since

$$\mathcal{O}((G_{\text{af}})_\eta) = \mathcal{O}((G_\eta)_{\text{af}}) = \mathcal{O}(G_\eta),$$

ρ_η is faithfully flat by [Per76], 4.2.5 (see also the addition VI_A, 6.6). If $\dim S = 1$, the same argument shows that ρ_s is faithfully flat, because the morphism

$$\mathcal{O}((G_{\text{af}})_s) \longrightarrow \mathcal{O}(G_s)$$

is injective by Remark 12.8. Lemma 12.7 now shows that ρ is faithfully flat. This proves (i) and (ii). In particular, N is flat over S .

Suppose now that G_η is affine. Since ρ_η coincides with the canonical morphism

$$G_\eta \longrightarrow (G_\eta)_{\text{af}},$$

its kernel N_η is the unit group. Let us prove that N is étale over S . Since N is flat over S , it remains to show that N_s is étale over $\kappa(s)$ for every point $s \neq \eta$ of S . There is nothing to prove when $S = \text{Spec}(\kappa(\eta))$, so we may suppose that $S = \text{Spec}(R)$, where R is a discrete valuation ring. Let s be the closed point of S , let K be the fraction field of R , and let π be a uniformizer.

Let $x \in N_s$, and let U_x be an affine open neighborhood of x in N . Since N is flat over S , the intersection $U_x \cap N_\eta$ is nonempty, and hence equals $\{\epsilon(\eta)\}$, where ϵ denotes the unit section. Thus

$$A = \mathcal{O}(U_x)$$

is a flat R -algebra such that $A_K = K$ and $\pi^{-1} \notin A$, since π belongs to the maximal ideal of $\mathcal{O}_{U_x, x}$. It follows that $A = R$, and hence the projection $U_x \rightarrow S$ is an isomorphism. Thus N is étale over S . If G is also separated over S , the inverse isomorphism $S \rightarrow U_x$ is the unit section, since the two coincide on the dense open subset $\{\eta\}$ of $S = \text{Spec}(R)$. Hence N is the unit group. This proves the proposition.

In particular, we obtain the following corollary, two other proofs of which may be found in [An73], Proposition 2.3.1, and [PY06], Proposition 3.1.

Corollary 12.10.1.— *Let R be a discrete valuation ring, K its fraction field, and G a separated, flat R -group scheme of finite type over R . If G_K is affine, then G is affine.*

Remarks 12.10.2.—

- (a) O. Gabber has pointed out to us examples in which G is a flat group of finite type over a discrete valuation ring, its generic fiber is an abelian variety, and the kernel N of $G \rightarrow G_{\text{af}}$ is not smooth.
- (b) M. Raynaud, on the other hand, gave an example, with S the affine plane of dimension 2 over a field k , of a smooth quasi-affine S -group scheme with affine, connected fibers that is not affine over S ; see [Ray70a], § VII.3, p. 116.

13. Flat affine groups over a regular base of dimension at most 2

Let us begin by observing¹³⁹ that the familiar argument showing that every affine algebraic group over a field k is linear, as well as Lemma 11.12, extends to a group scheme G that is affine, flat, and of finite type over a Noetherian regular base scheme S of dimension at most 2. For S of dimension at most 1, this proves Remark 11.11.1(b). The extension to $\dim S = 2$ rests on the following lemma, communicated to us by O. Gabber.

Lemma 13.1.— *Let S be a normal Noetherian scheme, let $\mathcal{A}$ be a flat quasi-coherent $\mathcal{O}_S$ -module, let $\mathcal{F}$ be a finite-type $\mathcal{O}_S$ -submodule of $\mathcal{A}$, let $\mathcal{F}^{**}$ be its bidual, and let U be the flatness locus of $\mathcal{F}$, that is, the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module.*

¹³⁹Editor's note: this section has been added.

(i) U is open in S , and

$$\mathcal{F}^{**} = j_* j^*(\mathcal{F}),$$

where $j : U \hookrightarrow S$ denotes the inclusion.*

(ii) For every quasi-coherent $\mathcal{O}_U$ -module $\mathcal{E}$, the canonical morphism

$$j_*(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_*(\mathcal{E} \otimes_{\mathcal{O}_U} j^*(\mathcal{A}))$$

is an isomorphism.

(iii) In particular,

$$\mathcal{V} = j_* j^*(\mathcal{F}) \subset \mathcal{A} = j_* j^*(\mathcal{A}),$$

and the canonical morphism

$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A} \longrightarrow j_* j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A})$$

is an isomorphism.

Proof. Replacing S by one of its connected components, we may suppose that S is integral. By EGA IV₂, 2.1.12, the flatness locus of $\mathcal{F}$, namely the set of points $s \in S$ such that $\mathcal{F}_s$ is a flat $\mathcal{O}_{S,s}$ -module, is an open subset U of S ; write $j : U \hookrightarrow S$ for its inclusion. Since $\mathcal{A}_s$ is flat, hence torsion-free, the same is true of $\mathcal{F}_s$. Thus U contains every point of codimension at most 1. Consequently, by [BAC], VII, § 4.2, Corollary to Theorem 1,

$$\mathcal{F}^{**} = j_* j^*(\mathcal{F}),$$

and hence there is a monomorphism

$$\mathcal{F}^{**} \longrightarrow j_* j^*(\mathcal{A}).$$

The proof of (ii) is analogous to that of EGA III, 1.4.15, recalled in 11.0. Since S is normal, the morphism

$$\mathcal{O}_S \longrightarrow j_* j^*(\mathcal{O}_S)$$

is an isomorphism (cf. EGA IV₂, 5.8.6 and 5.10.5). Applying (ii) to $\mathcal{E} = \mathcal{O}_U$ therefore gives

$$\mathcal{A} = j_* j^*(\mathcal{A}).$$

Finally, since

$$j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}) = j^*(\mathcal{F}) \otimes_{\mathcal{O}_U} j^*(\mathcal{A}),$$

the last assertion of (iii) follows from (ii) applied to $\mathcal{E} = j^*(\mathcal{F})$. This proves the lemma.

Recall also that a finitely generated R -module M is called *reflexive* if the canonical morphism from M to its bidual M^{**} is an isomorphism. If R is a Noetherian regular ring of dimension at most 2, this implies that M is projective. Indeed, for every finitely generated R -module N , choose a resolution

$$L_1 \longrightarrow L_0 \longrightarrow N \longrightarrow 0,$$

where L_0 and L_1 are finite free R -modules. There is then an exact sequence

$$0 \longrightarrow N^* \longrightarrow L_0^* \longrightarrow L_1^* \longrightarrow Q \longrightarrow 0,$$

* Source-reading note: the French re-edition prints $j : U \hookrightarrow X$ in the statement. The ambient scheme throughout the lemma is S , and its proof explicitly defines $j : U \hookrightarrow S$; the latter is used here. The Reinhold comparison lineage retains the printed X .

where Q is the cokernel of $L_0^* \rightarrow L_1^*$. Since R has homological dimension at most 2 ([BAC], X, § 4.2, Corollary 1 to Theorem 1), it follows that N^* is projective.

Proposition 13.2.— *Let S be a Noetherian regular scheme of dimension at most 2, let G be an affine flat S -group, and let $\mathcal{A}(G)$ be its affine algebra.*

(i) *If G is of finite type over S , it is isomorphic to a closed subgroup of*

$$H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank. If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d , so that $H = \mathrm{GL}_{d,S}$.

(ii) *$\mathcal{A}(G)$ is the filtered direct limit of flat finite-type $\mathcal{O}_S$ -sub-Hopf algebras.*

Proof. Let $\mathcal{B}$ be a finite-type $\mathcal{O}_S$ -subalgebra of $\mathcal{A}(G)$. Since every coherent module on an open subset of S extends to a coherent module on S (EGA I, 9.4.5), there is a coherent $\mathcal{O}_S$ -submodule $\mathcal{M}$ of $\mathcal{B}$ that generates $\mathcal{B}$ as an $\mathcal{O}_S$ -algebra (*loc. cit.*, 9.6.5). By 11.10.bis, $\mathcal{M}$ is contained in a coherent G -stable $\mathcal{O}_S$ -submodule $\mathcal{F}$. (N.B. Since G is affine over S , the proof of *loc. cit.* is simpler here: one may replace $f_*f^*(\mathcal{E})$ by $\mathcal{E} \otimes_{\mathcal{O}_S} \mathcal{A}(G)$, and so on.)

Let $j : U \hookrightarrow S$ be the inclusion, where U is the flatness locus of $\mathcal{F}$. By Lemma 13.1 and the preceding observation,

$$\mathcal{V} = j_*j^*(\mathcal{F})$$

is a locally free $\mathcal{O}_S$ -submodule of $\mathcal{A}(G)$, and the canonical morphism

$$\mathcal{V} \otimes_{\mathcal{O}_S} \mathcal{A}(G) \longrightarrow j_*j^*(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{A}(G))$$

is an isomorphism. Thus $\mathcal{V}$ is a G -submodule of $\mathcal{A}(G)$. The action of G on $\mathcal{V}$ induces a morphism of affine S -groups

$$\rho_{\mathcal{V}} : G \longrightarrow H = \mathrm{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

and therefore a morphism of $\mathcal{O}_S$ -Hopf algebras

$$\varphi_{\mathcal{V}} : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

Write $\mathcal{A}_{\mathcal{V}}$ for the image of $\varphi_{\mathcal{V}}$. It is the affine algebra of a closed subgroup $G_{\mathcal{V}}$ of H , the closed image of $\rho_{\mathcal{V}}$. We shall show that $\mathcal{A}_{\mathcal{V}}$ contains $\mathcal{B}$.

The question is local on S , so we may suppose that $S = \mathrm{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{V})$$

is a free R -module with basis $v_1, \dots, v_n$. Then $H \simeq \mathrm{GL}_{n,R}$, and $\mathcal{A}(H)$ is generated as an R -algebra by the “matrix coefficients” c_{ij} and by d^{-1} , where d denotes the determinant. Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of $A = \Gamma(S, \mathcal{A}(G))$. For $j = 1, \dots, n$, write

$$\Delta(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}.$$

Then $a_{ij} = \varphi(c_{ij})$ belongs to $\mathrm{Im}(\varphi)$. Since V is an R -submodule of A , the identity

$$(\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A$$

gives

$$v_j = \sum_i \varepsilon(v_i) a_{ij} \in \mathrm{Im}(\varphi).$$

As V contains a set of generators of $B = \Gamma(S, \mathcal{B})$, it follows that

$$B \subset \text{Im}(\varphi).$$

If G is of finite type over S , take $\mathcal{B} = \mathcal{A}(G)$. Then φ is surjective, so the morphism

$$G \longrightarrow H = \text{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

is a closed immersion.

If S is also affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G -module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\text{GL}_{d,S}$.

Finally, return to an arbitrary affine flat S -group G . By EGA I, 9.4.9, $\mathcal{A}(G)$ is the union of its coherent $\mathcal{O}_S$ -submodules $\mathcal{M}$, hence also of the $\mathcal{O}_S$ -sub-Hopf algebras $\mathcal{A}_{\mathcal{V}}$ constructed above. This proves (ii).

Example 13.3.— *Let R be a discrete valuation ring with uniformizer π and fraction field K . Consider the filtered projective system of R -groups*

$$\cdots \longrightarrow \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R} \xrightarrow{\times\pi} \mathbb{G}_{a,R}$$

corresponding to the direct system

$$R[X_0] \longrightarrow R[X_1] \longrightarrow R[X_2] \longrightarrow \cdots,$$

whose transition morphisms are given by $X_n = \pi X_{n+1}$. Its projective limit G is a flat affine R -group scheme, not of finite type, with trivial special fiber and generic fiber $\mathbb{G}_{a,K}$. The affine R -algebra $\mathcal{A}(G)$ is the subring of $K[X]$ consisting of the polynomials whose constant coefficient lies in R . (N.B. We already encountered this example in Remark 11.10.1.) Furthermore, G represents the functor assigning to every R -algebra B the set of sequences $(x_n)_{n \in \mathbb{N}}$ of elements of B such that

$$x_n = \pi x_{n+1} \quad \text{for every } n.$$

(In particular, every x_n is indefinitely π -divisible.)

Now let S be a Noetherian scheme such that every coherent $\mathcal{O}_S$ -module is a quotient of a locally free $\mathcal{O}_S$ -module of finite type. This holds, for example, when S is a separated Noetherian regular scheme; see SGA 6, Exposé II, 2.1.1 and 2.2.7. (One can show that every Noetherian regular scheme of dimension at most 1 also has this property. It does not hold, however, when S is the affine plane over a field k with the origin doubled; see *loc. cit.*, 2.2.7.2.)

Definition 13.4.— *Let G be an affine flat S -group. Following R. W. Thomason ([Th87], 2.1), we say that the pair (G, S) has the equivariant resolution property, or satisfies (RE), if, for every coherent G - $\mathcal{O}_S$ -module $\mathcal{F}$, there are a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of finite rank and a G -equivariant epimorphism*

$$\mathcal{E} \longrightarrow \mathcal{F}.$$

In *loc. cit.*, Theorem 3.1, Thomason proves the result below under the hypothesis that G be essentially free over S (see the remark below). Gabber pointed out to us the simpler proof given here, which does not use that hypothesis.

Proposition 13.5.— *Let S be a Noetherian scheme, and let G be an affine S -group, flat and of finite type, with affine algebra $\mathcal{A}(G)$. Suppose that (G, S) satisfies (RE). Then:*

(i) G is isomorphic to a closed subgroup of

$$H = \operatorname{Aut}_{\mathcal{O}_S}(\mathcal{V})$$

for some locally free $\mathcal{O}_S$ -module $\mathcal{V}$ of finite rank.

(ii) If, moreover, S is affine, one may take $\mathcal{V} = \mathcal{O}_S^{\oplus d}$ for some d , so that $H = \operatorname{GL}_{d,S}$.[†]

The proof is analogous to that of 13.2. As there, there is a coherent G -stable $\mathcal{O}_S$ -submodule $\mathcal{F}$ that generates $\mathcal{A} = \mathcal{A}(G)$ as an $\mathcal{O}_S$ -algebra. Replacing S by one of its connected components, we may suppose that S is connected. By (RE), there are a locally free G - $\mathcal{O}_S$ -module $\mathcal{E}$ of rank n and an epimorphism of $\mathcal{A}$ -comodules

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}.$$

Put $H = \operatorname{Aut}_{\mathcal{O}_S}(\mathcal{E})$. This is an S -group scheme locally isomorphic to $\operatorname{GL}_{n,S}$. The action of G on $\mathcal{E}$ induces a morphism of affine S -groups

$$\rho : G \longrightarrow H,$$

corresponding to a morphism of $\mathcal{O}_S$ -Hopf algebras

$$\varphi : \mathcal{A}(H) \longrightarrow \mathcal{A}(G).$$

We shall show that ρ is a closed immersion.

The question is local on S , so suppose that $S = \operatorname{Spec}(R)$ and that

$$V = \Gamma(S, \mathcal{E})$$

is a free R -module with basis $v_1, \dots, v_n$. Then $H \simeq \operatorname{GL}_{n,R}$, and

$$B = \Gamma(S, \mathcal{A}(H))$$

is generated as an R -algebra by the matrix coefficients c_{ij} and by d^{-1} , where d denotes the determinant. Let Δ (respectively, ε) be the comultiplication (respectively, augmentation) of

$$A = \Gamma(S, \mathcal{A}(G)),$$

let

$$\mu : V \longrightarrow V \otimes_R A$$

be the A -comodule structure on V , and put

$$F = \Gamma(S, \mathcal{F}).$$

For $j = 1, \dots, n$, write

$$\mu(v_j) = \sum_{i=1}^n v_i \otimes a_{ij}.$$

Then $\varphi(c_{ij}) = a_{ij}$. Since $\pi : V \rightarrow F$ is a morphism of A -comodules, one has

$$\sum_{i=1}^n \pi(v_i) \otimes a_{ij} = (\pi \otimes_R \operatorname{id}_A)(\mu(v_j)) = \Delta(\pi(v_j)),$$

[†] Source-reading note: the French statement prints “for some n ” after $\mathcal{O}_S^{\oplus d}$, while its conclusion is $H = \operatorname{GL}_{d,S}$. The same rank is denoted by d here. The Reinhold comparison lineage likewise uses d .

and therefore

$$\begin{aligned}\pi(v_j) &= (\varepsilon \otimes_R \text{id}_A) \Delta(\pi(v_j)) \\ &= \sum_{i=1}^n \varepsilon(\pi(v_i)) a_{ij} \\ &= \varphi \left(\sum_{i=1}^n \varepsilon(\pi(v_i)) c_{ij} \right).\end{aligned}$$

Thus $\varphi(B)$ contains $\pi(V) = F$, which generates A as an R -algebra. Hence φ is surjective, and ρ is a closed immersion. This proves (i).

If S is affine, there is a locally free $\mathcal{O}_S$ -module $\mathcal{V}'$ of finite rank such that

$$\mathcal{V} \oplus \mathcal{V}' = \mathcal{O}_S^{\oplus d}.$$

Regard $\mathcal{V}'$ as a trivial G -module and replace $\mathcal{V}$ by $\mathcal{O}_S^{\oplus d}$. This realizes G as a closed subgroup of $\text{GL}_{d,S}$ and completes the proof.

Remarks 13.6.—

- (a) For completeness, let us briefly sketch Thomason's argument ([Th87], Theorem 3.1), retaining the preceding notation. The G -equivariant epimorphism

$$\pi : \mathcal{E} \longrightarrow \mathcal{F}$$

induces a closed immersion

$$\tau : G \longrightarrow V(\mathcal{E})$$

such that

$$\tau(g'g) = \tau(g') \cdot g.$$

(N.B. G acts on the right on $V(\mathcal{E})$.) There is an isomorphism

$$H \simeq \text{Aut}_{\mathcal{O}_S}(V(\mathcal{E}))^{\text{op}}$$

compatible with the left action of G on $\mathcal{E}$ and its right action on $V(\mathcal{E})$. Let

$$N = \text{Transp}_H^{\text{str}}(\tau(G), \tau(G))$$

be the strict transporter. If G is essentially free over S , it follows from 6.2.4(e) that N is a closed subgroup scheme of H , hence affine over S . Moreover, ρ factors through a morphism of S -groups

$$\rho' : G \longrightarrow N.$$

For every $S' \rightarrow S$ and $h \in N(S')$, put

$$\pi(h) = \tau(1) \cdot h,$$

where 1 is the unit element of $G(S')$. This defines a morphism of S -schemes

$$\pi : N \longrightarrow \tau(G)$$

that is a retraction of ρ' , after identifying G with $\tau(G)$. Since N is separated over S , it follows that ρ' is a closed immersion.

- (b) The proof of [Th87], Theorem 3.1, appears to require the hypothesis that G be essentially free over S , although that hypothesis is not stated there (the author invokes instead the fact that H is essentially free). It is, however, satisfied when G is reductive (cf. Exposé XXII, 5.7.8), and hence in all cases considered in *loc. cit.*, Corollary 3.2. In particular, Thomason proves in *loc. cit.*, 2.5, that if S is separated, Noetherian, and regular of dimension at most 2, if G is affine and of finite presentation, and if $\mathcal{A}(G)$ is a locally projective $\mathcal{O}_S$ -module, then (G, S) satisfies (RE). By 13.5, this gives 13.2 under a slightly more restrictive hypothesis.

To conclude, let us point out that the proof of [Th87], 2.5, can be simplified slightly as follows. (For brevity, we work in the affine situation $S = \text{Spec}(R)$.)

Proposition 13.7.— *Let R be a Noetherian regular ring of dimension at most 2, let C be an R -coalgebra that is projective as an R -module, and let F be a C -comodule of finite type over R . Then F is a quotient of a C -comodule V that is projective and of finite type over R .*

Proof. Replacing $\text{Spec}(R)$ by one of its connected components, we may suppose that R is integral, with fraction field K . Let Δ (respectively, ε) denote the comultiplication (respectively, augmentation) of C , and let

$$\rho : F \longrightarrow F \otimes C$$

be the comodule structure on F . Let

$$\pi : W \longrightarrow F$$

be a surjective morphism, with W a finite free R -module. Give $W \otimes C$ the comodule structure defined by $\text{id}_W \otimes \Delta$, and do the same for $F \otimes C$. Then $\rho : F \rightarrow F \otimes C$ is a morphism of C -comodules with retraction $\text{id}_F \otimes \varepsilon$.

Let W' be the C -comodule defined by the following cartesian square:

$$\begin{array}{ccc} W' & \longrightarrow & F \\ \downarrow & & \downarrow \rho \\ W \otimes C & \xrightarrow{\pi \otimes \text{id}_C} & F \otimes C \end{array}$$

Thus W' identifies with the kernel of the morphism

$$W \otimes C \longrightarrow (F \otimes C)/\rho(F),$$

and the projection

$$\pi' : W' \longrightarrow F, \quad x \longmapsto (\pi \otimes \varepsilon)(x),$$

is surjective. Since F is a finitely generated R -module, there is a C -subcomodule V' of W' , finitely generated over R , such that

$$\pi'(V') = F.$$

Since $W \otimes C$ is R -torsion-free, so is V' . Replacing F by V' , we may therefore suppose from the outset that F is torsion-free.

Apply the preceding construction to this new F , obtaining V' as above. Consider the subcomodule V that is the kernel of the morphism

$$W \otimes C \longrightarrow E = \frac{W \otimes C \otimes K}{V' \otimes K}.$$

Then V contains V' , and

$$Q = (W \otimes C)/V$$

is an R -submodule of the K -vector space E . Put

$$Q' = E/Q.$$

Since $W \otimes C$ and E are flat R -modules, for every R -module N one obtains

$$\mathrm{Tor}_1^R(V, N) \simeq \mathrm{Tor}_2^R(Q, N) \simeq \mathrm{Tor}_3^R(Q', N).$$

Because R is regular of dimension at most 2, the right-hand term vanishes. Hence V is a flat R -module.

It remains to prove that V is finitely generated. Put

$$M = W \otimes C.$$

By construction,

$$V/V' \simeq (M/V')_{\mathrm{tors}},$$

the R -torsion submodule of M/V' . Since M is a projective R -module, there is a projective R -module P such that

$$M \oplus P = L$$

is free. Then

$$(M/V')_{\mathrm{tors}} \simeq (L/V')_{\mathrm{tors}}.$$

Since V' is finitely generated, there is a direct summand

$$L' \simeq R^n$$

of L containing V' . Hence

$$(L/V')_{\mathrm{tors}} \simeq (L'/V')_{\mathrm{tors}},$$

and the latter is finitely generated because L'/V' is. Thus V is a finitely generated flat R -module, hence a finitely generated projective R -module, since R is Noetherian. This proves Proposition 13.7.

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